Obstetrics & Gynecology NURSING
تمريض النساء والولادة

From pregnancy to newborn care

Foundations • antenatal care • labor • birth • postpartum



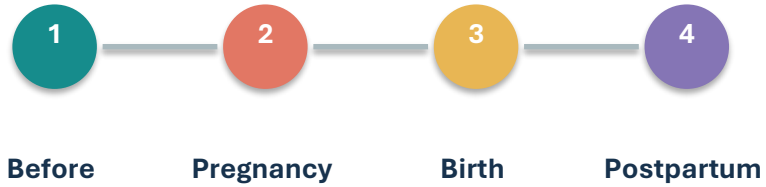
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2026

Course map and learning outcomes

- 22 units connect reproductive science to bedside nursing care.
- Dedicated units cover pregnancy, labor, birth, obstetric and gynecologic disorders, and family planning.
- Newborn transition remains included; local protocols and current evidence govern practice.
- Repeated source material was consolidated, with expanded teaching built from the supplied slides.

CONTINUUM



Timing and response depend on clinical context.

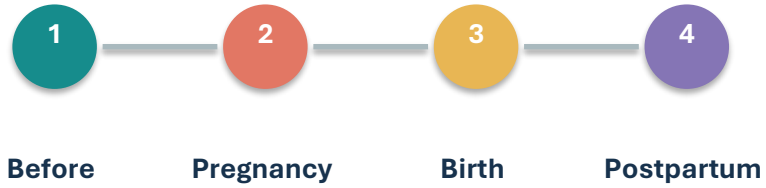


- **Maternity nursing** focuses on the care of childbearing women and their families through all stages of pregnancy childbirth, as well as the first 4 weeks after birth

The maternity nursing continuum

- Care begins before conception and continues through postpartum recovery.
- Nursing care protects health, supports informed choice, and responds to risk.
- Each family brings its own culture, strengths, needs, and goals.

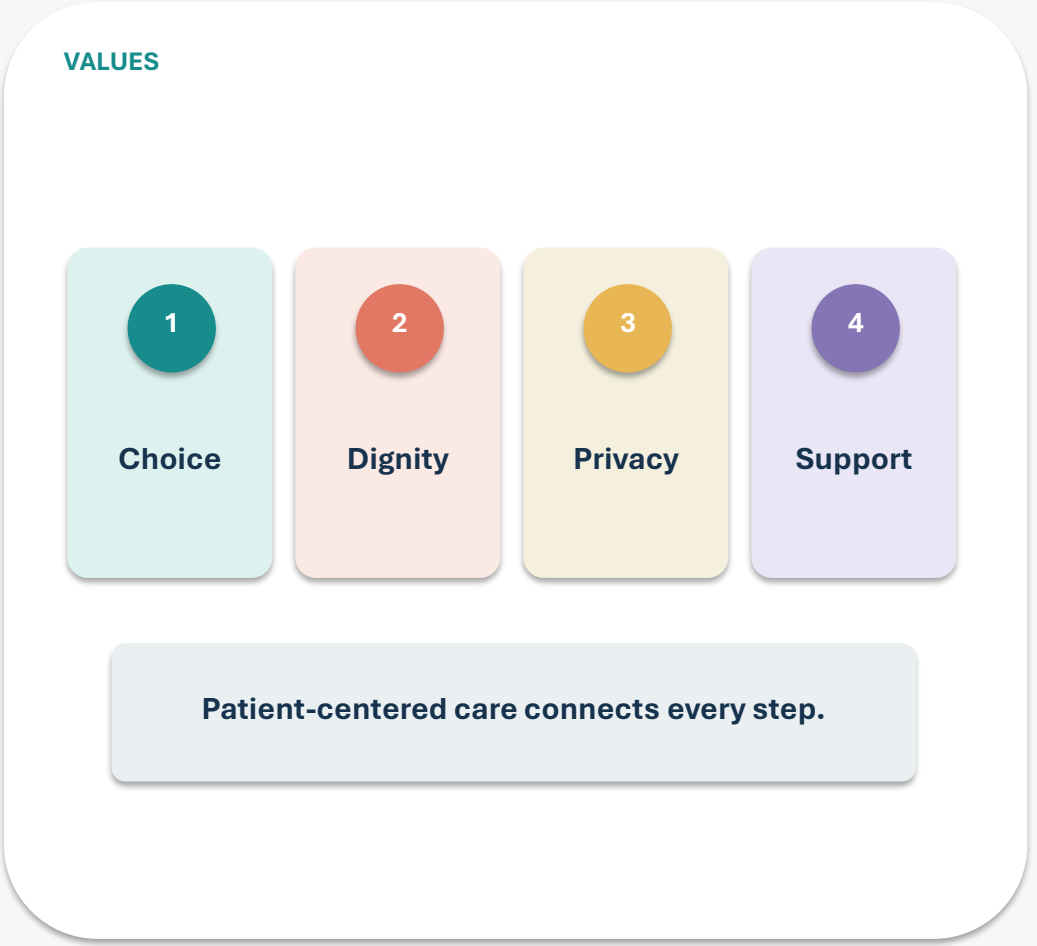
CONTINUUM



Timing and response depend on clinical context.

A family-centered, person-centered approach

- Ask how the patient wants to be addressed and who they want involved.
- Use shared decisions, consent, privacy, and trauma-informed communication.
- Recognize that pregnancy and birth experiences vary across identities and families.



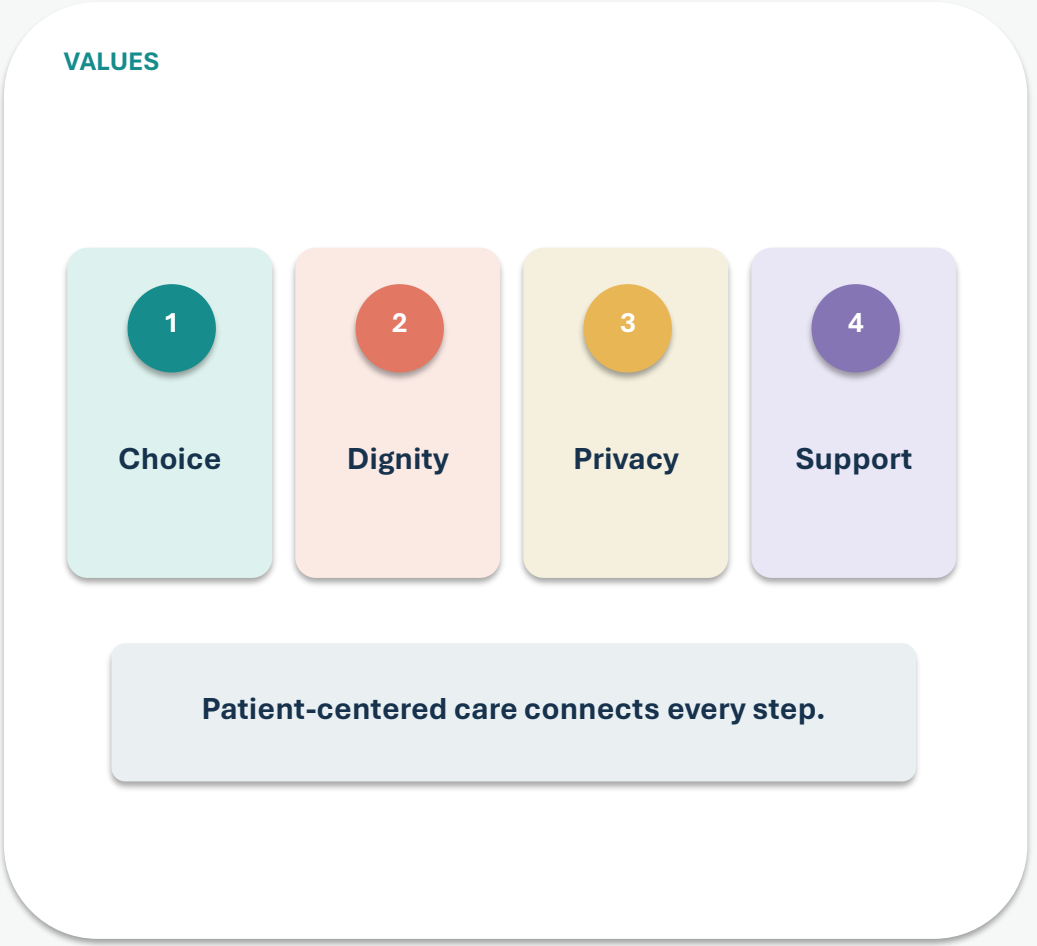
Safety and scope in maternity nursing

- Use current local protocols, emergency pathways, and medication references.
- Escalate deterioration early; do not wait for a complete diagnosis.
- Procedures require training, consent, and appropriate supervision.



Respectful maternity care

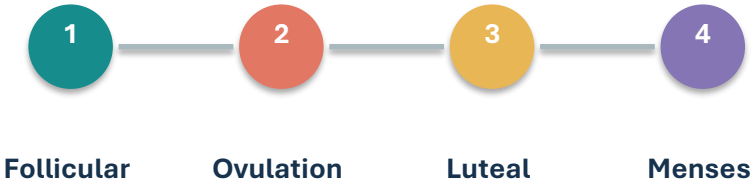
- Explain what you are doing before touching or examining.
- Offer choices when clinically appropriate and honor a patient’s refusal.
- Protect dignity, confidentiality, and continuous support during care.



The nursing process in maternity care

- Assess baseline health, current symptoms, preferences, and support needs.
- Identify priorities with the patient and interprofessional team.
- Evaluate maternal and newborn outcomes and adapt the plan.

CYCLE

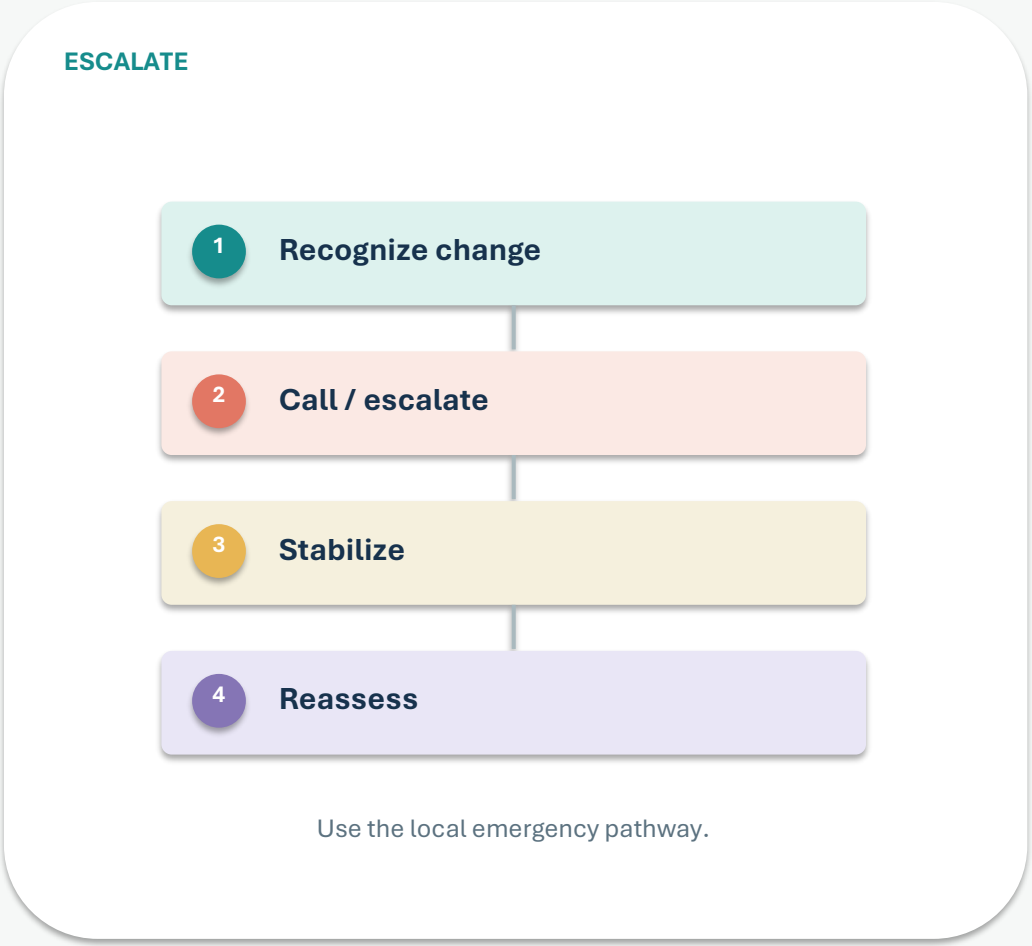


Timing and response depend on clinical context.

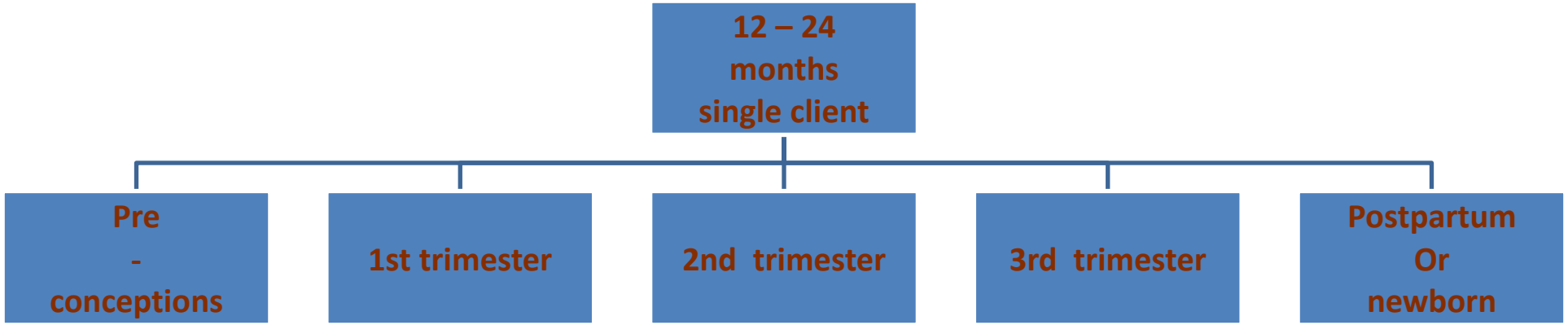
Sources: NANDA/WHO

Clinical reasoning: notice, interpret, respond

- Notice trends in vital signs, bleeding, pain, fetal status, and behavior.
- Interpret findings in context, including gestational age and baseline risk.
- Respond, reassess, and communicate using a structured handoff.



Perinatal continuum care



Reproductive foundations



10 focused lessons • assessment, teaching, and clinical reasoning

Reproductive Anatomy and Physiology

An understanding of the structure and function of the reproductive organs is necessary for the effective nursing care of women during and after their reproductive years and for couples who require family planning or infertility care.

SEXUAL DEVELOPMENT

- Sexual development begins at conception when the genetic sex is determined by the union of an ovum and a sperm.
- During childhood, the sex organs are inactive and then become active during puberty.

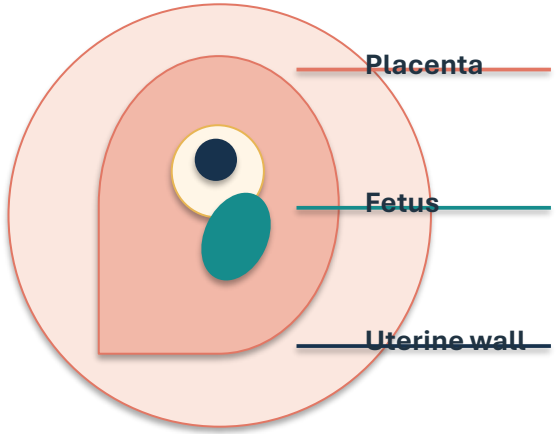
SEXUAL DEVELOPMENT

- The maturing child's hypothalamus gradually increases production of GnRH beginning at age 9 to 12.
- The level of GnRH increases slowly until it reaches a level adequate to stimulate the anterior pituitary to increase its production of (FSH) and (LH).
- The ovaries and testes increase production of sex hormones and begin maturing gametes (reproductive cells, an ovum in females and a sperm in males) in response to higher levels of FSH and LH.
- Sex hormones also induce development of secondary sex characteristics.

Pelvic and reproductive anatomy

- The uterus, cervix, ovaries, tubes, vagina, and pelvic floor support reproduction.
- The uterine cavity is the usual site for implantation and fetal growth.
- Anatomy varies; assess respectfully and avoid assumptions.

ANATOMY



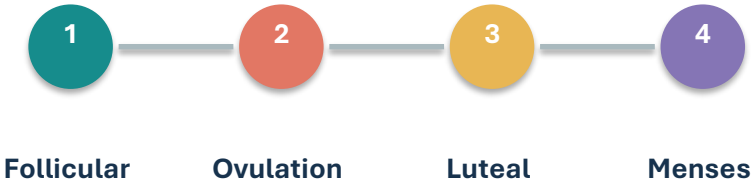
Schematic • not to scale

Sources: OpenStax/ACOG

Ovarian cycle

- FSH supports follicle growth; one dominant follicle usually matures.
- An LH surge triggers ovulation.
- The corpus luteum produces progesterone after ovulation.

CYCLE



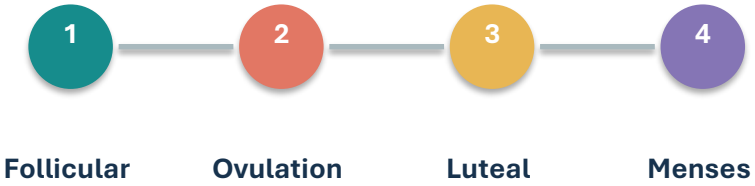
Timing and response depend on clinical context.

Sources: OpenStax

Endometrial cycle

- Estrogen drives endometrial proliferation after menstruation.
- Progesterone supports a secretory, implantation-ready lining.
- Hormone withdrawal leads to endometrial shedding when pregnancy does not occur.

CYCLE



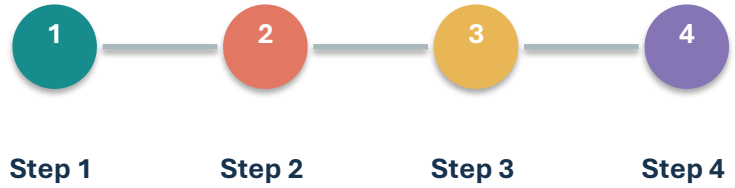
Timing and response depend on clinical context.

Sources: OpenStax

Fertilization and early transport

- Fertilization most often occurs in the ampulla of the uterine tube.
- Cleavage forms a morula and then a blastocyst.
- The blastocyst reaches the uterine cavity before implantation.

SEQUENCE



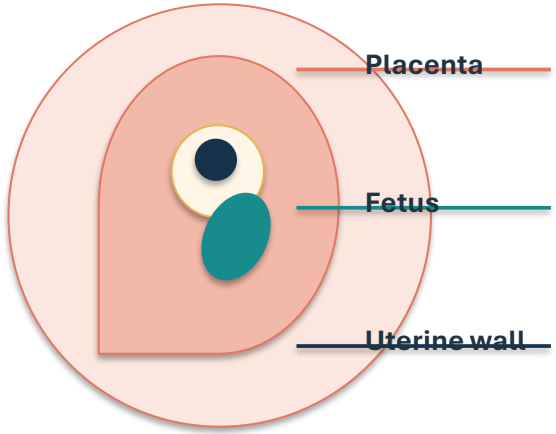
Timing and response depend on clinical context.

Sources: OpenStax

Implantation and early pregnancy signaling

- Implantation begins about 6–10 days after fertilization.
- Trophoblast tissue produces hCG, supporting the corpus luteum initially.
- Pregnancy tests detect hCG in urine or blood.

IMPLANT



Schematic • not to scale

Sources: ACOG

TABLE 3.1 Major Hormones in Reproduction

Produced By	Target Organs	Action in Female	Action in Male
Gonadotropin-Releasing Hormone			
Hypothalamus	Anterior pituitary	Stimulates release of FSH and LH, initiating puberty and sustaining female reproductive cycles; release is pulsatile.	Stimulates release of FSH and LH, initiating puberty; release is pulsatile.
Follicle-Stimulating Hormone			
Anterior pituitary	Ovaries (female) Testes (male)	Stimulates final maturation of follicle. Stimulates growth and maturation of graafian follicles before ovulation.	Stimulates Leydig cells of testes to secrete testosterone.
Luteinizing Hormone			
Anterior pituitary	Ovaries (female) Testes (male)	Stimulates final maturation of follicle. Surge of LH approximately 14 days before next menstrual period causes ovulation. Stimulates transformation of graafian follicle into corpus luteum, which continues secretion of estrogens and progesterone for about 12 days if ovum is not fertilized. If fertilization occurs, placenta gradually assumes this function.	Stimulates Leydig cells of testes to secrete testosterone.

Estrogens			
Ovaries and corpus luteum (female) Placenta (pregnancy) Formed in small quantities from testosterone in Sertoli cells of testes (male); other tissues, especially liver, produce estrogen in male	Internal and external reproductive organs Breasts (female) Testes (male)	Reproductive organs: a. Maturation at puberty b. Stimulation of endometrium before ovulation Breasts: Induce growth of glandular and ductal tissue; initiate deposition of fat at puberty. Stimulate growth of long bones but cause closure of epiphyses, limiting mature height. Pregnancy: Stimulate growth of uterus, breast tissue; inhibit active milk production; relax pelvic ligaments.	Necessary for normal sperm formation.
Progesterone			
Ovary, corpus luteum, placenta	Uterus, female breasts	Stimulates secretion of endometrial glands; causes endometrial vessels to become dilated and tortuous in preparation for possible embryo implantation. Pregnancy: Induces growth of cells of fallopian tubes and uterine lining to nourish embryo; decreases contractions of uterus; prepares breasts for lactation but inhibits prolactin secretion.	Not applicable.

Prolactin			
Anterior pituitary	Female breasts	Stimulates secretion of milk (lactogenesis); estrogen and progesterone from placenta have an inhibiting effect on milk production until after placenta is expelled at birth; suckling of newborn stimulates prolactin secretion to maintain milk production.	Not applicable.
Oxytocin			
Posterior pituitary	Uterus, female breasts	Uterus: Stimulates contractions during birth and stimulates postpartum contractions to compress uterine vessels and control bleeding. Stimulates let-down, or milk-ejection reflex, during breastfeeding.	Not applicable.
Testosterone			
Leydig cells of the testes (male) Adrenal glands (female) Ovaries (female)	Sexual organs (male) Male body conformation after puberty	Small quantities of androgenic (masculinizing) hormones from adrenal glands cause growth of pubic and axillary hair at puberty; most androgens, such as testosterone, are converted to estrogen.	Induces development of male sex organs in fetus. Induces growth and division of cells that mature sperm. Induces development of male secondary sex characteristics.

Anatomy & Physiology of the Reproductive Organs

- An understanding of the structure and function of the reproductive organs is necessary for the effective nursing care of women during and after their reproductive years and for couples who require family planning or infertility care.

Anatomy & Physiology of the Female Reproductive Organs

- External Genitalia: Collectively, the external female reproductive organs are called the vulva. These structures include the mons pubis, labia majora and minora, clitoris, structures of the vestibule, and perineum

- Internal Genitalia

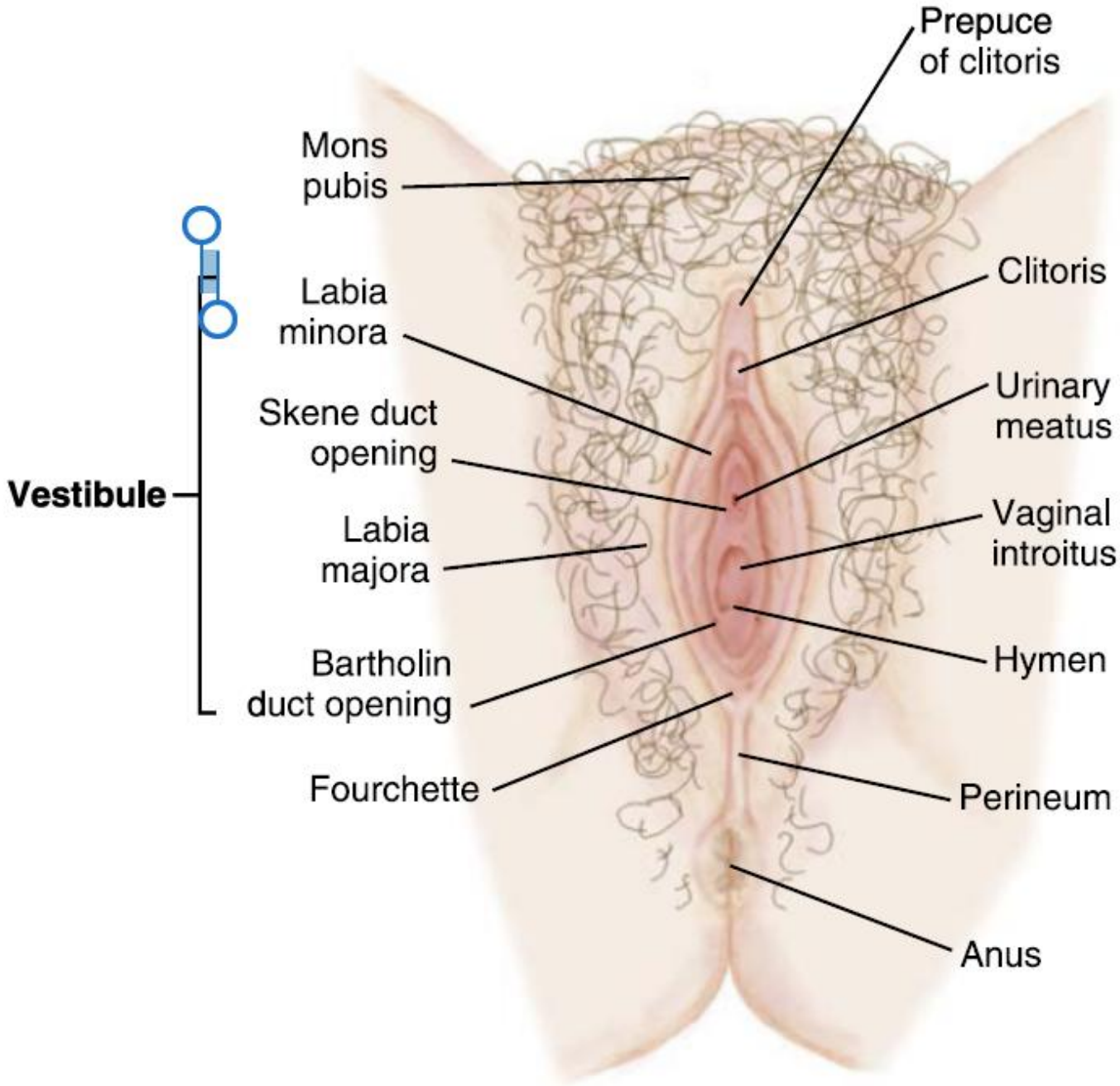
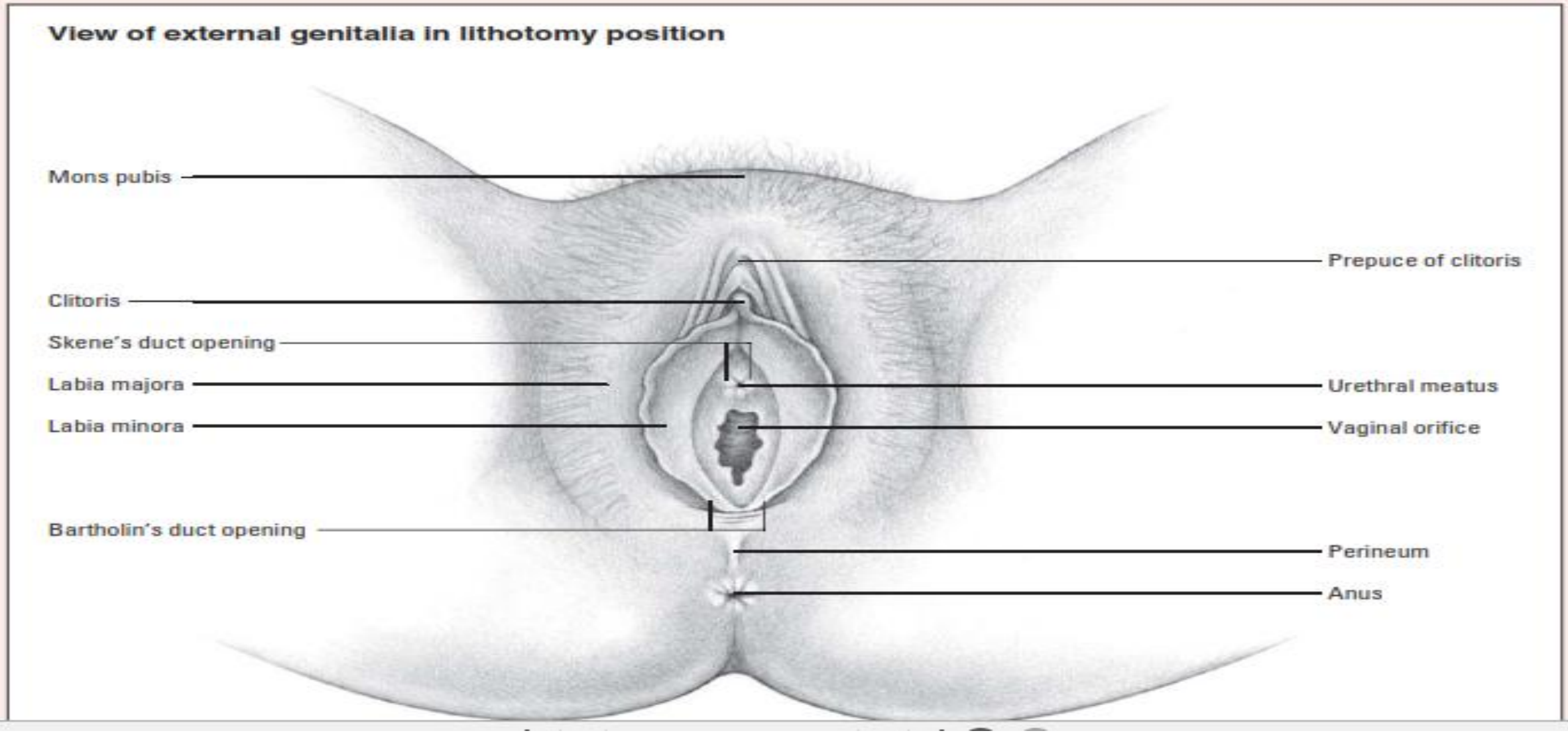


FIG. 3.1 External female reproductive structures.

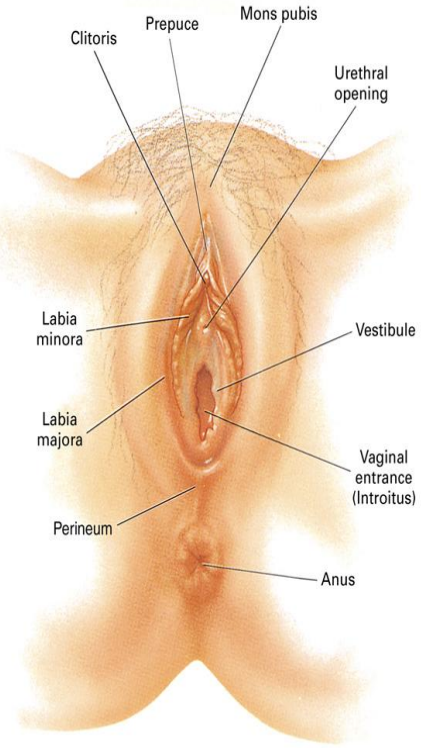
Structures of the female reproductive system

The female reproductive system consists of external and internal genitalia. These structures include the vagina, cervix, uterus, fallopian tubes, ovaries, and other structures. Reproductive, urinary, and gastrointestinal (GI) structures are housed in the female pelvis. These include the bladder, anus, and rectum.



External Genitalia

- Perineum
  - Muscular tissue that separates the vagina and the anus.
- Mons Pubis
  - Fatty layer of tissue over the pubic symphysis.
- Labia (Labia Maiora and Minora)
  - Structures that protect the vagina and the urethra.
  - The labia minora extend from the clitoris anteriorly and merge posteriorly to vaginal opening. The labia minora do not have pubic hair. They are highly vascular and respond to stimulation by becoming engorged with blood.
- Clitoris (a small projection at the anterior junction of the two labia minora)
  - Vascular erectile tissue that lies anterior to the labia minora.. This structure is composed of highly sensitive erectile tissue similar to that of the penis.
- Urethra
  - Drains the urinary bladder.



External Genitalia

- **Vestibule:** The vestibule refers to structures enclosed by the labia minora.
- The urinary meatus, vaginal introitus, and ducts of Skene and Bartholin glands lie within the vestibule.
- Skene, or periurethral, glands provide lubrication for the urethra.
- Bartholin glands provide lubrication for the vaginal introitus, particularly during sexual arousal.
- The vaginal introitus is surrounded by erectile tissue. During sexual stimulation, blood flows into the erectile tissue, allowing the introitus to tighten around the penis. This adds a massaging feeling that heightens the male's sexual sensations and encourages ejaculation.

Lateral view of internal genitalia

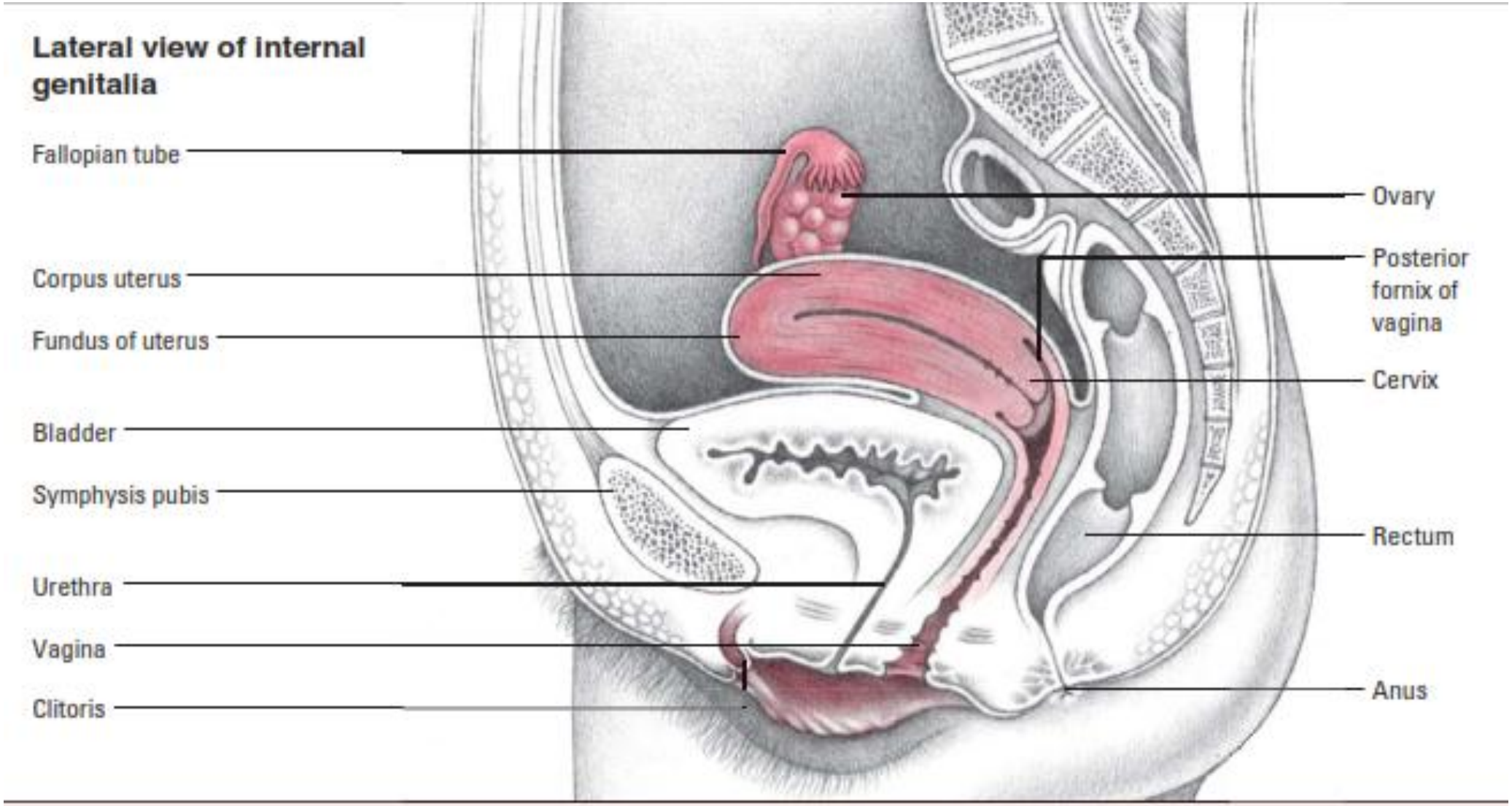


TABLE 3.2 Functions of Female Reproductive and Accessory Organs

Organ	Function
Vagina	1. Passageway for menstrual flow 2. Female organ for coitus; receives male penis during coitus 3. Passageway for fetus during birth
Uterus	1. Houses and nourishes fetus to sufficient maturity to function outside mother's body; propels fetus to outside
Fallopian tube	1. Passageway for ovum as it travels from ovary to uterus 2. Site of fertilization
Ovaries	1. Secrete estrogens and progesterone 2. Contain ova within follicles for maturation during woman's reproductive life
Breasts	
Alveoli	1. Secrete milk after childbirth (acinar cells within alveoli)
Lactiferous ducts	2. Collect milk from alveoli and conduct it to outside

Internal Genitalia

- Vagina
  - Female organ of copulation.
  - Birth canal.
  - Outlet for menstruation.

The vagina does not end abruptly at the uterine opening, but arches to form a pouch-like structure called the vaginal fornix.

- Uterus
  - Site of fetal development. The uterus measures approximately 7.5 x 5 x 2.5 cm
- Fallopian Tubes
  - Transports the egg from the ovary to the uterus.
  - Fertilization usually occurs here.
- Ovaries
  - Primary female gonads.

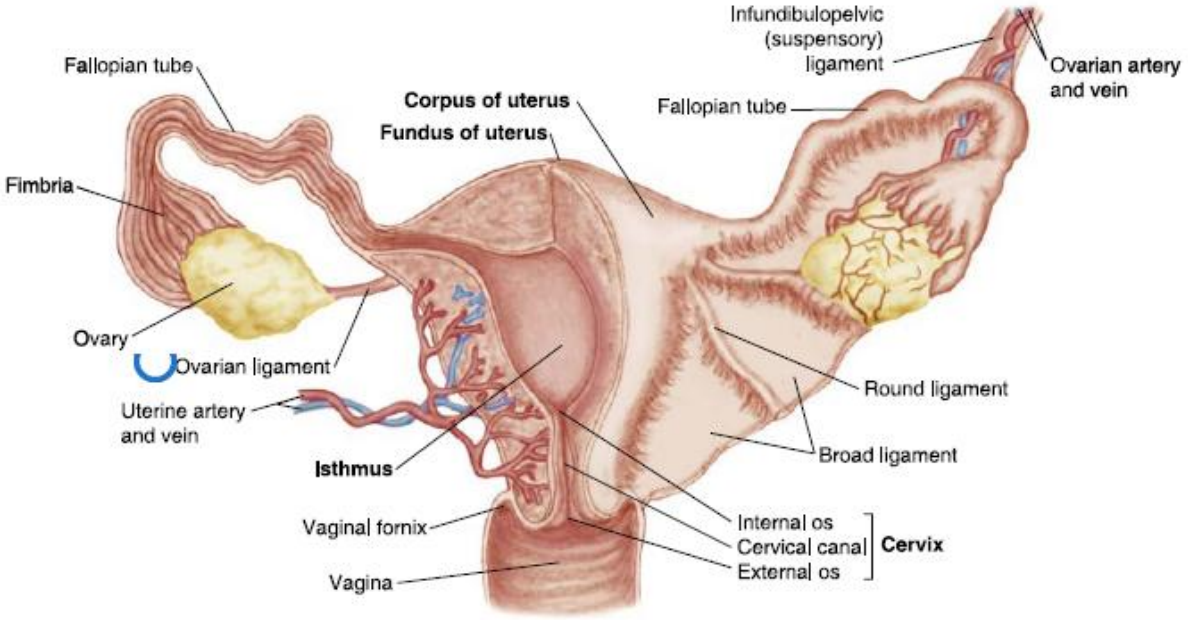


FIG. 3.2 Internal female reproductive structures, anterior view.

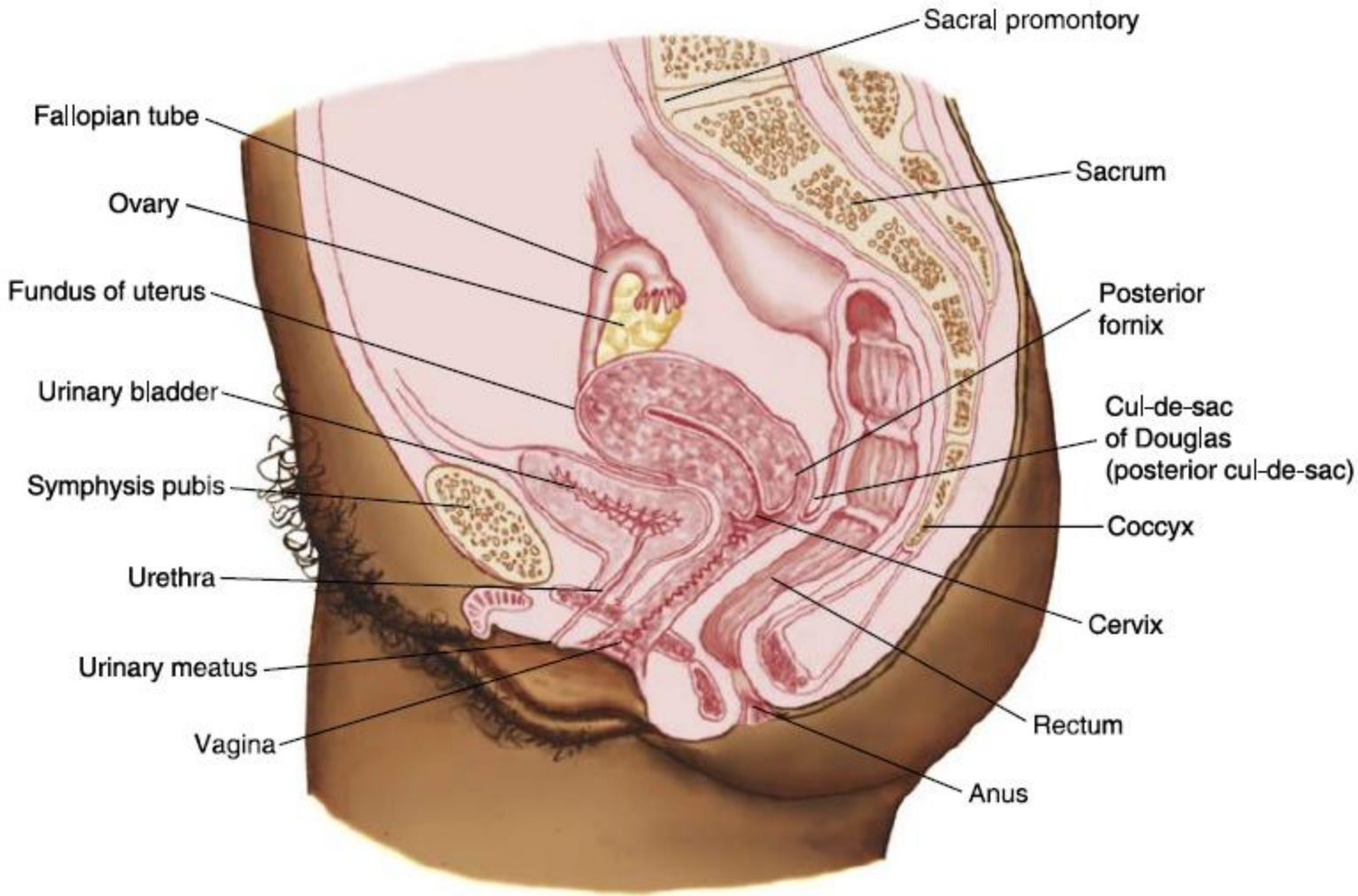


FIG. 3.3 Internal female reproductive structures, midsagittal view.

Layers of the Uterus.

- The uterus has three layers:
  - the perimetrium, the myometrium, and the endometrium.
- **Perimetrium**. The perimetrium is the outer peritoneal layer of serous membrane that covers most of the uterus.
- Laterally the perimetrium is continuous with the broad ligaments on both sides of the uterus.

Layers of the Uterus.

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Myometrium.

The myometrium is the middle layer of thick muscle. Most muscle fibers are concentrated in the upper uterus, and their number diminishes progressively toward the cervix.

The myometrium contains three types of smooth muscle fiber, each suited to specific functions in childbearing:

1. **Longitudinal fibers** are found mostly in the fundus and are designed to expel the fetus efficiently toward the pelvic outlet during birth.

2. **Interlacing fibers** constitute the middle layer: contract after birth to compress blood vessels that pass between them to limit blood loss.

3. **Circular fibers** form constrictions where prevent reflux of menstrual blood and tissue into the fallopian tubes, promote normal implantation of the fertilized ovum by controlling its entry into the uterus, and retain the fetus until the appropriate time of birth.

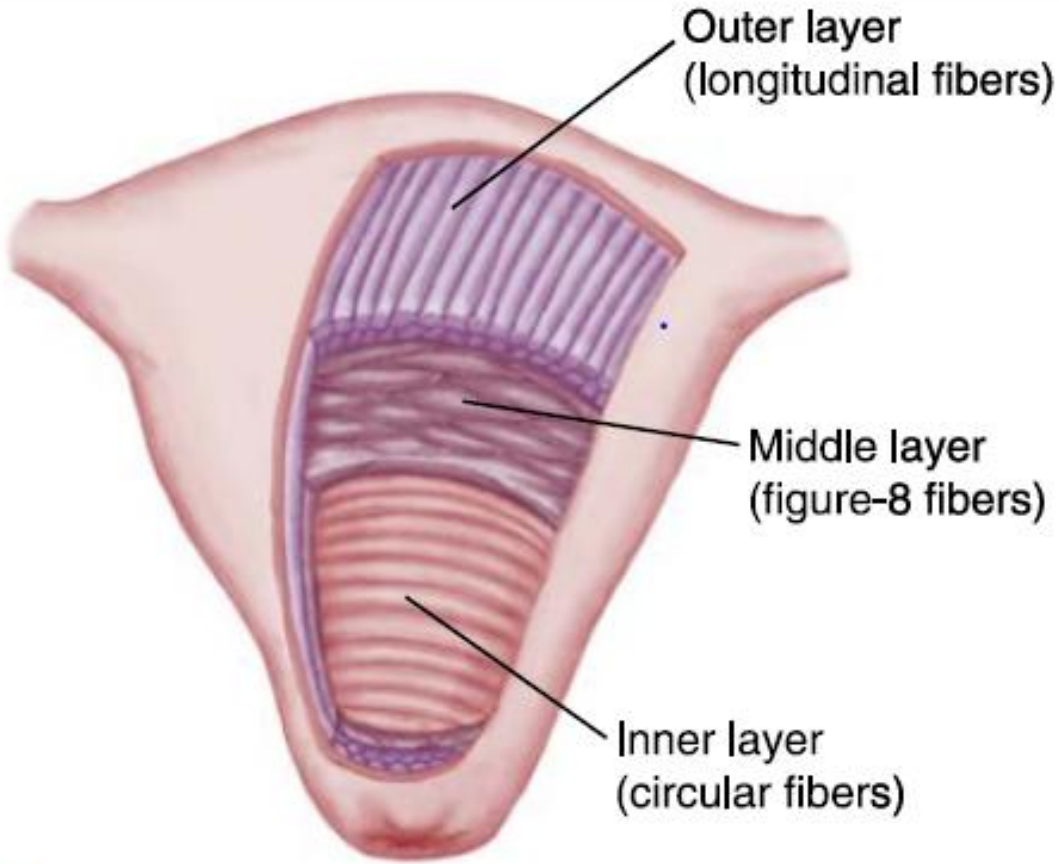


FIG. 3.4 Layers of the myometrium showing the three types of smooth muscle fiber.

Endometrium

The endometrium is the inner layer of the uterus. It responds to the cyclic variations of estrogen and progesterone during the female reproductive cycle.

The endometrium has two layers:

1. **The basal layer** is the area nearest the myometrium that regenerates the functional layer of the endometrium after each menstrual period and after childbirth.

2. **The functional layer** lies above the basal layer. Endometrial arteries, veins, and glands extend into the functional layer and are shed during each menstrual period and after childbirth in the lochia, the vaginal drainage after childbirth.

- Menstruation is the normal, predictable physiologic process whereby the inner lining of the uterus (endometrium) is expelled by the body.
- Menstruation has many effects on girls and women, emotional and self-image
- The average age at menarche (the start of menstruation in females) is 12.8 years, with a range between 8 and 18.
- Genetics is the most important factor in determining the age at which menarche starts, but geographic location, nutrition, weight, general health, nutrition, and psychological factors are also important
- Pubertal events preceding the first menses the appearance of pubic and then axillary hair, followed by a growth spurt (occurring about 2 years after the start of breast development).

Fallopian Tubes

The fallopian tubes, also called oviducts, are 8 to 14 cm long and quite narrow (2 to 3 mm at their narrowest and 5 to 8 mm at their widest).

They form a pathway for the ovum between the ovary and uterus. Fertilization occurs in the fallopian tubes.

The fallopian tubes are lined with folded epithelium containing hairlike processes called cilia that beat rhythmically toward the uterine cavity to propel the ovum through the tube.

The rough, folded surface of its lining and small diameter make the fallopian tube vulnerable to blockage from infection or scar tissue. Tubal blockage may result in sterility or a tubal (ectopic) pregnancy because the fertilized ovum cannot enter the uterus for proper implantation.

The fallopian tubes have four divisions:

1. The interstitial portion runs into the uterine cavity and lies within the uterine wall.
2. The isthmus is the narrow part adjacent to the uterus.
3. The ampulla is the wider area of the tube lateral to the isthmus, where fertilization occurs.
4. The infundibulum is the wide, funnel-shaped terminal end of the tube.

Fallopian Tubes

Fimbriae are finger-like processes that surround the infundibulum.

The fallopian tubes are not directly connected to the ovary.

The ovum is expelled into the abdominal cavity near the fimbriae at ovulation.

Wavelike motions of the fimbriae draw the ovum into the tube.

However, the tubal isthmus remains contracted until 3 days after conception to allow the fertilized ovum to develop within the tube. Initial growth of the fertilized ovum within the fallopian tube promotes its normal implantation in the upper uterus.

Ovaries

The ovaries are the female gonads, or sex glands. There are two functions of the ovary: (1) sex hormone production and (2) maturation of an ovum during each reproductive cycle.

The ovaries secrete estrogen and progesterone in varying amounts during a woman's reproductive cycle to prepare the uterine lining for pregnancy. Ovarian hormone secretion

gradually declines to very low levels during the climacteric.

At birth, the ovary contains all the ova it will ever have Approximately two million immature ova are present at birth.

Many of these degenerate during childhood, and at puberty approximately 200,000 to 400,000 viable ova remain.

Many ova begin the maturation process during each reproductive cycle, but most never reach maturity.

During a woman's reproductive life, only about 400 of the ova ever mature enough to be released and fertilized.

By the time a woman reaches the climacteric, almost all her ova have been released during ovulation or have regressed. The few remaining ova are unresponsive to stimulating hormones and do not mature

FEMALE REPRODUCTIVE CYCLE

The term female reproductive cycle refers to the regular and recurrent changes in the anterior pituitary secretions, ovaries, and uterine endometrium designed to prepare the body for pregnancy

The duration of the cycle is approximately 28 days, although it may range from 20 to 45 days

The female reproductive cycle is further divided into two cycles reflecting changes in the **ovaries** and uterine endometrium.

Ovarian Cycle

In response to GnRH from the woman's hypothalamus, the anterior pituitary secretes FSH and LH. These secretions

stimulate the ovaries to mature and release an ovum and secrete additional hormones to prepare the endometrium for implantation of a fertilized ovum.

The ovarian cycle consists of three phases: **follicular, ovulatory, and luteal.**

Follicular Phase

- The follicular phase is the period during which an ovum matures. It begins with the first day of menstruation and ends approximately 14 days later in a 28-day cycle.
- The length of this phase varies more among different women than do the lengths of the other two phases.
- The fall in estrogen and progesterone secretion by the ovary just before menstruation stimulates secretion of FSH and LH by the anterior pituitary.
- As the FSH and LH levels rise slightly, 6 to 12 graafian follicles (sacs within the ovary), each containing an immature ovum, begin to grow.
- Each follicle secretes fluid containing high levels of estrogen, which accelerates maturation by making the follicle more sensitive to the effects of FSH.
- Eventually, one follicle outgrows the others to reach maturity. The mature follicle secretes large amounts of estrogen, which depresses FSH secretion. The dip in FSH secretion just before ovulation blocks further maturation of the less-developed follicles.
- Occasionally, more than one follicle matures and releases its ovum, which can lead to a multifetal pregnancy. Women who take fertility drugs may release multiple mature ova that are used for assisted reproductive techniques

Ovulatory Phase

- Near the middle of a 28-day reproductive cycle and about 2 days before ovulation, LH secretion rises markedly. Secretion of FSH also rises but to a lesser extent than that of LH.
- These surges in LH and FSH levels cause a slight fall in follicular estrogen production and a rise in progesterone secretion, which stimulates final maturation of a single follicle and release of its ovum.
- Ovulation marks the beginning of the luteal phase of the female reproductive cycle and occurs 14 days before the next menstrual period.
- The mature follicle is a mass of cells with a fluid-filled chamber. At ovulation a blister-like projection called a stigma forms on the wall of the follicle, the follicle ruptures, and the ovum with its surrounding cells is released from the surface of the ovary, where it is picked up by the fimbriated end of the fallopian tube for transport to the uterus.

Luteal Phase

- After ovulation and under the influence of LH, the remaining cells of the old follicle persist for approximately 12 days as a corpus luteum.
- The corpus luteum secretes estrogen and large amounts of progesterone to prepare the endometrium for a fertilized ovum.
- During this phase, levels of FSH and LH decrease in response to higher levels of estrogen and progesterone.
- If the ovum is fertilized, it secretes a hormone (chorionic gonadotropin) that causes persistence of the corpus luteum to maintain an early pregnancy.
- If the ovum is not fertilized, FSH and LH fall to low levels, and the corpus luteum regresses. Decline of estrogen and progesterone levels along with corpus luteum regression results in menstruation as the uterine lining breaks down.
- The loss of estrogen and progesterone from the corpus luteum at the end of one cycle stimulates the anterior pituitary to again secrete more FSH and LH, initiating a new female reproductive cycle. The old corpus luteum is replaced by fibrous tissue called the corpus albiccms.

Endometrial Cycle

- The uterine endometrium responds to ovarian hormone stimulation with cyclic changes. Four phases mark the changes in the endometrium: proliferative, secretory, ischemic, and menstrual.

Proliferative Phase

- The proliferative phase occurs as the ovum matures and is released during the first half of the ovarian cycle. After completion of a menstrual period, the endometrium is very thin.
- The basal layer of endometrial cells remains after menstruation.
- These cells multiply to form new endometrial epithelium and endometrial glands under the stimulation of estrogen secreted by the maturing ovarian follicles.
- Endometrial spiral arteries and endometrial veins elongate to accompany thickening of the functional endometrial layer and nourish the proliferating cells.
- As ovulation approaches, the endometrial glands secrete thin/stringy mucus that aids entry of sperm into uterus.

Secretory Phase

- The secretory phase occurs during the last half of the ovarian cycle as the uterus is prepared to receive a fertilized ovum.
- The endometrium continues to thicken under the influence of estrogen and progesterone from the corpus luteum, reaching its maximum thickness of 5 to 6 mm.
- The blood vessels and endometrial glands become twisted and dilated.
- Progesterone from the corpus luteum causes the thick endometrium to secrete substances to nourish a fertilized ovum.
- Large quantities of glycogen, proteins, lipids, and minerals are stored within the endometrium, awaiting arrival of the ovum.

Ischemic and Menstrual Phases

- If fertilization does not occur, the corpus luteum regresses, and its production of estrogen and progesterone falls. About 2 days before the onset of menstruation, vasospasm of the endometrial blood vessels causes the endometrium to become ischemic and necrotic.
- The necrotic areas of endometrium separate from the basal layers, resulting in the menstrual flow.
- The durations of the menstrual phase, from the first day of flow to the last is approximately 5 days.
- During a menstrual period, women lose approximately 40 mL of blood. Because of the recurrent loss of blood, many women are mildly anemic during their reproductive years, especially if their diets are low in iron.

Ischemic and Menstrual Phases

- If fertilization does not occur, the corpus luteum regresses, and its production of estrogen and progesterone falls. About 2 days before the onset of menstruation, vasospasm of the endometrial blood vessels causes the endometrium to become ischemic and necrotic.
- The necrotic areas of endometrium separate from the basal layers, resulting in the menstrual flow.
- The durations of the menstrual phase, from the first day of flow to the last is approximately 5 days.
- During a menstrual period, women lose approximately 40 mL of blood. Because of the recurrent loss of blood, many women are mildly anemic during their reproductive years, especially if their diets are low in iron.

Changes in Cervical Mucus

- During most of the female reproductive cycle, the mucus of the cervix is scant, thick, and sticky.
- Just before ovulation, cervical mucus becomes thin, clear, and elastic to promote passage of sperm into the uterus and fallopian tube, where they can fertilize the ovum.
- Spinnbarkeit refers to the elasticity of cervical mucus. A woman may assess the elasticity of her cervical mucus to either avoid or promote conception

Menstrual Disorders

- Amenorrhea
- Dysmenorrhea
- dysfunctional uterine bleeding (DUB)
- premenstrual syndrome (PMS)
- premenstrual dysphoric disorder (PMDD)
- endometriosis.

Amenorrhea

- is the absence of menses during the reproductive years.
- Amenorrhea is normal in prepubertal, pregnant, and postmenopausal females.
- The two categories of amenorrhea are primary and secondary amenorrhea.
- Primary amenorrhea is defined as either: 1.Absence of menses by age 14, with absence of growth and development of secondary sexual characteristics, or 2. Absence of menses by age 16, with normal development of secondary sexual characteristics
- Secondary amenorrhea is the absence of menses for three cycles or 6 months in women who have previously menstruated regularly.

Etiology (causes of primary amenorrhea):

- Extreme weight gain or loss
- Congenital abnormalities of the reproductive system
- Stress from a major life event
- Excessive exercise
- Eating disorders (anorexia nervosa or bulimia)
- Cushing’s disease • Polycystic ovary syndrome • Hypothyroidism
- Turner syndrome • Imperforate hymen • Chronic illness • Pregnancy • Cystic fibrosis • Congenital heart disease (cyanotic) • Ovarian or adrenal tumors

Etiology (causes of secondary amenorrhea):

- Pregnancy, Breastfeeding
- Emotional stress • Pituitary, ovarian, or adrenal tumors • Depression • Hyperthyroid or hypothyroid conditions • Malnutrition • Hyperprolactinemia
- Rapid weight gain or loss • Chemotherapy or radiation therapy to the pelvic area • Vigorous exercise, such as long-distance running • Kidney failure • Colitis • Use of tranquilizers or antidepressants
- Postpartum pituitary necrosis (Sheehan syndrome) • Early menopause

Laboratory and Diagnostic Tests

Common laboratory tests that might be ordered to determine the cause of amenorrhea include:

- Karyotype (might be positive for Turner syndrome)
- Ultrasound to detect ovarian cysts
- Pregnancy test to rule out pregnancy
- Thyroid function studies to determine thyroid disorder
- Prolactin level (an elevated level might indicate a pituitary tumor)
- Follicle-stimulating hormone (FSH) level (an elevated level might indicate ovarian failure)
- Luteinizing hormone (LH) level (an elevated level might indicate gonadal dysfunction)
- 17-ketosteroids (an elevated level might indicate an adrenal tumor)

In Cushing's disease, increased circulating androgens produced by the adrenal gland and **cortisol** can suppress GnRH and therefore LH and FSH.

- Laparoscopy to detect polycystic ovary syndrome
- CT scan of head if a pituitary tumor is suspected

Therapeutic Management

- Therapeutic intervention depends on the cause of the amenorrhea.
- The treatment of primary amenorrhea involves the correction of any underlying disorders and estrogen replacement therapy to stimulate the development of secondary sexual characteristics.
- If a pituitary tumor is the cause, it might be treated with drug therapy, surgical resection, or radiation therapy.
- Surgery might be needed to correct any structural abnormalities of the genital tract. Therapeutic interventions for secondary amenorrhea may include: • Cyclic progesterone, when the cause is anovulation, or oral contraceptives • Bromocriptine to treat hyperprolactinemia • Nutritional counseling to address anorexia, bulimia, or obesity • Gonadotropin-releasing hormone (GnRH), when the cause is hypothalamic failure • Thyroid hormone replacement, when the cause is hypothyroidism

Therapeutic Management

- Bromocriptine to treat hyperprolactinemia
- Nutritional counseling to address anorexia, bulimia, or obesity
- Gonadotropin-releasing hormone (GnRH), when the cause is hypothalamic failure
- Thyroid hormone replacement, when the cause is hypothyroidism

Dysmenorrhea

Dysmenorrhea refers to painful menstruation. This condition has also been termed cyclic perimenstrual pain (CPPD) (Taylor, 2005).

It may affect more than half of menstruating women

Uterine contractions occur during all periods, but in some women these cramps can be frequent and very intense.

Dysmenorrhea is categorized as primary or secondary.

Etiology

Primary dysmenorrhea is caused by increased prostaglandin production by the endometrium in an ovulatory cycle. This hormone causes contraction of the uterus

These levels are highest during the first 2 days of menses.

This results in increased rhythmic uterine contractions from vasoconstriction of the small vessels of the uterine wall.

Etiology

Secondary dysmenorrhea is painful menstruation due to pelvic or uterine pathology. It may be caused by endometriosis, adenomyosis, fibroids, pelvic infection, an intrauterine device, cervical stenosis, or congenital uterine or vaginal abnormalities.

Adenomyosis involves the ingrowth of the endometrium into the uterine musculature.

Endometriosis involves ectopic implantation of endometrial tissue in other parts of the pelvis. It occurs most commonly in the third or fourth decades of life and affects 10% of women of reproductive age (Endometriosis is the most common cause).

Therapeutic Management

Therapeutic intervention is directed toward pain relief and building coping strategies that will promote a productive lifestyle. Treatment measures usually include treating infections if present; suppressing the endometrium if endometriosis is suspected by administering low-dose oral contraceptives; administering prostaglandin inhibitors to reduce the pain; administering Depo-Provera; and initiating lifestyle changes.

Dysfunctional uterine bleeding (DUB)

- is a disorder that occurs most frequently in women at the beginning and end of their reproductive years.
- Defined as irregular, abnormal bleeding that occurs with no identifiable anatomic pathology.
- it affects 33% to 50% of women. It is frequently associated with anovulatory cycles, which are common for the first year after menarche and later in life as women approach menopause.
- The pathophysiology of DUB is related to a hormone disturbance. With anovulation, estrogen levels rise as usual in the early phase of the menstrual cycle. In the absence of ovulation, a corpus luteum never forms and progesterone is not produced. The endometrium moves into a hyperproliferative state, ultimately outgrowing its estrogen supply. This leads to irregular sloughing of the endometrium and excessive bleeding.
- If the bleeding is heavy enough and frequent enough, anemia can result.

DUB is similar to several other types of uterine bleeding disorders and sometimes overlaps these conditions. They include:

- Menorrhagia (abnormally long, heavy periods)
- Oligomenorrhea (bleeding occurs at intervals of more than 35 days)
- Metrorrhagia (bleeding between periods)
- Menometrorrhagia (bleeding occurs at irregular intervals with heavy flow lasting more than 7 days)
- Polymenorrhea (too frequent periods)

The possible causes of DUB may include:

- Adenomyosis
- Pregnancy
- Hormonal imbalance
- Fibroid tumors
- Endometrial polyps or cancer
- Endometriosis
- Intrauterine device (IUD)
- Polycystic ovary syndrome
- Morbid obesity
- Steroid therapy
- Hypothyroidism
- Blood dyscrasias/clotting disorder

Therapeutic Management

- Treatment of DUB depends on the cause of the bleeding and the age of the client.
- Management of DUB might include medical care with pharmacotherapy or insertion of an IUD (levonorgestrel-releasing IUD decrease menstrual blood loss).
- The drug categories used in the treatment of DUB are:
- Estrogens: cause vasospasm of the uterine arteries to decrease bleeding
- Progestins: used to stabilize an estrogen-primed endometrium
- Oral contraceptives: regulate the cycle and suppress the endometrium
- NSAIDs: inhibit prostaglandins
- Levonorgestrel-20 Intrauterine System: suppresses endometrial growth
- Iron salts: replenish iron stores lost during heavy bleeding
- If the client does not respond to medical therapy, surgical intervention might include dilation and curettage (D&C), endometrial ablation, or hysterectomy.
- A thermal balloon or a laser is used to ablate the tissue

Premenstrual syndrome (PMS)

- describes a wide range of recurrent symptoms that occur during the luteal phase or last half of the menstrual cycle and resolve with the onset of menstruation
- The exact cause of PMS is not known.
- It is thought to be related to the interaction between hormonal events and neurotransmitter function, specifically serotonin.
- Not all women respond to serotonin reuptake inhibitors (SSRIs
- tension headache and a migraine, sometimes markedly interferes with work and school, or with social activities and relationships with others.

Therapeutic Management

- Treatment of PMS is often frustrating for both patients and health care providers.
- The management of PMS or PMDD requires a multidimensional approach because these conditions are not likely to have a single cause, and they appear to affect multiple systems within a woman's body
- Treatments may include vitamin supplements, diet changes, exercise, lifestyle changes, and medications (antidepressant and antianxiety drugs, diuretics, anti-inflammatory medications, analgesics, synthetic androgen agents, oral contraceptives, or GNrH agonists to regulate menses)

Endometriosis

- is one of the most common gynecologic diseases, affecting more than 5.5 million women in the United States. In this condition, bits of functioning endometrial tissue are located outside of their normal site, the uterine cavity. This endometrial tissue is commonly found attached to the ovaries, fallopian tubes, the outer surface of the uterus, the bowels, the area between the vagina and the rectum (rectovaginal septum), and the pelvic side wall .
- Endometrial tissue found outside the uterus responds to hormones released during the menstrual cycle in the same way as endometrial lining within the uterus.



Endometriosis

Etiology and Risk Factors

- It is not currently known why endometrial tissue becomes transplanted and grows in other parts of the body
- factors that increase a woman’s risk of developing endometriosis:
- Increasing age • Family history of endometriosis in a first-degree relative • Short menstrual cycle (less than 28 days) • Long menstrual flow (more than 1 week) • Young age of menarche (younger than 12) • Few (one or two) or no pregnancies

Therapeutic Management

Therapeutic management of the client with endometriosis needs to take into consideration the following factors:

- severity of symptoms
- desire for fertility
- degree of disease

The aim of therapy is to suppress levels of estrogen and progesterone, which cause the endometrium to grow. Treatment can include surgery or medication

MALE REPRODUCTIVE ANATOMY AND PHYSIOLOGY

External Male Reproductive Organs

- The male has two external organs of reproduction: the penis and scrotum.

Penis

- The penis has two functions. As part of the urinary tract, it carries urine from the bladder to the exterior during urination.
- As a reproductive organ, the penis deposits semen into the female vagina during coitus.
- The penis is composed mostly of erectile tissue, which is spongy tissue with many small spaces inside.
- The three areas of erectile tissue are the corpus spongiosum, which surrounds the urethra, and two columns of the corpus cavernosum on each side of the penis.
- The penis is flaccid most of the time because the small spaces within the erectile tissue are collapsed.
- During sexual stimulation, arteries within the penis dilate and veins are partly occluded, trapping blood in the spongy tissue.
- Entrapment of blood within the penis causes erection and enables penetration of the vagina during sexual intercourse.
- The glans is the distal end of the penis.
- The glans is very sensitive to tactile stimulation, which adds to the male’s sensation during coitus.

MALE REPRODUCTIVE ANATOMY AND PHYSIOLOGY

Scrotum

- The scrotum is a pouch of thin skin and muscle suspended behind the penis.
- The skin of the scrotum is somewhat darker than the surrounding skin and is covered with small ridges called rugae.
- The scrotum is divided internally by a septum.
- One of the male gonads (testicle) is contained within each pocket of the scrotum.
- The scrotum's main purpose is to keep the testes cooler than the core body temperature.
- A cremaster muscle is attached to each testicle. Contraction of the cremaster muscles draws the testicles closer to the body for warming, and relaxation of these muscles allows the testicles to move away from the body for cooling.

Internal Male Reproductive Organs

Testes

- The male gonads/testes, have two functions: they serve as endocrine glands, and they produce male gametes, or sperm, also called spermatozoa.
- Androgens (male sex hormones) are the primary endocrine secretions of the testes. Androgens are produced by Leydig cells of the testes. The primary androgen produced by the testes is testosterone.
- A small amount of testosterone is converted to estrogen in males and is necessary for sperm formation.
- Spermatogenesis occurs within tiny coiled tubes, the seminiferous tubules of the testes. Leydig cells are interstitial cells supporting seminiferous tubules and secrete testosterone, which necessary to form new cells that will mature into sperm.
- Sertoli cells within the seminiferous tubules respond to FSH secretion by nourishing and supporting sperm as they mature.

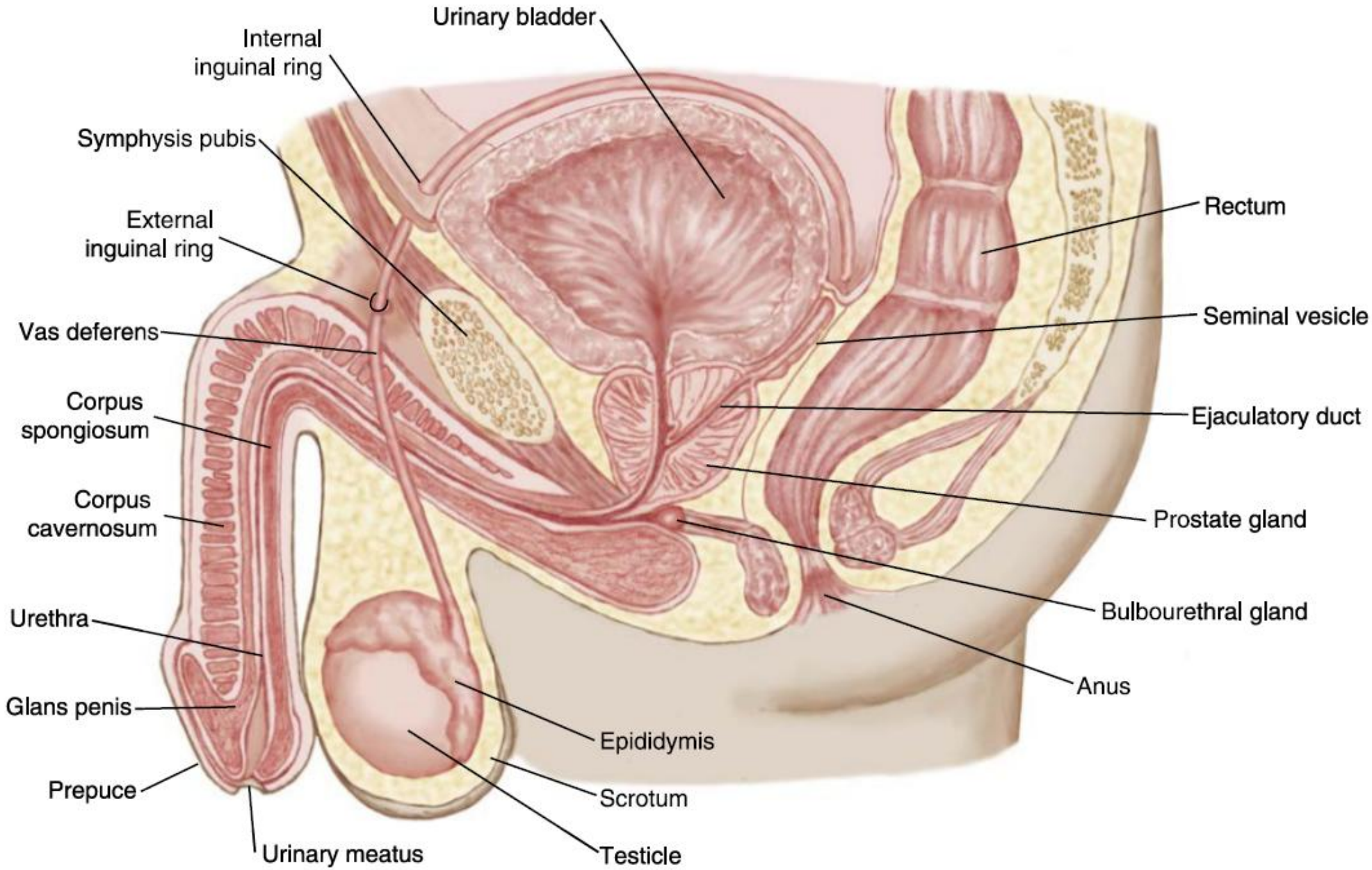


FIG. 3.9 Structures of the male reproductive system, midsagittal view.

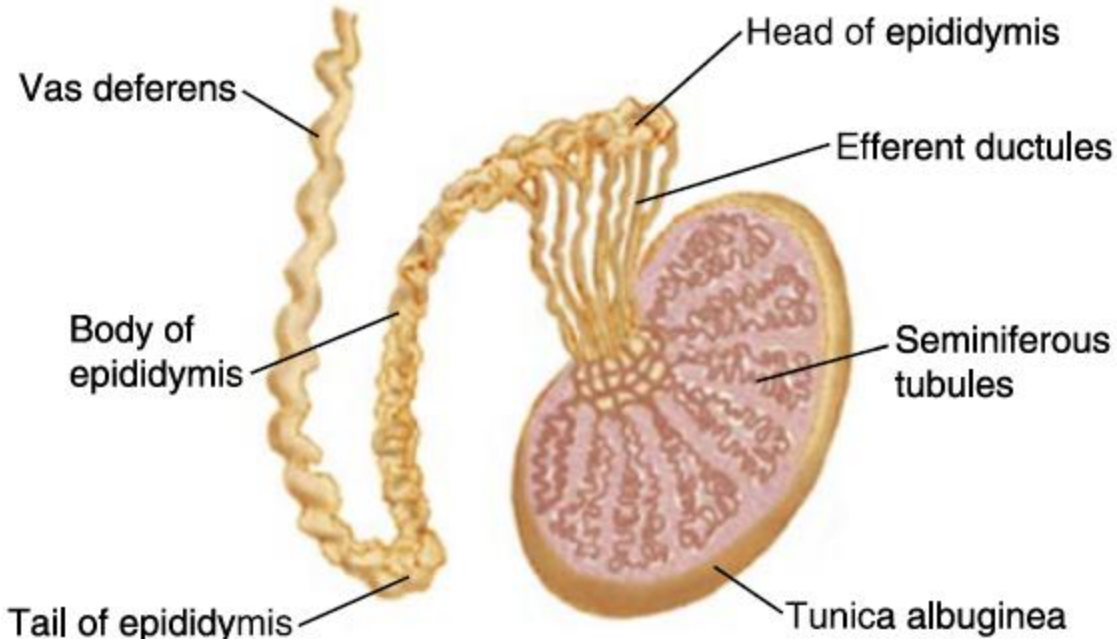
MALE REPRODUCTIVE ANATOMY AND PHYSIOLOGY

- At ejaculation, approximately 35 to 200 million sperm are deposited in the vagina.
- Only a few sperm ever reach the fallopian tube, where an ovum may be available for fertilization.
- When the first sperm penetrates the ovum, changes within the ovum prevent other sperm from also fertilizing it.

TABLE 3.3 Functions of Male Reproductive and Accessory Organs

Organ	Function
Penis	1. Conduit for urine from bladder 2. Male organ of sexual intercourse
Scrotum	1. Housing of testes and maintenance of their temperature at a level cooler than trunk of the body, thus promoting normal sperm formation
Testes	1. Endocrine glands that secrete primary male hormone (testosterone) 2. Sperm formation
Seminiferous tubules	1. Location of spermatogenesis within testes
Epididymis	1. Storage of some sperm 2. Final sperm maturation 3. Location where sperm develop ability to be motile
Vas deferens	1. Storage of sperm 2. Conduction of sperm from epididymis to urethra
Seminal vesicles, prostate, and bulbourethral glands	1. Secretion of seminal fluids that carry sperm and provide the following: • Nourishment of sperm • Protection of sperm from hostile acidic environment of vagina • Enhancement of motility of sperm • Washing of all sperm from urethra

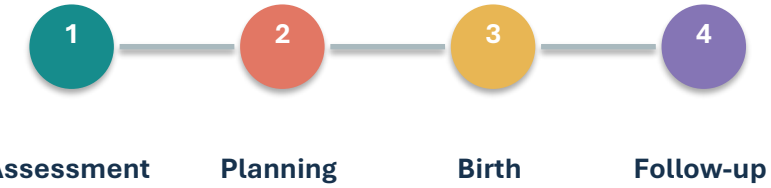
Three glands—the seminal vesicles, the prostate, and bulbourethral gland—secrete seminal fluids that carry sperm into the vagina during intercourse.



Gestational age and due date

- Gestational age is counted from the first day of the last menstrual period.
- Fertilization usually occurs about two weeks after that date.
- Ultrasound dating may refine estimates; the due date is not an exact birth date.

TIMELINE

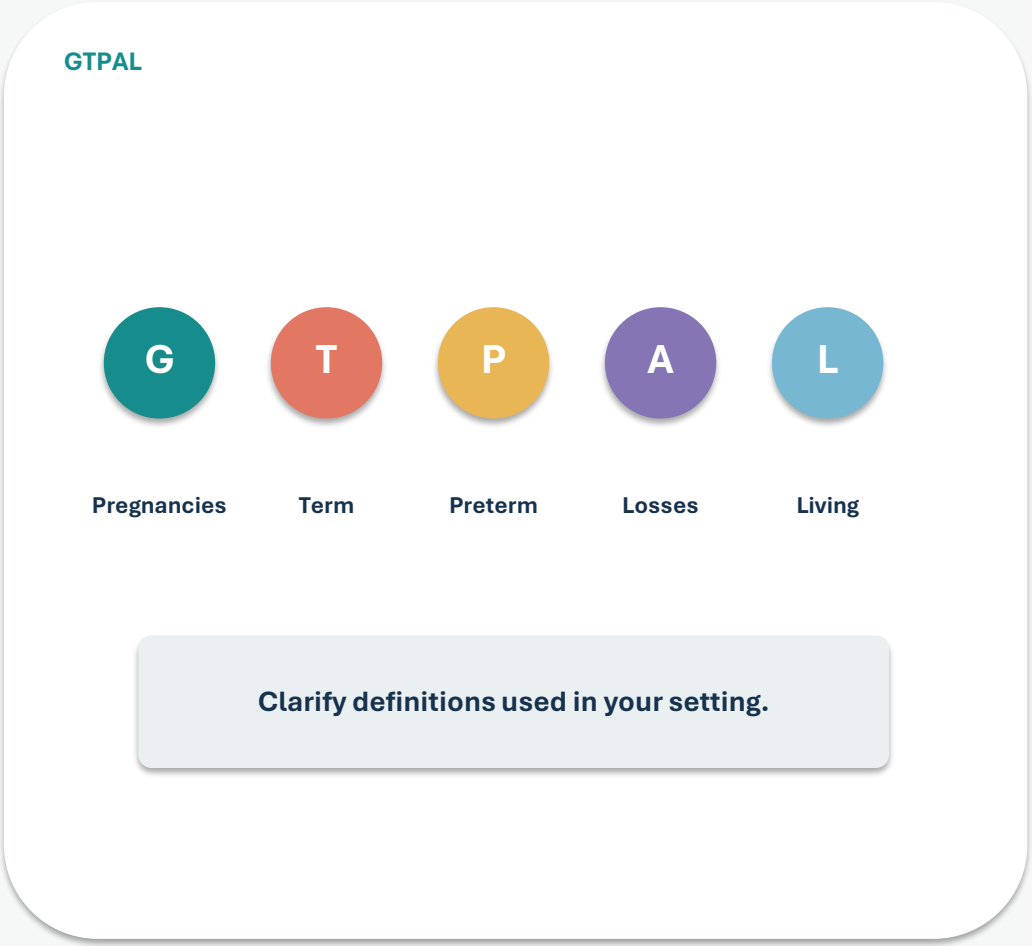


Timing and response depend on clinical context.

Sources: ACOG

Gravida, para, and GTPAL

- Gravida counts pregnancies, including the current pregnancy.
- GTPAL records term, preterm, abortion, and living outcomes.
- Twins count as one pregnancy but may increase the number of living children.



Sources: ACOG

Conception and fetal development

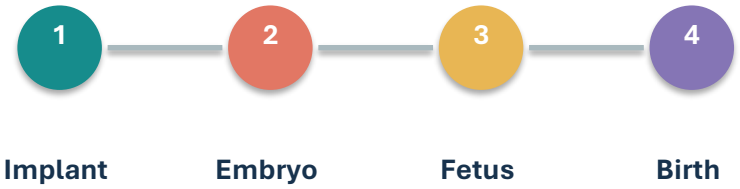


10 focused lessons • assessment, teaching, and clinical reasoning

Three developmental periods

- Pre-embryonic development: fertilization through implantation.
- Embryonic period: organ formation, with high sensitivity to teratogens.
- Fetal period: growth and functional maturation through birth.

DEVELOPMENT

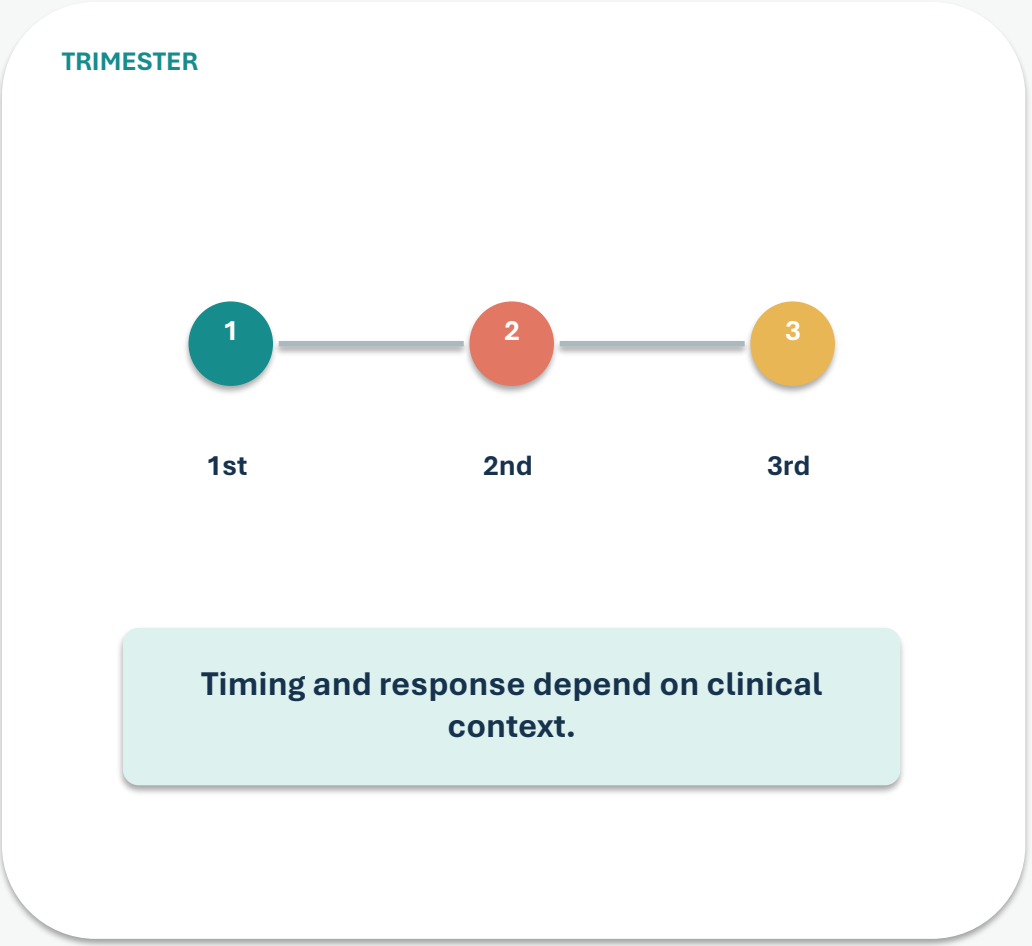


Timing and response depend on clinical context.

Sources: OpenStax

First trimester: foundations form

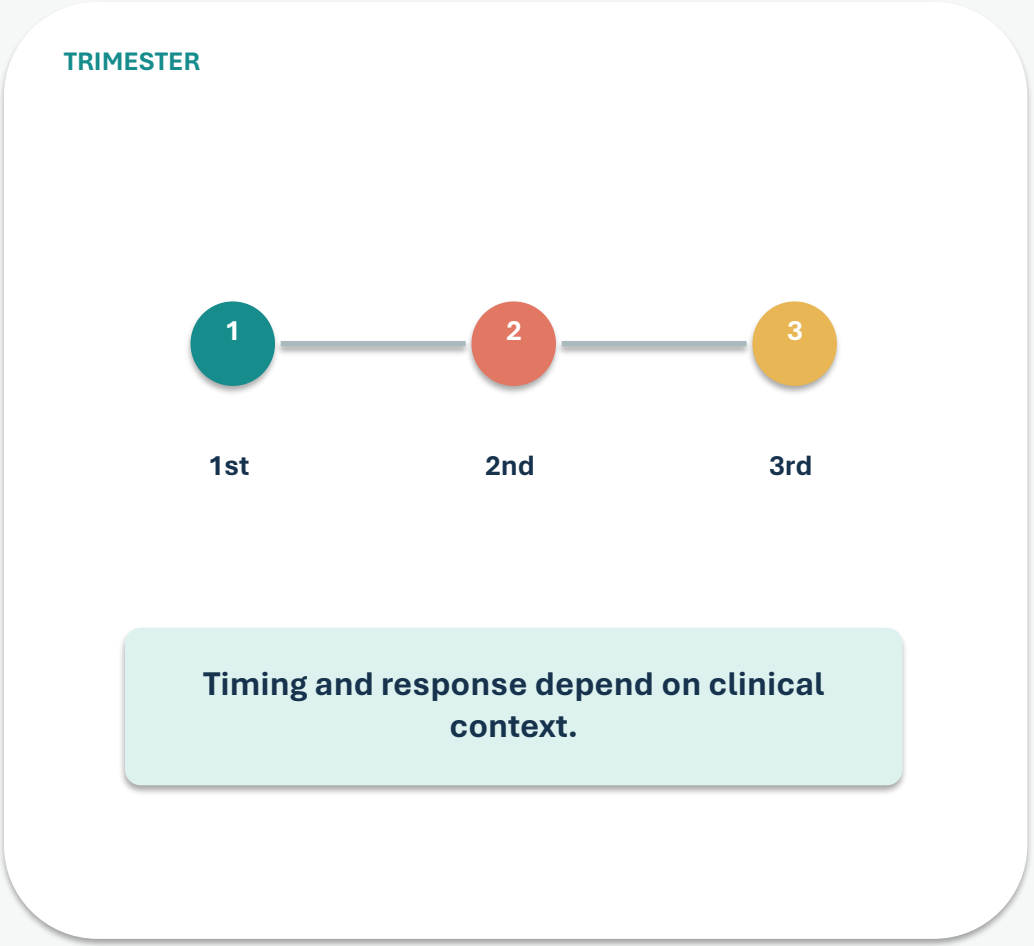
- Major organs and body structures begin to form early in pregnancy.
- The heart, neural tube, limbs, and facial structures develop rapidly.
- Early prenatal care supports risk review and timely screening.



Sources: ACOG

Second trimester: growth and movement

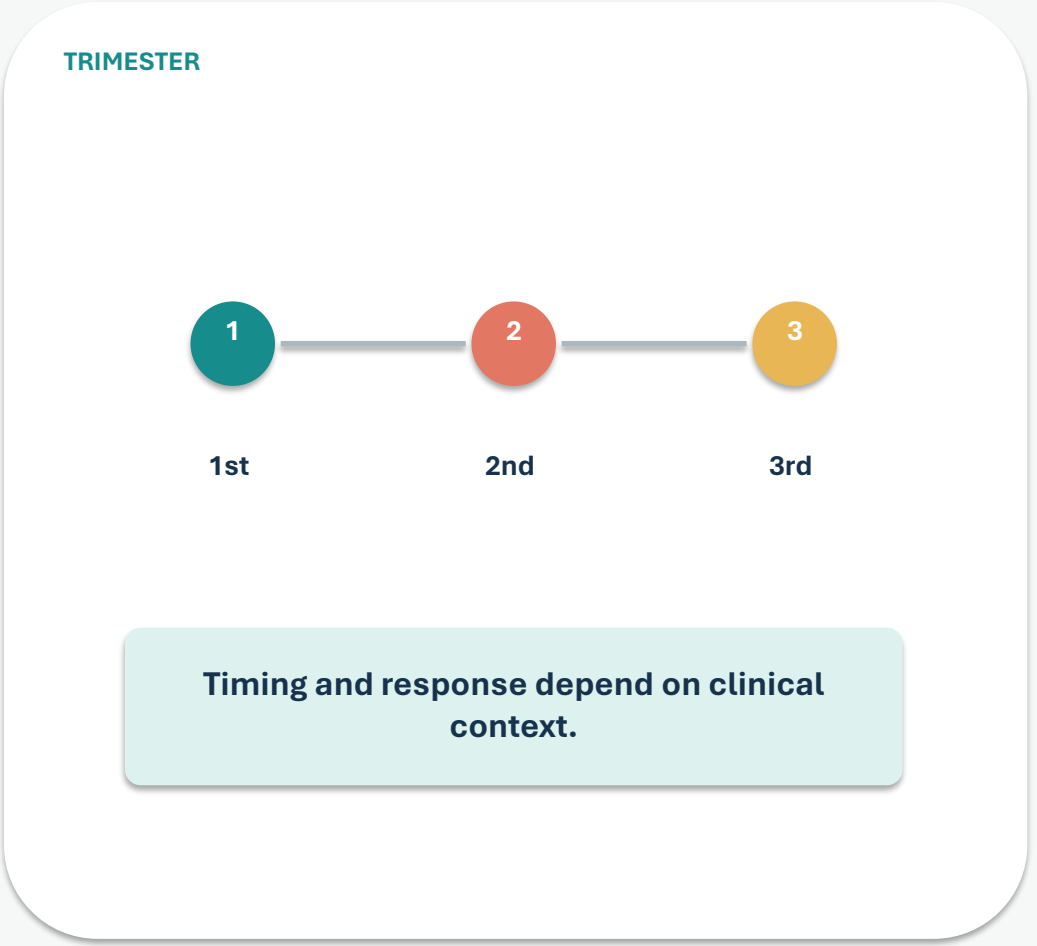
- Fetal movement is often first perceived between about 16 and 24 weeks.
- Anatomy ultrasound commonly occurs around 18–22 weeks, depending on setting.
- Viability thresholds and care options vary by neonatal resources and law.



Sources: ACOG

Third trimester: maturation and preparation

- Brain, lungs, fat stores, and feeding coordination continue to mature.
- Fetal position and growth surveillance depend on clinical context.
- Birth preparation includes warning signs, preferences, and transport planning.

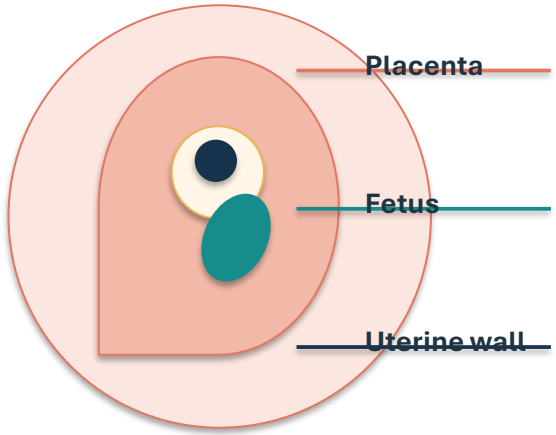


Sources: ACOG

Placenta: exchange and endocrine roles

- Maternal and fetal blood normally remain in separate circulations.
- The placenta transfers oxygen, nutrients, waste products, and selected antibodies.
- It produces hormones that support pregnancy and prepare for lactation.

PLACENTA



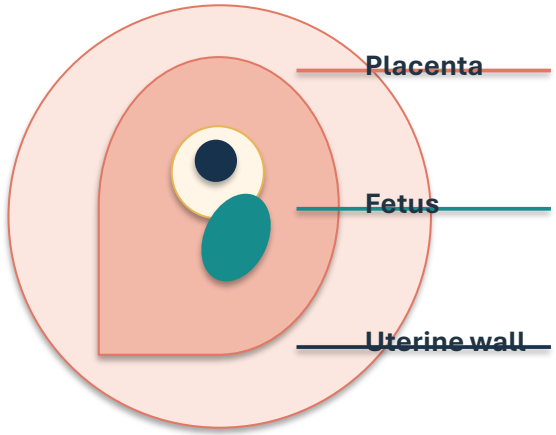
Schematic • not to scale

Sources: OpenStax

Umbilical cord and fetal membranes

- The cord normally contains two arteries and one vein within Wharton’s jelly.
- Amnion and chorion surround the fetus and amniotic fluid.
- Cord compression or membrane rupture can change intrapartum risk.

CORD



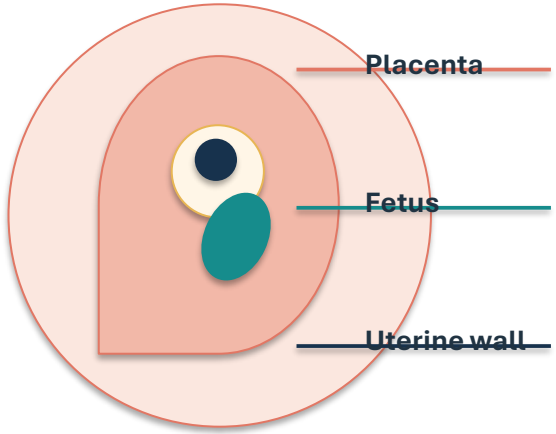
Schematic • not to scale

Sources: OpenStax

Amniotic fluid

- Fluid cushions the fetus and allows movement and lung development.
- Volume changes can reflect fetal, placental, or maternal conditions.
- Assess fluid appearance after membrane rupture and report abnormalities.

FLUID

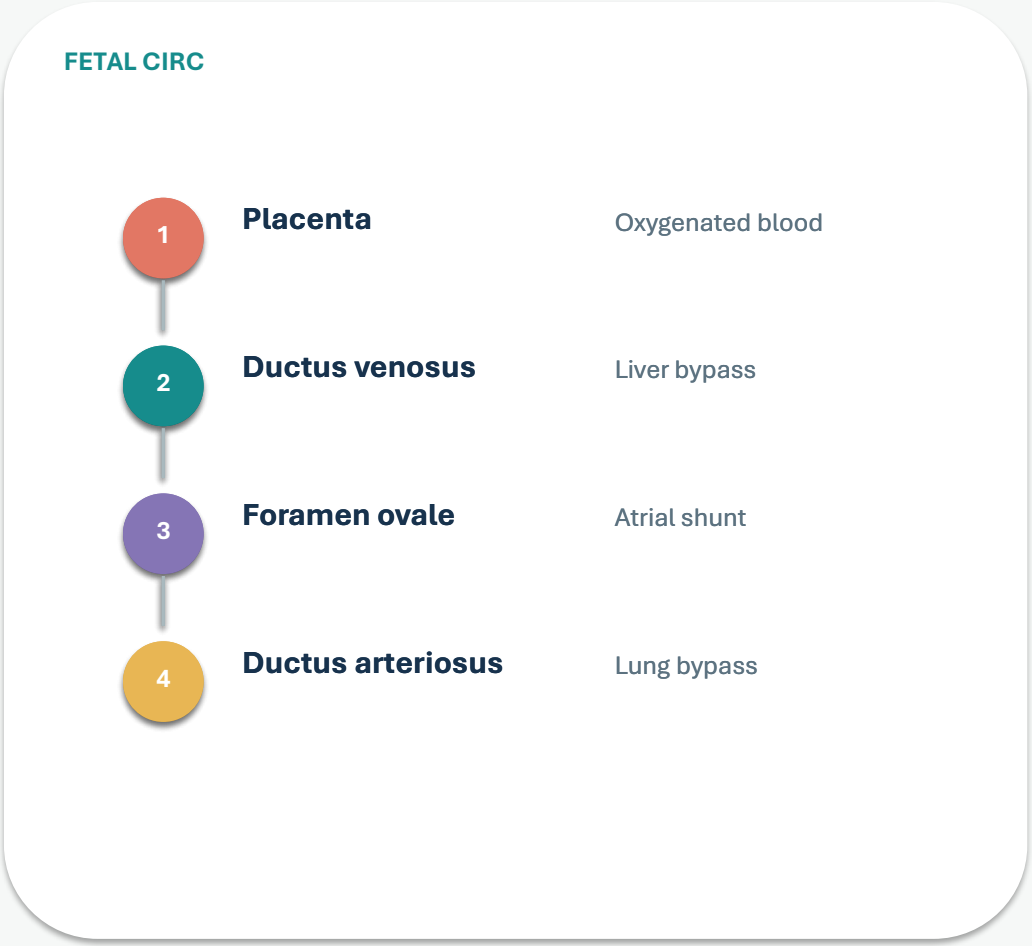


Schematic • not to scale

Sources: ACOG

Fetal circulation

- The ductus venosus, foramen ovale, and ductus arteriosus route blood around fetal lungs.
- Placental gas exchange supplies oxygen before birth.
- Cord clamping and the first breaths trigger major circulatory transitions.

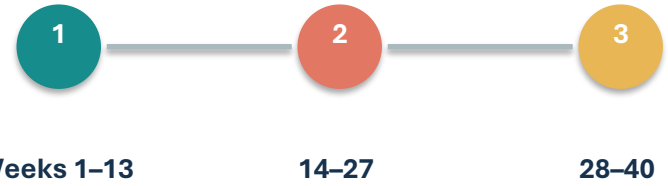


Sources: OpenStax

Fetal growth and surveillance

- Growth is assessed with serial measurements and ultrasound when indicated.
- Fundal height is a screening measure in singleton pregnancies.
- Interpret growth using gestational age, parental size, and clinical context.

GROWTH



Timing and response depend on clinical context.

Sources: ACOG

Pregnancy

GAMETOGENESIS

- Gametogenesis is the development of ova in the woman and sperm in the man.
- Production of gametes (reproductive or germ cells) requires a process different from formation of somatic cells or other body cells.
- Somatic cells reproduce by a process called mitosis.
- Each somatic cell has 46 paired chromosomes: 22 pairs of autosomes (non-sex chromosomes) and 1 pair of sex chromosomes (X or Y chromosomes).
- During mitosis, the cell divides into two new cells, each having 46 chromosomes like the parent cell.
- Gametogenesis requires a special reduction division called meiosis.

Oogenesis

- Oogenesis is the formation of female gametes within the ovary. Oogenesis begins during prenatal life when primitive ova (oogonia) multiply by mitosis.
- Each oogonium contains 46 chromosomes. Before birth, these oogonia enlarge to form primary oocytes, each surrounded by a layer of follicular cells. These are called primary follicles.
- The primary oogonium begins its first meiotic division during fetal life but does not complete the process until puberty.
- By the 30th week of gestation, the female fetus has all the ova she will ever have. Many of these ova regress during childhood.
- When a girl's reproductive cycles begin at puberty, some of the primary follicles present at birth begin maturing.

GAMETOGENESIS

Spermatogenesis

- Spermatogenesis (formation of sperm, or male gametes, in the testes) begins during puberty in the male and requires approximately 70 days to be completed.
- Primitive sperm cells, or spermatogonia, develop during the prenatal period and begin multiplying by mitosis during puberty.
- Although male fertility gradually declines with age, men can father children in their fifties, sixties, and beyond.
- Preparation for Conception in the Female Before ovulation, several oocytes begin to mature under the influence of follicle-stimulating hormone (FSH) and luteinizing hormone (LH) from the woman's anterior pituitary gland.

Ejaculation

When a male ejaculates during sexual intercourse, 35 to 200 million sperm are deposited in the upper vagina and over the cervix. The sperm are suspended in 2 to 5 milliliters (mL) of seminal fluid, which nourishes and protects the sperm from the acidic environment of the vagina. Many sperm are lost as the ejaculate drips from the vaginal introitus. Other sperm are inactivated by acidic vaginal secretions or digested by vaginal enzymes and phagocytes.

Many sperm are relatively immobile for approximately 15 to 30 minutes until other seminal enzymes dissolve the coagulated fluid and allow the sperm to begin moving upward through the cervix

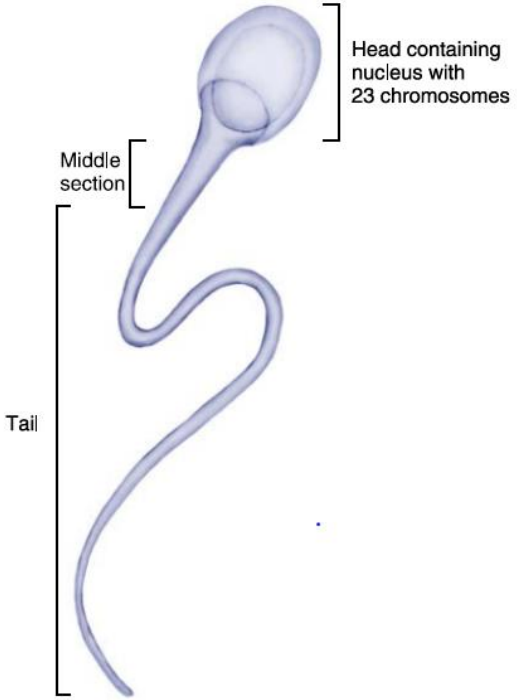


FIG. 5.2 Mature sperm.

- The whiplike movement of the tails of spermatozoa propels them through the cervix, uterus, and fallopian tubes.
- Uterine contractions induced by prostaglandins in the seminal fluid enhance movement ofthe sperm toward the ovum.
- Only sperm cells enter the cervix. The seminal fluid remains in the vagina.
- Sperm are not immediately ready to fertilize the ovum when they are ejaculated, the sperm undergo changes enabling one of them to penetrate the protective layers surrounding the ovum, a process called **capacitation**.
- During capacitation, a glycoprotein coat and seminal proteins are removed from the acrosome, which is the tip of the sperm head.
- After capacitation, the sperm look the same but are more active and can better penetrate the **corona radiata** and **zona pellucida** surrounding the ovum.
- The sperm that reach the ovum release **hyaluronidase and acrosin** to digest a pathway through the corona radiata and zona pellucida.
- Their tails beat to propel them toward the center of the ovum. Eventually, one spermatozoon penetrates the ovum.
- Fertilization is complete, and cell division can begin when the nuclei of the sperm and ovum unite.
- The preembryonic period is the first 2 weeks after conception. Around the fourth day after conception, the fertilized ovum, now called a zygote, enters the uterus.

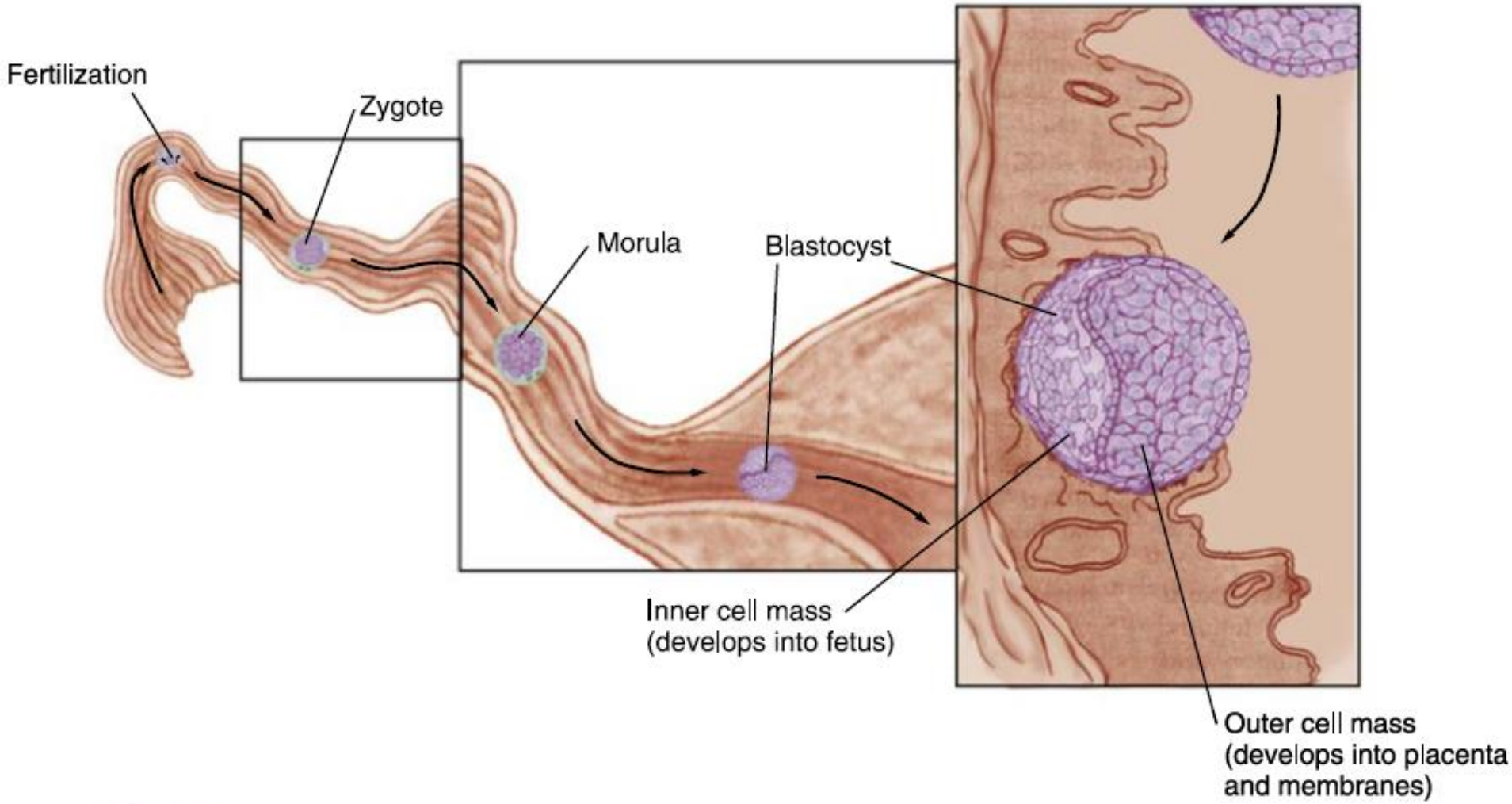


FIG. 5.4 Prenatal Development from Fertilization through Implantation of the Blastocyst. Implantation gradually occurs from day 6 through day 10. Implantation is complete on day 10.

Implantation in the Decidua

- Implantation, or nidation, is a gradual process that occurs between days 6 and 10 after conception.
- The zygote secretes human chorionic gonadotropin (hCG) to signal the woman's body that a pregnancy has begun. Production of hCG by the conceptus causes the corpus luteum to persist and continue secretion of estrogen and progesterone until the placenta takes over this function.
- During the embryonic period, structures are vulnerable to damage from teratogens because these structures are developing rapidly. Normal development of one structure often requires normal and properly timed development of another structure.

Weekly Developments

- Prenatal development proceeds in patterns that continue after birth:
  . Cephalocaudal (head to toe)
  . Central-to-peripheral direction (from center outward),
  . Simple to complex (early cells may become any cell of the then specialized into specific structures with specific functions)
  . General to specific (upper extremities begin as limb buds before detailed development of bones, joints, muscles, ligaments, and fingers)
- Full term ranges from 36 to 40 weeks of fertilization age calculated from date of conception or 38 to 42 weeks of gestational
- The embryo from the beginning of the third week through the eighth week after conception
- The fetal period begins 9 weeks after conception and ends with birth.

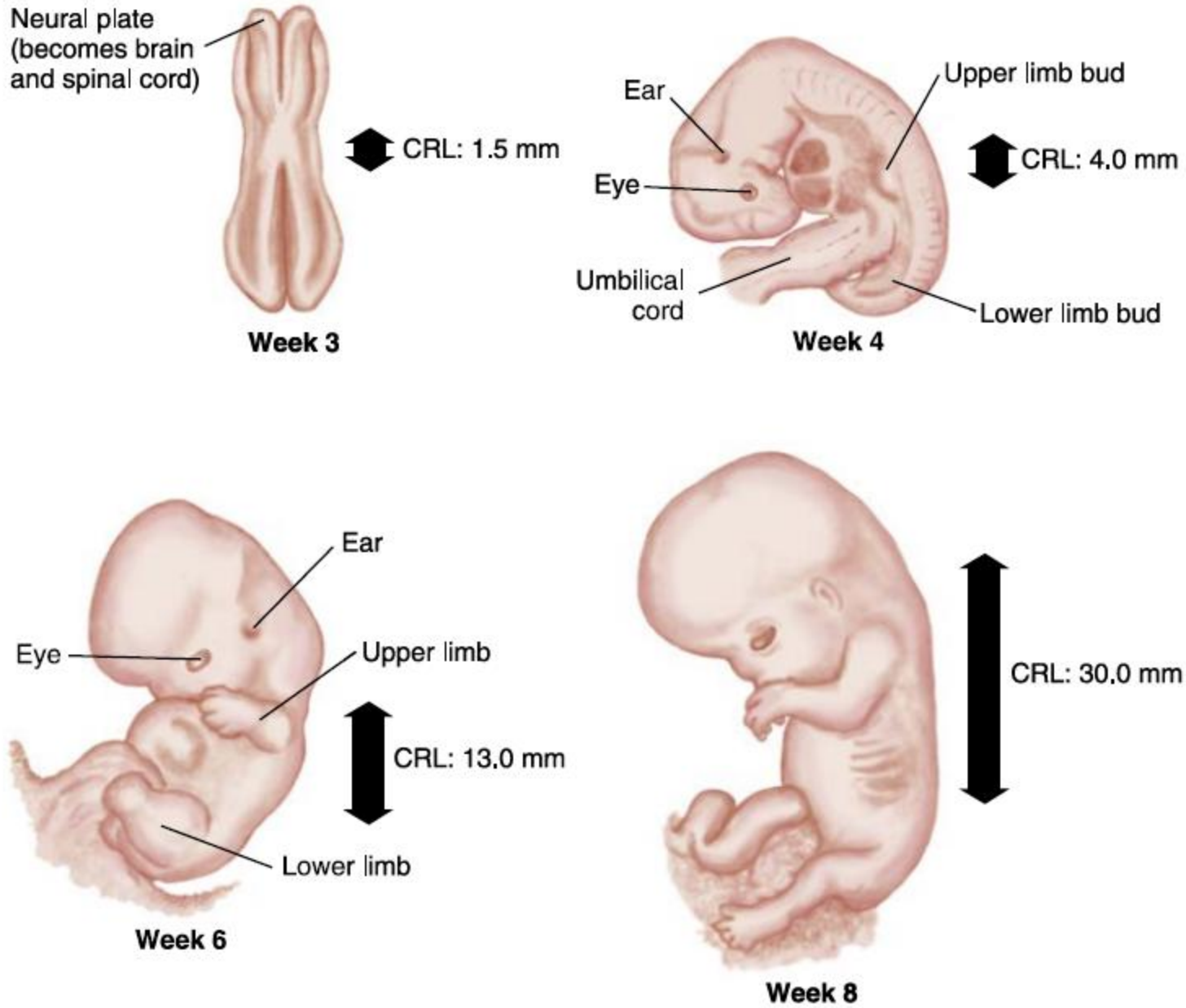

Neural plate
(becomes brain
and spinal cord)
CRL: 1.5 mm
Week 3
Ear
Upper limb bud
Eye
CRL: 4.0 mm
Umbilical
cord
Lower limb bud
Week 4
Ear
Eye
Upper limb
CRL: 13.0 mm
Lower limb
Week 6
CRL: 30.0 mm
Week 8

TABLE 5.2 Timetable of Prenatal Development Based on Fertilization Age*

Nervous/ Sensory System	Cardiorespiratory System	Digestive System	Genitourinary System	Musculoskeletal System	Integumentary System
3 Weeks: 1.5-mm CRL†					
Flat neural plate begins closing to form neural tube. Neural tube still open at each end.	Heart consists of two parallel tubes that fuse into a single tube. Contractions of heart tube begin. Chorionic villi of early placenta connect with heart.	Endoderm (inner germ layer) will become digestive tract.		Paired, cube-shaped swellings (somites) appear and will form most of head and trunk skeleton. Muscle, bone, and cartilage develop from mesoderm.	Epidermis (outer skin layer) will develop from ectoderm (outer germ layer). Dermis (deep skin layer) and connective tissue will develop from mesoderm (middle germ layer).
4 Weeks: 4-mm CRL					
Neural tube closed at each end. Cranial end of neural tube will form brain; caudal end will form spinal cord. Eye development begins as an outgrowth of forebrain. Nose development begins as two pits. Inner ear begins developing from hindbrain.	Heart begins partitioning into four chambers and begins beating. Blood circulating through embryonic vessels and chorionic villi. Tracheal development begins as a bud on upper gut and branches into two bronchial buds.	Development of primitive gut as embryo folds laterally. Stomach begins as a widening of tube-shaped primitive gut. Liver, gallbladder, and biliary ducts begin as a bud from primitive gut.	Primordial germ (reproductive) cells are present on embryonic yolk sac.	Upper limb buds are present and look like flippers. Lower limb buds appear.	Mammary ridges that will develop into mammary glands appear.

6 Weeks: 13-mm CRL

- Development of pituitary gland and cranial nerves.
- Head sharply flexed because of rapid brain growth.
- Eyelid development beginning.
- External ear development begins in neck region as six swellings.
- Blood formation primarily in liver.
- Three right and two left lung lobes develop as outgrowths of right and left bronchi.
- Partitioning of heart into four chambers completed.
- Most intestines are contained within umbilical cord because liver and kidneys occupy most of abdominal cavity.
- Stomach nearing final form.
- Development of upper and lower jaws.
- Kidneys are near bladder in pelvis.
- Kidneys occupy much of abdominal cavity.
- Primordial germ cells incorporated into developing gonads.
- Male and female gonads are identical in appearance.
- Arms are paddle shaped, and fingers webbed.
- Feet and toes develop similarly, but a few days later than arms and hands.
- Bones cartilaginous, but ossification of skull begins.
- Mammary glands begin development.
- Tooth buds for primary (deciduous) teeth begin developing.

8 Weeks: 30-mm CRL

- Spinal cord stops at end of vertebral column.
- Taste buds begin developing.
- Eyelids fuse.
- Ears have final form but are low-set.
- Heart partitioned into four chambers.
- Heartbeat detectable with ultrasound.
- Additional branching of bronchi.
- Stomach has reached final form.
- Lips are fused.
- Intestines remain in umbilical cord.
- Testes begin developing under influence of Y chromosome.
- Ovaries will develop if a Y chromosome is not present.
- External genitalia begin to differentiate but still appear quite similar.
- Fingers and toes still webbed, but distinct by end of week 8.
- Bones begin to ossify.
- Joints resemble those of adults.
- Auricles of ear low-set but beginning to assume final shape.

Nervous/ Sensory System	Cardiorespiratory System	Digestive System	Genitourinary System	Musculoskeletal System	Integumentary System
10 Weeks: 61-mm CRL; weight, 14 g†					
Head flexion still present, but straighter. Eyelids closed and fused. Top of external ear slightly below eye level.	May be possible to detect heartbeat with Doppler transducer. Blood produced in spleen and lymphatic tissue.	Intestines contained within abdominal cavity as growth of this cavity catches up with digestive system development. Digestive tract patent from mouth to anus.	Kidneys in their adult position. Male and female external genitalia have different appearance but are still easily confused.	Toes distinct; soles face each other.	Fingernails begin developing. Tooth buds for permanent teeth begin developing below those for primary teeth.
12 Weeks: 87-mm CRL; weight, 45 g					
Surface of brain is smooth, without sulci (grooves) or gyri (convolutions). Nasal septum and palate complete development.	Heartbeat should be detected with Doppler transducer.	Sucking reflex present. Bile formed by liver.	Kidneys begin producing urine. Male and female external genitalia can be distinguished by appearance.	Limbs are long and thin. Involuntary muscles of viscera develop.	Downy lanugo begins developing at end of this week.
16 Weeks: 140-mm CRL; weight, 200 g					
Face is human-looking because eyes face forward rather than to side.	Pulmonary vascular system developing rapidly.	Fetus swallows amniotic fluid and produces meconium (bowel contents).	Urine excreted into amniotic fluid.	Lower limbs reach final relative length, longer than upper limbs. A woman who has been pregnant before may	External ears have enough cartilage to stand away from head somewhat Blood vessels easily visible through delicate skin.

20 Weeks: 160-mm CRL; weight, 460 g

Myelination of nerves begins and continues through first year of postnatal life.	Heartbeat should be detectable with regular fetoscope.	Peristalsis well developed.	More than 40% of nephrons are mature and functioning. Testes contained in abdomen but begin descent toward scrotum.	Fetal movements felt by mother and may be palpable by an experienced examiner.	Skin is thin and covered with vernix caseosa. Brown fat production complete. Nipples begin development.

24 Weeks: 230-mm CRL; weight, 820 g

Spinal cord ends at level of first sacral vertebra because of more rapid growth of vertebral canal.	Primitive thin-walled alveoli (air sacs) have developed and are surrounded by capillary network. Surfactant production begins in lungs. Respiration possible, but many fetuses die or have disabilities if born at this time.		Testes descending toward inguinal rings.	Fetus is active. Fetal movements become progressively more noticeable to both mother and examiner.	Body appearance lean. Skin wrinkled and red. Fingerprints and footprints developed. Fingernails present. Eyebrows and lashes present.

28 Weeks: 270-mm CRL; weight, 1300 g

Major sulci and gyri are present. Eyelids no longer fused after 26 weeks.	Erythrocyte formation completely in bone marrow. Sufficient alveoli, surfactant, and capillary		Testes descended through inguinal canal into scrotum by end of week 26.		Skin slightly wrinkled but smoothing out as subcutaneous fat is deposited under it.

32 Weeks: 300-mm CRL; weight, 2100 g

Maturation of parasympathetic nears that of sympathetic nervous system, resulting in fetal heart rate variability on electronic fetal monitor tracing.

Surfactant production nears mature levels.
Respiratory distress still possible if born at 32 weeks.
Fetal heart rate variability gradually increases toward full term.

Skin smooth and pigmented.
Large vessels visible beneath skin.
Fingernails reach fingertips.
Lanugo disappearing.

38 Weeks: 360-mm CRL, weight, 3400 g

Sulci and gyri developed.
Visual acuity ~20/600 at birth.

Newborn infant has ~⅛ to ⅙ number of alveoli of an adult.
Well-developed ability to exchange gas.

Both testes usually palpable in scrotum at birth.
The newborn girl's ovaries contain ~1 million follicles. No new ones are formed after birth; their numbers continue to decline after birth.

Fetus plump, and skin smooth.
Vernix caseosa present in major body creases.
Lanugo present on shoulders and upper back only.
Fingernails extend beyond fingertips.
Ear cartilage firm.

The embryonic disc develops three layers, called germ layers, which, in turn, give rise to major organ systems of the body.

The three germlayers are the ectoderm, the mesoderm, and the endoderm.

TABLE 5.3 Derivatives of the Three Germ Layers

Ectoderm	Mesoderm	Endoderm
Brain and spinal cord	Cartilage	Lining of gastrointestinal and respiratory tracts
Peripheral nervous system	Bone	Tonsils
Pituitary gland	Connective tissue	Thyroid
Sensory epithelium of eye, ear, and nose	Muscle tissue	Parathyroid
Epidermis	Heart	Thymus
Hair	Blood vessels	Liver
Nails	Blood cells	Pancreas
Subcutaneous glands	Lymphatic system	Lining of urinary bladder and urethra
Mammary glands	Spleen	Lining of ear canal
Tooth enamel	Kidneys	
	Adrenal cortex	
	Ovaries	
	Testes	
	Reproductive system	
	Lining membranes (pericardial, pleural, and peritoneal)	

Fetal Membranes and Amniotic Fluid

- The two fetal membranes are the amnion (inner membrane) and the chorion (outer membrane).
- membranes rupture in labor, amnion and chorion usually rupture together, releasing amniotic fluid within the sac.
- Amniotic fluid protects the growing fetus and promotes normal prenatal development.
- Amniotic fluid protects the fetus by the following actions:
  - . Cushioning against impacts to the maternal abdomen
  - . Maintaining a stable temperature
  - . Promoting normal prenatal development by the following actions:
  - . Allowing symmetric development as the major body surfaces fold toward the midline
  - . Preventing the membranes from adhering to developing fetal parts
  - . Allowing room and buoyancy for fetal movement
- Amniotic fluid is derived from fetal urine and fluid transported from the maternal blood across the amnion.

CONFIRMATION OF PREGNANCY

- Although many women undergo early ultrasonography to confirm the pregnancy, the diagnosis of pregnancy has traditionally been based on symptoms experienced by the woman as well as signs observed by a physician, nurse-midwife, or nurse practitioner.
- The signs and symptoms of pregnancy are grouped into three classifications: presumptive, probable, and positive indications.
- A diagnosis of pregnancy cannot be made solely on the presumptive or probable signs because they may have other causes, such as Chronic diseases, GE, ..etc..
- A definitive diagnosis of pregnancy can be based only on positive signs.

Presumptive Indications of Pregnancy

- Most, but not all, presumptive indications are subjective changes experienced and reported by the woman.
- These changes are the least reliable indicators of pregnancy because they can be caused by conditions other than pregnancy.

1. Amenorrhea

- Absence of menstruation (amenorrhea) in a sexually active woman who menstruates regularly is one of the first changes noted and strongly suggests that conception has occurred.
- A small amount of bleeding from implantation of the blastocyst may cause the woman to think she is having a period. This occurs in 30% to 40% of pregnant women

2. Nausea and Vomiting

- Approximately 60% to 80% of women experience nausea and vomiting that begin at 4 to 8 weeks of pregnancy and resolve at 10 to 12 weeks
- Nausea and vomiting are believed to be caused by the increased hormones (such as hCG and estrogen) and decreased gastric motility (an effect of progesterone).

3. Fatigue

- Many pregnant women experience fatigue and drowsiness during the first trimester. The direct cause is unknown, but it may be from changes in hormones such as progesterone.

4. **Urinary Frequency**

- Urinary frequency begins in the first few weeks of pregnancy and results from hormonal and fluid volume changes.
- It continues later as the expanding uterus exerts pressure on the bladder. Late in the third trimester, the fetus settles into the pelvic cavity and causes more frequency and urgency of urination as the uterus presses against the bladder.

5. Breast and Skin Changes

- Breast changes begin by the 6th week of pregnancy (breast tenderness, tingling, feelings of fullness, and increased size and pigmentation of the areolae).
- Pigmentation of the skin (such as melasma, linea nigra, darkening of the areolae of the breasts) during pregnancy. These skin changes are the result of estrogen and progesterone on melanocytes

6. Vaginal and Cervical Color Change

7. Fetal Movement

- fetal movement (quickening) is not perceived until the second trimester which gradually increase in intensity between 16 and 20 weeks of gestation

TABLE 6.2 Indications of Pregnancy and Other Possible Causes

Sign	Other Possible Causes
Presumptive Indications	
Amenorrhea	Emotional stress, strenuous physical exercise, endocrine problems, chronic disease, early menopause, anovulation, low body weight
Nausea and vomiting	Gastrointestinal virus, food poisoning, emotional stress
Fatigue	Illness, stress, sudden changes in lifestyle
Urinary frequency	Urinary tract infection
Breast and skin changes	Premenstrual changes, use of oral contraceptives
Chadwick's sign	Infection or hormonal imbalance causing pelvic congestion
Quickening	Intestinal gas, peristalsis, or pseudocyesis (false pregnancy)

Probable Indications of Pregnancy

- Probable indications of pregnancy are objective findings that can be documented by an examiner. They are primarily related to physical changes in the reproductive organs. Although these signs are stronger indicators of pregnancy, a positive diagnosis cannot be made because they may have other causes.

1. Abdominal Enlargement

- Enlargement of the abdomen during the childbearing years is generally a reliable indication of pregnancy,
- Pregnancy is even more likely when abdominal enlargement is accompanied by amenorrhea.

2. Cervical Softening

- In the early weeks of pregnancy, the cervix softens because of pelvic vasocongestion (Goodell's sign). Cervical softening is noted during pelvic examination

3. Changes in Uterine Consistency

- About 6 to 8 weeks after the last menses, the lower uterine segment is so soft it can be compressed to the thinness of paper. This is called the Hegar's sign. The body of the uterus can be easily flexed against the cervix.

4. Ballottement

Near midpregnancy, a sudden tap on the cervix during vaginal examination may cause the fetus to rise in the amniotic fluid and then rebound to its original position

This movement, called ballottement, is a strong indication of pregnancy, but it may also be caused by other factors such as uterine or cervical polyps.

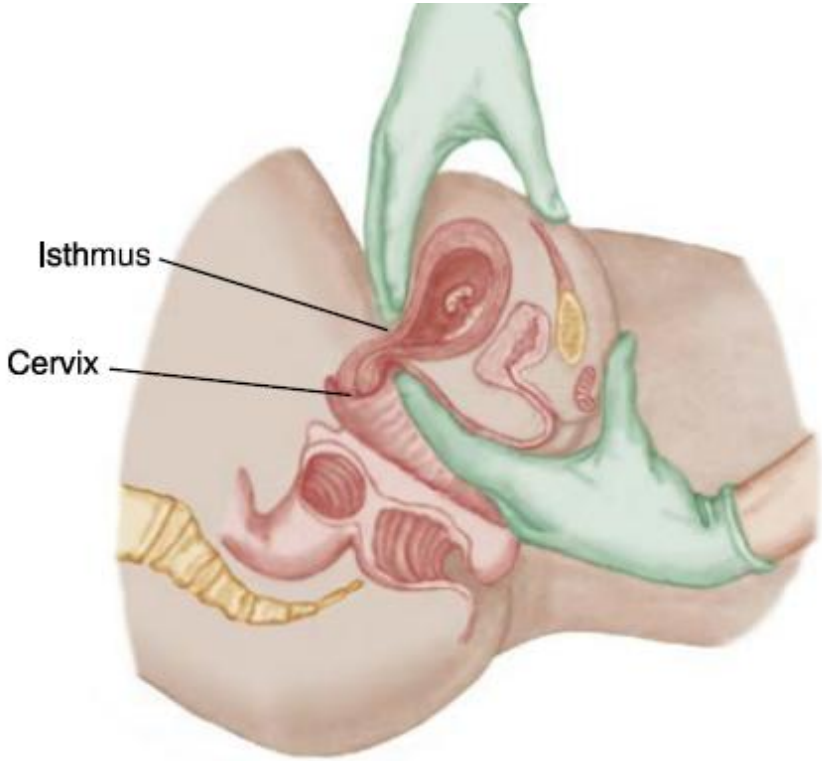
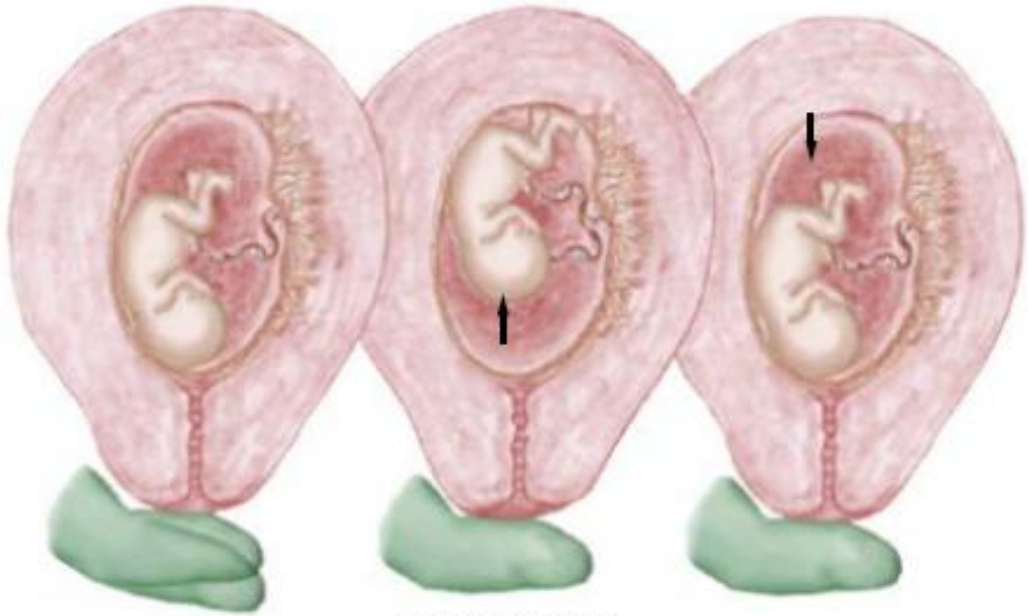


FIG. 6.9 Hegar's sign demonstrates softening of the isthmus of the cervix.



Ballottement

FIG. 6.10 When the cervix is tapped, the fetus floats upward in the amniotic fluid. A rebound is felt by the examiner when the fetus falls back.

Probable Indications of Pregnancy

5. Braxton Hicks Contractions

- Irregular, painless contractions occur at the third trimester.
- It is important to differentiate Braxton Hicks contractions from the contractions of preterm labor.

6. Palpation of the Fetal Outline

- An experienced practitioner can palpate the outlines of the fetal body by the middle of pregnancy. Outlining the fetus becomes easier as the uterine walls thin to accommodate the growing fetus.

7. Uterine Souffle

- Late in pregnancy, the uterine souffle—a soft, blowing sound—may be auscultated over the uterus. This is the sound of blood circulating through the dilated uterine vessels, and it corresponds to the maternal pulse.
- Therefore to identify the uterine souffle, the rate of the maternal pulse should be checked simultaneously. Uterine souffle differs from funic souffle—the soft, whistling sound heard over the umbilical cord and corresponding to the fetal heart rate.

8. Pregnancy Tests

- Pregnancy tests detect hCG which is secreted by the placenta and present in maternal blood and urine shortly after conception.
- They can detect hCG at very low concentrations and are positive as early as 3 to 7 days after conception
- The first morning void is preferred because it is most concentrated.
- Inaccurate Pregnancy Test Results:
  - When pregnancy test results are reported as negative and the woman is in fact pregnant, the results are called false-negative. False-negative results may occur when the instructions are not followed properly, it is too early in the pregnancy, the urine is too dilute, or the woman is taking drugs such as diuretics.
  - Hematuria, proteinuria, or some tumors may cause false-positive results, in which the test indicates a pregnancy when the woman is not pregnant. Some anticonvulsants, antiparkinsonian drugs, tranquilizers, and hypnotics also may cause a false-positive result

Probable Indications	
Abdominal enlargement	Abdominal or uterine tumors
Goodell's sign	Hormonal contraceptives or imbalance
Hegar's sign	Hormonal imbalance
Ballottement	Uterine or cervical polyps
Braxton Hicks contractions	Intestinal gas
Palpation of fetal outline	Large leiomyoma that feels like the fetal head, small soft leiomyoma that simulates fetal body parts
Uterine souffle	Confusion with the mother's pulse
Positive pregnancy test	Hematuria, proteinuria, some medications, tumors that produce human chorionic gonadotropin, some drugs

Positive Indications of Pregnancy

- Only three signs are accepted as positive confirmation of pregnancy: auscultation of fetal heart sounds, fetal movement detected by an examiner, and visualization of the embryo or fetus.
- **Auscultation of Fetal Heart Sounds** can be heard with a fetoscope by 18 to 20 weeks of gestation
- The electronic Doppler, detects heart motion and makes an audible sound by as early as 6 weeks of gestation
- Visualization of the Embryo or Fetus Confirmation of pregnancy makes it possible to view the embryo or fetus and observe the fetal heartbeat very early in pregnancy. Positive confirmation of pregnancy is possible by transvaginal ultrasonography as early as 3 to 4 weeks of gestation

Positive Indications
Auscultation of fetal heart sounds
Fetal movements detected by an examiner
Visualization of embryo or fetus

- The objective of antepartum care is to promote optimum health of the mother and infant. Prenatal care involves early and continuing risk assessments, health education, counseling, and social support.
- Ideally, antepartum care begins before conception and continues on a regular basis until birth
- Ideally, the first visit takes place before conception.
- Preconception care (the period of time prior to conception) and interconception care (the period of time between pregnancies) are important to identify problems or risk factors that might harm the mother or the infant once pregnancy occur

The following are the primary objectives of the first antepartum visit:

- . Establish trust and rapport with the childbearing family.
- . Verify or rule out pregnancy.
- . Evaluate the pregnant woman's physical and psychological health relevant to childbearing.
- . Assess the growth and health of the fetus.
- . Establish baseline data for comparison with future observations.
- . Evaluate the psychosocial needs of the woman and her family.
- . Assess the need for counseling or teaching.
- . Negotiate a plan of care to promote optimal health of the mother and baby.

- Gravida refers to a woman who is or has been pregnant,regardless of the length of the pregnancy.
- A primigravida is a woman pregnant for the first time.
- A multigravida has been pregnant more than once.
- Para refers to the number of pregnancies that have ended at 20 or more weeks, regardless of whether the infant was born alive or stillborn.
- A multiple gestation pregnancy, such as twins or triplets, is still one pregnancy and therefore one para.
- A nullipara is a woman who has never been pregnant or has not completed a pregnancy of 20 weeks or more.
- A primipara has delivered one pregnancy of at least 20 weeks.
- A multipara has delivered two or more pregnancies of at least 20 weeks.

There are two methods of summarizing a woman's obstetric history (gravida and para).

1. The first uses two digits, gravida (the number of times she has been pregnant) and para (the number of completed pregnancies of 20 weeks or more).
2. The second method, described by the acronym GTPAL, is more informative because it uses five digits: G = pregnancies or gravida, T 2 term pregnancies delivered, P = preterm pregnancies delivered, A = abortions (spontaneous and induced), and L 2 living children

- Carpal Tunnel Syndrome. Carpal tunnel syndrome is believed to result when edema compresses the median nerve at the point where it passes through the carpal tunnel of the wrist.
- Symptoms include pain, burning, numbness, or tingling of the hand and wrist. Splinting of the wrist at night may provide improvement. It resolves by 3 months after childbirth for most women



FIG. 7.1 Uterine measurements include the distance between the upper border of the symphysis pubis and the top of the fundus.

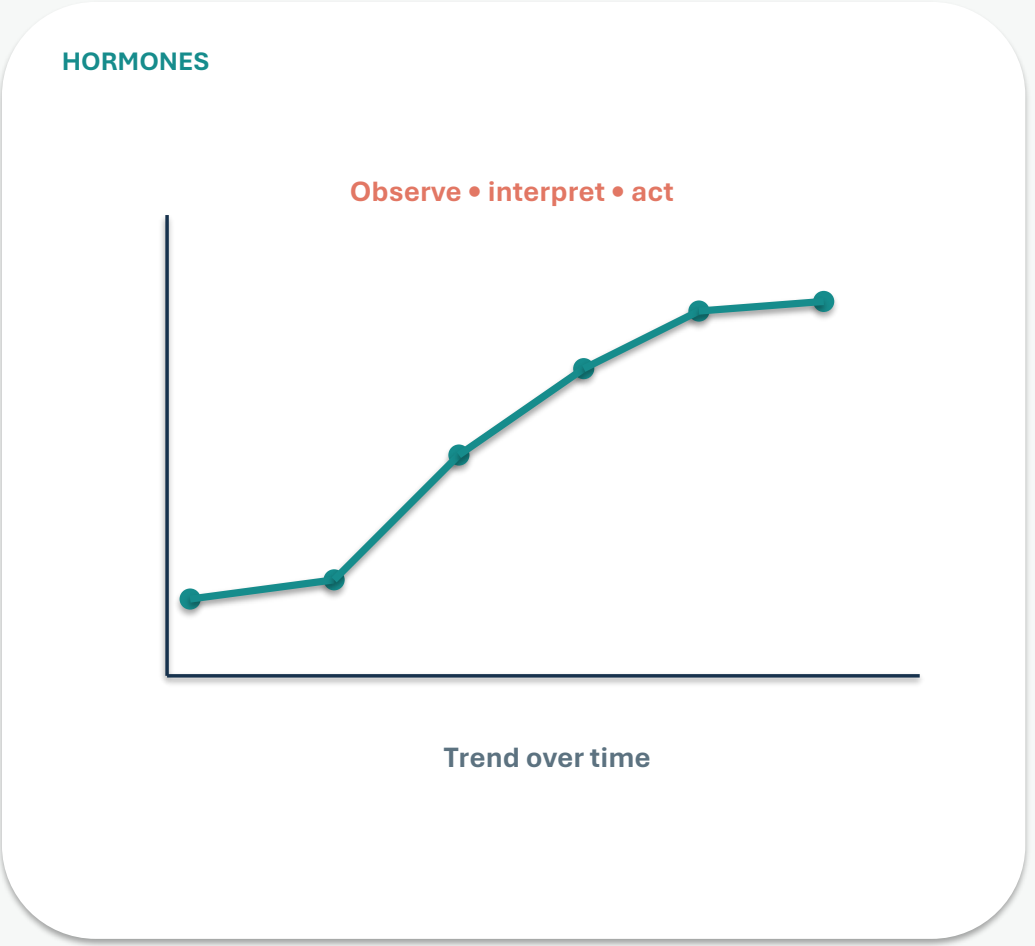
Maternal adaptations



10 focused lessons • assessment, teaching, and clinical reasoning

Hormonal adaptation

- hCG supports early pregnancy; progesterone helps maintain the uterine environment.
- Estrogen supports uterine and breast growth.
- Relaxin and other mediators influence connective tissue and vascular tone.

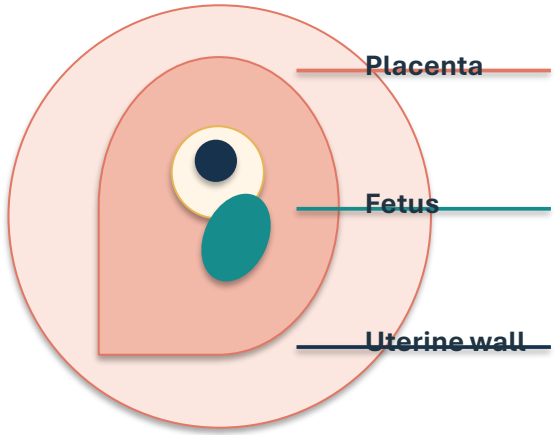


Sources: OpenStax

Uterus, cervix, and vagina

- The uterus enlarges and becomes more vascular as pregnancy advances.
- The cervix softens and forms a mucus plug.
- Increased vaginal discharge can be normal; odor, pain, or itching needs assessment.

REPRO ADAPT



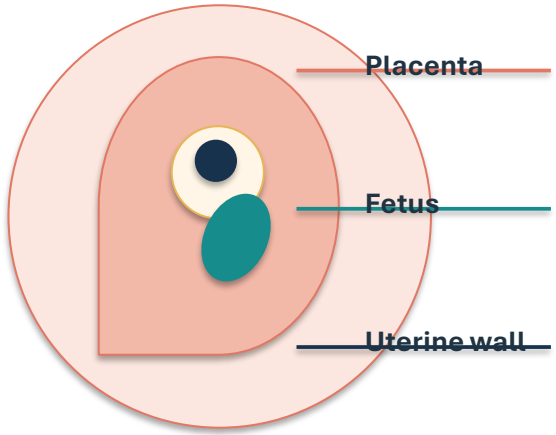
Schematic • not to scale

Sources: OpenStax

Breast changes in pregnancy

- Breast fullness, tenderness, and areolar changes are common.
- Colostrum may appear before birth but is not required for breastfeeding success.
- Assess lumps, skin changes, or focal pain rather than dismissing symptoms.

BREAST



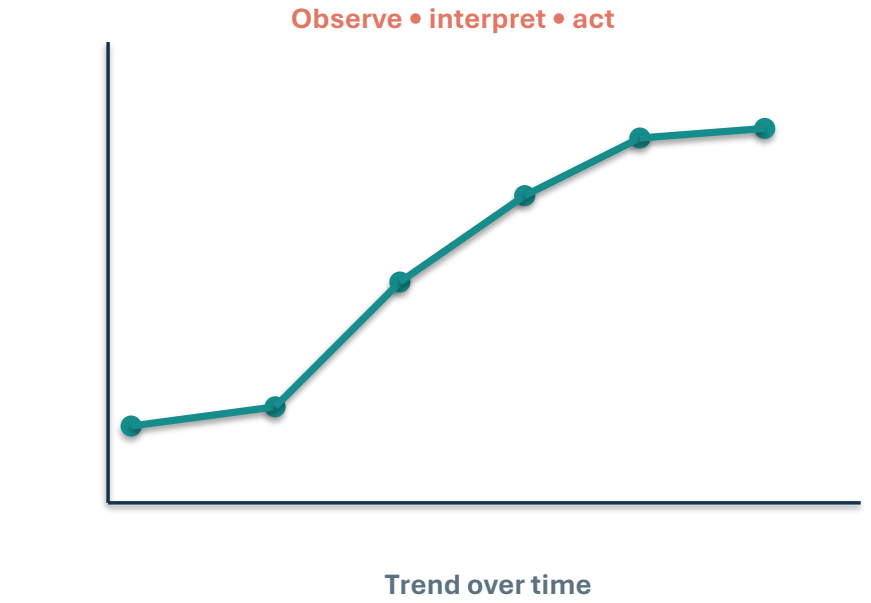
Schematic • not to scale

Sources: ACOG

Cardiovascular changes

- Plasma volume and cardiac output rise to support maternal-fetal circulation.
- Blood pressure may fall mid-pregnancy; supine hypotension can occur later.
- Pregnancy increases venous stasis and clotting tendency.

CARDIO



Sources: OpenStax

Hematologic changes

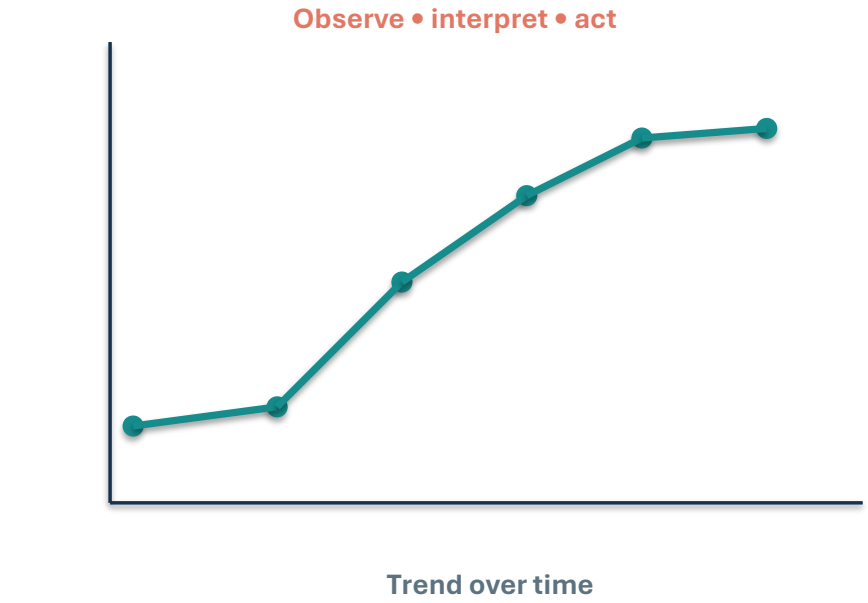
- Plasma expansion can lower hemoglobin concentration: physiologic hemodilution.
- Iron needs increase as maternal red cell mass and fetal stores grow.
- Thromboembolism risk rises during pregnancy and postpartum.



Sources: ACOG

Respiratory changes

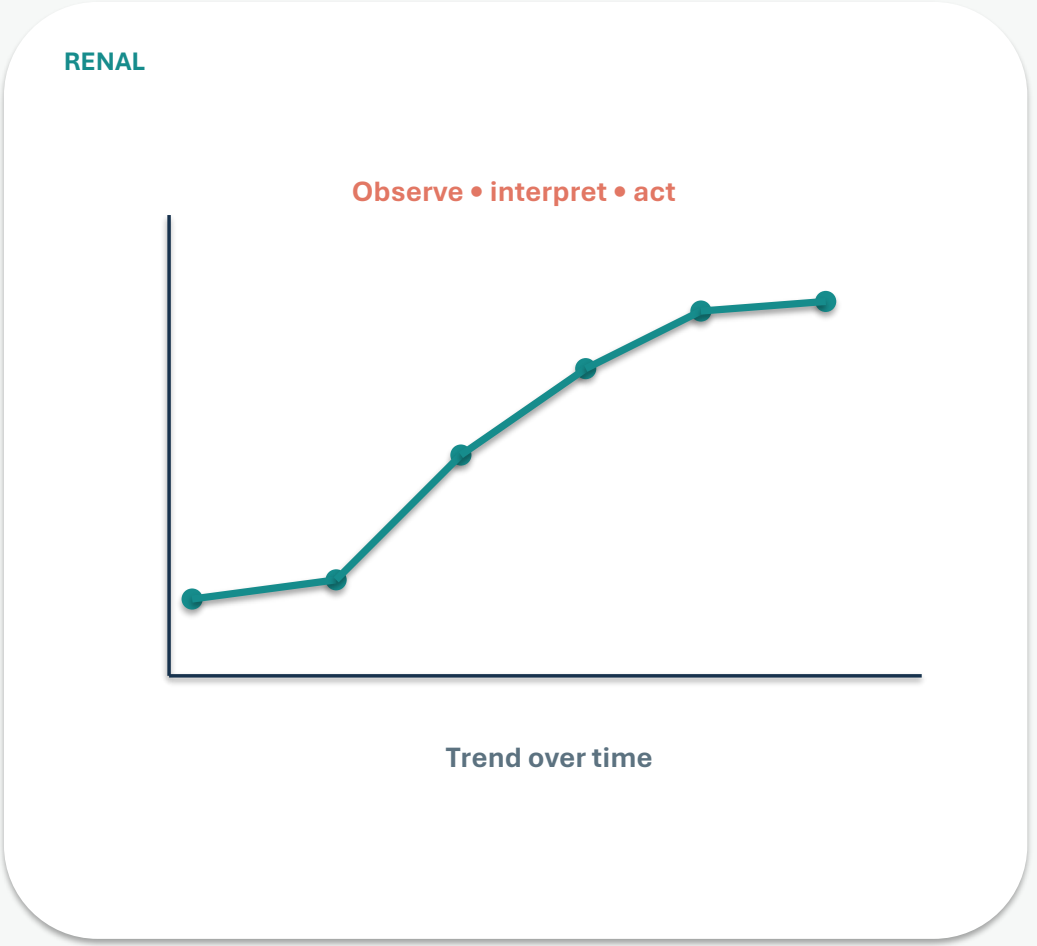
- Oxygen demand and minute ventilation increase in pregnancy.
- Mild breathlessness can occur; sudden or severe dyspnea is urgent.
- The diaphragm rises as the uterus enlarges, while chest dimensions adapt.



Sources: OpenStax

Renal and urinary changes

- Renal blood flow and filtration increase.
- Urinary frequency is common, but dysuria or fever suggests infection.
- Urine culture screening and treatment follow local recommendations.



Sources: ACOG

Gastrointestinal changes

- Progesterone slows motility, contributing to reflux and constipation.
- Nausea and vomiting are common early symptoms.
- Persistent vomiting, dehydration, or weight loss needs prompt evaluation.



Sources: ACOG

Musculoskeletal and skin changes

- Posture, gait, and joint mobility adapt to a growing uterus.
- Linea nigra, melasma, and striae can occur.
- New severe headache, one-sided swelling, or focal pain requires assessment.

CLINICAL IMAGE



Supplied course photo: pregnancy skin changes

Sources: OpenStax

Physiologic Adaptations During Pregnancy

- Every system of a woman's body changes during pregnancy to accommodate the needs of the growing fetus, and with startling rapidity.
- These changes are necessary to support and nourish the fetus, prepare the woman for childbirth and lactation, and maintain the woman's healthThe physical changes of pregnancy can be uncomfortable.
- Many adaptations are reversible after the woman gives birth, but some persist for life.

Reproductive System Adaptations (uterus)

- Uterus During the first few months of pregnancy, estrogen stimulates uterine growth, and the uterus undergoing increase in size, weight, length, width, depth, volume, and overall capacity throughout pregnancy.

- The weight of the uterus increases from 70 g to about 1100 g at term; its capacity increases from 10 mL to 5000 mL or more at term

- Blood vessels elongate, enlarge, dilate, and sprout new branches to support and nourish the growing muscle tissue and accommodate the growing fetus

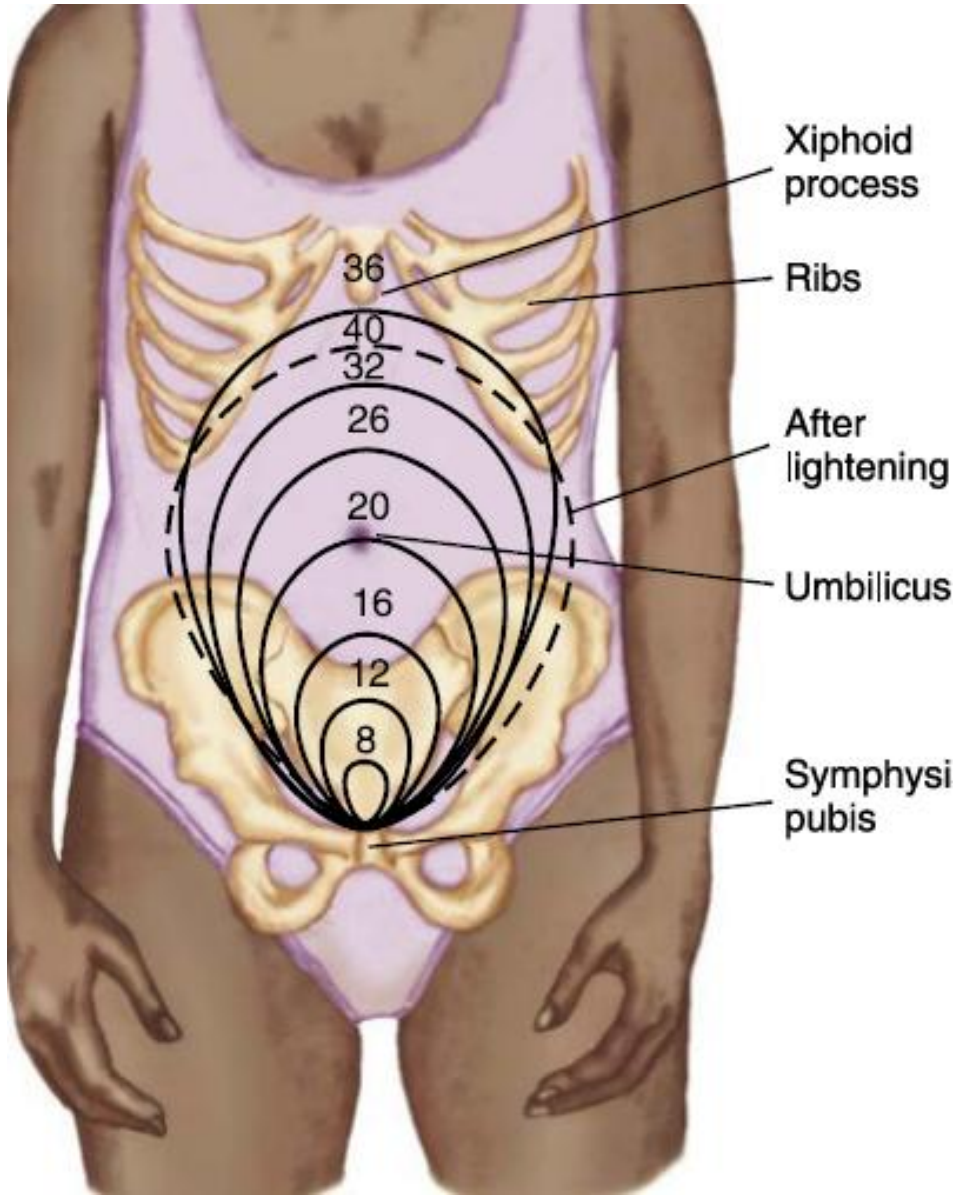
Reproductive System Adaptations (uterus)

- By 20 weeks' gestation, the fundus, or top of the uterus, is at the level of the umbilicus and measures 20 cm.

- The fundus reaches its highest level, at the xiphoid process, at approximately 36 weeks.

- Between 38 to 40 weeks, fundal height drops as the fetus begins to descend and engage into the pelvis

- By 40 weeks, the fetal head begins to descend and engage in the pelvis, which is termed lightening


Xiphoid process
Ribs
After lightening
Umbilicus
Symphysi pubis
36
40
32
26
20
16
12
8

Reproductive System Adaptations (Cervix)

- 6 and 8 wk of pregnancy, the cervix begins to soften (Goodell's sign) due to vasocongestion.
- Endocervical glands increase in size and number and produce more cervical mucus.
- By the influence of progesterone, a thick mucus plug is formed that blocks the cervical os and protects the opening from bacterial invasion.
- Increased vascularization of the cervix causes Chadwick's sign.
- Cervical ripening (softening, effacement, and increased distensibility) begins about 4 weeks before birth. Ripening is an inflammatory process and is not dependent on uterine contractions

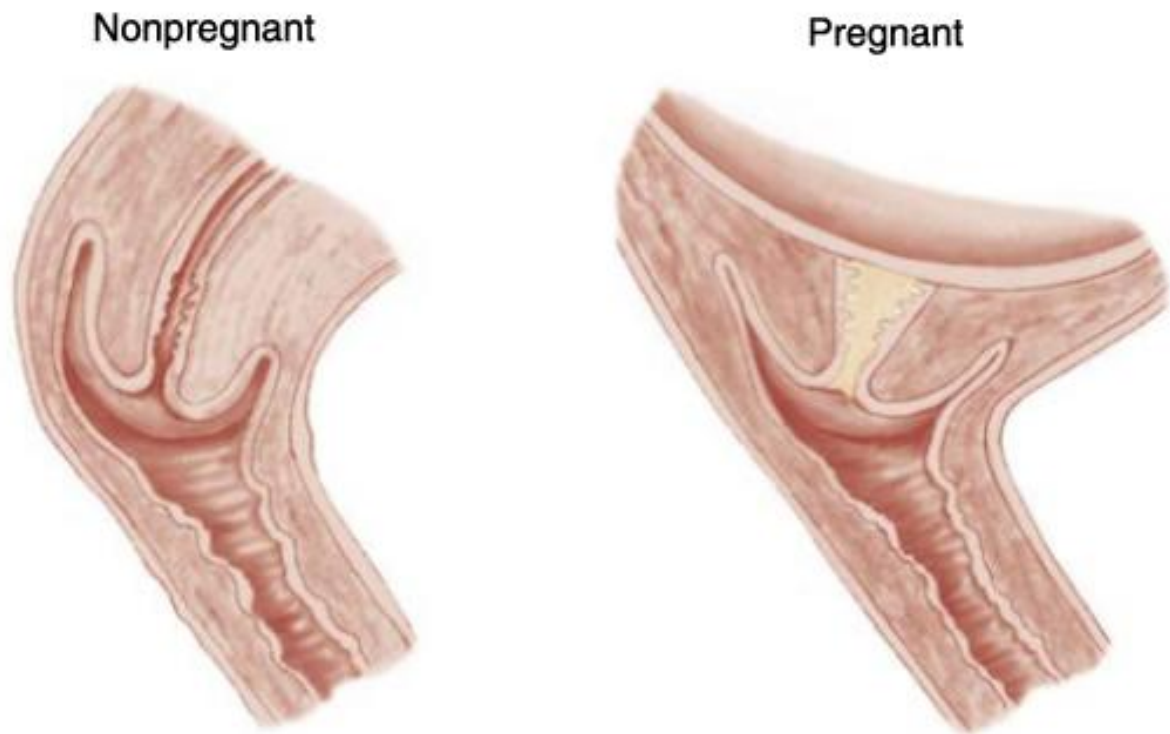


FIG. 6.2 Cervical Changes That Occur during Pregnancy. Note the thick mucous plug filling the cervical canal.

Reproductive System Adaptations (Vagina)

- The influences of estrogen, resulting in pelvic congestion and hypertrophy of the vagina in preparation for the distention needed for birth.
- The vaginal mucosa thickens, the connective tissue begins to loosen, the smooth muscle begins to hypertrophy, and the vaginal vault begins to lengthen.
- Vaginal secretions become more acidic, white, and thick.
- Most women experience an increase in a whitish vaginal discharge, called leukorrhea.

Reproductive System Adaptations (Vagina)

- This is normal except when it is accompanied by itching and irritation, possibly suggesting Candida albicans, a monilial vaginitis, which is a very common occurrence in this glycogen-rich environment.
- Monilial vaginitis is a benign fungal condition that is uncomfortable for the woman, but it can be transmitted from an infected mother to her newborn at birth.
- Neonates develop an oral infection known as thrush, which presents as white patches on the mucous membranes of their mouths.
- It is self-limiting and is treated with local antifungal agents

Reproductive System Adaptations (Ovaries**)**

- The increased blood supply to the ovaries causes them to enlarge until approximately the 12th to 14th week of gestation.
- The ovaries are not palpable after that time because the uterus fills the pelvic cavity.
- Ovulation ceases during pregnancy because of the elevated levels of estrogen and progesterone, which block secretion of FSH and LH from the anterior pituitary.
- The ovaries are very active in hormone production to support the pregnancy until about weeks 6 to 7, when the corpus luteum regresses and the placenta takes over the major production of progesterone.

Reproductive System Adaptations (Breasts**)**

- The breasts increase in fullness, become tender, and grow larger under the influence of estrogen and progesterone.
- The breasts become highly vascular, and veins become visible under the skin.
- The nipples become larger and more erect and the areola become deeply pigmented
- Changes that occur in the connective tissue of the breasts, along with the tremendous growth, lead to striae (stretch marks) in approximately half of all pregnant women .
- Creamy, yellowish breast fluid called colostrum can be expressed by the third trimester. This fluid provides nourishment for the breastfeeding newborn during the first few days of life

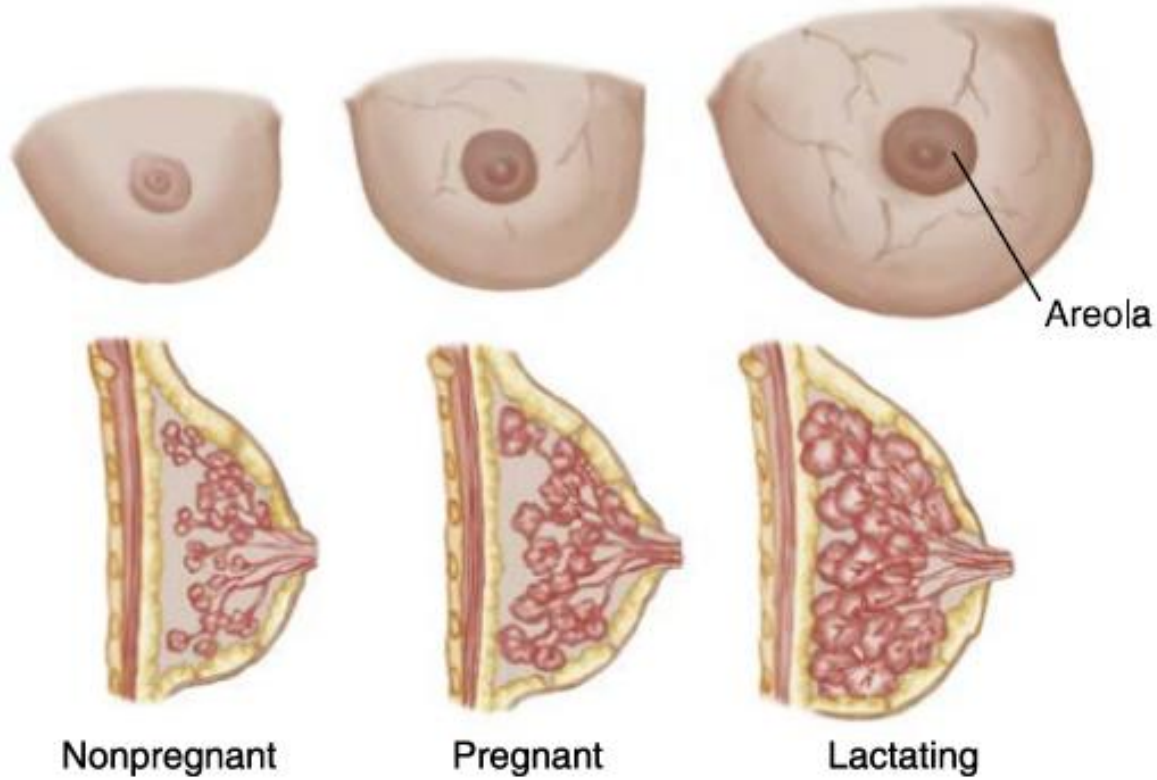


FIG. 6.3 Breast Changes That Occur during Pregnancy. The breasts increase in size and become more vascular, the areolae become darker, and the nipples become more erect.

- During pregnancy, the gums become hyperemic, swollen, and friable and tend to bleed easily. This change is influenced by estrogen and increased proliferation of blood vessels and circulation to the mouth.
- In addition, the saliva produced in the mouth becomes more acidic. Some women complain about excessive salivation, termed ptyalism, which may be caused by the decrease in unconscious swallowing by the woman when nauseated.
- Dental plaque, calculus, and debris deposits increase and are all associated with gingivitis.

- Elevated progesterone levels cause smooth muscle relaxation, which results in delayed gastric emptying and decreased peristalsis.
- Transition time of food throughout the GI tract may be so much slower that more water than normal is reabsorbed, leading to bloating and constipation.
- Constipation can also result from low-fiber food choices, reduced fluid intake, use of iron supplements, decreased activity level, and intestinal displacement secondary to a growing uterus. Constipation, increased venous pressure, and the pressure of the gravid uterus contribute to the formation of hemorrhoids.

- The slowed gastric emptying combined with relaxation of the cardiac sphincter allows reflux, which causes heartburn. Antacids will usually relieve the symptoms
- The emptying time of the gallbladder is prolonged secondary to the smooth muscle relaxation from progesterone. Hypercholesterolemia can follow, increasing the risk of gallstone formation.
- Nausea and vomiting, better known as morning sickness, plagues about 50% to 80% of pregnant womens is between 6 and 12 weeks. The physiologic basis for morning sickness is still debatable. It has been linked to the high levels of hCG, high levels of circulating estrogens, reduced stomach acidity, and the lowered tone and motility of the digestive tract

Blood Volume

- Blood volume increases by approximately 1,500 mL
- It begins at weeks 10 to 12, peaks at weeks 32 to 34, and decreases slightly at week 40.
- This increase in blood volume is needed to provide adequate hydration of fetal and maternal tissues, to supply blood flow to perfuse the enlarging uterus, and to provide a reserve to compensate for blood loss at birth and during postpartum.
- This increase is also necessary to meet the increased metabolic needs of the mother and to meet the need for increased perfusion of other organs, especially the woman’s kidneys, because she is excreting waste products for herself and the fetus.

Cardiac Output and Heart Rate

- Cardiac output is the product of stroke volume and heart rate.
- It increases from 30% to 50% over the nonpregnant rate by the 32nd week of pregnancy and declines to about a 20% increase at 40 weeks' gestation. The increase in cardiac output is associated with an increase in venous return and greater right ventricular output
- HR increases by 10-15 bpm between 14 and 20 weeks of gestation, and this persists to term.
- Blood pressure, especially the diastolic pressure, declines slightly during pregnancy as a result of peripheral vasodilation caused by progesterone

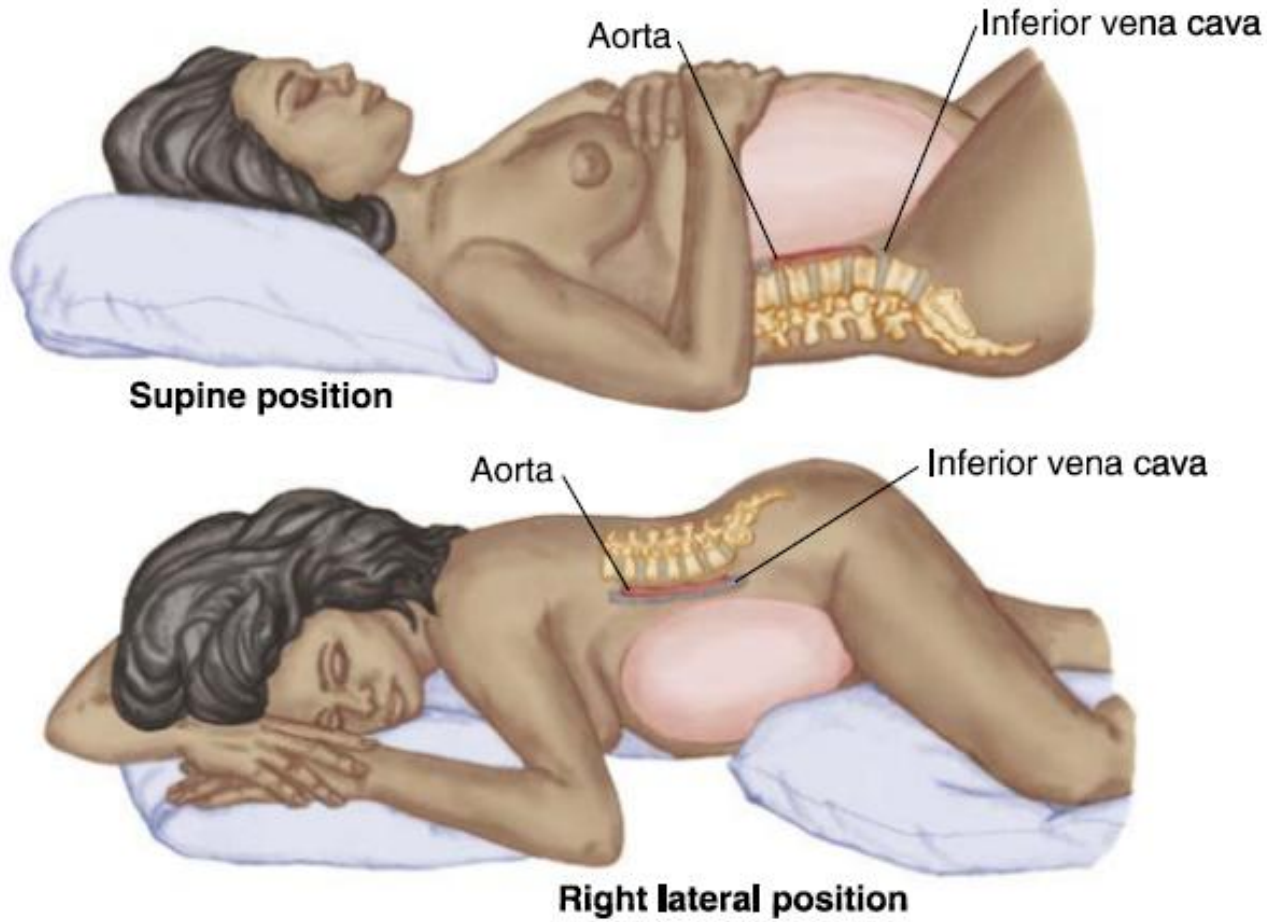


FIG. 6.4 Supine Hypotensive Syndrome. When the pregnant woman is in the supine position, the weight of the uterus partially occludes the vena cava and the aorta. The side-lying position corrects supine hypotension.

- The number of RBCs also increases throughout pregnancy to a level 25% to 33% higher than nonpregnant values, depending on the amount of iron available. This increase is necessary to transport the additional oxygen required during pregnancy.
- Changes in RBC volume are due to increased circulating erythropoietin and accelerated RBC production. The rise in erythropoietin in the last two trimesters is stimulated by progesterone, prolactin, and human placental lactogen.
- Iron requirements during pregnancy increase because of the demands of the growing fetus and the increase in maternal blood volume.

- Both fibrin and plasma fibrinogen levels increase, along with various blood-clotting factors.
- These factors make pregnancy a hypercoagulable state. These changes, coupled with venous stasis secondary to venous pooling, which occurs during late pregnancy after long periods of standing in the upright position with the pressure exerted by the uterus on the large pelvic veins, contribute to slowed venous return, pooling, and dependent edema.
- These factors also increase the woman's risk for venous thrombosis.

- The growing uterus and the increased production of the hormone progesterone cause the lungs to function differently during pregnancy.
- The growing uterus does change the size and shape of the thoracic cavity, but diaphragmatic excursion increases, chest circumference increases by 2 to 3 in
- A pregnant woman breathes faster and more deeply because she and the fetus need more oxygen.
- Increased vascularity of the respiratory tract is influenced by increased estrogen levels, leading to congestion. This congestion gives rise to nasal and sinus stuffiness, epistaxis

- Changes in renal structure occur from hormonal influences of estrogen and progesterone, pressure from an enlarging uterus, and an increase in maternal blood volume.
- Changes in kidney function occur to accommodate a heavier workload while maintaining a stable electrolyte balance and blood pressure.
- As more blood flows to the kidneys, the glomerular filtration rate (GFR) increases, leading to an increase in urine flow and volume, substances delivered to the kidneys,

- The activity of the kidneys normally increases when a person lies down and decreases on standing. This difference is amplified during pregnancy, which is one reason a pregnant woman feels the need to urinate frequently while trying to sleep.
- Late in the pregnancy, the increase in kidney activity is even greater when the woman lies on her side rather than her back. Lying on the side relieves the pressure that the enlarged uterus puts on the vena cava carrying blood from the legs. Subsequently, venous return to the heart increases, leading to increased cardiac output. Increased cardiac output results in increased renal perfusion and glomerular filtration.

- Pregnancy is characterized by changes in posture and gait.
- The relaxation of the joints peaks by the beginning of the third trimester. The purpose of these changes is to increase the size of the pelvic cavity and to make delivery easier.
- The postural changes of pregnancy—an increased swayback and an upper spine extension to compensate for the enlarging abdomen
- Relaxation and increased mobility of joints occur because of the hormones progesterone and relaxin, which lead to the characteristic “waddle gait”

- The skin of pregnant women undergoes hyperpigmentation primarily as a result of estrogen, progesterone, and melanocyte-stimulating hormone levels.
- These changes are mainly seen on the nipples, areola, umbilicus, perineum, and axilla.
- Many pregnant women express stretch marks, skin color changes, and hair loss.
- Unfortunately, little is known about how to avoid these changes.
- The increased pigmentation that occurs on the breasts and genitalia also develops on the face to form the “mask of pregnancy,” or facial melasma.
- There is a genetic predisposition toward melasma, which is exacerbated by the sun.

- The skin in the middle of the abdomen may develop a pigmented line called linea nigra, which extends from the umbilicus to the pubic area.
- Striae gravidarum, or stretch marks, are irregular reddish streaks that appear on the abdomen, breasts, and buttocks in about half of pregnant women. Striae are most prominent by 6 to 7 months. They result from reduced connective tissue strength resulting from the elevated adrenal steroid levels and stretching of the structures secondary to growth. They are more common in younger women, women with larger infants, and women with higher body mass indices. Nonwhites and women with a history of breast or thigh striae or a family history of striae gravidarum also are at higher risk



FIG. 6.6 **Striae Gravidarum.** Lineal tears that may occur in connective tissue.

facial melasma





Linea nigra.

Integumentary System



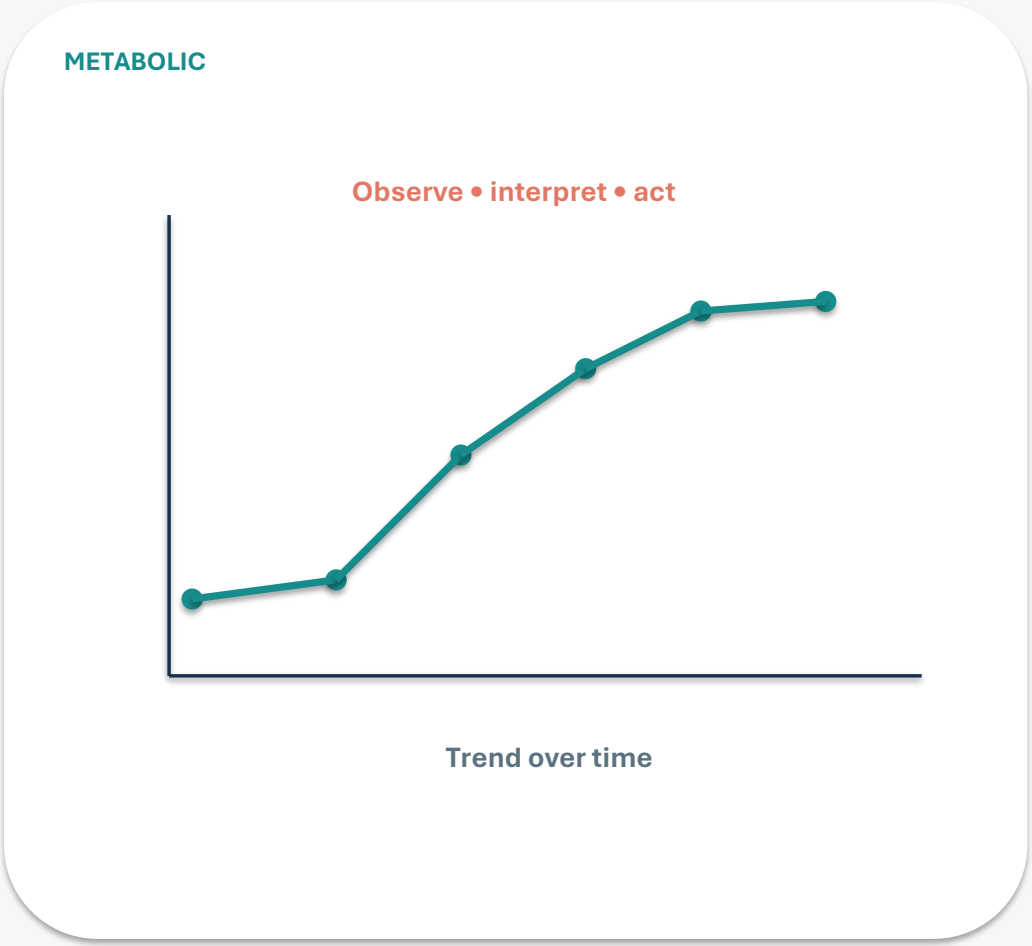
Palmar erythema



vascular spiders

Endocrine and metabolic changes

- Insulin resistance rises later in pregnancy to preserve glucose for the fetus.
- The thyroid and adrenal systems adapt to changing hormone demands.
- Screen for gestational diabetes according to local guidance.



Sources: ACOG

Maternal Weight Gain

- Currently, the American College of Obstetricians & Gynecologists (2005) recommends a 25- to 35-pound weight gain during pregnancy.
- Any woman who has a prepregnancy BMI of less than 19.8 is considered to be high risk and should be referred to a nutritionist. These women are encouraged to gain 28 to 40 pounds during the pregnancy
- Encourage the woman to eat snacks that are high in calories such as nuts, peanut, butter, milkshakes, cheese, fruit, yogurt, and ice cream.
- **Vegetarian diets,** Suggestions include:
  - For protein: substitute soy foods, beans, lentils, nuts, grains, and seeds.
  - For iron: eat a variety of meat alternatives, along with vitamin C-rich foods.
  - For calcium: substitute soy, calcium-fortified orange juice, and tofu.
  - For vitamin B12: eat fortified soy foods and a B12 supplement.

TABLE 11.6 Normal Distribution of Weight Gain During Pregnancy

Component	Weight
Fetus	7.5–8.5 pounds
Blood	4 pounds
Uterus	2 pounds
Breasts	1 pound
Placenta and umbilical cord	1.5 pounds
Fat and protein stores	7.5 pounds
Tissue fluids	2.7 pounds
Amniotic fluid	1.8 pounds
Approximate total weight gain	29 pounds

Sources: ACOG, 2005; Blackburn, 2007; Dudek, 2006.

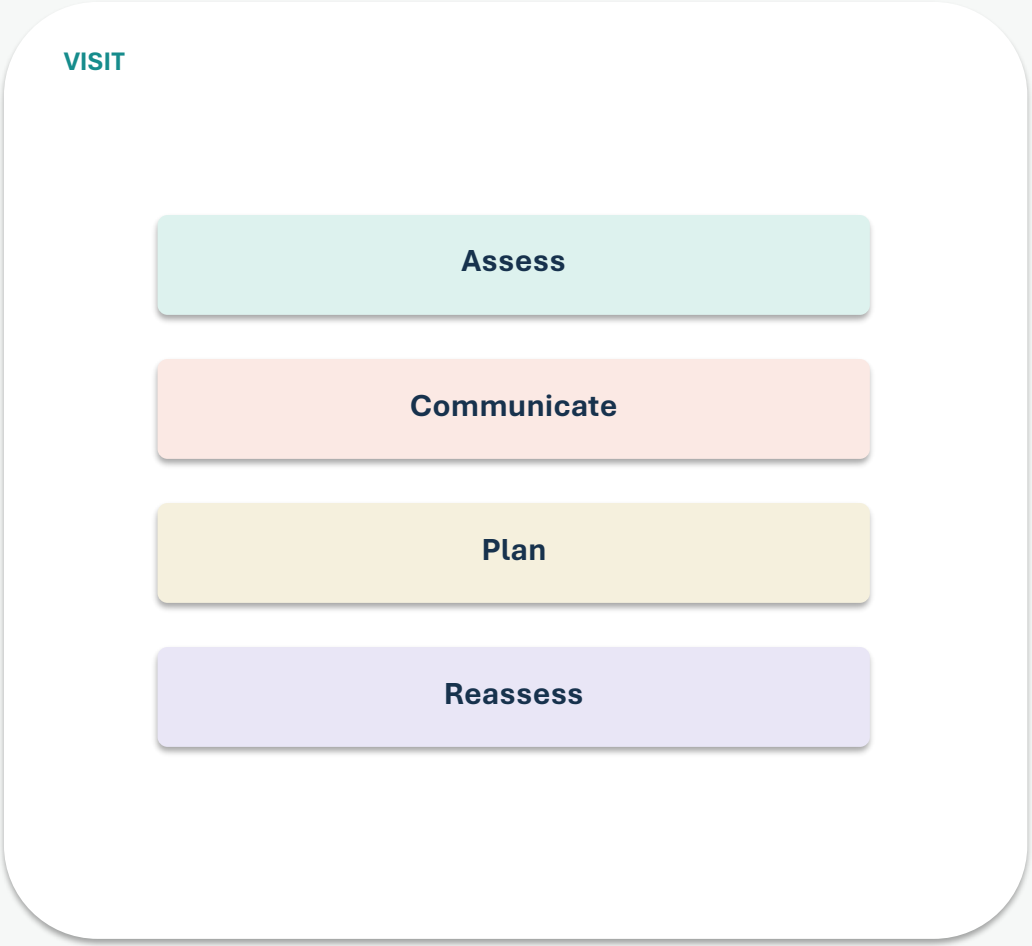
Prenatal care and assessment



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Preconception and first prenatal visit

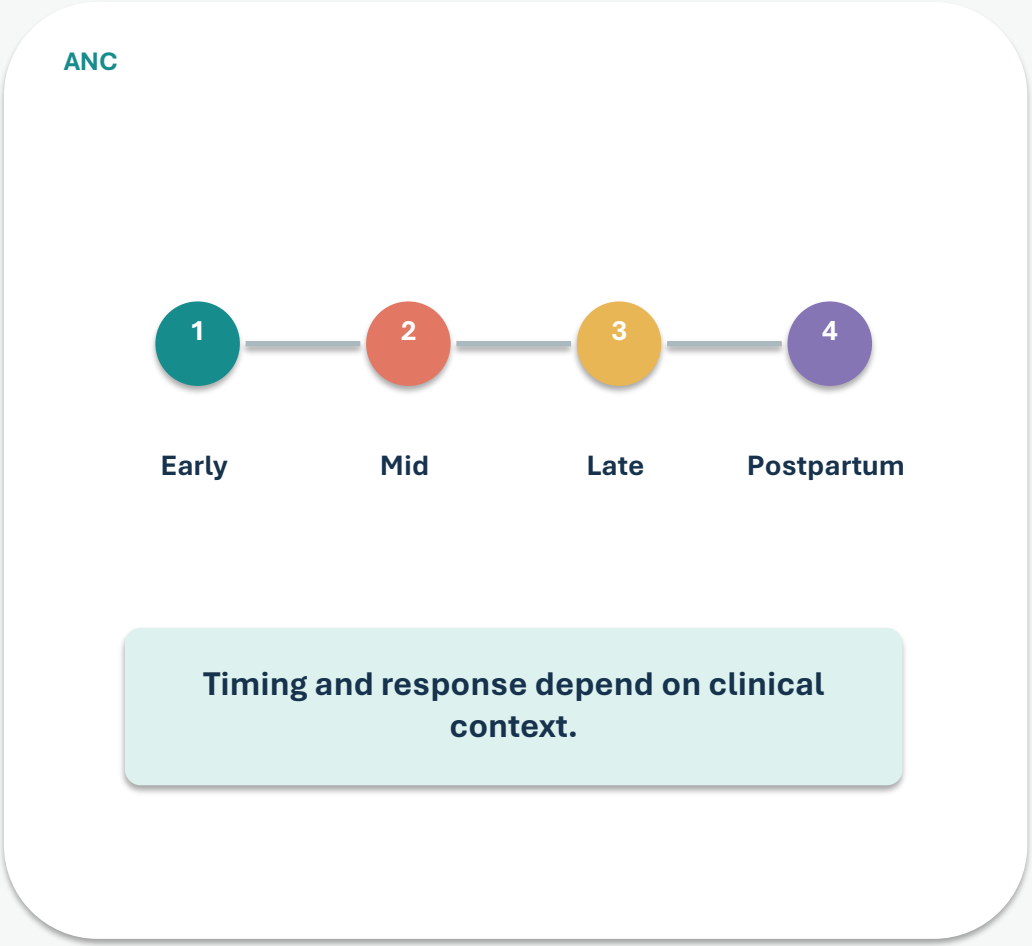
- Review health conditions, medicines, immunizations, and prior pregnancy outcomes.
- Estimate gestational age and establish baseline observations.
- Screen for safety, mental health, nutrition, and social needs.



Sources: ACOG/WHO

Antenatal visit schedule and continuity

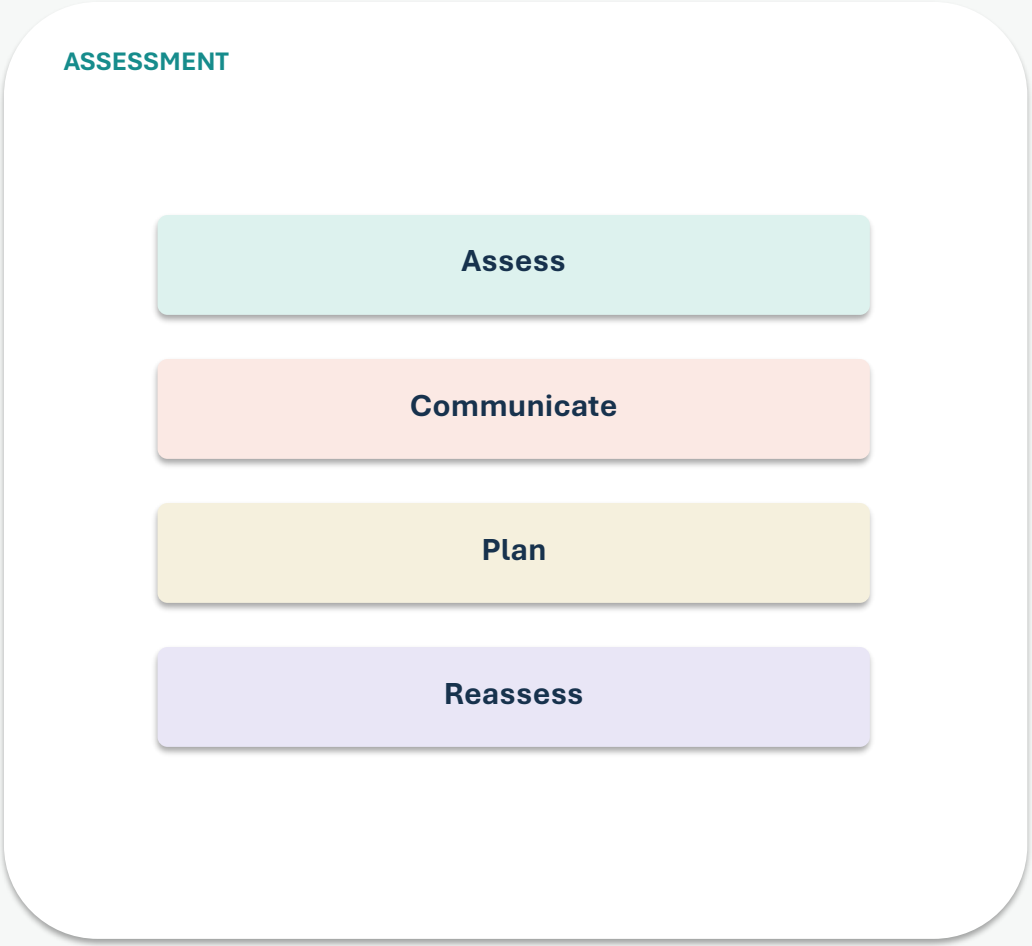
- WHO recommends at least eight antenatal contacts for routine care.
- Timing and visit content should adapt to risk, setting, and patient preference.
- Continuity supports trust, education, and timely escalation.



Sources: WHO 2016

Initial history and physical assessment

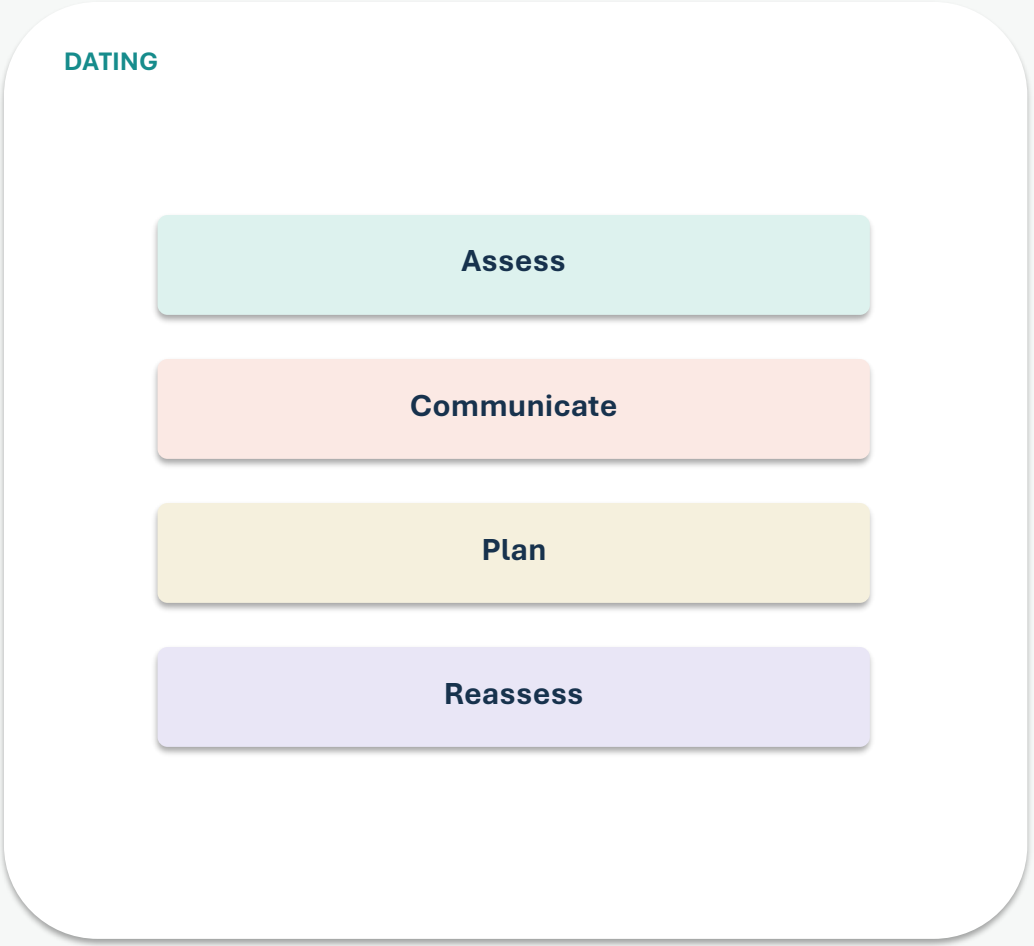
- Ask about symptoms, menstrual dates, pregnancy history, and family history.
- Assess vital signs, weight, medications, and relevant examination findings.
- Document patient goals, supports, and preferred communication.



Sources: ACOG

Pregnancy confirmation and dating

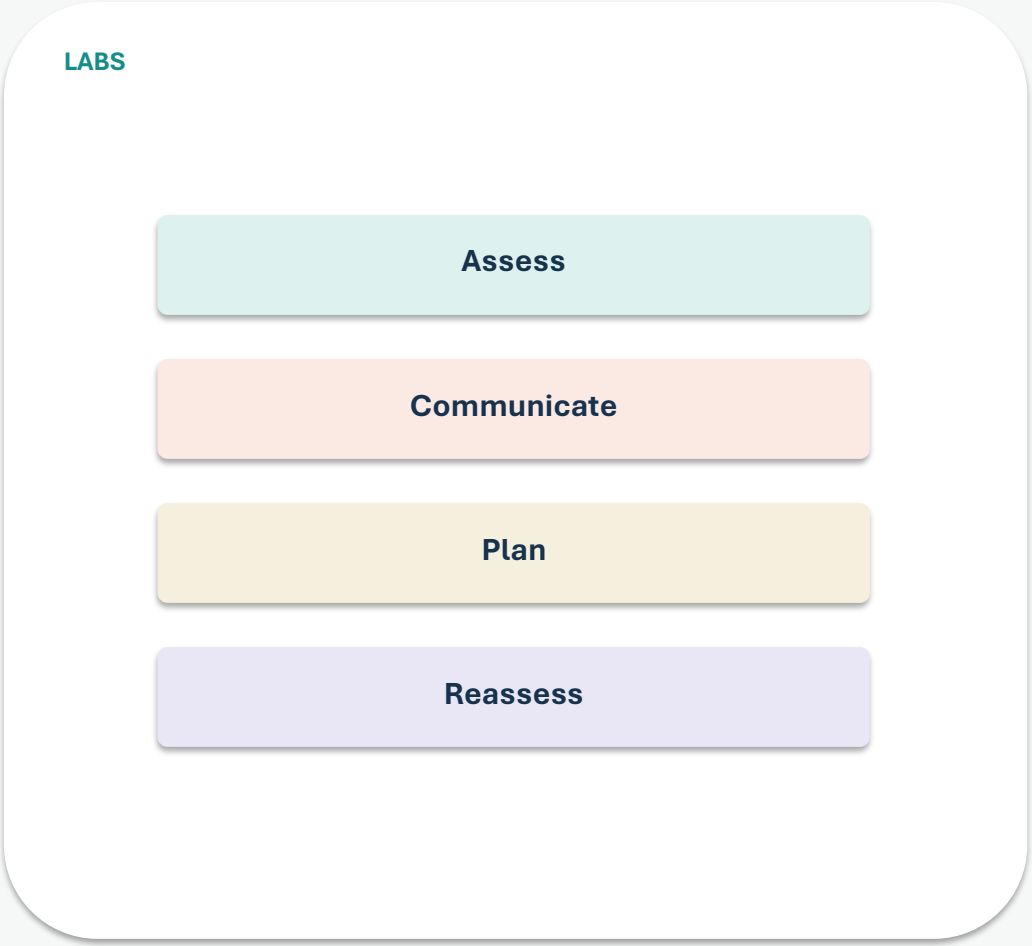
- Urine or serum hCG confirms biochemical pregnancy.
- Ultrasound confirms location, viability, number, and dating when indicated.
- A positive test with pain or bleeding requires assessment for ectopic pregnancy.



Sources: ACOG

Routine prenatal laboratory screening

- Common tests include blood group/Rh, antibody screen, CBC, and infection screening.
- Screening panels vary with national guidance and individual risk.
- Explain purpose, consent, results, and follow-up for each test.

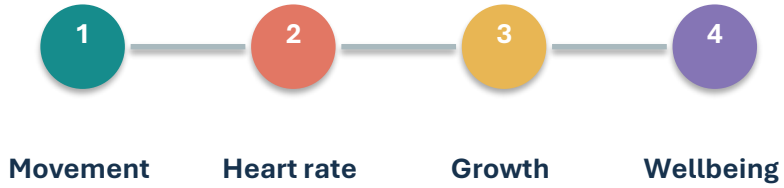


Sources: ACOG

Ultrasound and fetal assessment

- Ultrasound can assess dating, anatomy, placenta, fluid, and growth.
- Doppler, nonstress testing, or biophysical profile is used for selected indications.
- Choose tests based on the clinical question and gestational age.

FETAL ASSESS

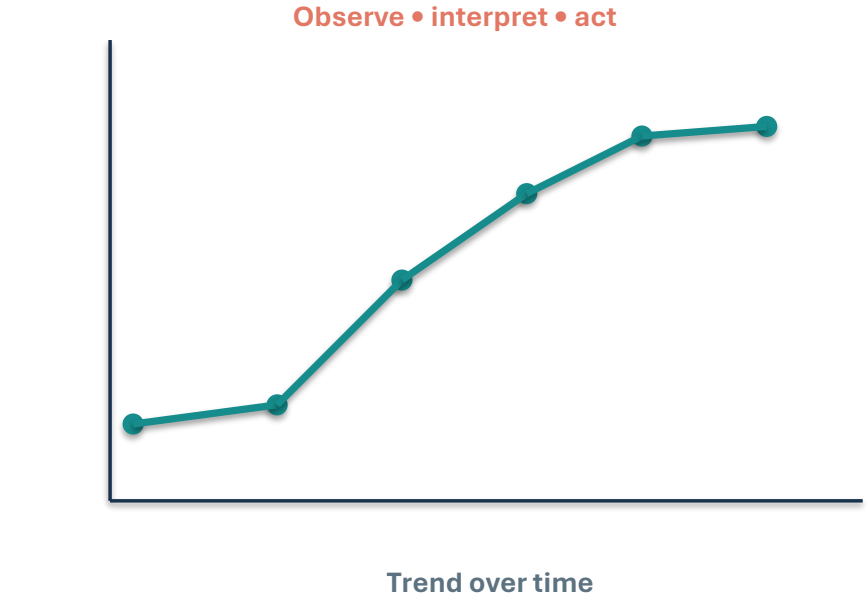


Timing and response depend on clinical context.

Sources: ACOG

Fundal height and fetal movement

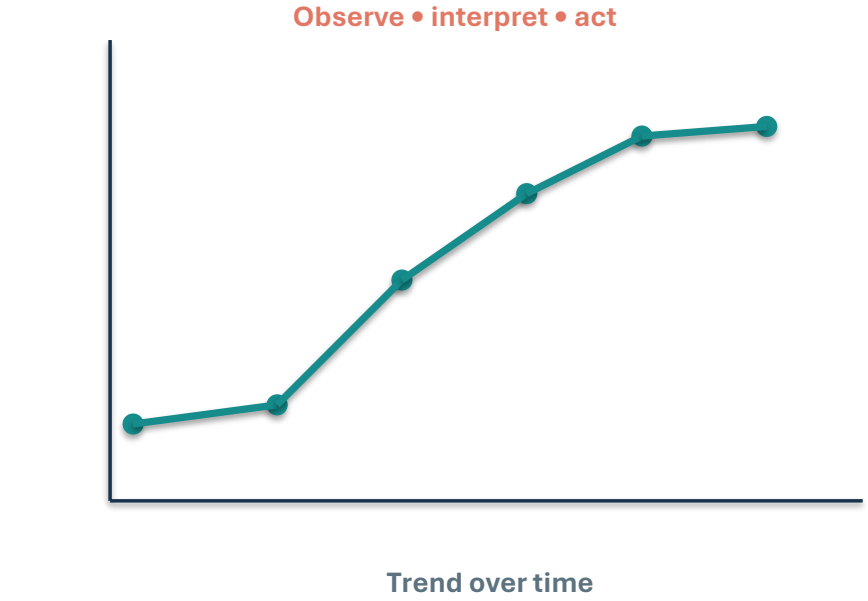
- Measure fundal height consistently after the uterus rises above the pelvis.
- A discrepancy prompts review of dating, growth, fluid, and maternal factors.
- Teach patients to report reduced fetal movement using local advice.



Sources: ACOG

Blood pressure and preeclampsia screening

- Measure blood pressure with correct cuff size and positioning.
- Ask about severe headache, vision change, epigastric pain, and sudden swelling.
- Proteinuria is not required when other severe features establish the diagnosis.



Sources: ACOG

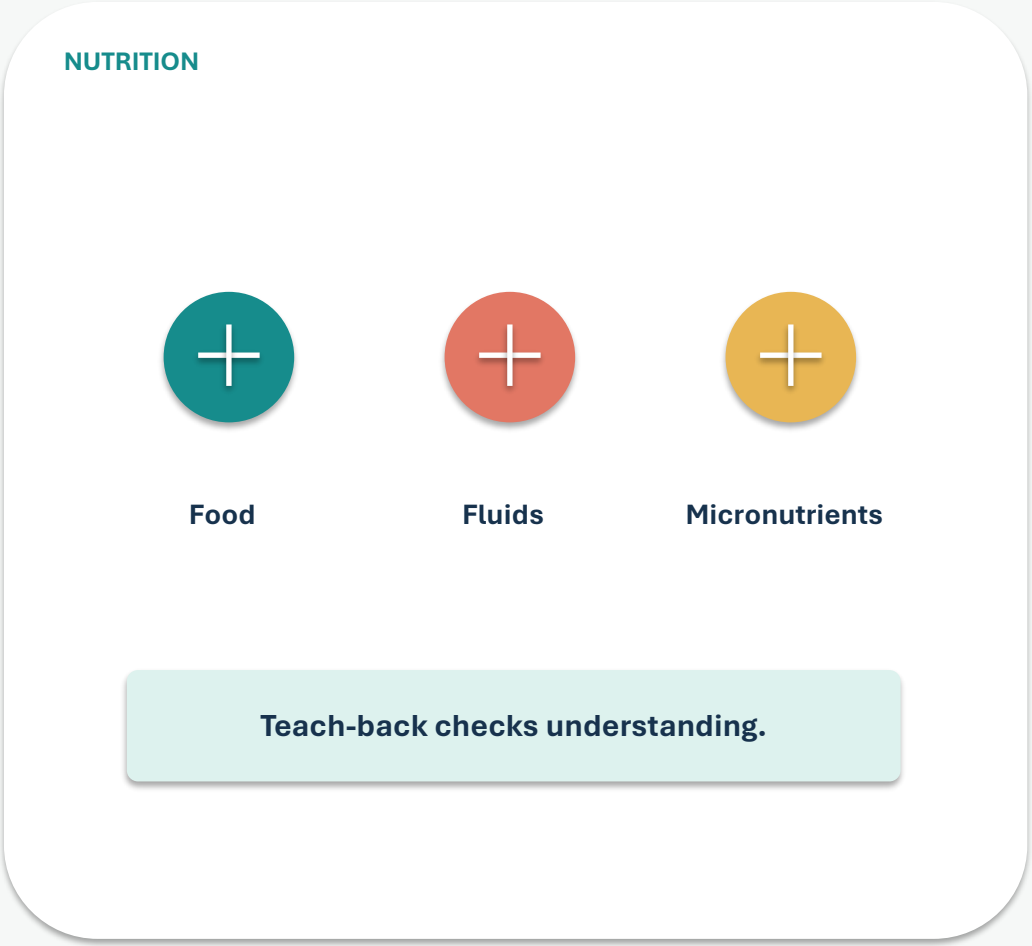
Health promotion and teaching



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Nutrition and healthy eating

- Emphasize varied, nutrient-dense foods and food safety.
- Needs vary with health, culture, food access, and multiple pregnancy.
- Refer for dietetic support when intake, growth, or medical needs warrant.



Sources: ACOG

Weight gain in pregnancy

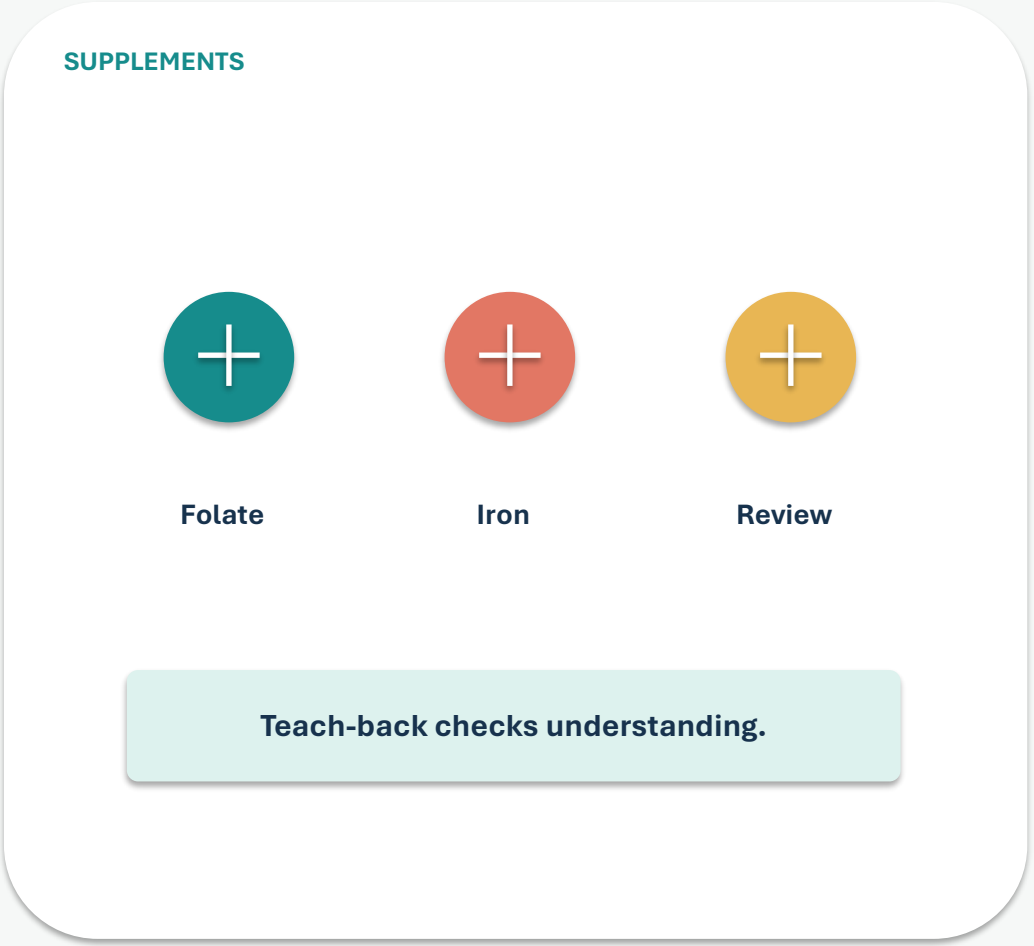
- Recommended gain depends on prepregnancy BMI and number of fetuses.
- Use current national guidance; avoid one target for every patient.
- Discuss trends without stigma and assess nutrition, edema, and fetal growth.



Sources: CDC/ACOG

Folic acid and iron

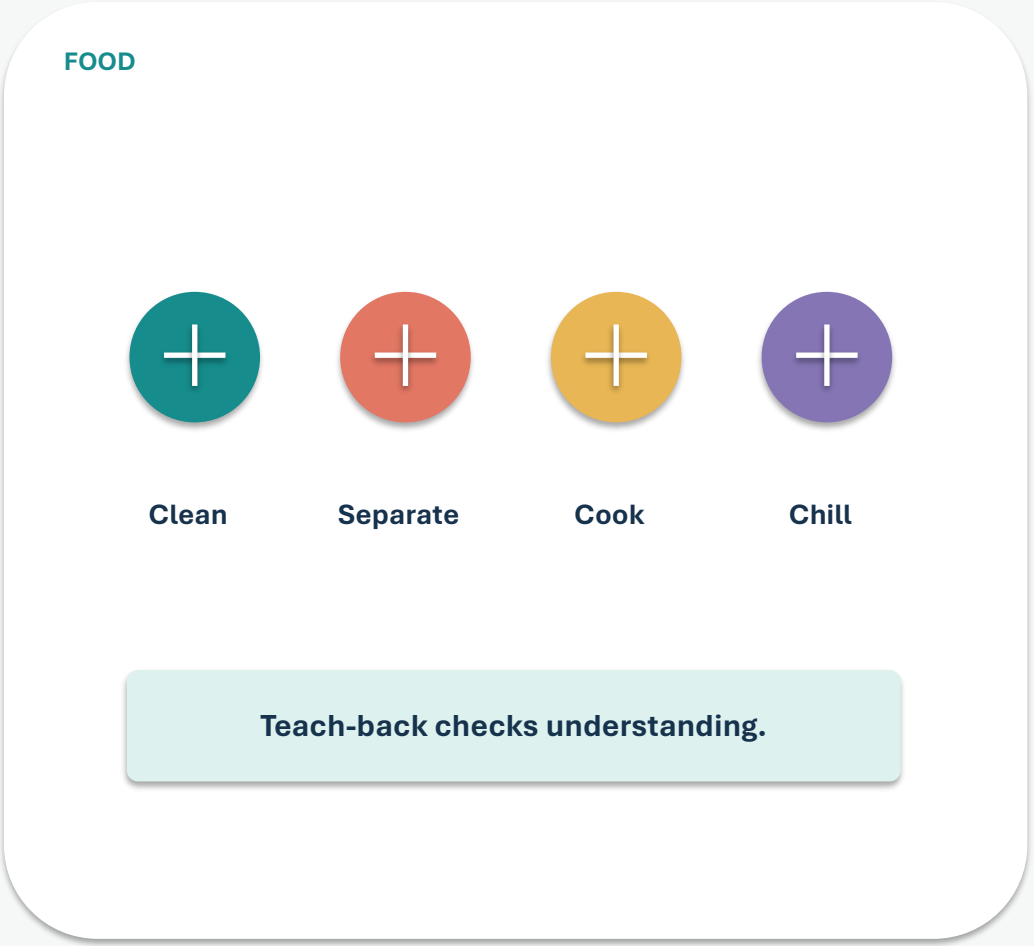
- Folic acid before conception and early pregnancy lowers neural tube defect risk.
- Iron supplementation may be recommended based on diet, labs, and local policy.
- Review dose and tolerability with a qualified clinician.



Sources: CDC/ACOG

Food safety and infection prevention

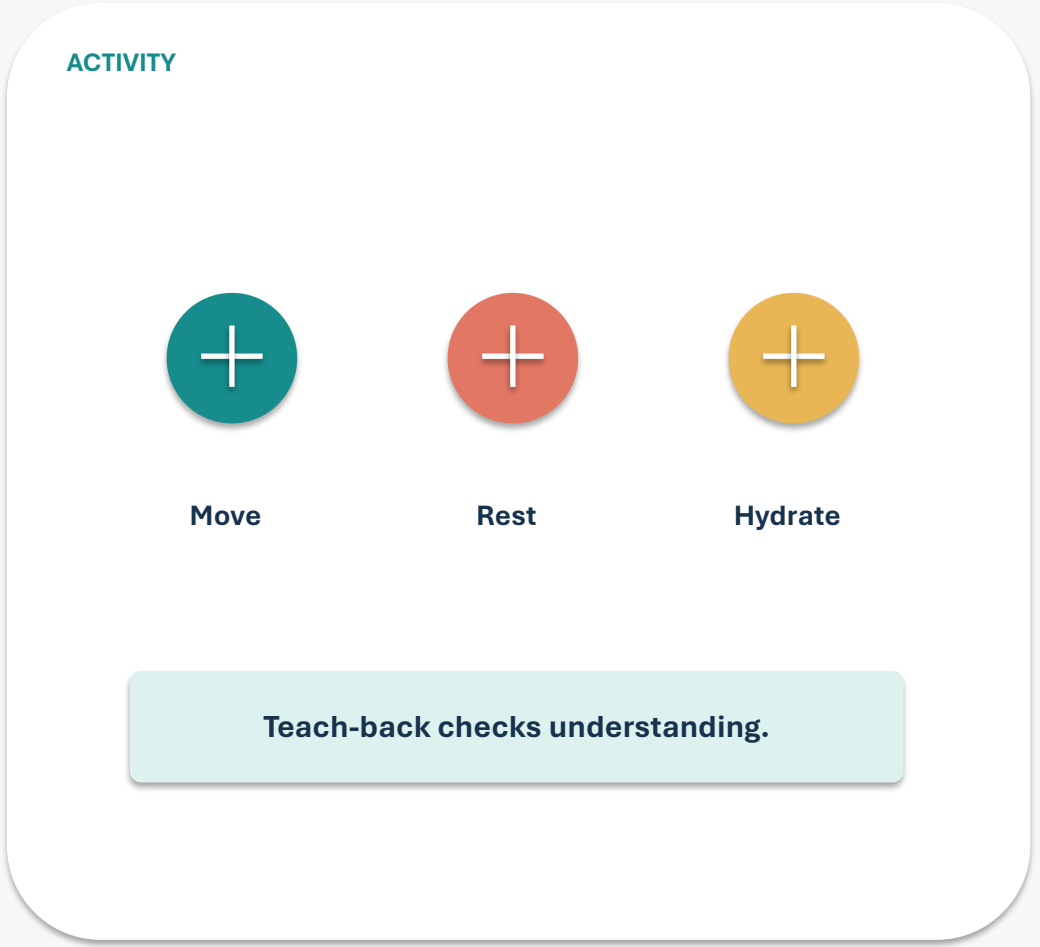
- Avoid high-risk unpasteurized foods and undercooked animal products.
- Wash produce and follow local fish-mercury guidance.
- Hand hygiene and safe food storage reduce infection risk.



Sources: CDC

Physical activity, rest, and work

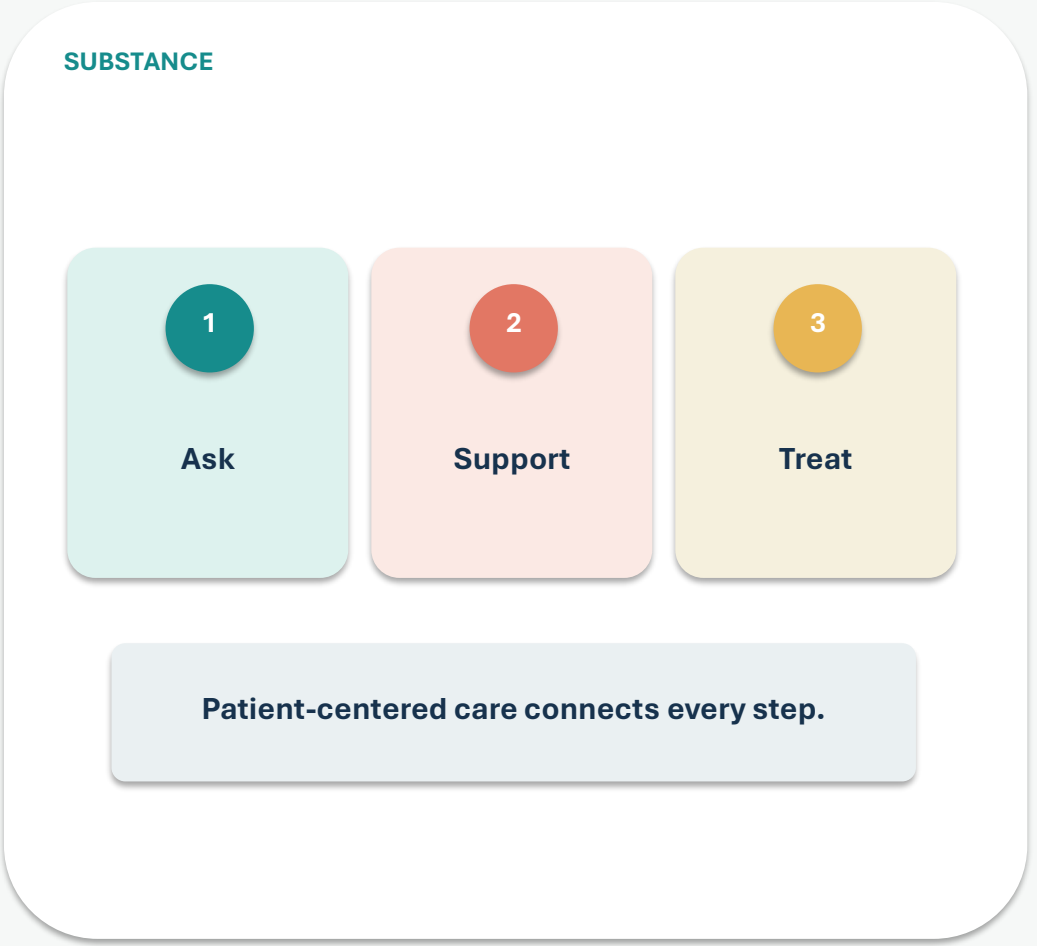
- Encourage regular moderate activity when no contraindication exists.
- Discuss hydration, heat, fall risk, and occupational exposures.
- Tailor guidance for complications, baseline fitness, and patient goals.



Sources: ACOG

Substance use and brief intervention

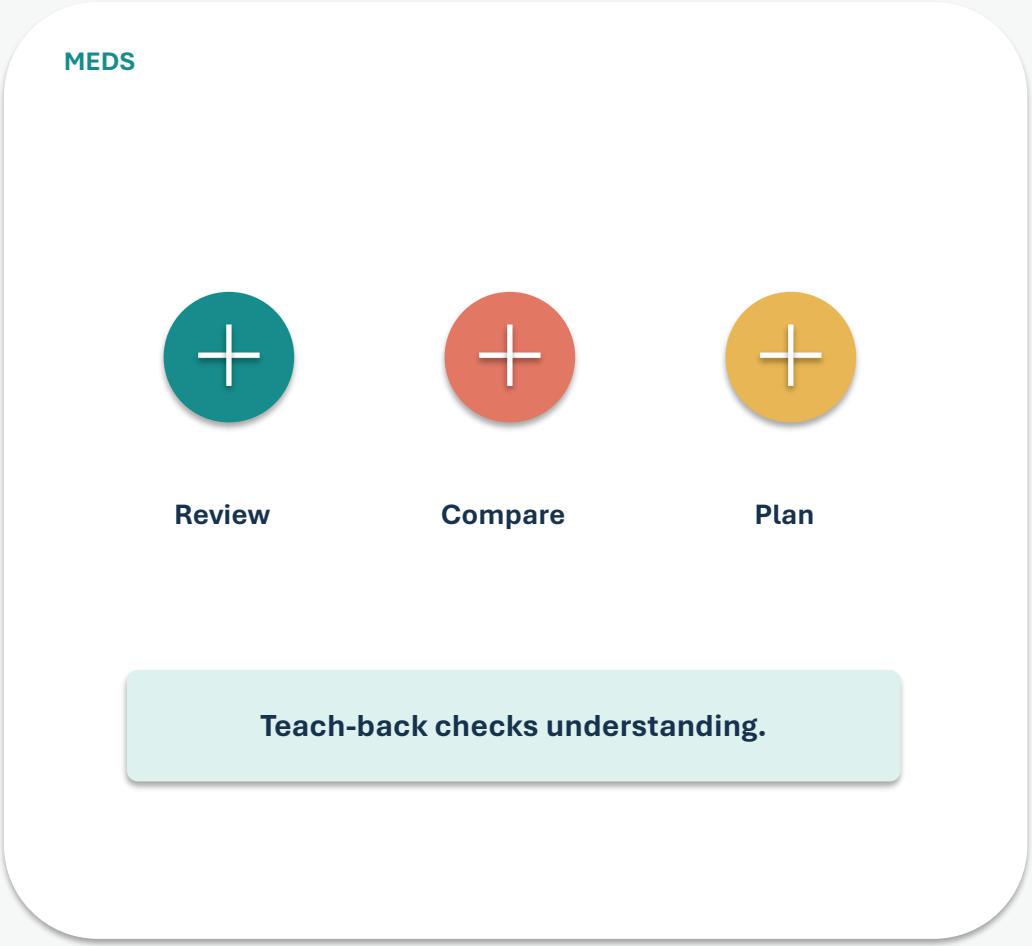
- Ask privately about tobacco, alcohol, cannabis, opioids, and other substances.
- Use supportive, non-punitive language and offer evidence-based treatment.
- Any alcohol exposure carries avoidable fetal risk; cessation support matters.



Sources: CDC/ACOG

Medication and vaccine review

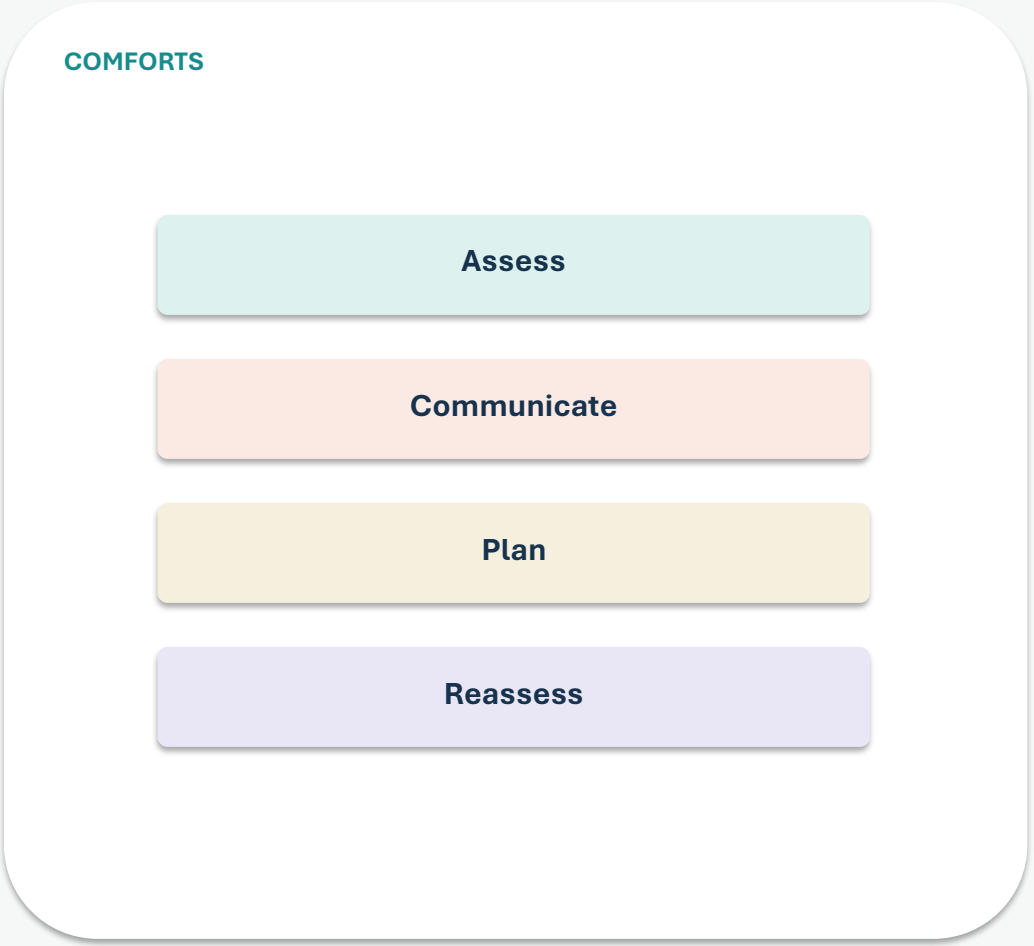
- Review medicines and supplements at every transition in care.
- Do not stop essential treatment without discussing risks and alternatives.
- Vaccine recommendations are seasonal and jurisdiction-specific.



Sources: CDC 2026

Common discomforts: symptom care

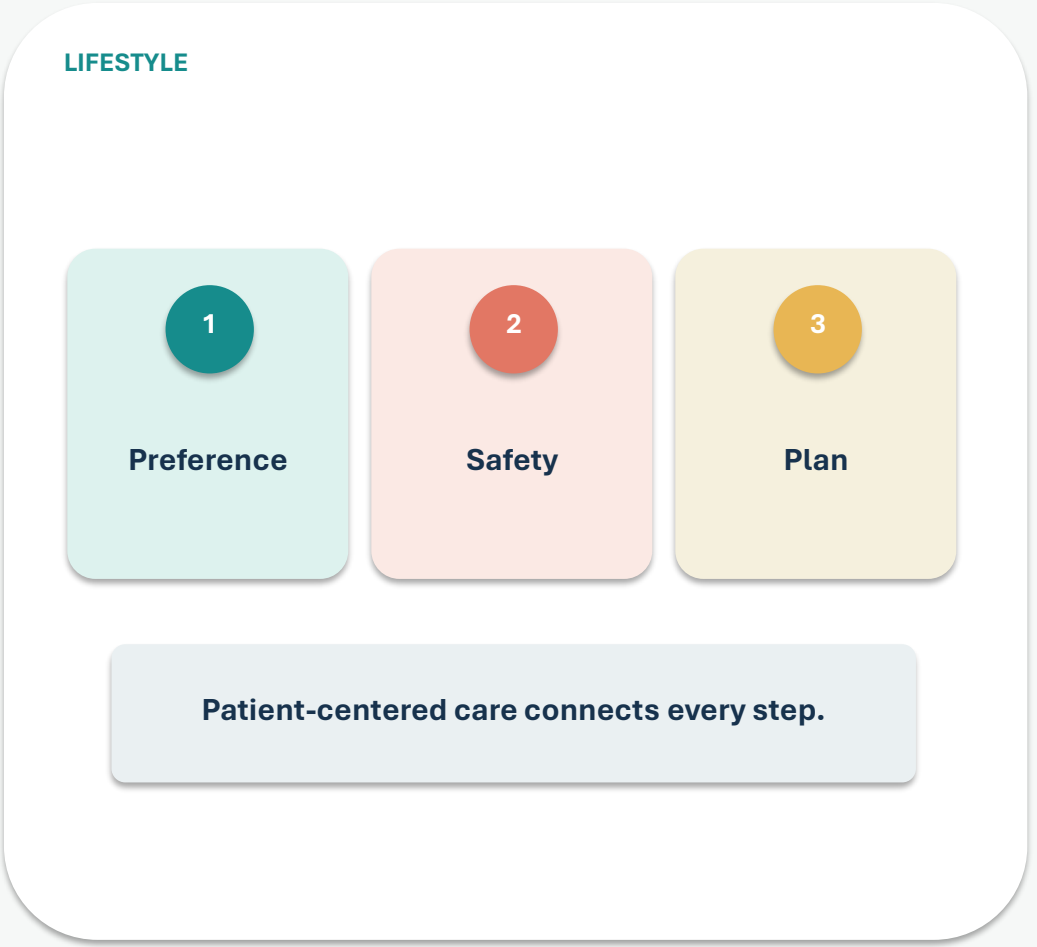
- Small frequent meals may ease reflux or nausea for some patients.
- Movement, hydration, and fiber may help constipation when safe.
- Assess red flags before giving routine self-care advice.



Sources: ACOG

Sleep, travel, and sexual health

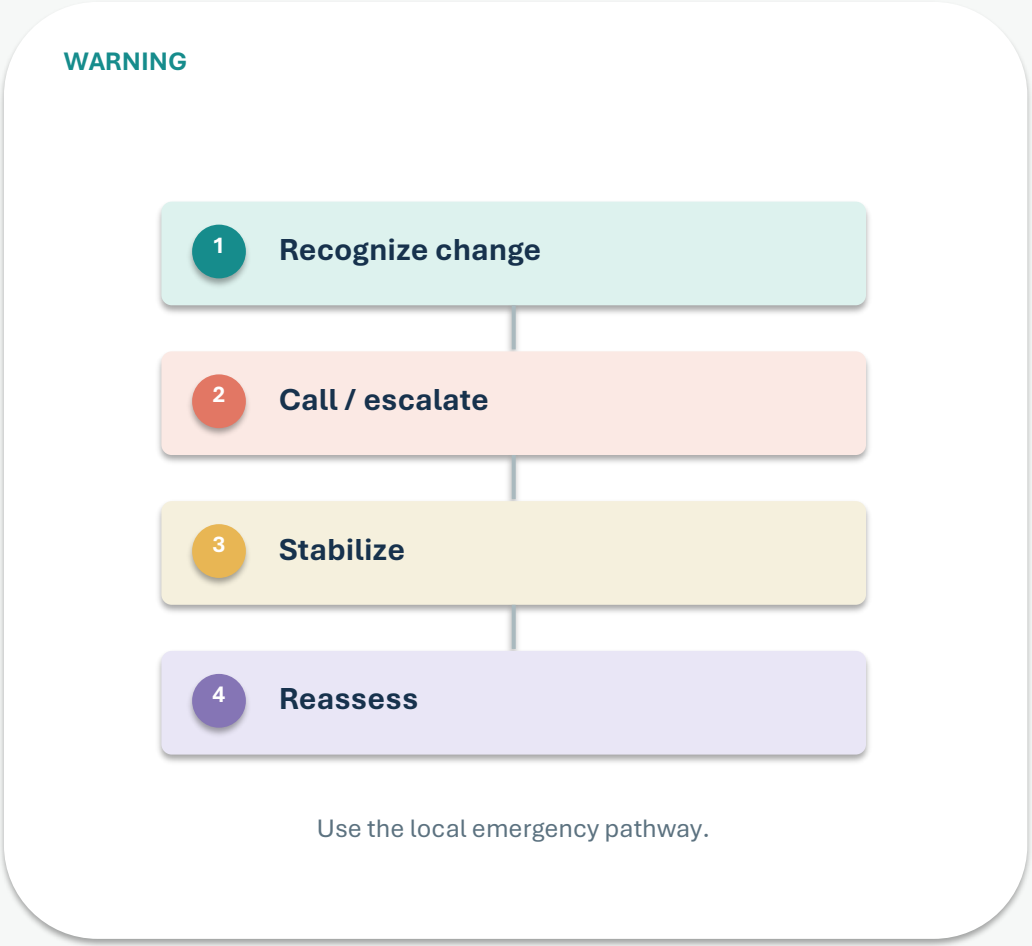
- Discuss sleep positioning for comfort and individual risk.
- Travel advice depends on gestational age, medical history, and destination.
- Sex is usually safe in uncomplicated pregnancy; bleeding or fluid loss needs evaluation.



Sources: ACOG

Warning signs: when to seek care

- Urgent symptoms include heavy bleeding, severe pain, fainting, or trouble breathing.
- After mid-pregnancy, severe headache, vision change, or reduced movement needs prompt contact.
- Use the local maternity triage number and emergency service when symptoms are severe.



Sources: ACOG

Pregnancy complications



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High risk pregnancy

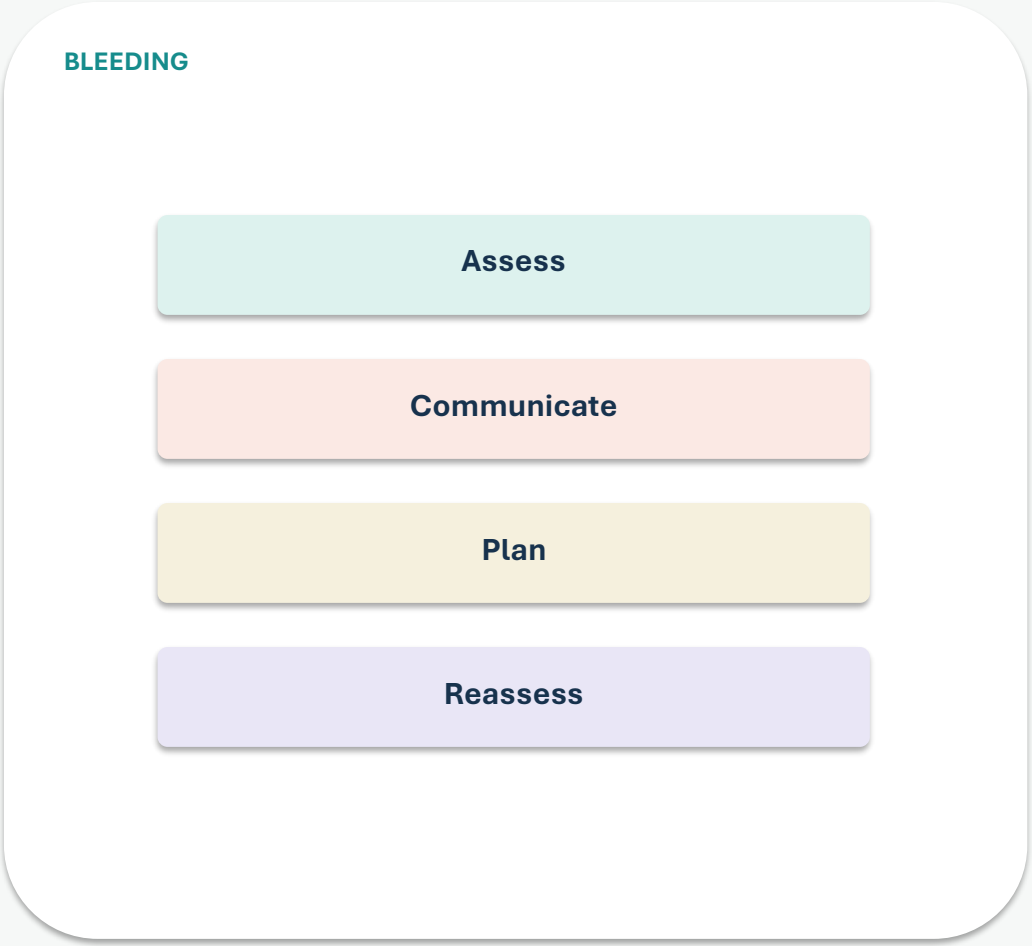
Nursing Management of Pregnancy at Risk: Pregnancy-Related Complications

KEY TERMS

abortion
abruptio placentae
ectopic pregnancy
gestational hypertension
gestational trophoblastic disease (GTD)
high-risk pregnancy
hyperemesis gravidarum
multiple gestation
oligohydramnios
polyhydramnios
placenta previa
premature rupture of membranes (PROM)
preterm premature rupture of membranes (PPROM)

Early pregnancy bleeding: first response

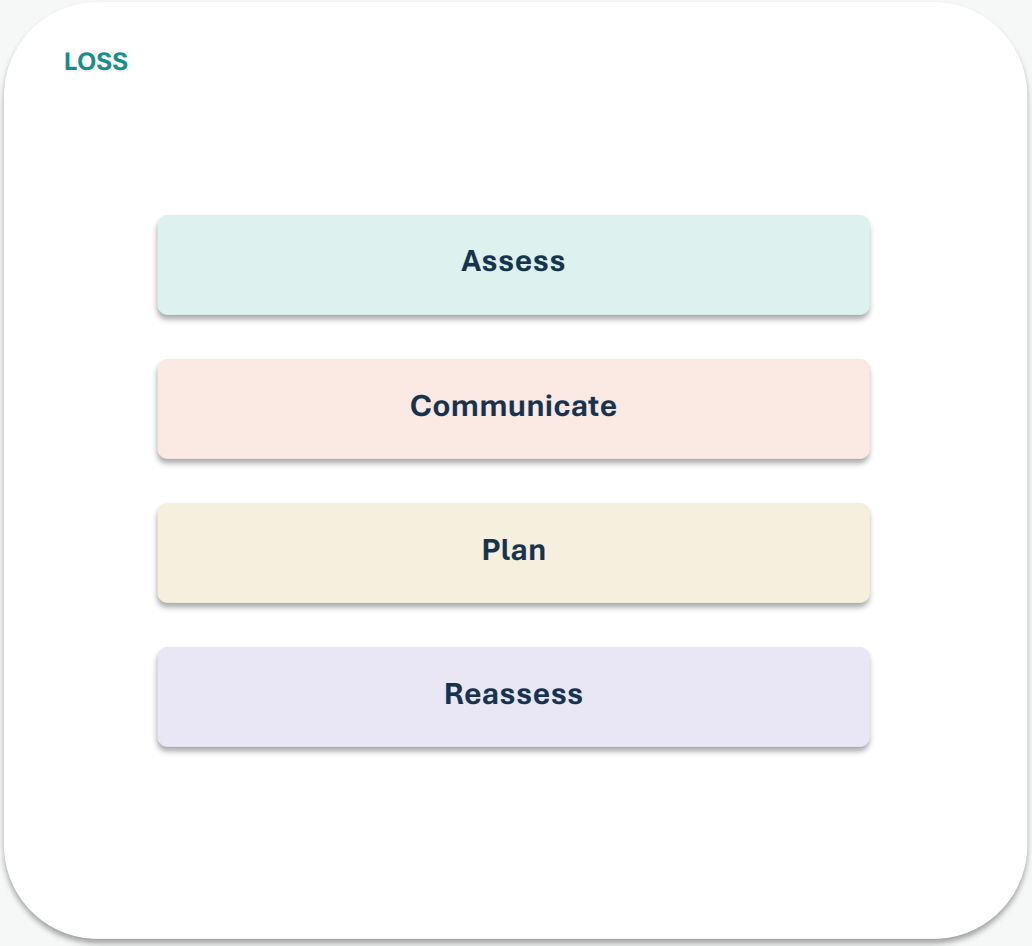
- Assess stability, bleeding amount, pain, and gestational age.
- Consider miscarriage, ectopic pregnancy, and other causes.
- Escalate hemodynamic instability or severe pain immediately.



Sources: ACOG

Miscarriage and pregnancy loss

- Use clear, compassionate language and ask what the patient understands.
- Management may be expectant, medical, or procedural depending on context.
- Offer pain support, follow-up, grief resources, and reproductive counseling.

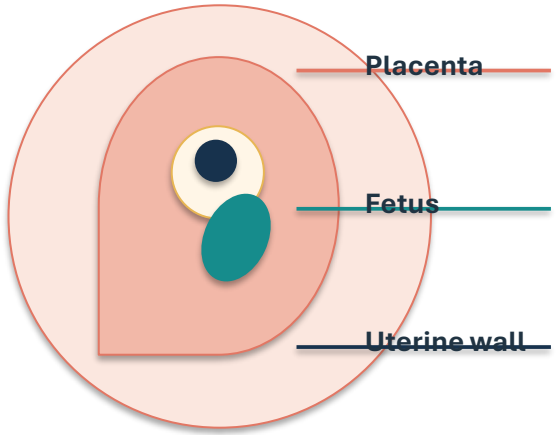


Sources: ACOG

Ectopic pregnancy

- Possible signs include unilateral pain, bleeding, shoulder pain, or syncope.
- Risk includes prior ectopic pregnancy, tubal surgery, and some infections.
- Suspected rupture is an emergency; rapid escalation is essential.

ECTOPIC

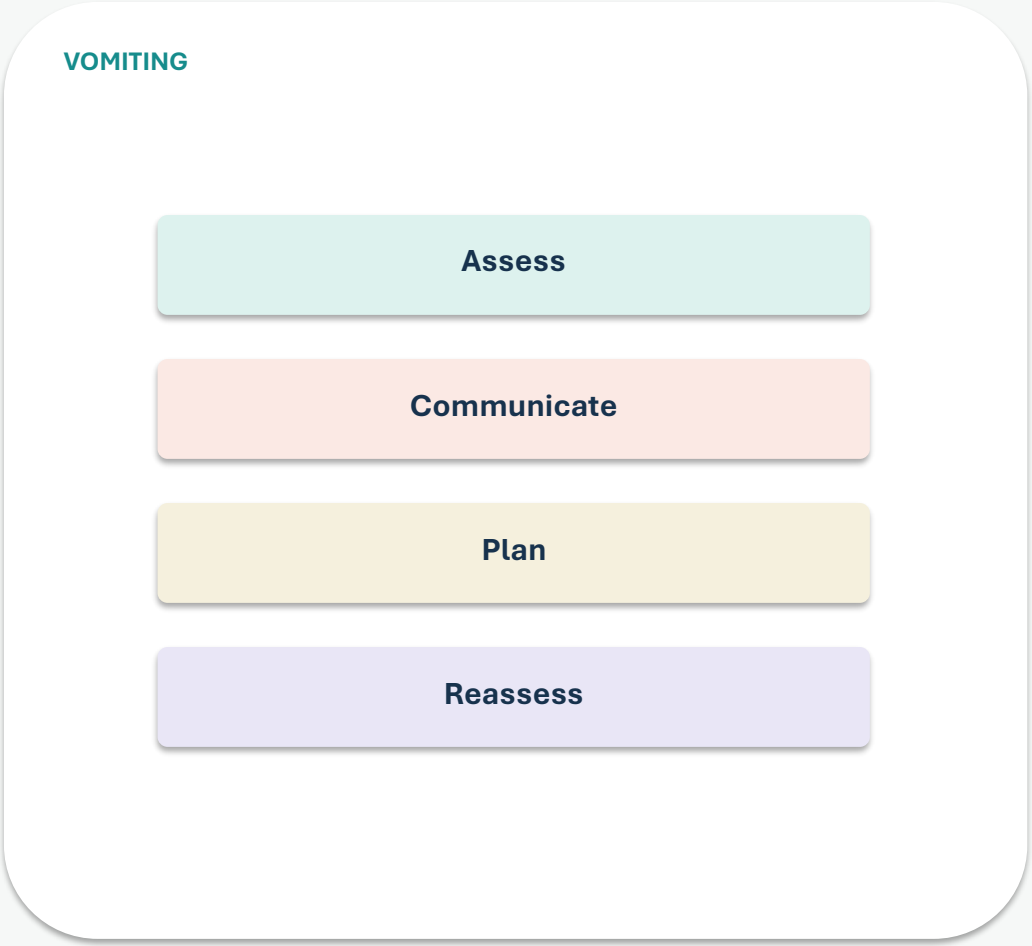


Schematic • not to scale

Sources: ACOG

Hyperemesis gravidarum

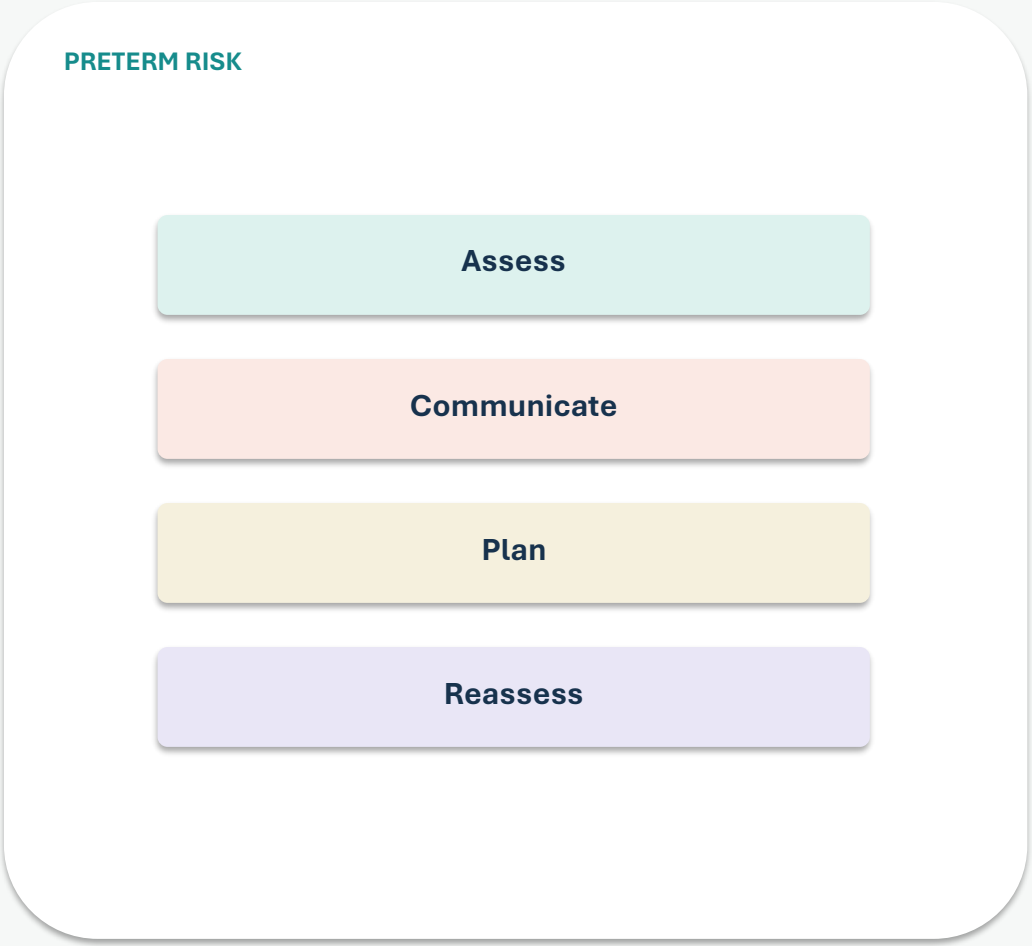
- Persistent vomiting can cause dehydration, electrolyte disturbance, and weight loss.
- Assess intake, urine output, vital signs, labs, and response to treatment.
- Management may require IV fluids, antiemetics, nutrition support, and follow-up.



Sources: ACOG

Cervical insufficiency and preterm birth risk

- Painless cervical shortening or dilation can precede early birth.
- Prior history and ultrasound findings guide surveillance and interventions.
- Teach symptoms of preterm labor and the local pathway for urgent assessment.

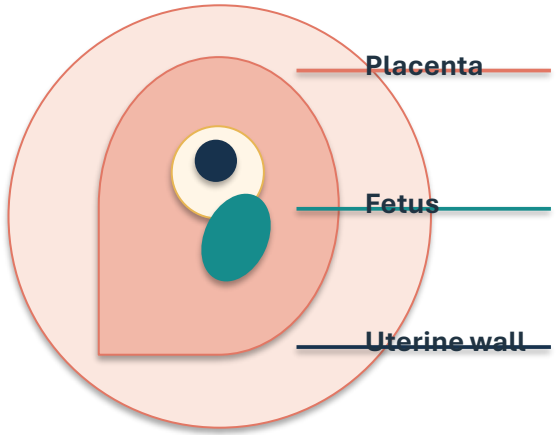


Sources: ACOG

Placenta previa and antepartum bleeding

- Placenta previa may cause painless bleeding later in pregnancy.
- Avoid digital vaginal examination until placental location is known.
- Maternal stabilization and fetal assessment take priority.

PREVIA



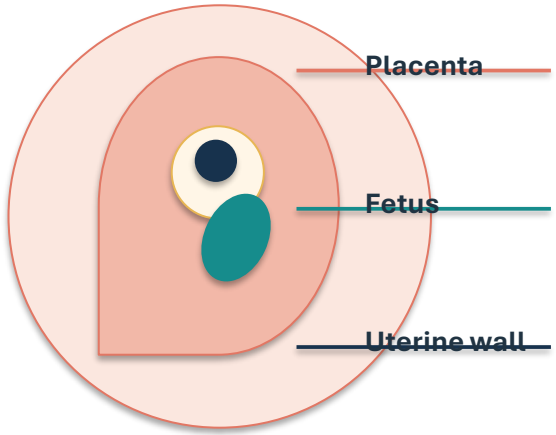
Schematic • not to scale

Sources: ACOG

Placental abruption

- Abruption may present with pain, uterine tenderness, bleeding, or fetal compromise.
- Visible blood loss may underestimate severity.
- Activate urgent obstetric response and prepare for resuscitation.

ABRUPTION

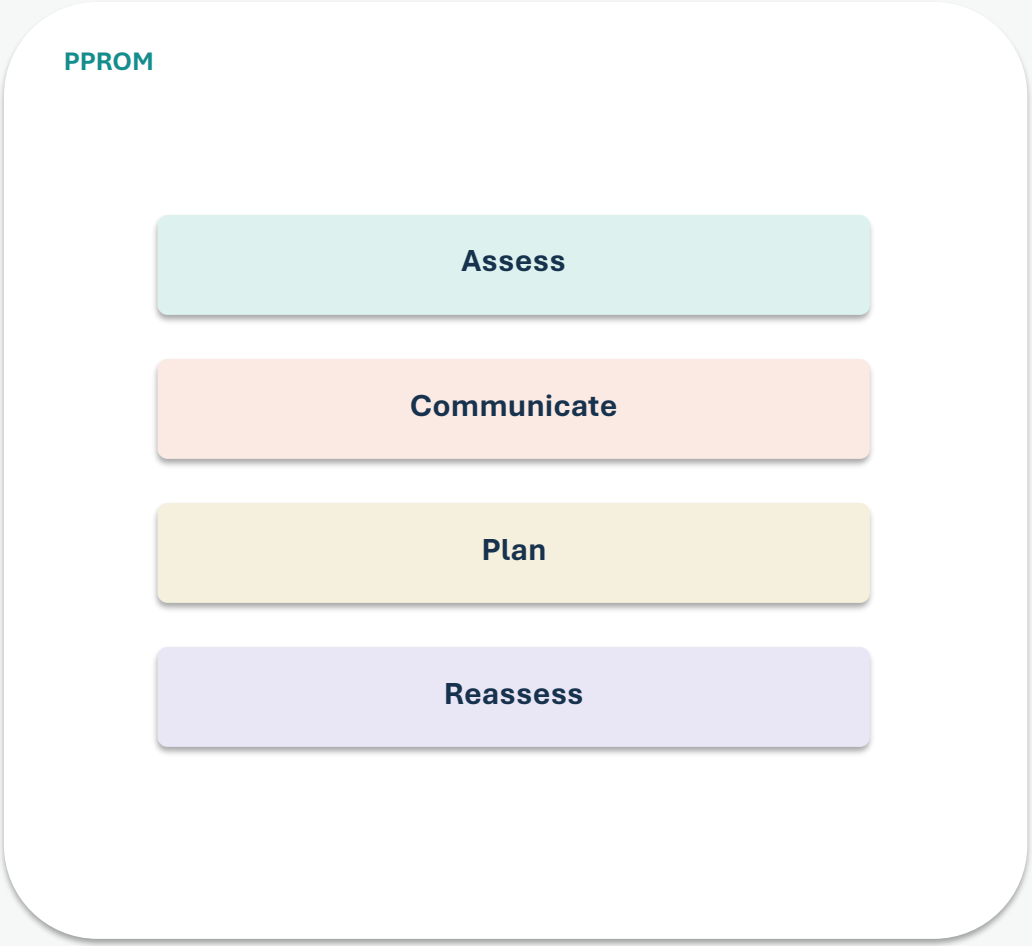


Schematic • not to scale

Sources: ACOG

Preterm prelabor rupture of membranes

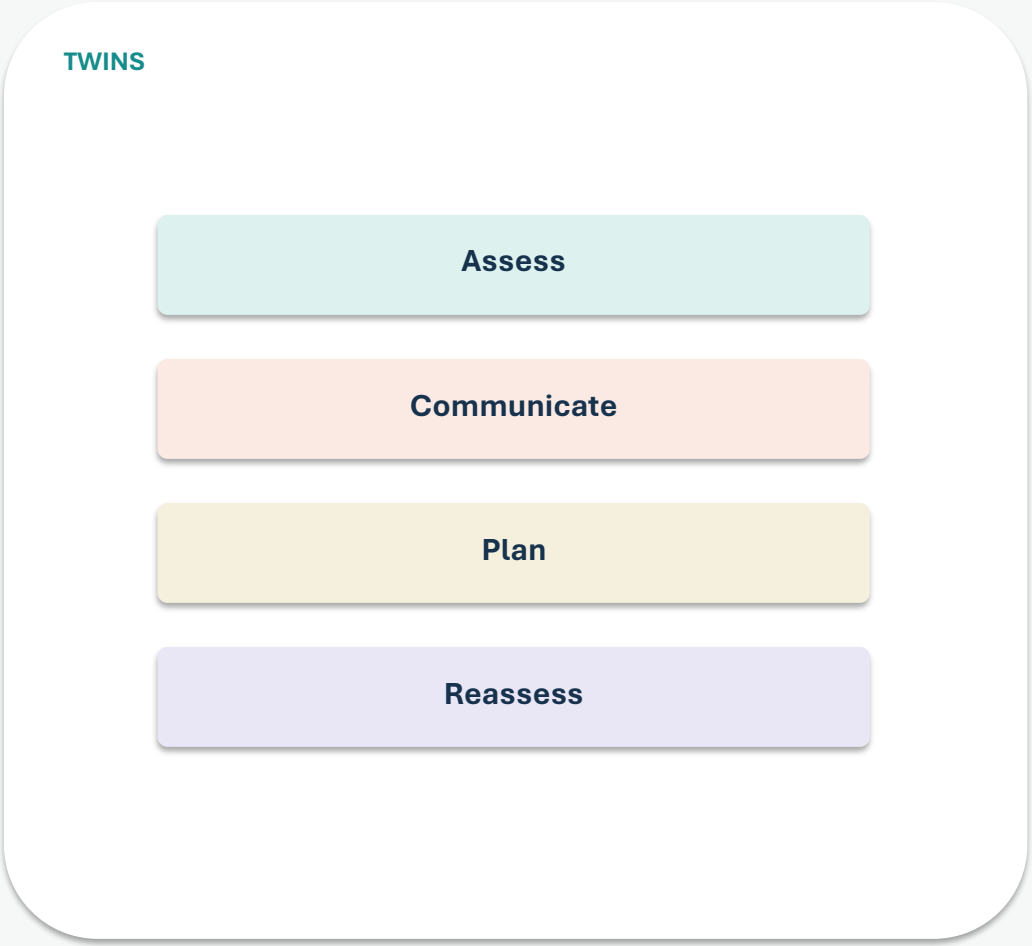
- PPROM is membrane rupture before 37 weeks and before labor.
- Assess infection, fetal status, gestational age, and fluid loss.
- Antibiotics, corticosteroids, and timing decisions follow specialist guidance.



Sources: ACOG 2025

Multifetal pregnancy

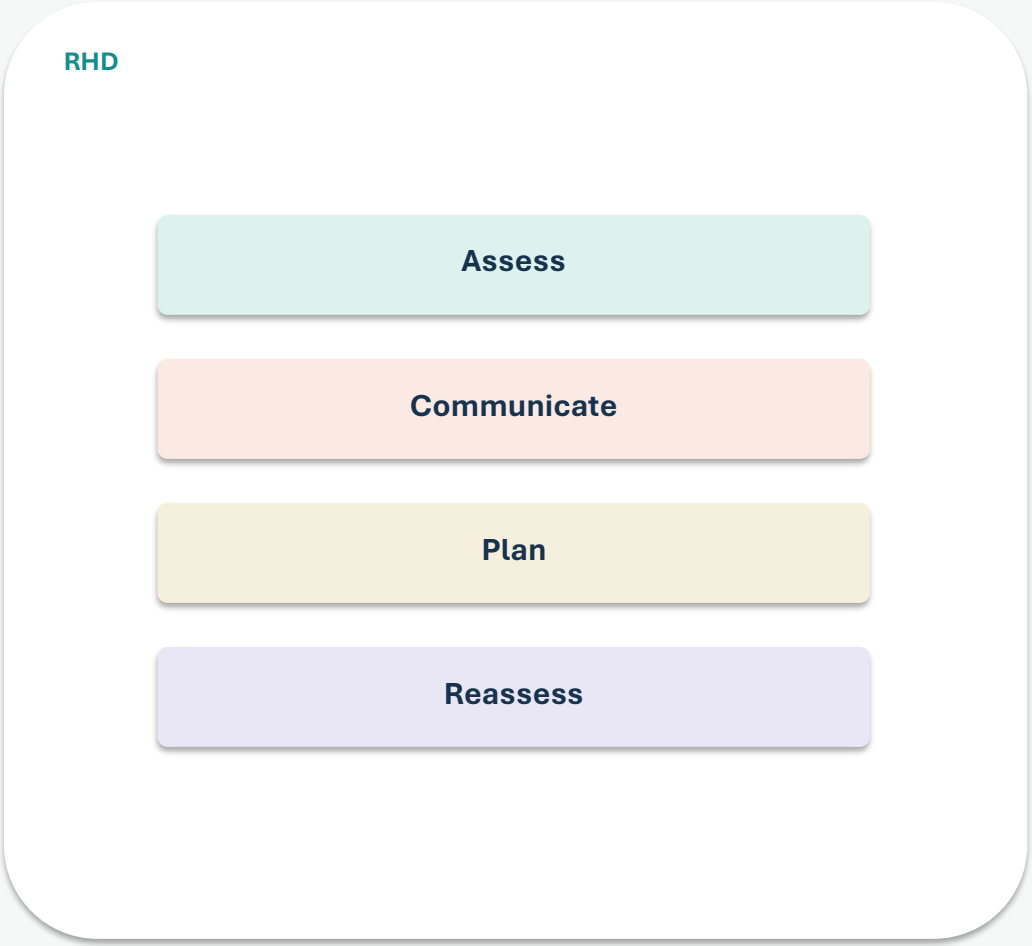
- Twin chorionicity and amnionicity shape surveillance and birth planning.
- Risks include preterm birth, growth discordance, and hypertensive disorders.
- Coordinate care with maternal-fetal and neonatal teams.



Sources: ACOG

RhD alloimmunization

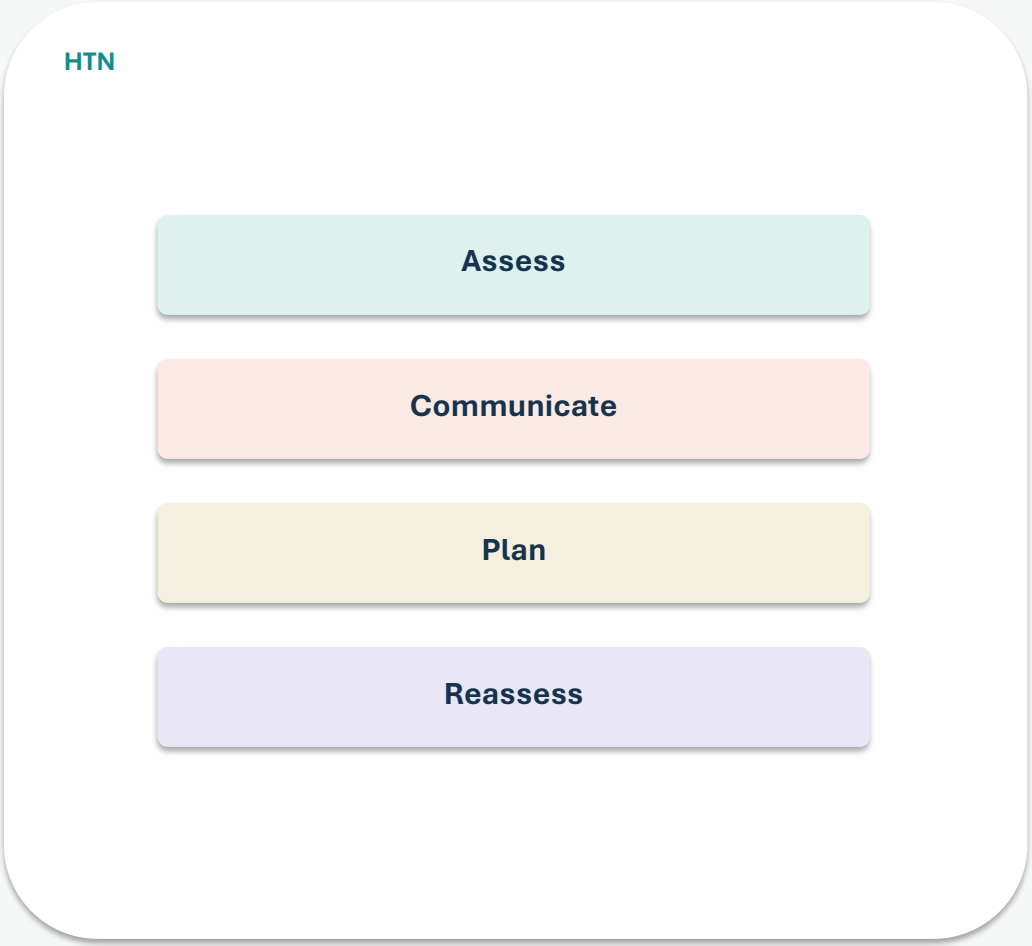
- Determine RhD status and antibody screen early in pregnancy.
- Anti-D immunoglobulin is offered when indicated by local protocols.
- Established antibodies require fetal risk assessment and specialist follow-up.



Sources: ACOG

Hypertension in pregnancy: categories

- Chronic hypertension begins before pregnancy or before 20 weeks.
- Gestational hypertension develops after 20 weeks without severe features.
- Preeclampsia includes hypertension with proteinuria or end-organ features.



Sources: ACOG

Preeclampsia pathophysiology and assessment

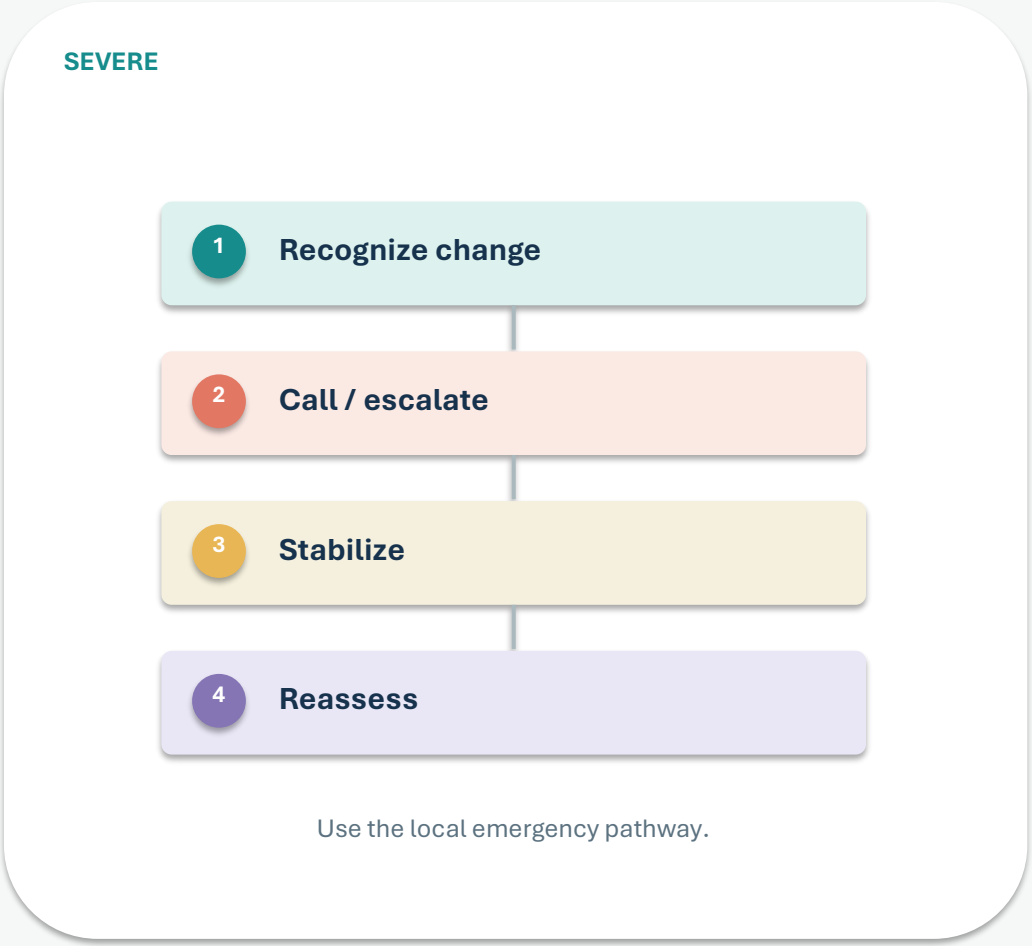
- Placental dysfunction contributes to systemic endothelial injury.
- Assess BP trend, symptoms, platelets, renal and liver function, and fetal status.
- Proteinuria helps diagnosis but its absence does not exclude preeclampsia.



Sources: ACOG

Severe features and escalation

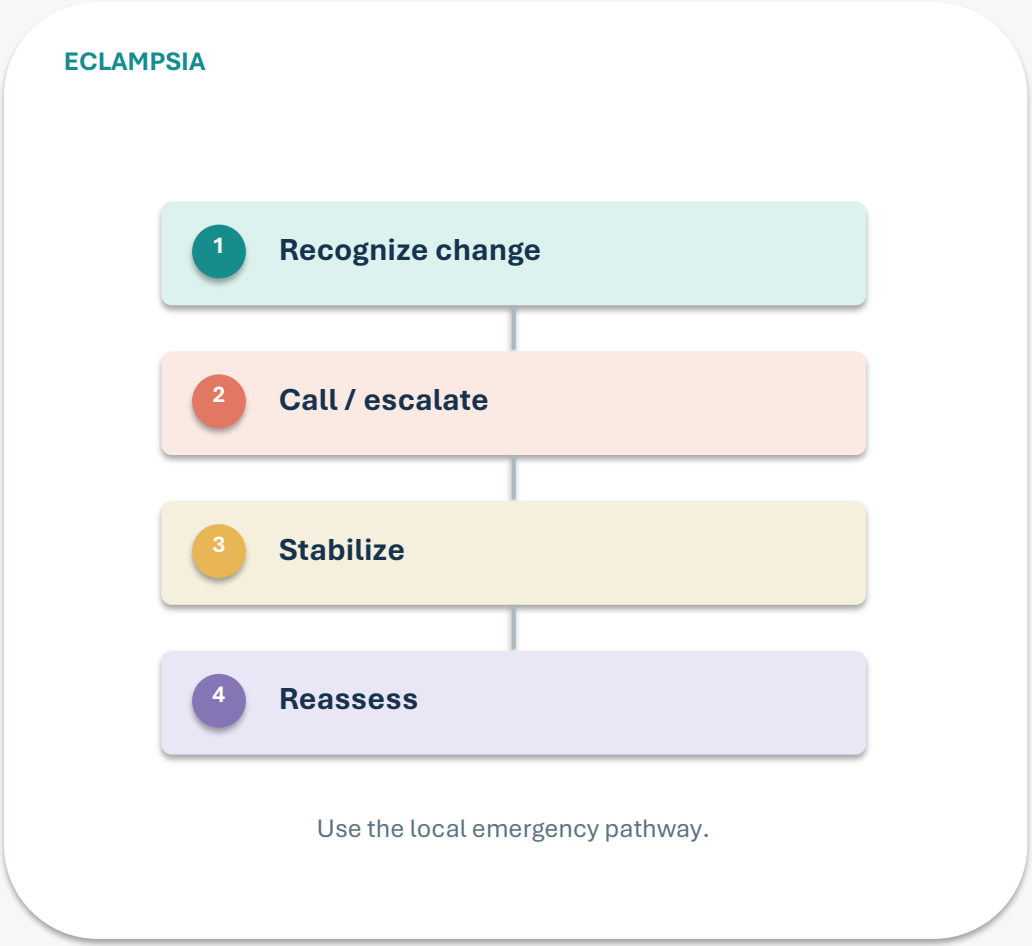
- Severe-range blood pressure requires urgent treatment under protocol.
- Persistent headache, visual symptoms, pulmonary edema, or RUQ pain are concerning.
- Escalate promptly and prepare for maternal stabilization and birth planning.



Sources: ACOG

Eclampsia and seizure response

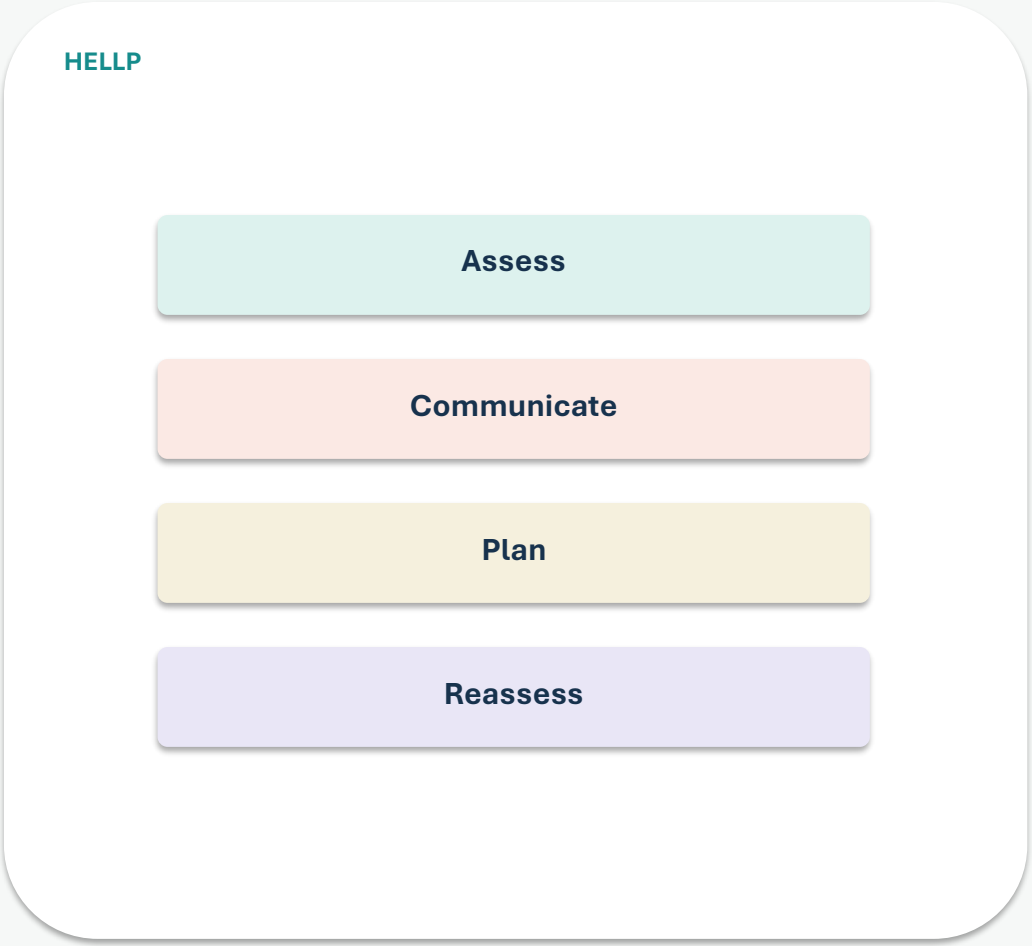
- Protect airway and prevent injury; summon the obstetric emergency team.
- Magnesium sulfate is commonly used for seizure treatment and prevention.
- Treat severe hypertension and assess maternal-fetal status after stabilization.



Sources: ACOG

HELLP syndrome

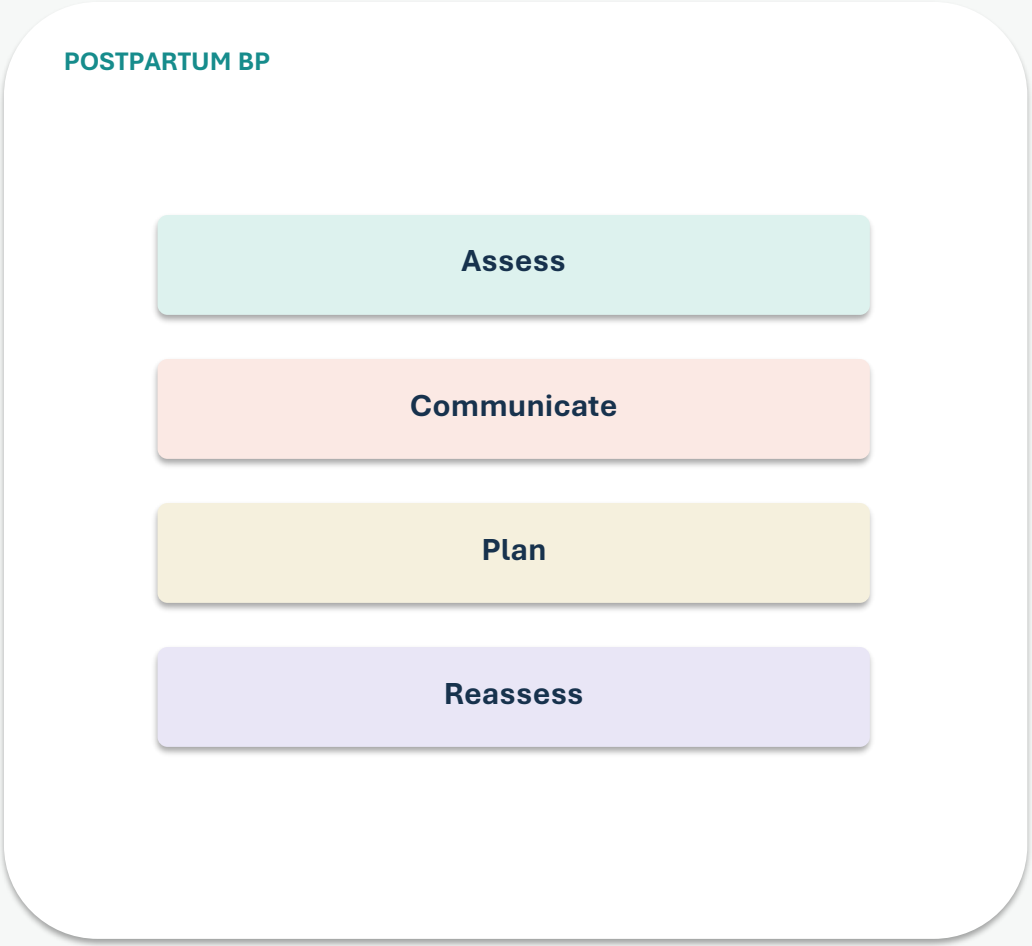
- Hemolysis, elevated liver enzymes, and low platelets can signal severe disease.
- Symptoms may include malaise, nausea, RUQ pain, or headache.
- Urgent evaluation is needed even when blood pressure is not markedly elevated.



Sources: ACOG

Postpartum hypertension and preeclampsia

- Hypertensive disease may first appear or worsen after birth.
- Teach warning symptoms and ensure an explicit BP follow-up plan.
- Severe headache, visual changes, dyspnea, or severe BP needs urgent care.

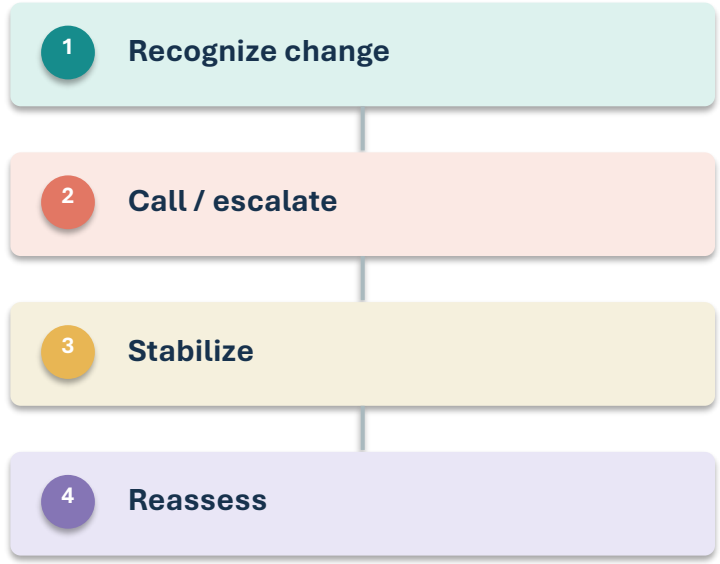


Sources: ACOG

Case: headache at 34 weeks

- A patient reports persistent headache and blurred vision.
- Assess BP, symptoms, reflexes, labs, and fetal status; escalate immediately.
- Do not reassure based on one normal urine dipstick.

CASE



Use the local emergency pathway.

Sources: ACOG

Prevention and long-term health

- Identify high-risk history and review preventive options with the clinician.
- Preeclampsia increases later cardiovascular risk.
- Connect postpartum patients to primary care and risk-factor follow-up.



Sources: ACOG

A high-risk pregnancy

is one in which a condition exists that jeopardizes the health of the mother, her fetus, or both.

The condition may result from the pregnancy, or it may be a condition that was present before the woman became pregnant.

Approximately one in four pregnant women is considered to be at high risk or diagnosed with complications

The risk status of a woman and her fetus can change during the pregnancy, with a number of problems occurring during labor, birth, or afterward

BOX 19.1 Factors Placing a Woman at Risk During Pregnancy

Biophysical Factors
- Genetic conditions
- Chromosomal abnormalities
- Multiple pregnancy
- Defective genes
- Inherited disorders
- ABO incompatibility
- Large fetal size
- Medical and obstetric conditions
- Preterm labor and birth
- Cardiovascular disease
- Chronic hypertension
- Incompetent cervix
- Placental abnormalities
- Infection
- Diabetes
- Maternal collagen diseases
- Pregnancy-induced hypertension
- Asthma
- Postterm pregnancy
- Hemoglobinopathies
- Nutritional status
- Inadequate dietary intake
- Food fads
- Excessive food intake
- Under- or overweight status
- Hematocrit value less than 33%
- Eating disorder

Psychosocial Factors
- Smoking
- Caffeine
- Alcohol
- Drugs
- Inadequate support system
- Situational crisis
- History of violence
- Emotional distress
- Unsafe cultural practices

Sociodemographic Factors
- Poverty status
- Lack of prenatal care
- Age younger than 15 years or older than 35 years
- Parity—all first pregnancies and more than five pregnancies
- Marital status—increased risk for unmarried
- Accessibility to health care
- Ethnicity—increased risk in nonwhite women

Environmental Factors
- Infections
- Radiation
- Pesticides
- Illicit drugs
- Industrial pollutants
- Second-hand cigarette smoke
- Personal stress (Gilbert, 2007; Blackburn, 2007)

Abortion

- Abortion is defined as the expulsion of an embryo or fetus before it is viable
- An abortion is the loss of an early pregnancy, usually before week 20 gestation.
- Abortion can be spontaneous or induced.
- A spontaneous abortion refers to the loss of a fetus resulting from natural causes—that is, not elective or therapeutically induced by a procedure.

- The term miscarriage is often used by nonmedical people to denote an abortion that has occurred spontaneously. A miscarriage can occur during early pregnancy & many women who miscarry may not even be aware that they are pregnant.
- Abortion can be a medical or surgical procedure.
- About 80% of spontaneous abortions occur within the first trimester.

Pathophysiology

- The causes of spontaneous abortion are varied and often unknown.
- The most common cause for first-trimester abortions is fetal genetic abnormalities, usually unrelated to the mother.

- Those occurring during the second trimester are more likely related to maternal conditions, such as incompetent cervix, congenital or acquired anomaly of the uterine cavity, hypothyroidism, diabetes mellitus, chronic nephritis, use of crack cocaine, inherited and acquired thrombophilias, lupus, and acute infection such as rubella virus, cytomegalovirus, herpes simplex virus, bacterial vaginosis, and toxoplasmosis.

Types of miscarriages



Threatened

- Vaginal bleeding
- Closed cervical os
- Fetal cardiac activity



Missed

- No vaginal bleeding
- Closed cervical os
- No fetal cardiac activity or empty sac



Inevitable

- Vaginal bleeding
- Dilated cervical os
- Products of conception may be seen or felt at or above cervical os



Incomplete

- Vaginal bleeding
- Dilated cervical os
- Some products of conception expelled & some remain



Complete

- Vaginal bleeding or none
- Closed cervical os
- Products of conception completely expelled

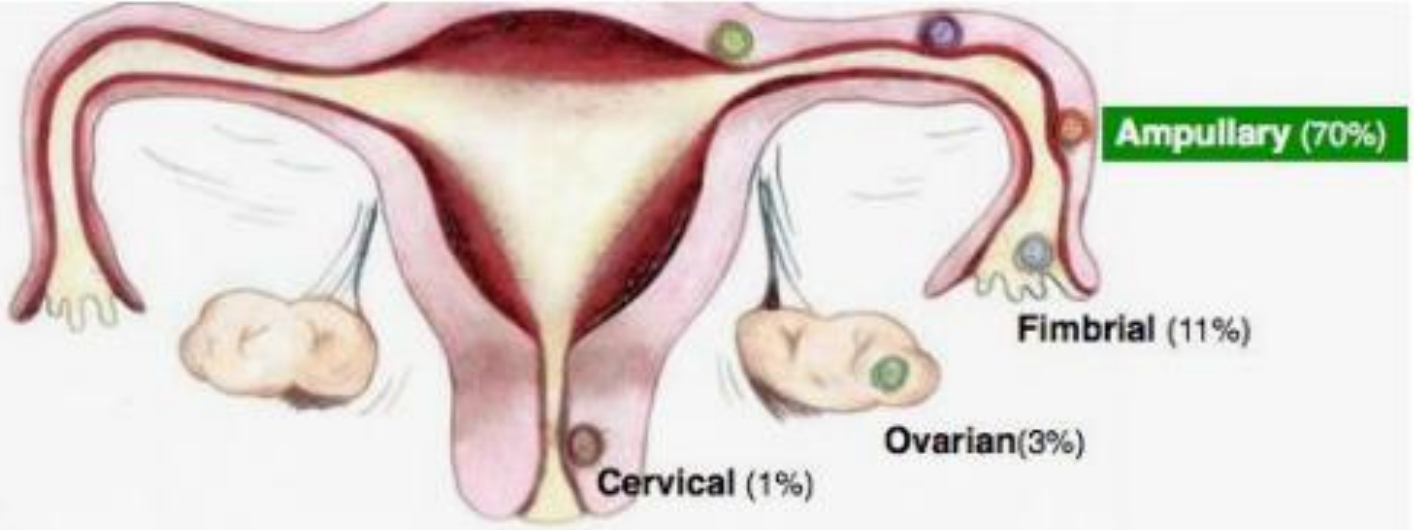
World, LLC

ECTOPIC PREGNANCY

- is any pregnancy in which the fertilized ovum implants outside the uterine cavity.
- With an ectopic pregnancy, rupture and hemorrhage occur due to the growth of the embryo.
- It is a potentially life-threatening condition and involves pregnancy loss.
- It is the leading cause of maternal mortality in the first trimester and accounts for 10% to 15% of all pregnancy related deaths.
- The most common site for implantation is the fallopian tubes, but some ova may implant in the cornua of the uterus, the ovary, the cervix, or the abdominal cavity.

ECTOPIC PREGNANCY

- It result from conditions that obstruct or slow the passage of the fertilized ovum through the fallopian tube to the uterus or a physical blockage in the tube, or failure of the tubal epithelium to move the zygote. In the general population, most cases are the result of tubal scarring secondary to pelvic inflammatory disease. Organisms such as Neisseria gonorrhoeae and Chlamydia trachoma at is preferentially attack the fallopian tubes, producing silent infections. A recent study reported a twofold increased risk for ectopic pregnancy in women with a history of a chlamydia infection



Risk Factors for Ectopic Pregnancy
Previous ectopic pregnancy
Tubal surgery
Pelvic inflammatory disease
Tobacco use
Advanced maternal age
Prior spontaneous abortion
Medically induced abortion
History of infertility
Intrauterine device

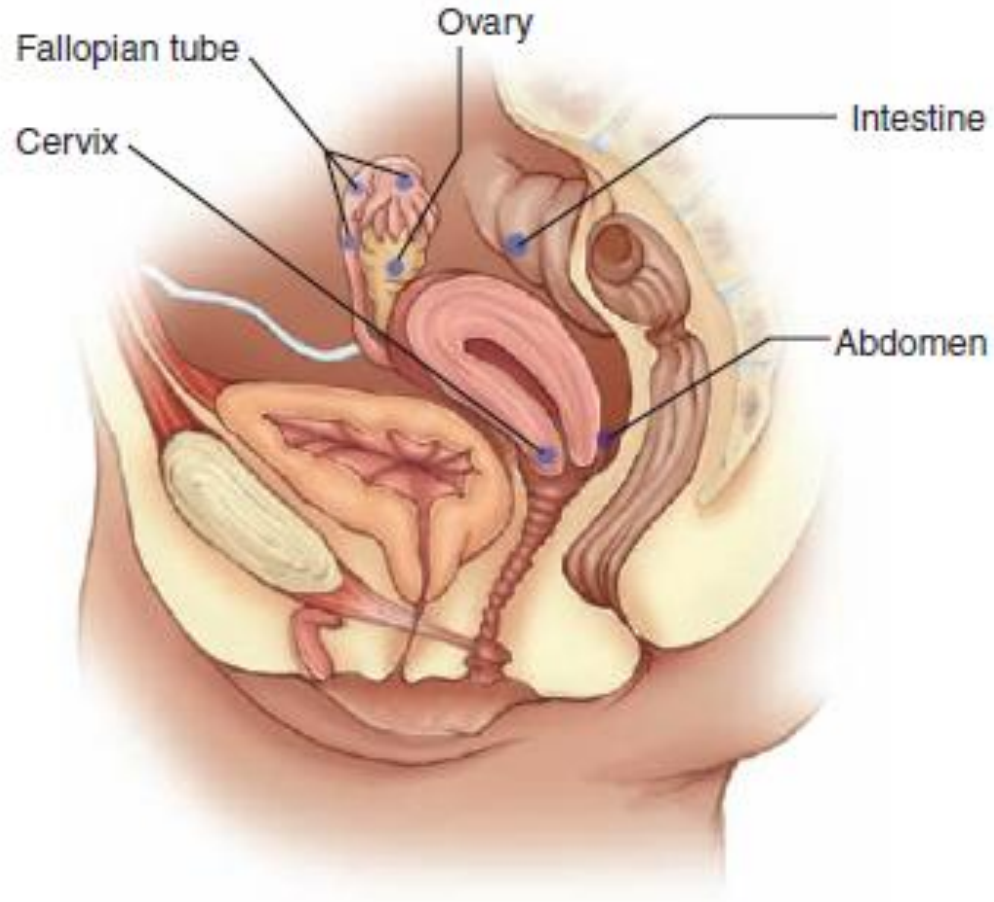


FIGURE 19.1 Possible sites for implantation with an ectopic pregnancy.

Therapeutic Management

- The therapeutic management of ectopic pregnancy depends on whether the tube is intact or has ruptured.
- Historically, the treatment of ectopic pregnancy was limited to surgery, but medical therapy is currently available.
- If the fallopian tube is still intact, medical management becomes an option.
- The potential advantages include avoidance of surgery, the preservation of tubal patency and function, and a lower cost. Methotrexate, prostaglandins, misoprostol, and actinomycin have all been used in the medical (nonsurgical) management

Therapeutic Management

- The agent most commonly used, is a folic acid antagonist that inhibits cell division in the developing embryo.
- Adverse effects associated with methotrexate include nausea, vomiting, stomatitis, diarrhea, gastric upset, increased abdominal pain, and dizziness.

- Surgical management for the unruptured fallopian tube might involve a linear salpingostomy to preserve the tube—an important consideration for the woman wanting to preserve her future fertility. With a ruptured ectopic pregnancy, surgery is necessary as a result of possible uncontrolled hemorrhage.
- All Rh-negative unsensitized clients are given Rh immunoglobulin to prevent isoimmunization in future pregnancies.

CERVICAL INSUFFICIENCY

- Cervical insufficiency, also called premature dilatation of the cervix, a weak, structurally defective cervix resulting in the loss of the pregnancy.
- The exact mechanism contributing to cervical insufficiency is not known.
- The incompetent cervix may have less elastin, less collagen, and greater amounts of smooth muscle than the normal cervix.
- Cervical insufficiency is may be congenital cervical hypoplasia, in utero diethylstilbestrol (DES) exposure that caused cervical hypoplasia, trauma to the cervix (conization, amputation, obstetric laceration, or forced cervical dilatation).
- Previous precipitous birth, a prolonged second stage of labor, or increased uterine volume (multiple gestation, hydramnios) are associated with cervical insufficiency.
- Short cervical length has associated with cervical insufficiency and PTB.

Therapeutic Management

- Cervical insufficiency may be treated in bed rest; pelvic rest; avoidance of heavy lifting; or surgically, via a procedure of a cervical cerclage in the second trimester.
- Cervical cerclage involves using a heavy pursestring suture to secure and reinforce the internal os of the cervix.
- Suture displacement, rupture of membranes, and chorioamnionitis are the most common complications associated with cerclage placement, and their incidence varies widely in relation to the timing and indications for the cerclage



PLACENTA PREVIA

- is a bleeding condition that occurs during the last two trimesters of pregnancy.
- the placenta implants over the cervical os.
- It complicate 1 in every 200 pregnancies and is associated with potentially serious consequences from hemorrhage, abruption (separation) of the placenta, cesarean birth.
- Pathophysiology The exact cause of placenta previa is unknown.
- It is initiated by implantation of the embryo in the lower uterus.
- Placental vascularization is defective, allowing the placenta to attach directly to the myometrium (accreta), invade the myometrium (increta), or penetrate the myometrium (percreta).

PLACENTA PREVIA

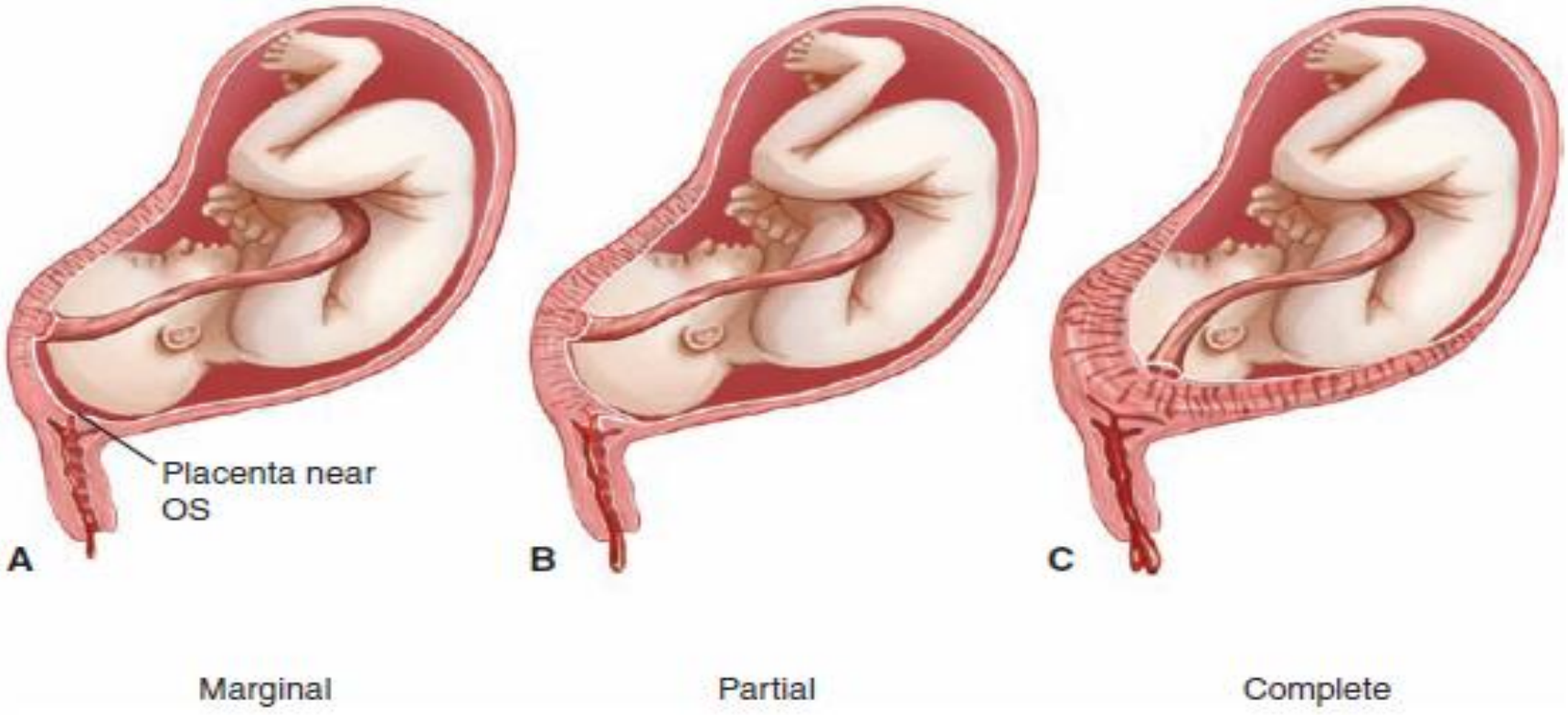
- Placenta previa is generally classified according to the degree of coverage or proximity to the internal os, as follows:
  - Total placenta previa: the internal cervical os is completely covered by the placenta
  - Partial placenta previa: the internal os is partially covered by the placenta
  - Marginal placenta previa: the placenta is at the margin or edge of the internal os
  - Low-lying placenta previa: the placenta is implanted in the lower uterine segment and is near the internal os but does not reach it

Therapeutic management depends on the extent of bleeding, the amount of placenta over the cervical os,

If the mother and fetus are both stable, therapeutic management may involve expectant (“wait-and-see”) care.

PLACENTA PREVIA

If there is no active bleeding and the client has readily available access to reliable transportation, can maintain bed rest at home, and can comprehend instructions, expectant care at home is appropriate. However, if the client requires continuous care and monitoring and cannot meet the home care requirements, the antepartal unit is the best environment.



Risk factors:

- Advancing maternal age (more than 35 years)
- Previous cesarean birth
- Multiparity
- Uterine insult or injury or Previous myomectomy to remove fibroids
- Cocaine use and Smoking
- Prior placenta previa
- Infertility treatment
- Asian ethnic background (incidence is increased in Asian cultural groups)
- Multiple gestations
- Previous induced surgical abortion

Health History

- Problems associated with bleeding, now or in the recent past.
- The classical clinical presentation is painless, bright-red vaginal bleeding occurring during the second or third trimester.
- The first episode of bleeding occurs (on average) at 27 to 32 weeks' gestation.
- If implantation in the upper uterus, minor disruptive placental attachment is not a problem, because there is a larger volume of myometrial tissue able to contract and constrict bleeding vessels.
- Palpate the uterus; typically it is soft and nontender on examination.
- distress is usually absent but may occur when cord problems arise, such as umbilical cord prolapse or cord compression, or when the client has experienced blood loss to the extent that maternal shock or placental abruption has occurred.

Laboratory and Diagnostic

- Testing To validate the position of the placenta
- a transvaginal ultrasound is done.
- magnetic resonance imaging may be ordered when preparing for delivery because it allows identification of placenta accreta (placenta abnormally adherent to the myometrium), increta (placenta accreta with penetration of the myometrium), or percreta (placenta accreta with invasion of the myometrium to the peritoneal covering, causing rupture of the uterus) in addition to placenta previa.
- These placental abnormalities, although rare, carry a very high morbidity and mortality rate, possibly necessitating a hysterectomy at delivery.

Nursing Management

Whether the care setting is in the client's home or in the health care facility, the nurse focuses on:

- monitoring the maternal–fetal status, including assessing for signs and symptoms of vaginal bleeding and fetal distress
- providing support and education to the client and her family
- For the majority of women, a cesarean birth will be planned.

Abruptio placentae

- is the separation of a normally located placenta after the 20th week of gestation and prior to birth that leads to hemorrhage.
- It is a significant cause of third-trimester bleeding, with a high mortality rate.
- It occurs in about 1% of all pregnancies throughout the world (1 in 120 pregnancies).
- The overall fetal mortality rate for placental abruption is 20% to 40%, depending on the extent of the abruption. Maternal mortality is approximately 6% in abruptio placentae and is related to cesarean birth and/or hemorrhage/coagulopathy.

Pathophysiology

- unknown; degenerative changes in the small maternal arterioles, resulting in thrombosis, degeneration of the decidua, and possible rupture of a vessel.
- Bleeding from the vessel forms a retroplacental clot causes increased pressure behind the placenta and results in separation
- Abruptio placentae is classified according to the extent of separation and the amount of blood loss from the maternal circulation. Classifications include:
  - Mild (grade 1): minimal bleeding (less than 500 mL), marginal separation (10% to 20%), tender uterus, no coagulopathy, no signs of shock, no fetal distress
  - Moderate (grade 2): moderate bleeding (1,000 to 1,500 mL), moderate separation (20% to 50%), continuous abdominal pain, mild shock
  - Severe (grade 3): absent to moderate bleeding (more than 1,500 mL), severe separation (more than 50%), profound shock, abd. pain, and disseminated intravascular coagulopathy (DIC)

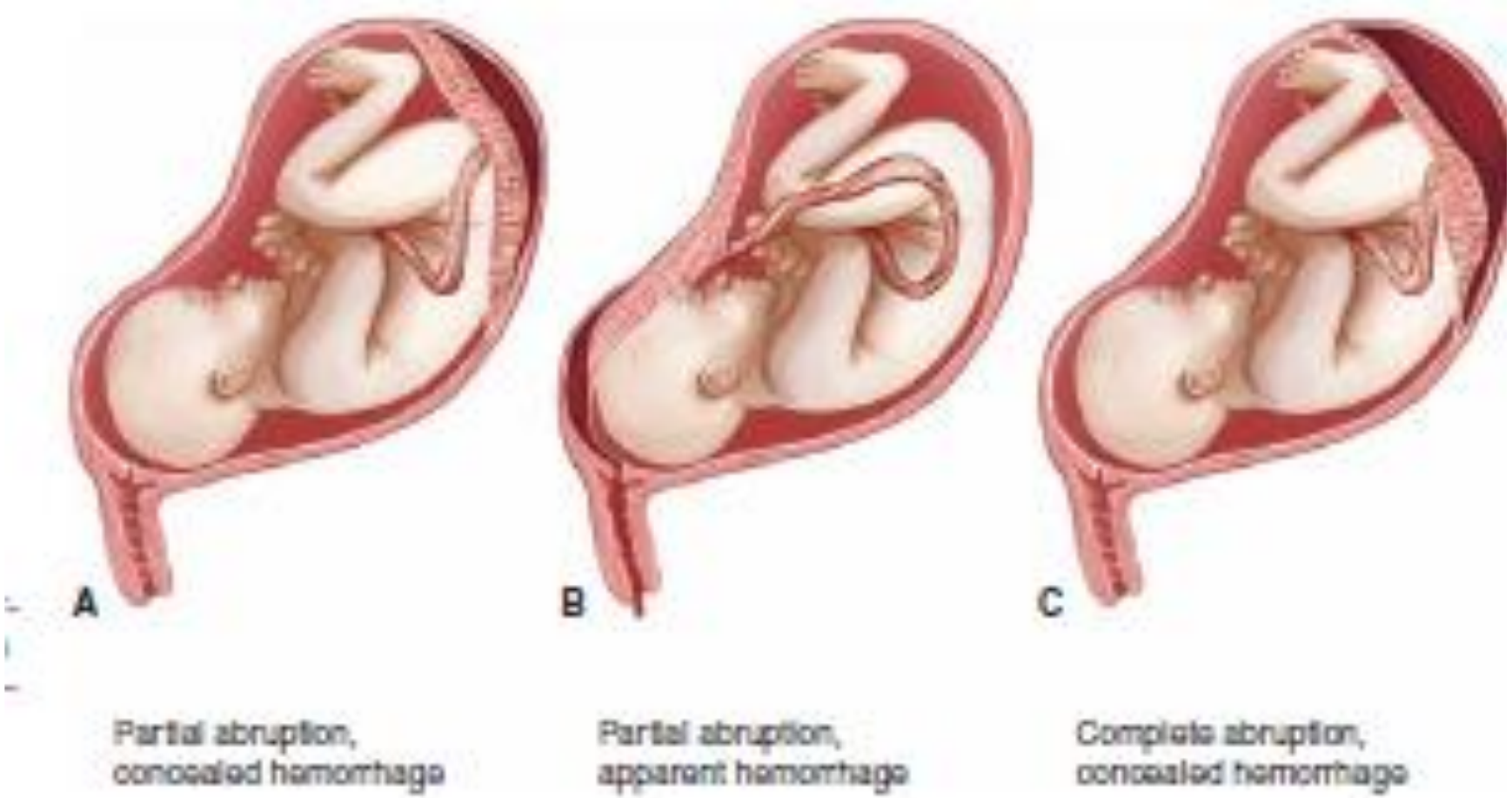
Therapeutic Management

- Abruptio placentae may be classified as partial or complete, by the degree of separation. it can be classified as concealed or apparent, by the type of bleeding.
- Treatment of abruptio placentae is designed to assess, control, and restore the amount of blood lost; to provide a positive outcome for both mother and newborn; and to prevent coagulation disorders, such as DIC.
- Emergency measures include starting two large-bore IV lines with normal saline or lactated Ringer's solution to combat hypovolemia, obtaining blood specimens for evaluating hemodynamic status values and for typing and cross-matching, and frequently monitoring fetal and maternal well-being.
- After the severity of abruption is determined and appropriate blood and fluid replacement is given, cesarean birth is done immediately if fetal distress is evident.

Therapeutic Management

- If the fetus is not in distress, close monitoring continues, with delivery planned at the earliest signs of fetal distress.
- Because of the possibility of fetal blood loss through the placenta, a neonatal intensive care team should be available during the birth process to assess and treat the newborn immediately for shock, blood loss, and hypoxia.
- If the woman develops DIC, treatment focuses on determining the underlying cause of DIC and correcting it. Replacement therapy of the coagulation factors is achieved by transfusion of fresh-frozen plasma along with cryoprecipitate to maintain the circulating volume and provide oxygen to the cells of the body.
- Anticoagulant therapy (heparin), packed red cells, platelet concentrates, antithrombin, and plasma protein fraction or albumin

Therapeutic Management



Hyperemesis Gravidarum

- At least 80% of women experience nausea and vomiting during their pregnancy.
- The term morning sicknessis often used to describe this condition when symptoms are mild disappear after the first trimester & affects the quality of life of the woman and her family results in dehydration, electrolyte imbalance, and the need for hospitalization.
- Unlike morning sickness, hyperemesis gravidarum is a complication of pregnancy characterized by persistent, uncontrollable nausea and vomiting that persists beyond the 20th week of pregnancy, causing weight loss of more than 5% of prepregnancy body weight, dehydration, metabolic acidosis from starvation, alkalosis from loss of hydrochloric acid, and hypokalemia. Hyperemesis is estimated to occur in approximately 5 per 1,000 pregnancies.
- The prevalence increases in molar pregnancies and multiple gestations.

Pathophysiology

– Although the exact cause of nausea and vomiting is unknown, It is likely that multiple factors contribute to it.

  - Elevated levels of hCG
  - Endocrine theory—high levels of hCG and estrogen during pregnancy
  - Metabolic theory—vitamin B6 deficiency
  - Psychological theory—psychological stress increases the symptoms

Therapeutic Management

– This usually focuses on dietary and lifestyle changes or hospitalization

– assess dehydration, electrolyte imbalance, ketosis, and malnutrition.

– lactated Ringer's with vitamins (5% dextrose, pyridoxine [B6]) and electrolytes added, Antiemetics, total parenteral nutrition to prevent malnutrition. Promethazine (Phenergan) and prochlorperazine (Compazine) dansetron (Zofran).

– Acupressure, massage, touch, ginger, and the wearing of Seabands to prevent nausea and vomiting.



Gestational Hypertension

- characterized by hypertension without proteinuria after 20 weeks of gestation and a return of the blood pressure to normal postpartum. Previously, gestational hypertension was known as pregnancy induced hypertension or toxemia of pregnancy, but these terms are no longer used.
- blood pressure of 140/90 mm Hg and G HTN is the second leading cause of maternal death in the United States after thromboembolism.
- The highest rates are in women younger than 20 or older than 40
- The classification of hypertensive disorders in pregnancy currently consists of four categories:
  - Gestational hypertension: blood pressure elevation (140/90 mm Hg) identified after midpregnancy without proteinuria
  - Chronic hypertension: hypertension prior to the pregnancy or diagnosed before week 20 of gestation
  - Transient hypertension: hypertension without preeclampsia at the time of birth, during labor, or within 24 hours postpartum; resolution within 12 weeks postpartum to normotensive blood pressure levels
  - Chronic hypertension with preeclampsia: chronic hypertension with new proteinuria or exacerbation of previously controlled blood pressure, or proteinuria, thrombocytopenia, or elevated liver enzymes.

Pathophysiology

- Gestational hypertension in some can be classified as preeclampsia or eclampsia .
- Gestational hypertension remains an enigma for the pathologic changes that result in pulmonary edema, oliguria, seizures, thrombocytopenia, and abnormal liver enzyme.
- Decreased liver perfusion leads to impaired liver function and subcapsular hemorrhage. This is demonstrated by epigastric pain and elevated liver enzymes in the maternal serum.
- Decreased brain perfusion leads to small cerebral hemorrhages and symptoms of arterial vasospasm such as headaches, visual disturbances, blurred vision
- Decreased kidney resulting in decreased urine output and increased serum levels of sodium, BUN, uric acid, and creatinine, further increasing extracellular fluid and edema.
- Poor placental perfusion resulting from prolonged vasoconstriction helps to contribute to intrauterine growth restriction, premature separation of the placenta (abruptio placentae), persistent fetal hypoxia, and acidosis.

Mild Preeclamp

- Mild elevations in blood pressure may be placed on bed rest at home as much as possible in the lateral recumbent position to improve uteroplacental blood flow, reduce her blood pressure, and promote diuresis. In addition, antepartal visits and diagnostic testing—such as CBC, clotting studies, liver enzymes, and platelet levels
- monitor her blood pressure daily (every 4 to 6 hours while awake), measure the amount of protein found in urine, weight gain, fetal movement counts also are implemented.
- A balanced, nutritional diet with no sodium restriction is advised.
- admission to the hospital is warranted based on the severity, the woman with mild preeclampsia is monitored closely for signs and symptoms of severe preeclampsia or impending eclampsia (e.g., persistent headache, hyperreflexia).
- Blood pressure, daily weights, edema, nonstress testing, and serial ultrasounds to evaluate fetal growth and amniotic fluid volume to confirm fetal well-being.

Severe preeclampsia

- Severe preeclampsia may develop suddenly and bring with it high blood pressure of more than 160/110 mmHg, proteinuria of more than 5 g in 24 hours, oliguria of less than 400 mL in 24 hours, cerebral and visual symptoms, and rapid weight gain, so immediate hospitalization is needed. Treatment is highly individualized and based on disease severity and fetal age. Therapy focuses on controlling hypertension, preventing seizures, preventing long-term morbidity, and preventing maternal, fetal, or newborn death.
- The woman in labor with severe preeclampsia typically receives oxytocin to stimulate uterine contractions and magnesium sulfate to prevent seizure activity, PGE2 gel may be used to ripen the cervix. A cesarean birth may be performed if the client is seriously ill.

Management of Eclampsia

- develops an eclamptic seizure, the convulsive activity begins with facial twitching, followed by generalized muscle rigidity. Respirations cease for the duration of the seizure, resulting from muscle spasms, coma usually follows the seizure activity, with respiration resuming.
- Eclamptic seizures are life-threatening emergencies and require immediate treatment to decrease maternal morbidity and mortality. As with any seizure, the initial management is to clear the airway and administer adequate oxygen.
- Positioning the woman on left side and protecting her from injury during the seizure are key.
- Suction equipment must be readily available to remove secretions from her
- IV fluids are administered after the seizure at a rate to replace urine output and additional insensible losses.
- Magnesium sulfate is administered IV to prevent further seizures. Hypertension is controlled with antihypertensive medications.

HELLP

- HELLP is an acronym for hemolysis, elevated liver enzymes, and low platelets.
- HELLP syndrome occurs in about 20% of severe preeclampsia.
- It has been reported as early as 17 weeks' gestation (22 and 36 weeks' gestation).
- HELLP syndrome leads to an increased maternal risk for developing liver hematoma or rupture, stroke, cardiac arrest, seizure, pulmonary edema, DIC, subendocardial hemorrhage, adult respiratory distress syndrome, renal damage, sepsis, hypoxic encephalopathy, and maternal or fetal death.
- Pathophysiology: The hemolysis that occurs is termed microangiopathic hemolytic anemia, when red blood cells become fragmented as they pass through small, damaged blood vessels. Elevated liver enzymes are the result of reduced blood flow to the liver secondary to obstruction from fibrin deposits. Hyperbilirubinemia and jaundice result from liver impairment. Low platelets result from vascular damage, the result of

Therapeutic Management

- The treatment for HELLP syndrome is based on the severityof the disease, the gestational age of the fetus, and the condition of the mother and fetus.
- The client should be admitted or transferred to a tertiary center with a neonatal intensive care unit.
- Additional treatments include magnesium sulfate, antihypertensives (hydralazine or labetalol), and correction of the coagulopathies that accompany HELLP syndrome (fresh-frozen plasma, packed red cells, or platelets).
- After this syndrome is diagnosed and the woman’s condition is stable, birth of the infant is indicated
- Birth may be delayed up to 96 hours so that betamethasone or dexamethasone can be given to stimulate lung maturation in the preterm fetus

POLYHYDRAMNIOS

- Polyhydramnios, also called hydramnios, is a condition in which there is too much amniotic fluid (more than 2,000 mL) surrounding the fetus between 32 and 36 weeks. It occurs in approximately 3% of all pregnancies and is associated with fetal anomalies of development.
- It is associated with poor fetal outcomes because of the increased incidence of preterm births, fetal malpresentation, and cord prolapse. (Dx:AFI>20cm, Fundus hight)
- Polyhydramnios can be caused by fetal malformations, multiple gestation, maternal diabetes, and various fetal disorders, too much fluid is being produced, fluid being taken up, or both. It can be associated with maternal disease and fetal anomalies, or idiopathic in nature.
- Therapeutic Management include close monitoring if the hydramnios is mild to moderate. In severe cases in which the woman is in pain and experiencing shortness of breath, an amniocentesis or artificial rupture of the membranes is done to reduce the fluid and the pressure. A noninvasive treatment may involve the use of a prostaglandin synthesis inhibitor (indomethacin) to decrease amniotic fluid volume by decreasing fetal urinary output.

Polyhydramnios



1 ↓ Swallowing

- TE Fistula
- Duodenal Atresia
- Esophageal Fistula

2 ↑ Urination

- Maternal DM
- Hyperglycemia baby

Oligohydramnios



1 N/A

2 ↓ Urination

- Bladder outlet obstruction
- PCKD

Etiology

Polyhydramnios

- Maternal DM
- Multiple Gestation
- Isoimmunization
- Pulmonary Abnormalities
- Fetal anomalies
  - Duodenal atresia/TE fistula
  - Anencephaly
- Twin-Twin transfusion

Oligohydramnios

- Fetal UT problems
  - Renal agenesis (Potter's)
  - GU obstruction
- Uteroplacental insufficiency
- ROM



Complications

Polyhydramnios

- Preterm labor
- Fetal malpresentation
- Cord prolapse

Oligohydramnios

- ↑Mortality
- Musculoskeletal abnormalities
- Pulmonary hypoplasia
- Potter's syndrome
- Umbilical cord compression
- IUGR

OLIGOHYDRAMNIOS

- Oligohydramnios is a decreased amount of amniotic fluid (less than 500 mL) between 32 weeks and 36 weeks' gestation. It occurs in 5% to 8% of all pregnancies.
- Oligohydramnios may result from any condition that prevents the fetus from making urine or blocks it from going into the amniotic sac. (Dx:AFI<5cm, Fundus hight)
- Reduction in amniotic fluid reduces the ability of the fetus to move freely without risk of cord compression, which increases the risk for fetal death and intrapartal hypoxia.
- The woman with oligohydramnios can be managed on an outpatient basis with serial ultrasounds and fetal surveillance through nonstress testing and biophysical profiles.
- birth is planned along with amnioinfusion (the transvaginal infusion of crystalloid fluid to compensate for the lost amniotic fluid). The fluid is introduced into the uterus through an intrauterine pressure catheter
- Amnioinfusion is thought to improve abnormal fetal heart rate patterns, decrease cesarean births, and possibly minimize the risk of neonatal meconium aspiration syndrome

Gestational diabetes mellitus

- Glucose intolerance with its onset during pregnancy or first detected in pregnancy
- During pregnancy, diabetes typically is categorized into two groups: pregestational diabetes (alteration in carbohydrate metabolism identified before conception), which includes women with type 1 or type 2 disease; and gestational diabetes, which develops during pregnancy.
- Gestational diabetes occurs in approximately 8% of all pregnant women, it is associated with either neonatal complications such as macrosomia, hypoglycemia, and birth trauma or maternal complications such as preeclampsia and cesarean birth.
- chronic metabolic disorder remains a high-risk condition during pregnancy. A favorable outcome requires commitment on the woman's part to comply with frequent prenatal visits, dietary restrictions, self-monitoring of blood glucose levels, frequent laboratory tests, intensive fetal surveillance, and perhaps hospitalization

Pathophysiology With diabetes

- In pregnancy, placental hormones cause insulin resistance at a level that tends to parallel the growth of the fetoplacental unit. As the placenta grows, more placental hormones are secreted. Human placental lactogen (hPL) and growth hormone (somatotropin) increase in direct correlation with the growth of placental tissue, rising throughout the last 20 weeks of pregnancy and causing insulin resistance. Subsequently, insulin secretion increases to overcome the resistance of these two hormones. In the nondiabetic pregnant woman, the pancreas can respond to the demands for increased insulin production to maintain normal glucose levels throughout the pregnancy
- With this increased demand on the pancreas in late pregnancy, women with diabetes or glucose intolerance cannot accommodate the increased insulin demand; glucose levels rise as a result of insulin deficiency, leading to hyperglycemia. Subsequently, the mother and her fetus can experience major problems

Therapeutic Management

- Preconception counseling is key for the woman with pregestational diabetes (the alteration in carbohydrate metabolism is identified before conception) to ensure that her disease state is stable.
- A glycosylated hemoglobin level of less than 7% indicates good control
- Preconception counseling also is important in helping to reduce the risk of congenital malformation (The most common malformations associated with diabetes occur in the renal, cardiac, skeletal, and central nervous systems. Since these defects occur by the eighth week of gestation)
- maintaining a fasting blood glucose level below 95 mg/dL, with postprandial levels below 140 mg/dL and 2-hour postprandial levels below 120 mg/dL.
- Key aspects of treatment include nutritional management, insulin regimens, and close maternal and fetal surveillance.
- Nutritional management focuses on maintaining balanced glucose levels and providing enough energy and nutrients for the pregnant woman, while avoiding ketosis

Group B streptococcus

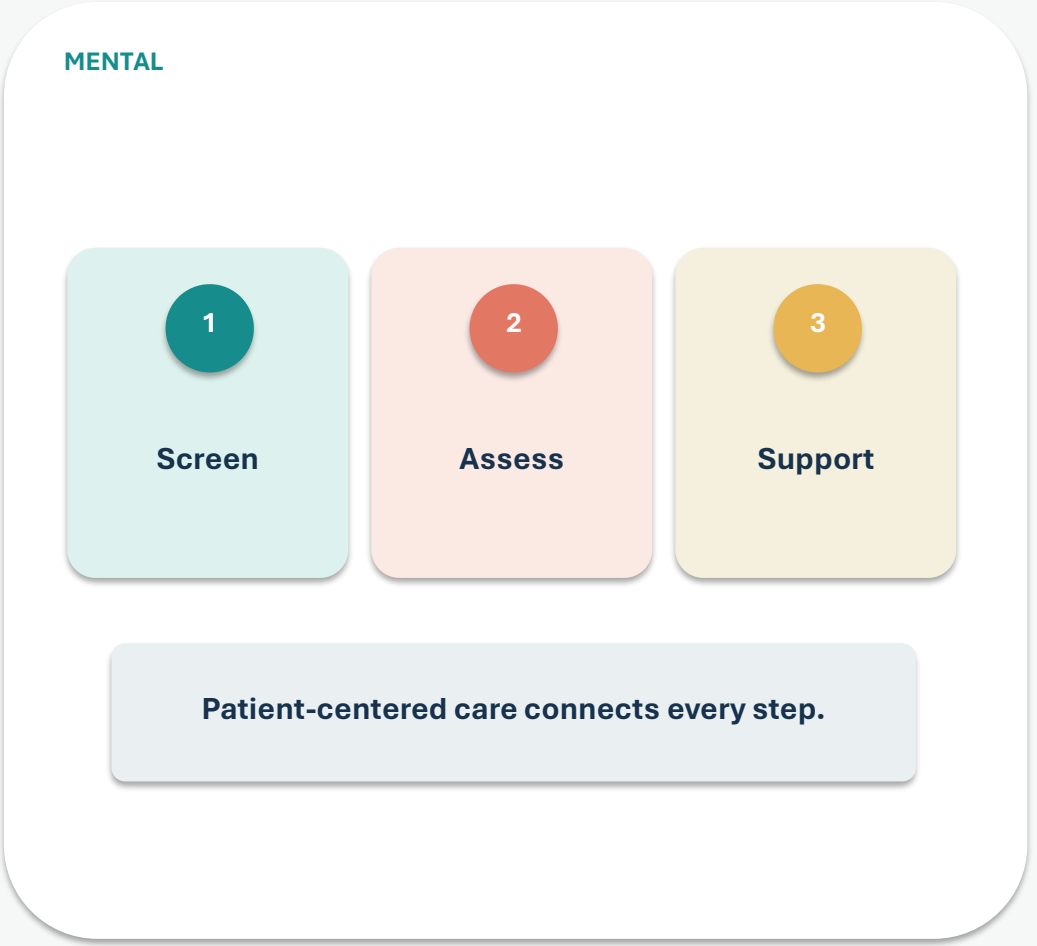
- Screening timing and intrapartum prophylaxis follow local protocols.
- Ask about allergy details when prophylaxis is indicated.
- Document status and communicate it at labor admission.



Sources: CDC/ACOG

Mental health and perinatal mood disorders

- Screen for depression, anxiety, trauma, and suicidality using validated tools.
- Normalize support-seeking and offer timely treatment pathways.
- Postpartum psychosis, suicidal intent, or inability to maintain safety is urgent.



Sources: ACOG

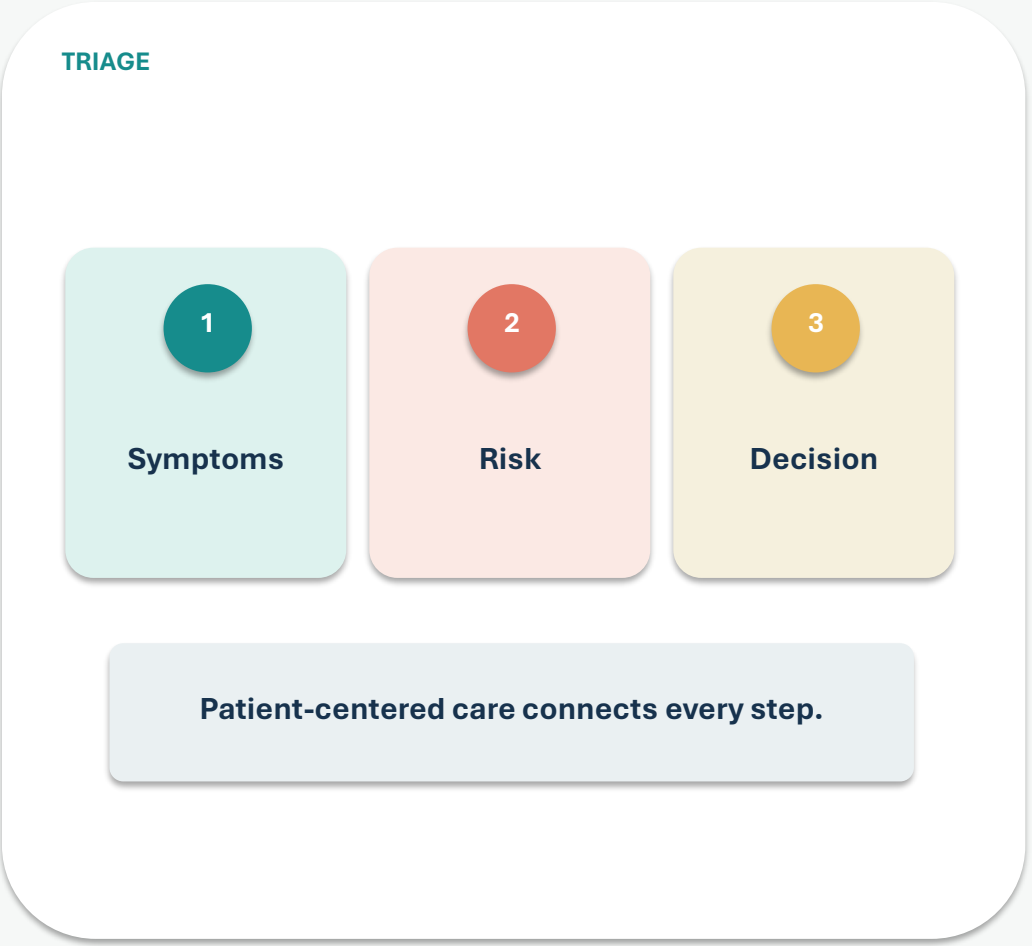
The labor process



10 focused lessons • assessment, teaching, and clinical reasoning

Signs of labor and triage

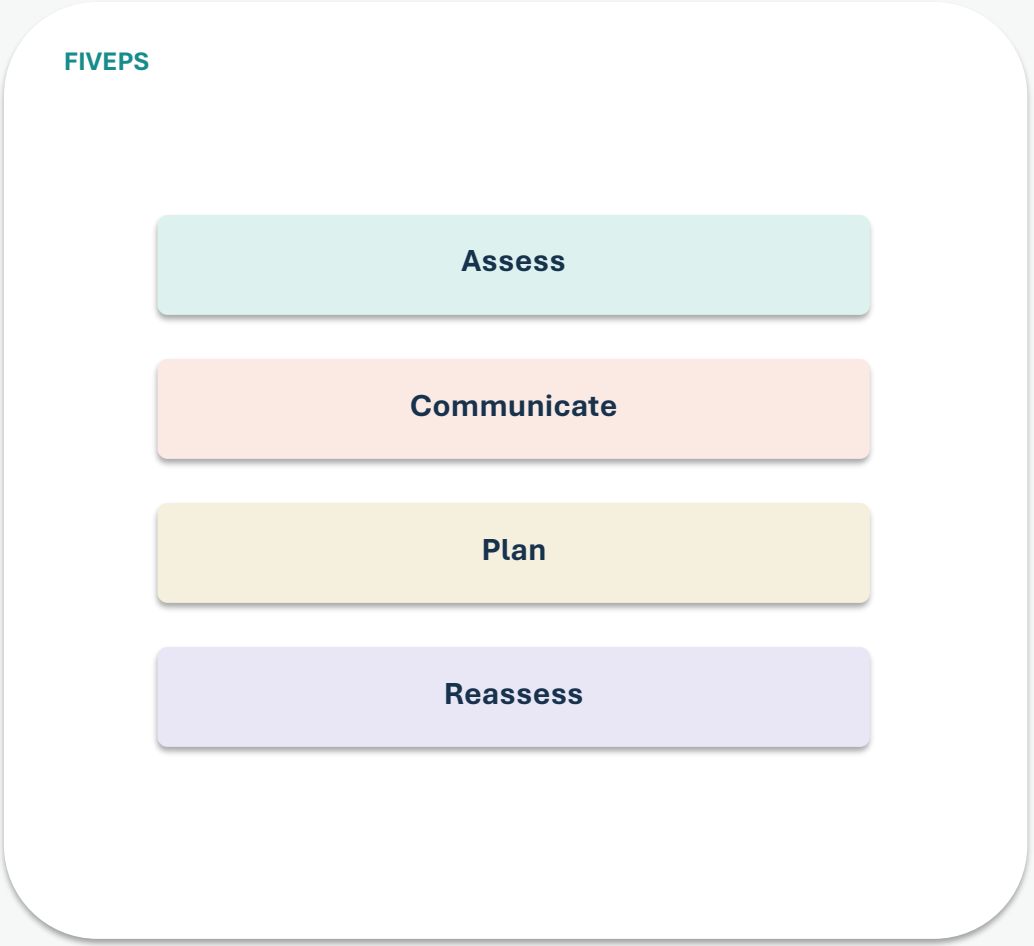
- Ask about contraction pattern, fluid loss, bleeding, fetal movement, and risk factors.
- True labor involves progressive cervical change with contractions.
- Triage decisions follow gestational age, history, and local policy.



Sources: NICE 2023

The five Ps

- Passenger: fetus, placenta, lie, presentation, attitude, and position.
- Passageway: pelvis and soft tissues; powers: contractions and pushing.
- Position and psychological response also influence labor progress.

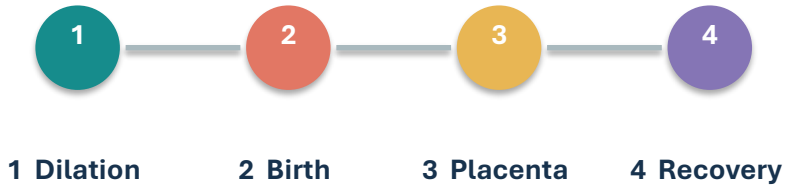


Sources: Course source

Stages of labor

- First stage: labor onset to complete cervical dilation.
- Second stage: complete dilation to birth of the newborn.
- Third stage: birth to delivery of placenta; fourth stage: early recovery.

STAGES



Timing and response depend on clinical context.

Sources: NICE 2023

LABOR AND BIRTH PROCESS

- Labor is a complex, multifaceted interaction between the mother and fetus.
- It is a series of processes by which the fetus is expelled from the uterus.
- labor is influenced by a combination of factors, including uterine stretch, progesterone withdrawal, increased oxytocin sensitivity, and increased release of prostaglandins.
- Theory suggests that labor is initiated by a change in the estrogen/progesterone ratio.
- During the last trimester of pregnancy, estrogen levels increase and progesterone levels decrease. This change leads to an increase in the number of myometrium gap junctions. Gap junctions are proteins that connect cell membranes and facilitate coordination of uterine contractions and myometrial stretching.

- Although physiologic evidence for the role of oxytocin in the initiation of labor is inconclusive, the number of oxytocin receptors in the uterus increases at the end of pregnancy.
- Estrogen, the levels of which are also rising, increases myometrial sensitivity to oxytocin.
- Oxytocin also aids in stimulating prostaglandin synthesis through receptors in the decidua. Prostaglandins lead to additional contractions, cervical softening, gap junction induction, and myometrial sensitization, thereby leading to a progressive cervical dilation (the opening or enlargement of the external cervical os).
- Prostaglandins are produced in the decidua and fetal membranes and have a central role in the initiation of labor.
- Prostaglandin levels increase late during pregnancy secondary to elevated estrogen levels. Prostaglandins stimulate smooth muscle contraction of the uterus. An increase in prostaglandins leads to myometrial contractions and a reduction in cervical resistance. Subsequently, the cervix softens, thins out, and dilates during labor

Premonitory Signs of Labor

- Cervical Changes
  - cervical softening and possible cervical dilation with descent of the presenting part into the pelvis occurs.
  - These changes can occur 1 month to 1 hour before actual labor begins.
  - These changes occur secondary to the effects of prostaglandins and pressure from Braxton Hicks contractions.
- Lightening
  - Lightening occurs when the fetal presenting part begins to descend into the maternal pelvis.
  - The woman usually notes that her breathing is much easier. However, she may complain of increased pelvic pressure, cramping, and low back pain.
  - She may notice an increase in vaginal discharge and more frequent urination.
  - Also, edema of the lower extremities may occur as a result of the increased stasis of pooling blood. In primiparas, lightening can occur 2 weeks or more before labor begins; among multiparas it may not occur until labor .

Premonitory Signs of Labor

- Increased Energy Level
  - Some women report a sudden increase in energy before labor.
  - This is sometimes referred to as nesting, because many women will focus this energy toward childbirth preparation by cleaning, cooking, preparing the nursery, and spending extra time with other children in the household.
  - It is as a result of an increase in epinephrine release caused by a decrease in progesterone
- Bloody Show
  - At the onset of labor, the mucous plug that fills the cervical canal during pregnancy is expelled as a result of cervical softening and increased pressure of the presenting part. These ruptured cervical capillaries release a smal amount of blood that mixes with mucus, resulting in the pink-tinged secretions known as bloody show

Premonitory Signs of Labor

- Braxton Hicks Contractions
  - Braxton Hicks contractions are typically felt as a tightening or pulling sensation of the top of uterus.
  - They occur primarily in the abdomen and groin and gradually spread downward before relaxing. In contrast, true labor contractions are more commonly felt in the lower back.
  - These contractions aid in moving the cervix from a posterior position to an anterior position. They also help in ripening and softening the cervix.
  - The contractions are irregular and can be decreased by walking, voiding, eating, increasing fluid intake, or changing position.
  - Braxton Hicks contractions usually last about 30 seconds but can persist for as long as 2 minutes.
  - if the contractions last longer than 30 seconds and occur more often than four to six times/ hour, she can be evaluated for possible preterm labor, especially if she is less than 38 weeks pregnant.

Premonitory Signs of Labor

- Spontaneous Rupture of Membranes
  - One in four women experience spontaneous rupture of the membranes before the onset of labor.
  - The rupture of membranes can result in either a sudden gush or a steady leakage of amniotic fluid.
  - After the amniotic sac has ruptured, the barrier to infection is gone and an ascending infection is possible. In addition, there is a danger of cord prolapse if engagement has not occurred with the sudden release of fluid and pressure with rupture.

TABLE 13.1 Differences Between True and False Labor

Parameters	True Labor	False Labor
Contraction timing	Regular, becoming closer together, usually 4–6 minutes apart, lasting 30–60 seconds	Irregular, not occurring close together
Contraction strength	Become stronger with time, vaginal pressure is usually felt	Frequently weak, not getting stronger with time or alternating (a strong one followed by weaker ones)
Contraction discomfort	Starts in the back and radiates around toward the front of the abdomen	Usually felt in the front of the abdomen
Any change in activity	Contractions continue no matter what positional change is made	Contractions may stop or slow down with walking or making a position change
Stay or go?	Stay home until contractions are 5 minutes apart, last 45–60 seconds, and are strong enough so that a conversation during one is not possible—then go to the hospital or birthing center.	Drink fluids and walk around to see if there is any change in the intensity of the contractions; if the contractions diminish in intensity after either or both—stay home.

Sources: Arenson & Drake, 2007; Blackburn, 2007; Cunningham et al., 2005.

Factors Affecting the Labor Process

- Critical factors that affect the process of labor and birth are outlined as the “five P’s”:
  - 1. Passageway (birth canal)
  - 2. Passenger (fetus and placenta)
  - 3. Powers (contractions)
  - 4. Position (maternal)
  - 5. Psychological response
- Five additional “P’s” can also affect the labor process:
  - 1. Philosophy (low-tech, high-touch) 2. Partners (support caregivers)
  - 3. Patience (natural timing) 4. Patient preparation (childbirth knowledge base)
  - 5. Pain control (comfort measures)

Passageway

- The birth passageway is the route through which the fetus must travel to be born vaginally.
- The passageway consists of the maternal pelvis and soft tissues.
- Typically the pelvis is assessed and measured during the first trimester, often at the first visit to the health care provider, to identify any abnormalities that might hinder a successful vaginal birth.
- As the pregnancy progresses, the hormones relaxin and estrogen cause the connective tissues to become more relaxed and elastic and cause the joints to become more flexible to prepare the mother’s pelvis for birth.

Passageway

- Bony Pelvis The maternal bony pelvis can be divided into the true and false portions.
- The false (or greater) pelvis is composed of the upper flared parts of the two iliac bones with their concavities and the wings of the base of the sacrum.
- The false pelvis lies above imaginary line (linea terminalis); the true pelvis lies below it.
- The true pelvis is the bony passageway through which the fetus must travel. It is made up of three planes: the inlet, the mid-pelvis (cavity), and the outlet.
- Pelvic Inlet The pelvic inlet allows entrance to the true pelvis. It is bounded by the sacral prominence in the back, the ilium on the sides, and the superior aspect of the symphysis pubis in the front. The pelvic inlet is wider in the transverse aspect (sideways) than it is from front to back.

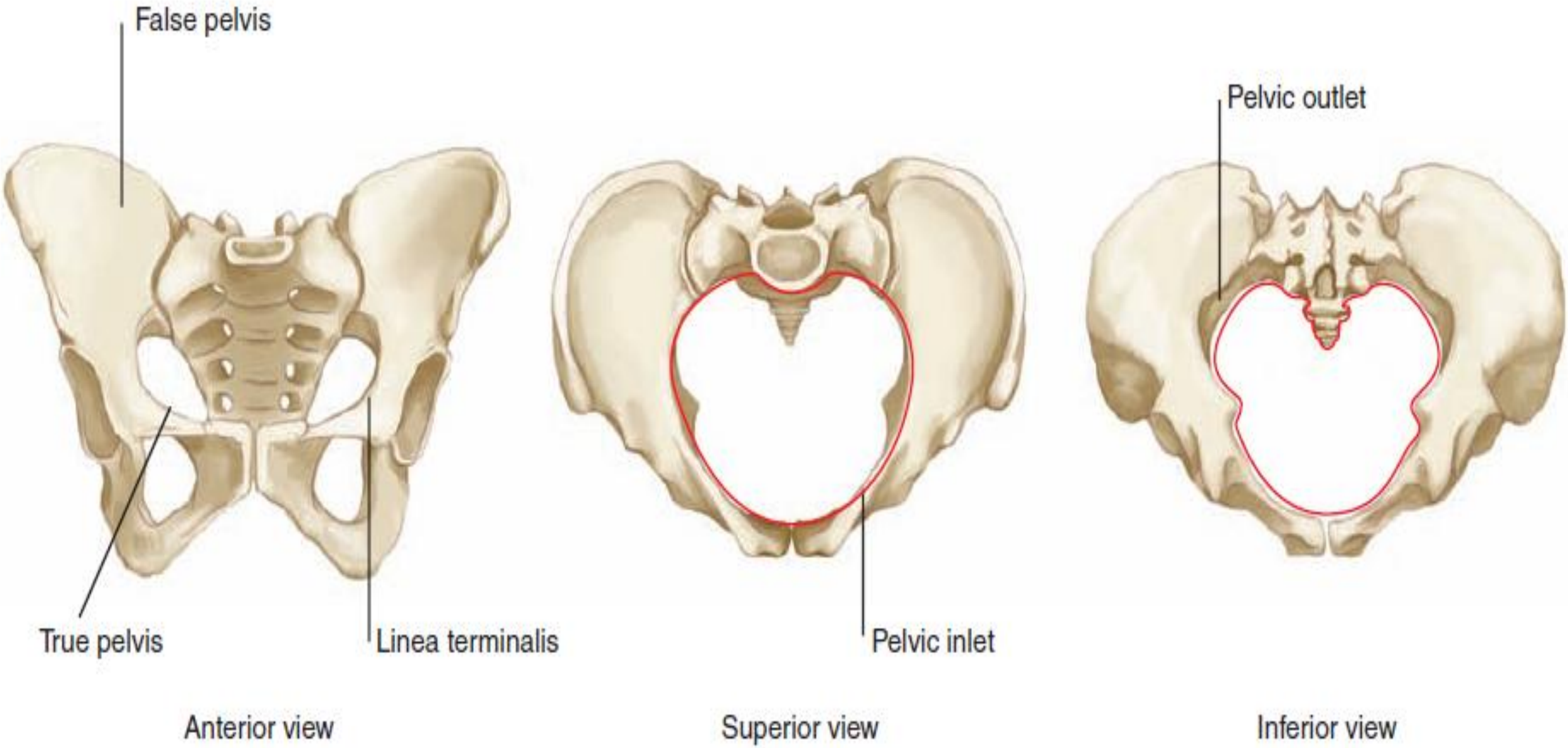
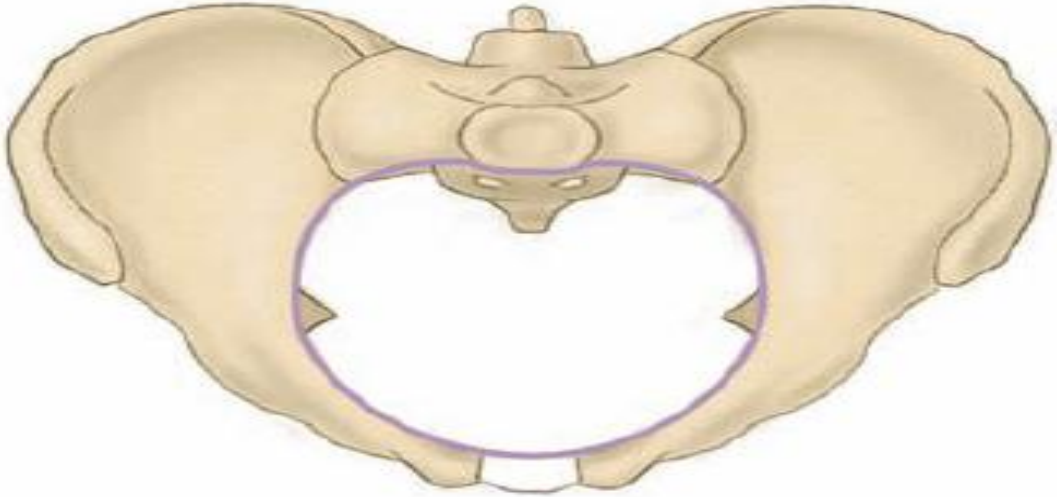


FIGURE 13.1 The bony pelvis.

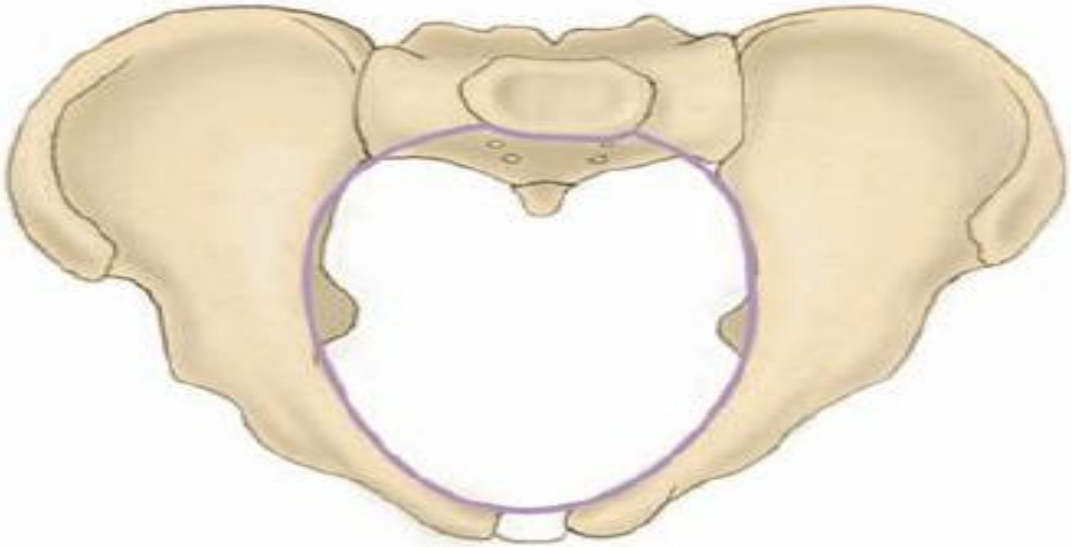
Pelvic Shape

- The shape of a woman's pelvis is a determining factor for a vaginal birth.
- The pelvis is divided into four main shapes: gynecoid, anthropoid, android, and platypelloid.
- **The gynecoid pelvis** is considered the true female pelvis, occurring in about 50% of all women; it is less common in men. Vaginal birth is most favorable with this type of pelvis because the inlet is round and the outlet is roomy. This type of pelvis allows early and complete fetal internal rotation during labor.
- **The anthropoid pelvis** is common in men and occurs in 20% to 30% of women. The pelvic inlet is oval. Vaginal birth is more favorable with this pelvic shape compared to the android or platypelloid shape
- **The android pelvis** is male-shaped pelvis and is characterized by a funnel shape. It occurs in approx 20% of women. The prognosis for labor is poor, subsequently leading to cesarean birth.
- **The platypelloid** or flat pelvis is the least common type of pelvic structure among men and women. It is not favorable for a vaginal birth and usually require cesarean birth.

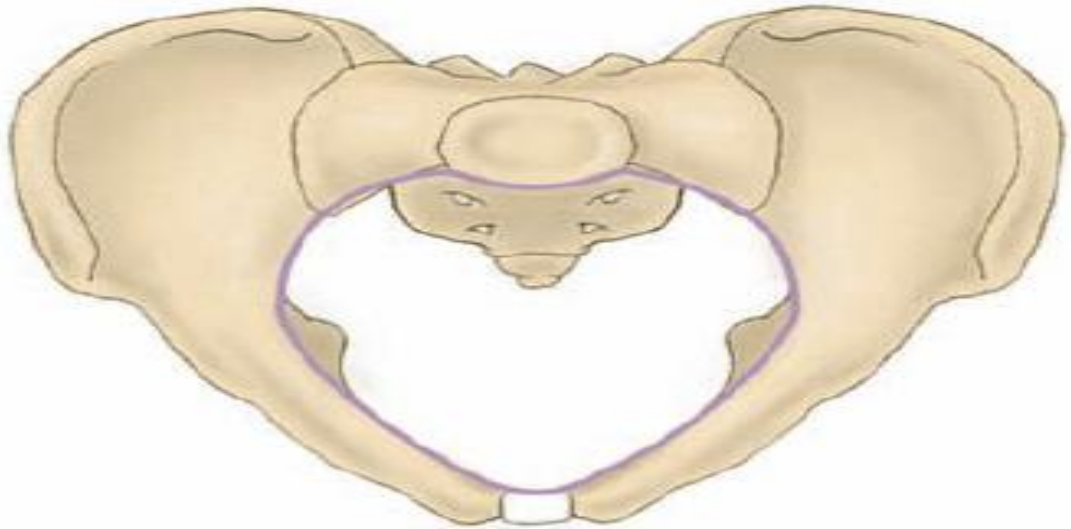
Pelvic Shape



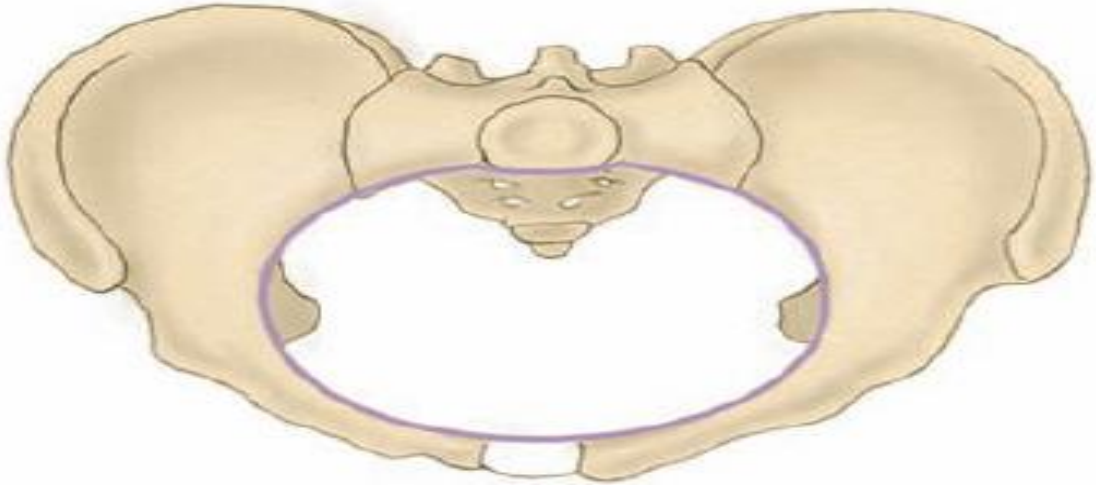
A Gynecoid



C Anthropoid



B Android



D Platypelloid

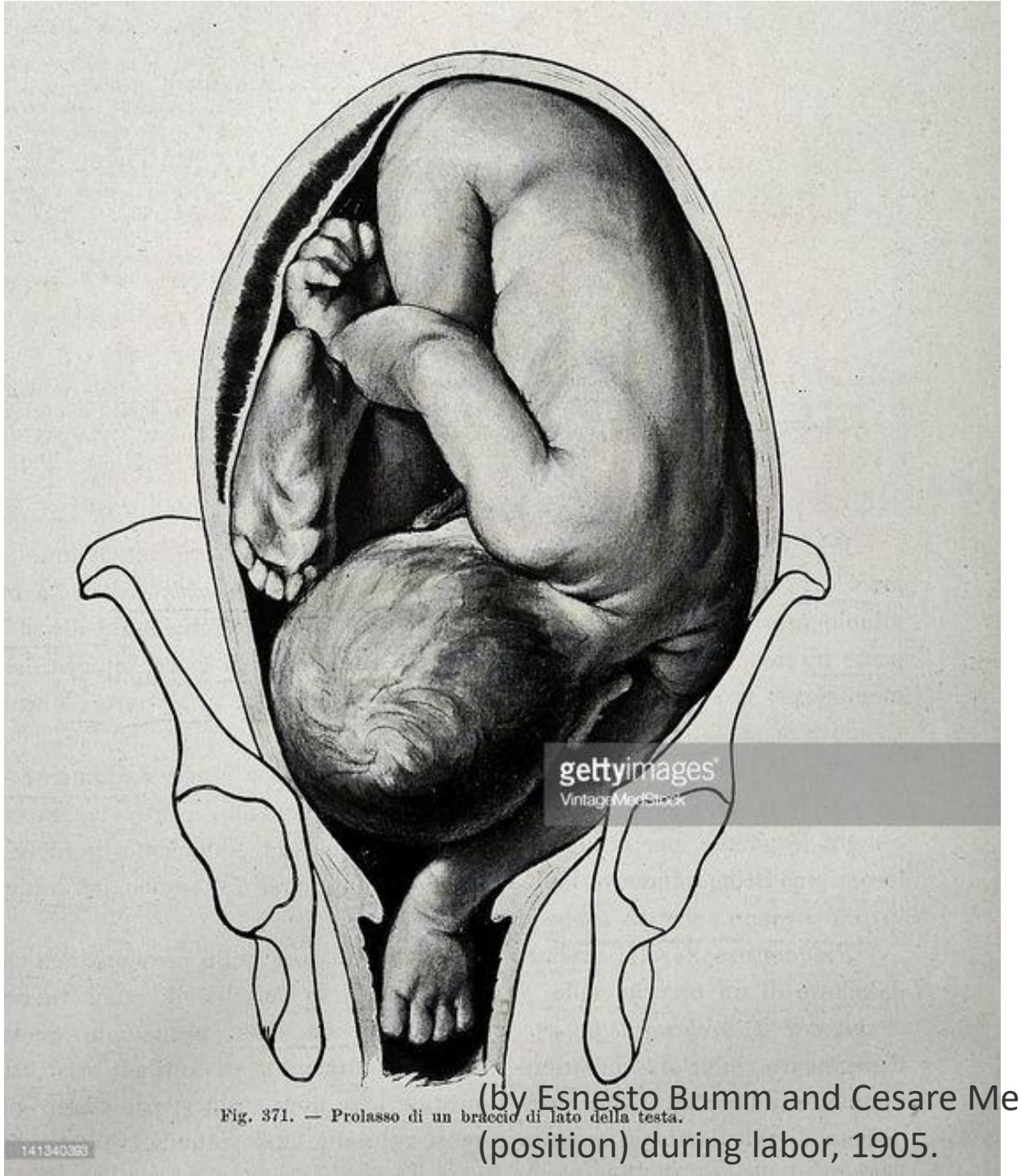


Fig. 371. — Prolasso di un braccio di lato della testa.

(by Esnesto Bumm and Cesare Merletti) illustrates the fetal attitude (position) during labor, 1905.

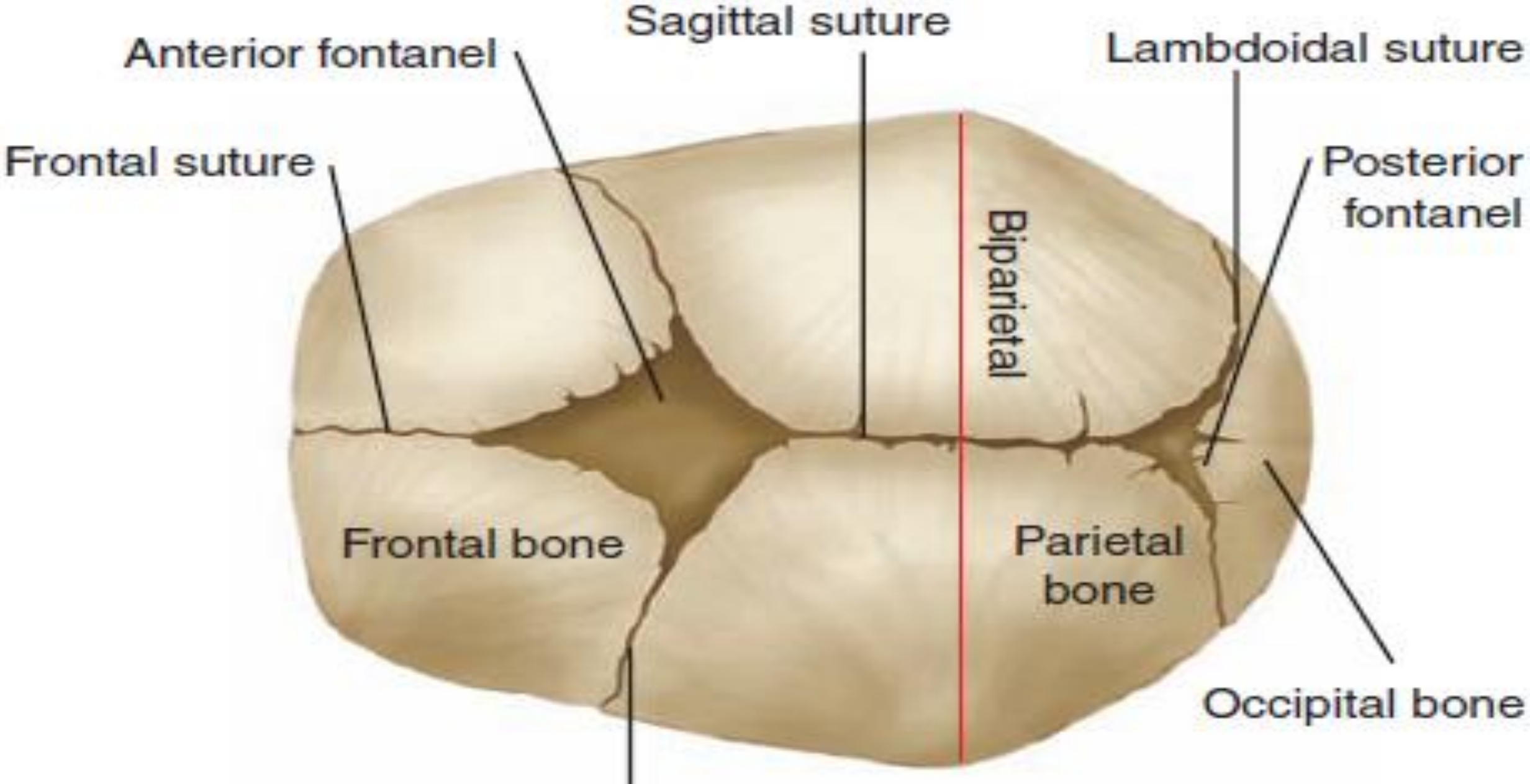
Passenger

- The fetus (with placenta) is the passenger.
- The fetal head (size and presence of molding),
- fetal attitude (degree of body flexion),
- fetal lie (relationship of body parts),
- fetal presentation (first body part),
- fetal position (relationship to maternal pelvis),
- fetal station, and fetal engagement are all important factors that have an impact on the ultimate outcome in the birthing process.

Fetal Head

- The fetal head is large in proportion to the rest of the body, ¼ body surface area.
- Frontal bones, parietal bones, and the occipital bone are not fused
- The gaps are membranous spaces between the cranial bones, are called sutures
- The intersections of these sutures are called fontanelles.
- The overlapping of the cranial bones is known as molding. Along with molding, fluid can also collect in the scalp (caput succedaneum disappears within 3 to 4 days) or blood can collect beneath the scalp (cephalohematoma reabsorbed over the next 4 to 6 weeks)
- Sutures also play a role in helping to identify the position of the fetal head during a vaginal examination.

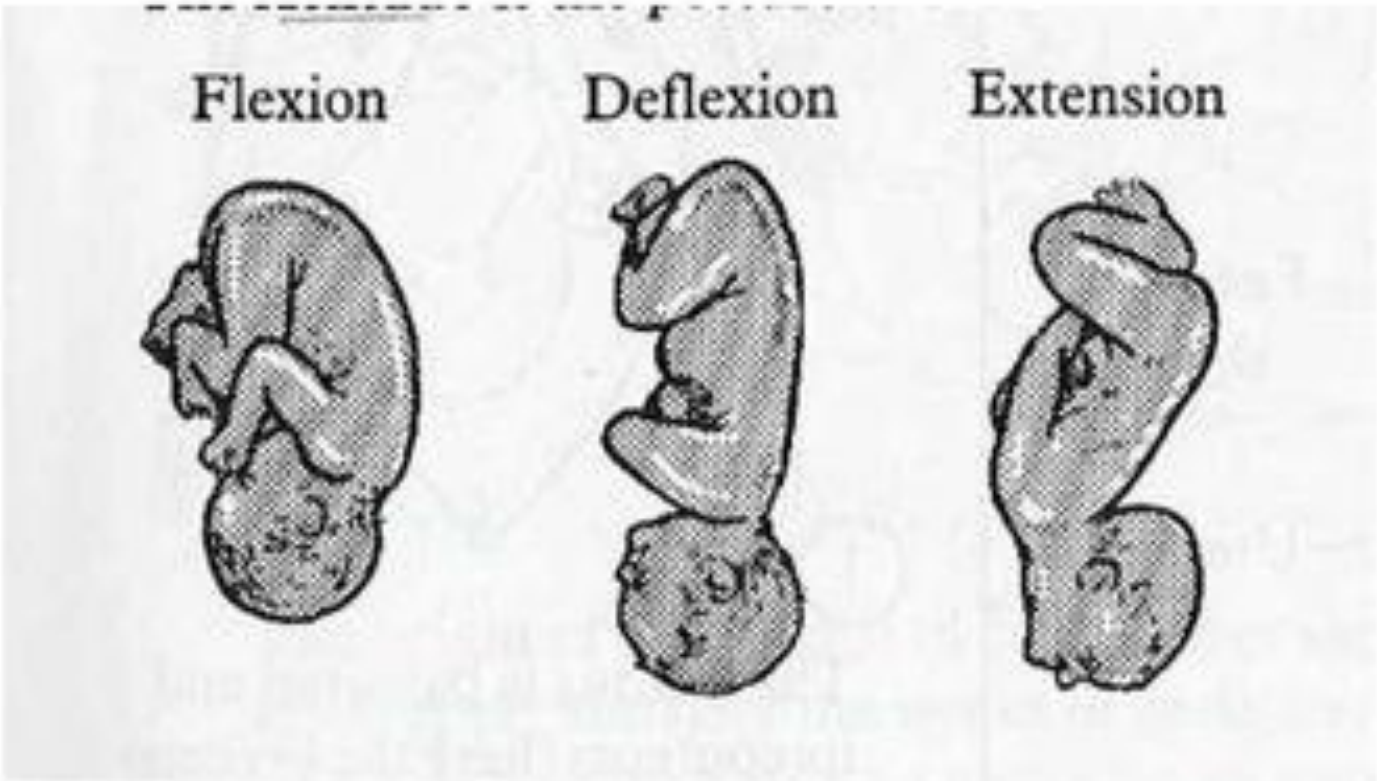
Fetal Head


Sagittal suture
Anterior fontanel
Lambdoidal suture
Frontal suture
Posterior fontanel
Biparietal
Frontal bone
Parietal bone
Occipital bone

Fetal Attitude

- Fetal attitude refers to the posturing (flexion or extension) of the joints and the relationship of fetal parts to one another.
- The most common fetal attitude when labor begins is with all joints flexed—the fetal back is rounded, the chin is on the chest, the thighs are flexed on the abdomen, and the legs are flexed at the knees.
- This normal fetal position is most favorable for vaginal birth, presenting the smallest fetal skull diameters to the pelvis. When the fetus presents to the pelvis with abnormal attitudes (no flexion or extension), increasing the difficulty of birth.

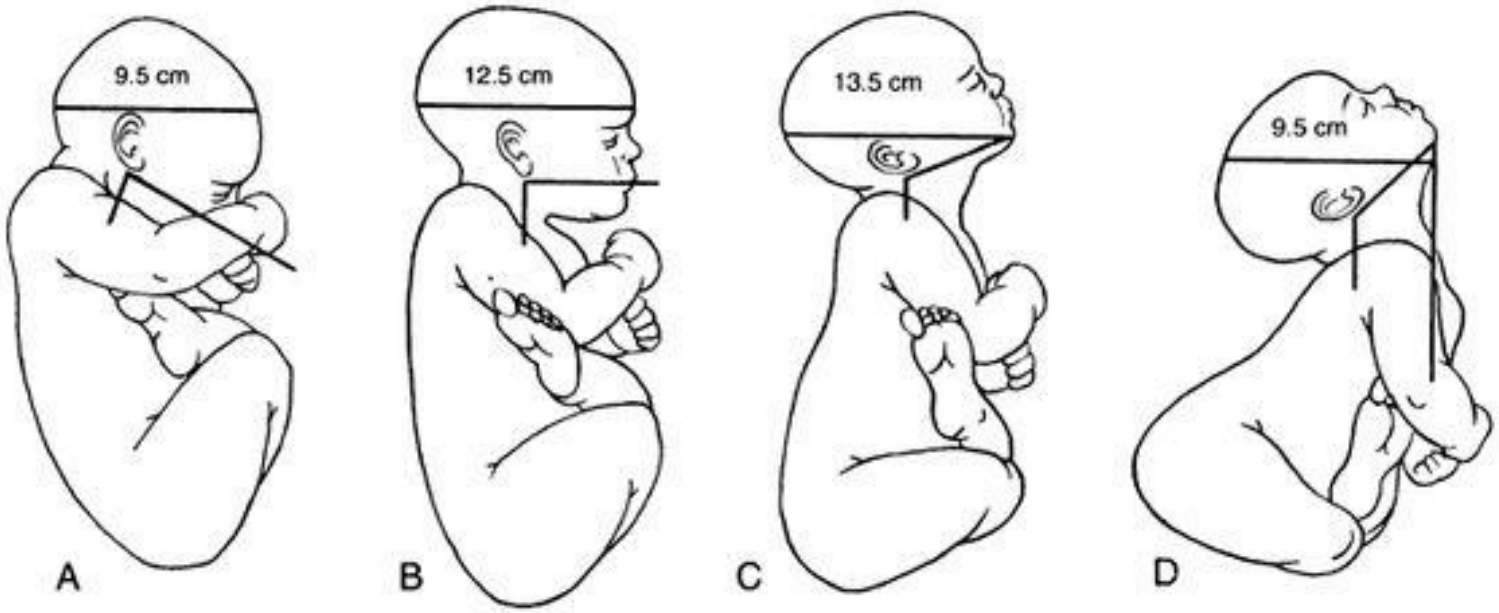
- ***The attitude***: is the posture of the fetus (flexion, deflexion, extension)



Fetal Attitude



FIGURE 13.5 Fetal attitude: Full flexion. Note that the smallest diameter presents to the pelvis.



Fetal Attitude

Importance of cranial flexion is emphasized by noting the increased diameters presented to the birth canal with progressive deflection. A.Flexed head. B. Military position. C, D. Progressive...

Fetal Lie

- Fetal lie refers to the relationship of the long axis (spine) of the fetus to the long axis (spine) of the mother.
- There are two primary lies:
  - longitudinal (which is the most common) and transverse. A longitudinal lie occurs when the long axis of the fetus is parallel to that of the mother (fetal spine to maternal spine side-by-side).
  - A transverse lie occurs when fetus spine lies across the maternal abdomen and crosses her spine. A fetus in a transverse lie position cannot be delivered vaginally



A. Longitudinal lie



B. Transverse lie

FIGURE 13.6 Fetal lie.

Fetal Presentation

- Fetal presentation refers to the body part of the fetus that enters the pelvic inlet or the cervical os first (the “presenting part”).
- The three main fetal presentations are
  - cephalic (head first, 95%), usually the occiput portion
  - breech (pelvis first, 3%)
  - shoulder (scapula first, 2%).
- Primarily, the largest part of the fetus (skull) is born last and may become “hung up” or stuck in the pelvis.

Fetal Presentation

- In addition, the umbilical cord can become compressed between the fetal skull and the maternal pelvis
- The buttocks are soft and are not as effective as a cervical dilator during labor compared with a cephalic presentation.
- Finally, there ılt of the lack of oppor

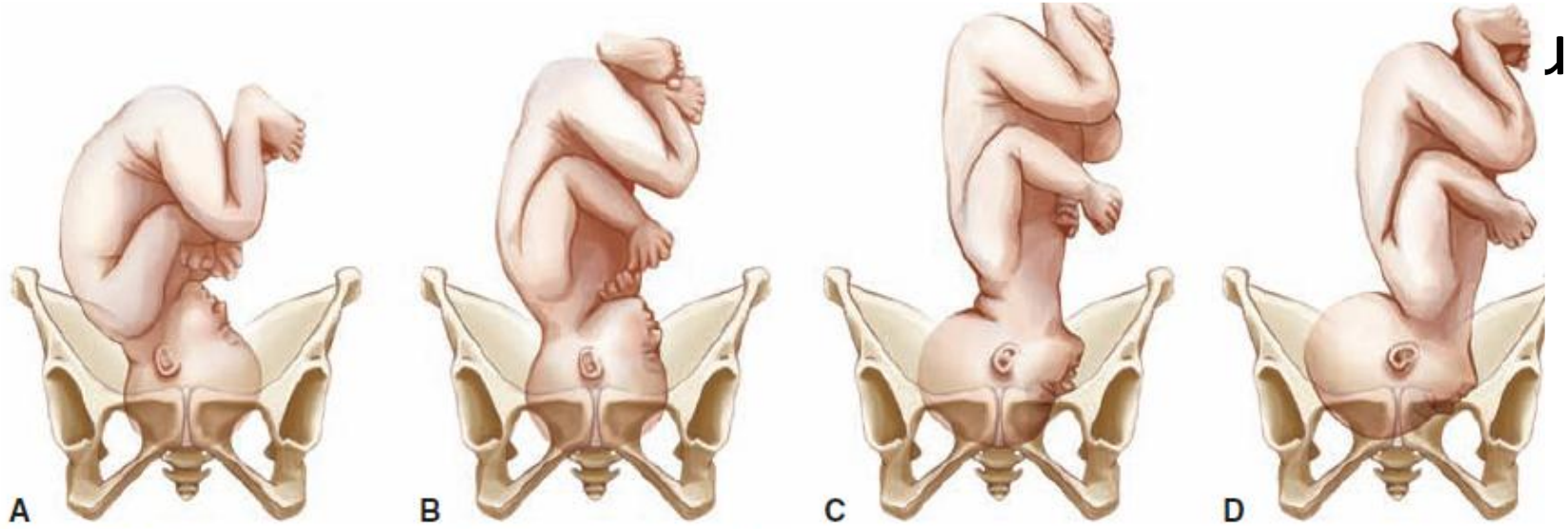


FIGURE 13.7 Fetal presentation: cephalic presentations. (A) Vertex. (B) Military. (C) Brow. (D) Face.



Face presentation

Brow presentation



Vertex presentation



Breech presentation



Shoulder presentation

Fetal Presentation

- The types of breech presentations are determined by the positioning of the fetal legs. In a frank breech (50% to 70%), the buttocks present first with both legs extended up toward the face.
- In a full or complete breech (5% to 10%), the fetus sits crossed-legged above the cervix.
- In a footling or incomplete breech (10% to 30%), one or both legs are presenting.
- Breech presentations are associated with prematurity, placenta previa, multiparity, uterine abnormalities (fibroids), and some congenital anomalies such as hydrocephaly.

Fetal Presentation

- A frank breech can result in a vaginal birth, but complete, footling, and incomplete breech presentations generally necessitate a cesarean birth.

- A shoulder presentation occurs when the fetal shoulders present first, with the head tucked inside. Odds of a shoulder presentation are one in 1,000
- The fetus is in a transverse lie with the shoulder as the presenting part.
- Conditions associated with shoulder presentation include placenta previa, multiple gestation, or fetal anomalies. A cesarean birth is typically necessary.

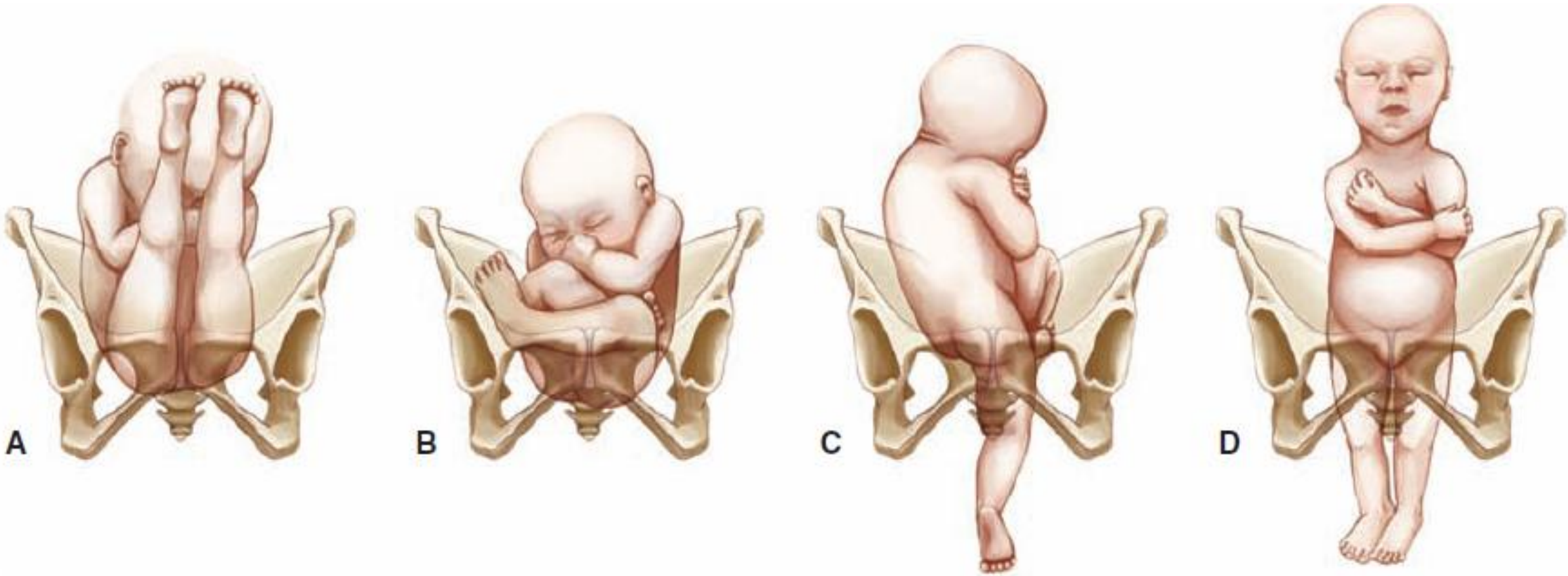
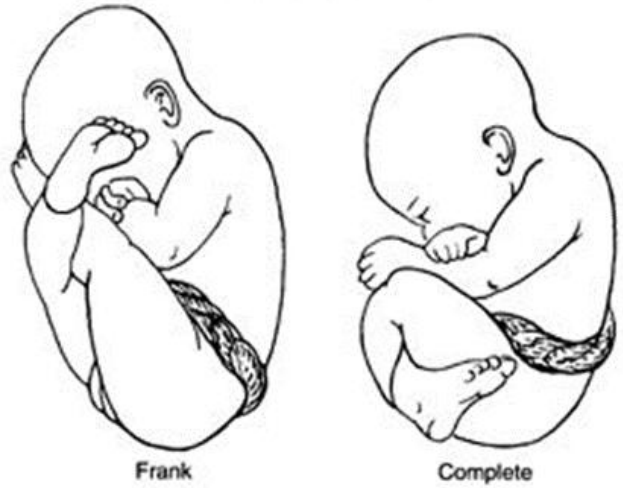


FIGURE 13.8 Breech presentations. (**A**) Frank breech. (**B**) Complete breech. (**C**) Single footling breech. (**D**) Double footling breech.

"Full" Varieties



Incomplete Varieties

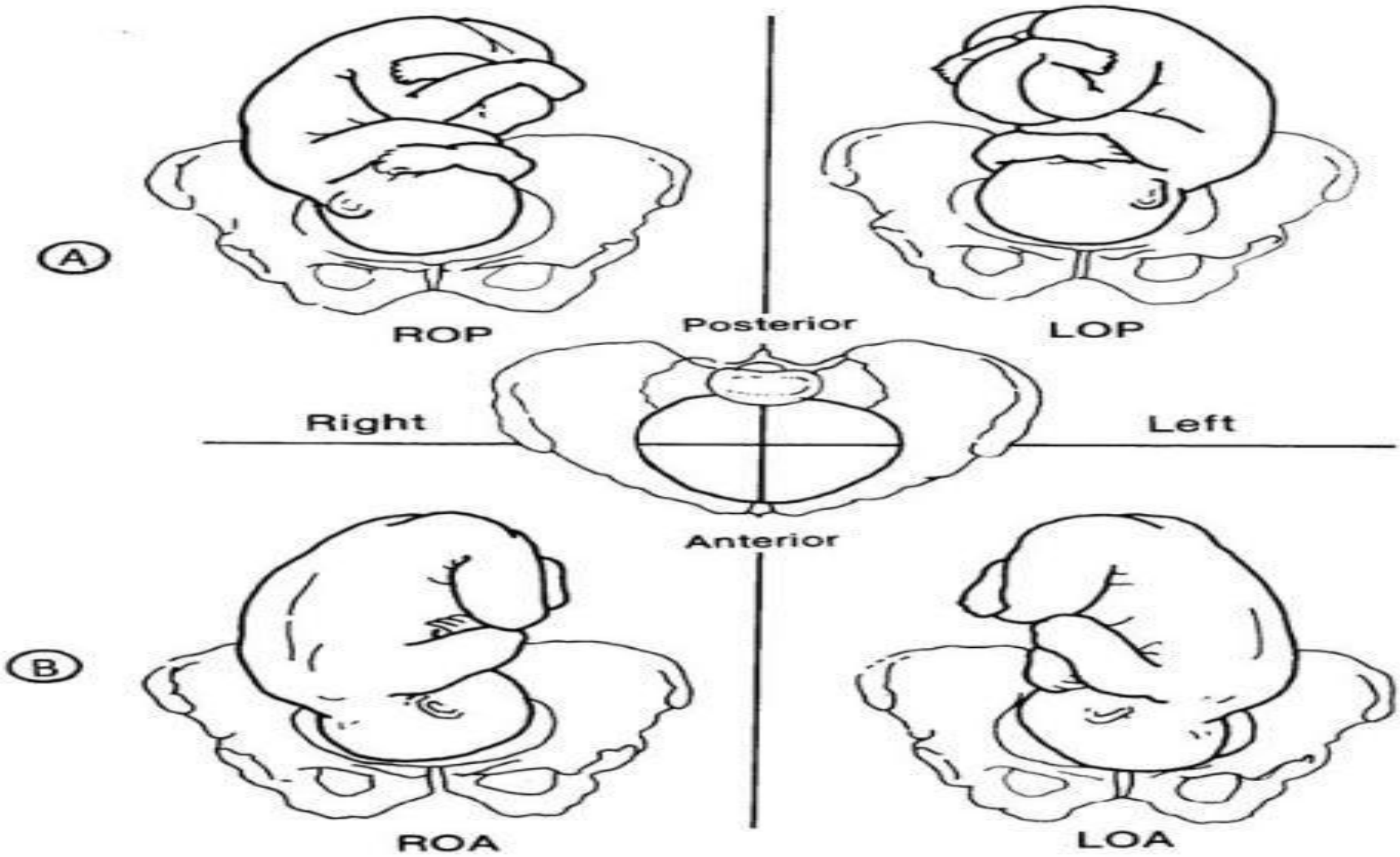


Abnormal Fetal Lie and Presentation

Fetal Position

- Fetal position describes the relationship of a given point on the presenting part of the fetus to a designated point of the maternal pelvis.
- The landmark fetal presenting parts include the occipital bone (O), which designates a vertex presentation; the chin (mentum [M]), which designates a face presentation; the buttocks (sacrum [S]), which designate a breech presentation; and the scapula(acromion process [A]), which designates a shoulder presentation.

- In addition, the maternal pelvis is divided into four quadrants: right anterior, left anterior, right posterior, and left posterior.


A
ROP
Posterior
LOP
Right
Left
Anterior
B
ROA
LOA

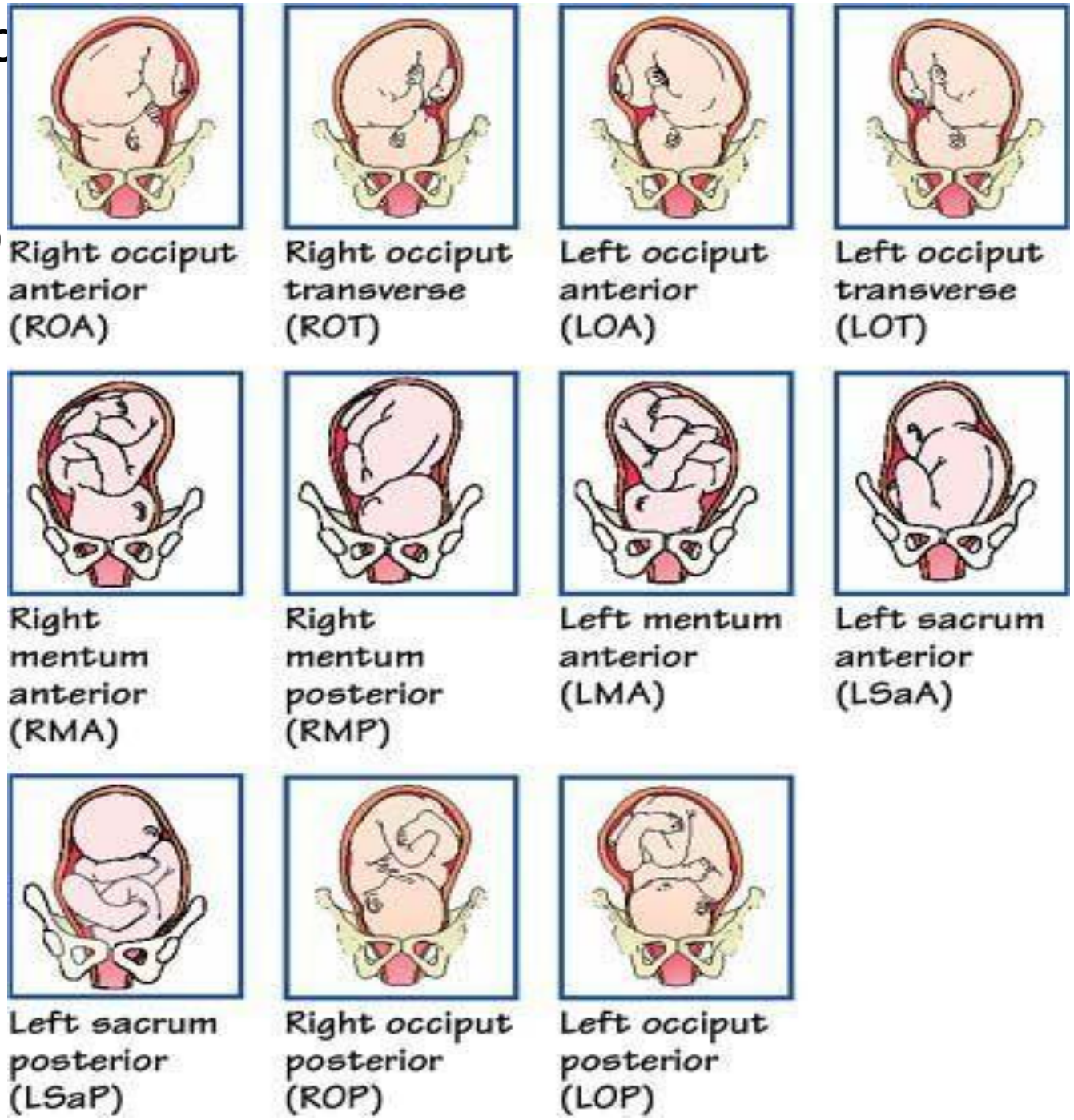
Fetal Position

- These quadrants designate whether the presenting part is directed toward the front, back, left, or right side of the pelvis.
- Fetal position is determined first by identifying the presenting part and then the maternal quadrant the presenting part is facing.
- Position is indicated by a three-letter abbreviation as follows:
  - The first letter defines whether the presenting part is tilted toward the left (L) or the right (R) side of the maternal pelvis.
  - The second letter represents the particular presenting part of the fetus: O for occiput, S for sacrum (buttocks), M for mentum (chin), A for acromion process, and D for dorsal (refers to the fetal back) when denoting the fetal position in shoulder presentations

Fetal Po

The third letter defines the location of the presenting part in relation to the anterior (A) portion of the maternal pelvis or the posterior (P) portion of the maternal pelvis. If the presenting part is directed to the side of the maternal pelvis, the fetal presentation is designated as transverse (T).

LOA is the most common (most favorable), followed by right occipitoanterior (ROA).

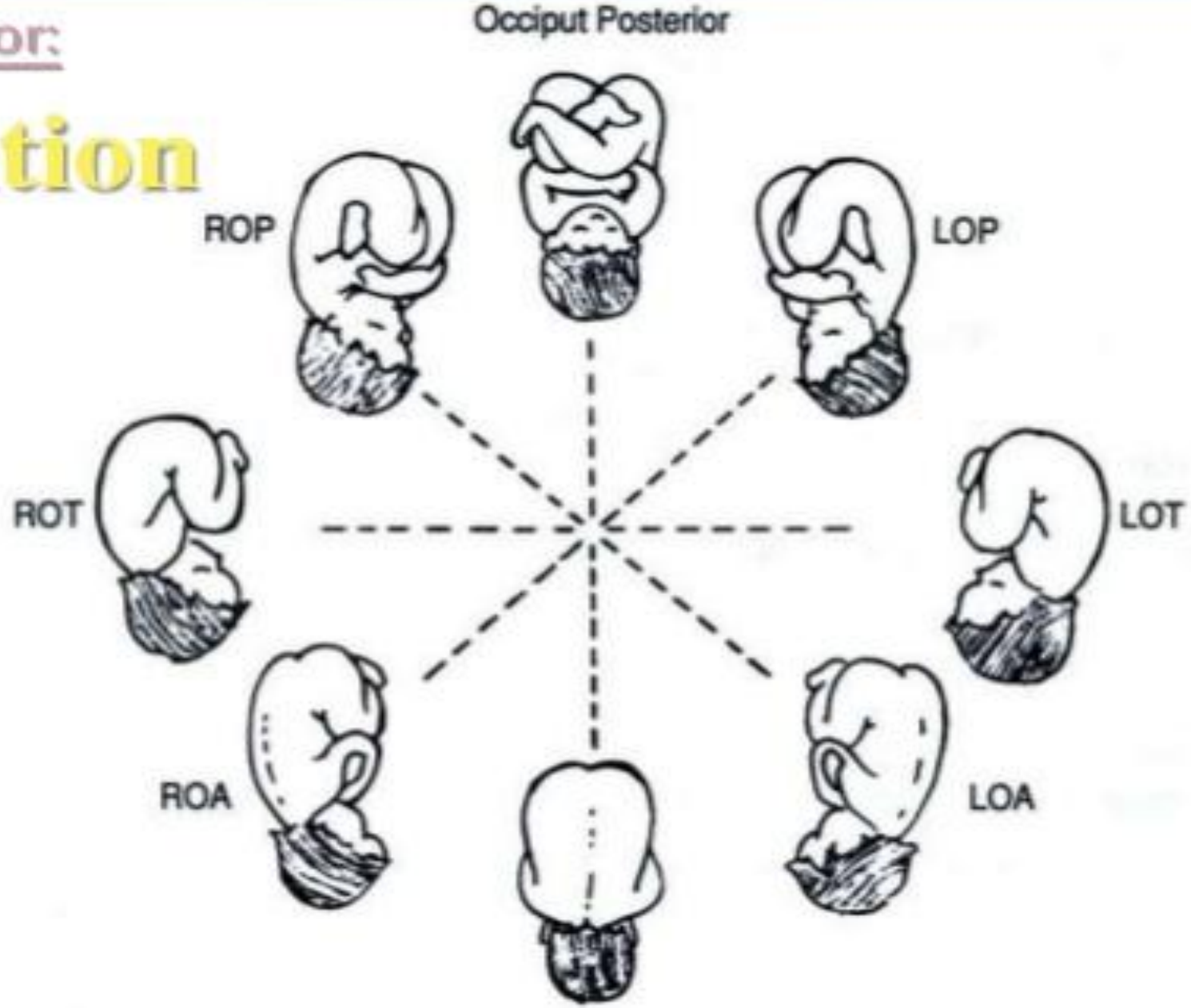


Mechanism of Labor:

Fetal Position



Chapter 22: Normal Labor. William's Obstetrics 24th Edition.
https://www.pinterest



A

Fetal Station

- Station refers to the relationship of the presenting part to the level of the maternal pelvic ischial spines.
- Fetal station is measured in centimeters and is referred to as a minus or plus, depending on its location above or below the ischial spines.
- Ischial spines are the narrowest part of the pelvis and are the natural measuring point for the birth progress.
- Zero (0) station is designated when the presenting part is at the level of the maternal ischial spines. When the presenting part is above the ischial spines, the distance is recorded as minus stations. When the presenting part is below the ischial spines, the distance is recorded as plus stations.

Fetal Station

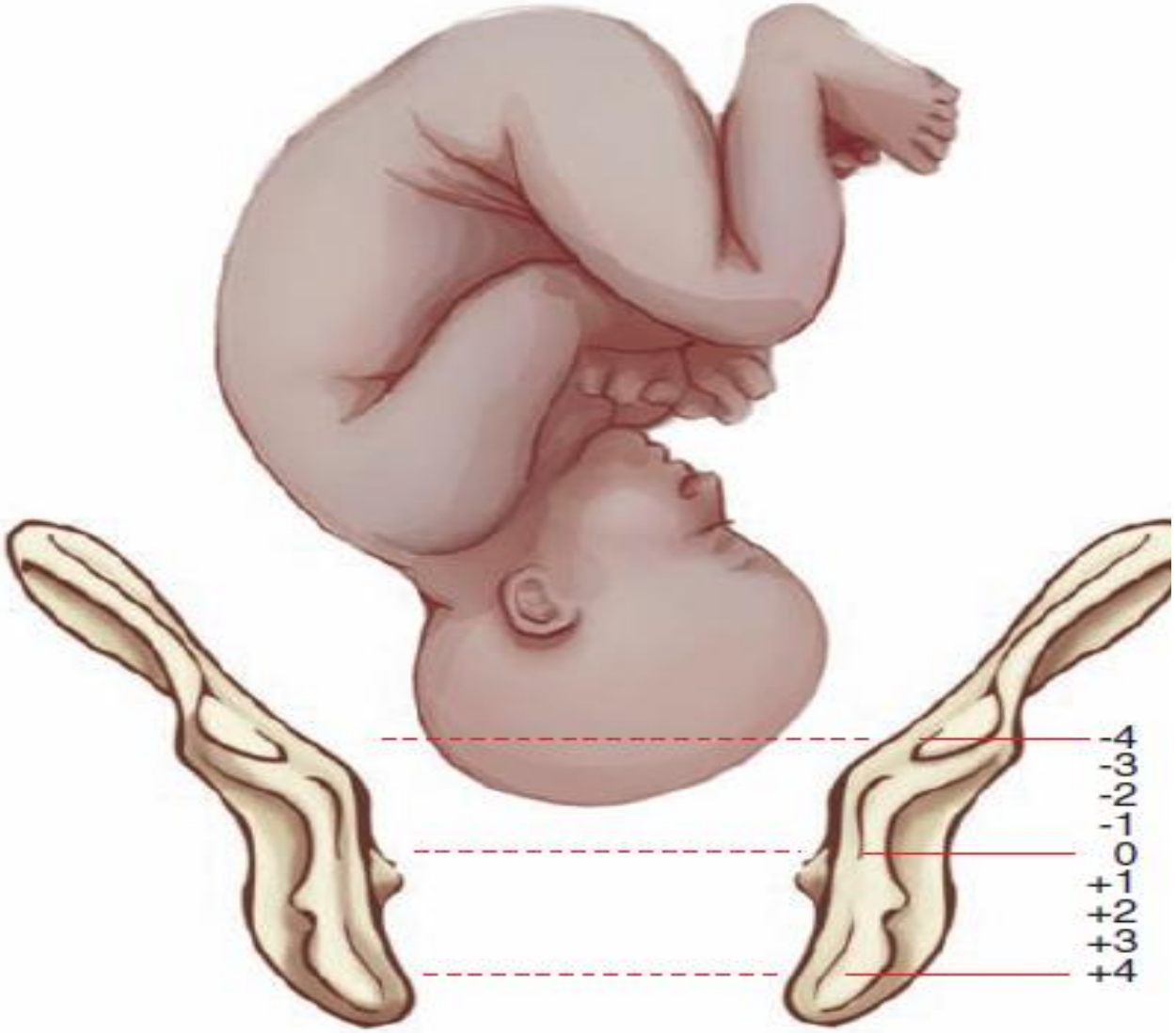
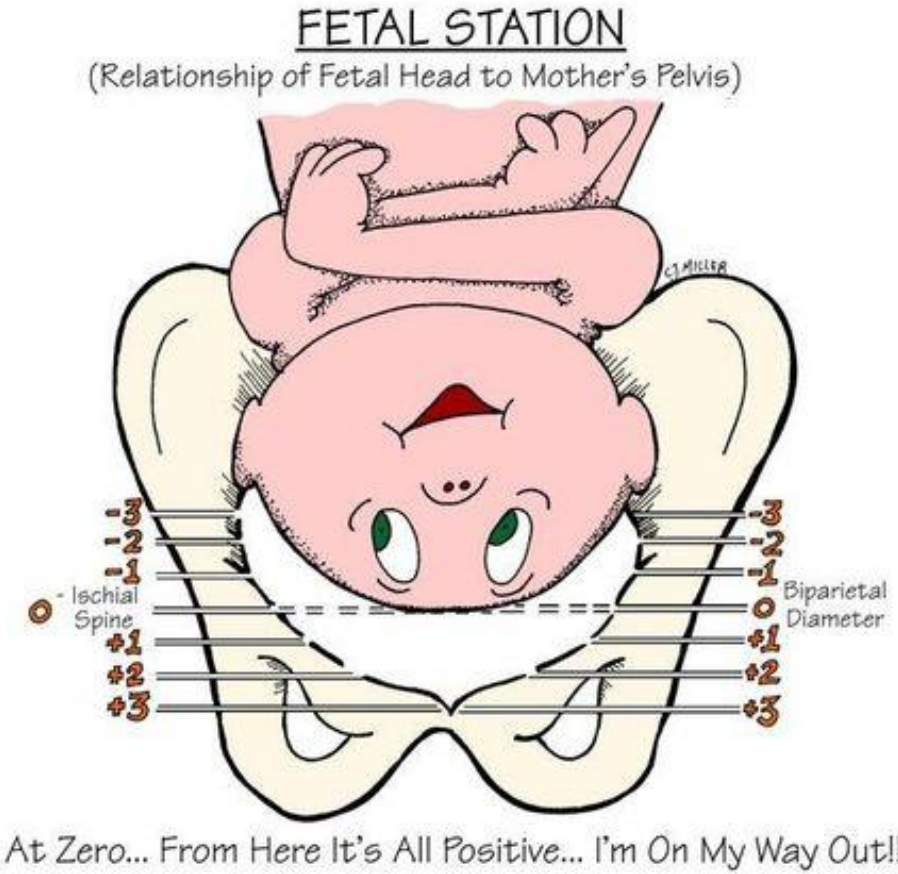


FIGURE 13.10 Fetal stations.

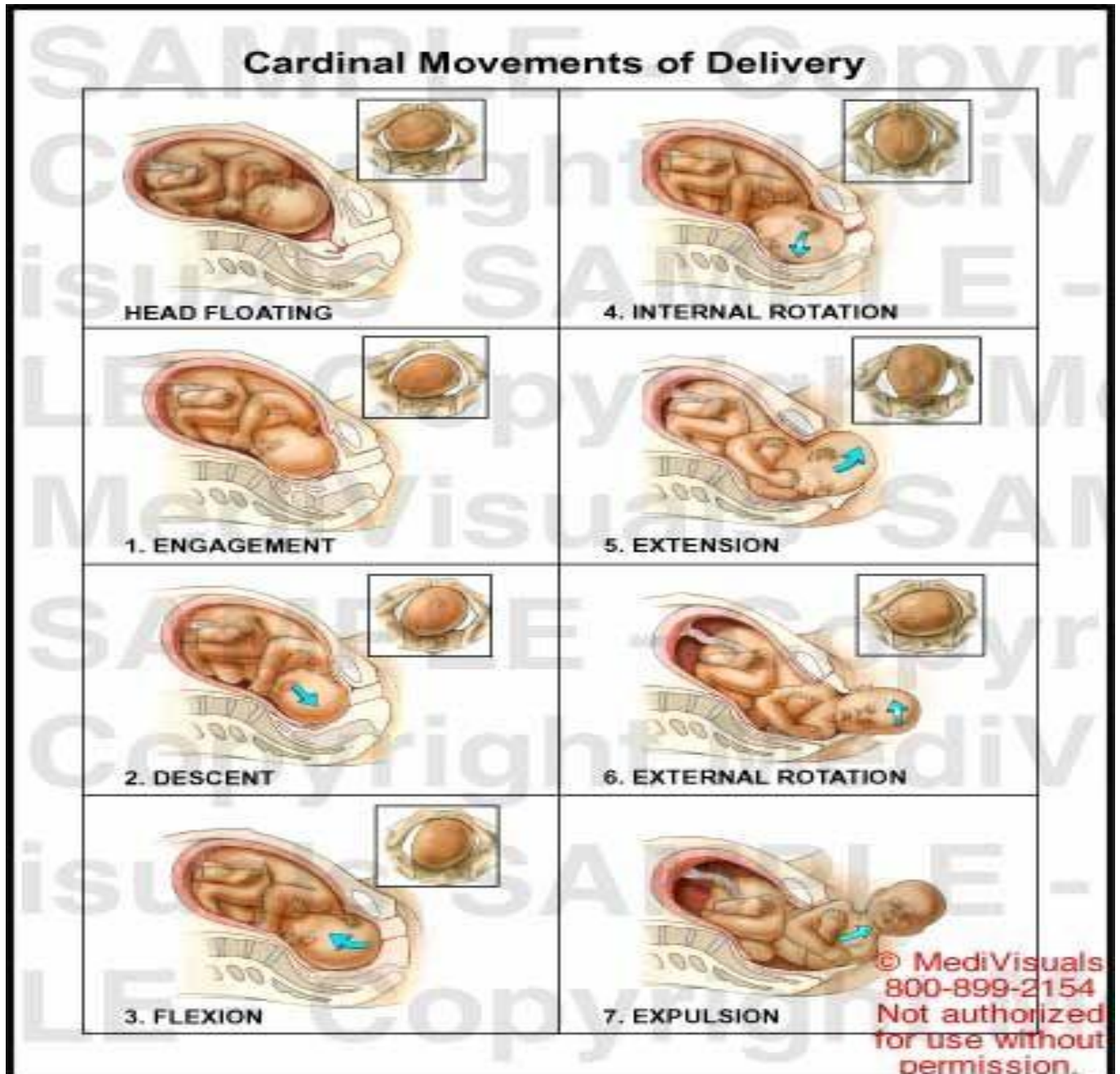


Cardinal Movement

- Cardinal Movements of Labor refers to the fetus goes through many positional changes as it travels through the passageway.
- Engagement occurs when the greatest transverse diameter of the head in vertex (biparietal diameter) passes through the pelvic inlet (usually 0 station).
- Descent is the downward movement of the fetal head until it is within the pelvic inlet.
- Descent occurs intermittently with contractions and is brought about by one or more of the following forces:
  - Pressure of the amniotic fluid
  - Contractions of the abdominal muscles
  - Direct pressure of the fundus
  - Extension and straightening of the fetal body
- Descent occurs throughout labor, ending with birth. (the first requisite for birth)

- Flexion occurs as the vertex meets resistance from the cervix, the walls of the pelvis, or the pelvic floor. As a result, the chin is brought into contact with the fetal thorax and the presenting diameter is changed from occipitofrontal to suboccipitobregmatic
- Internal Rotation: As a result of resistance from one side of the pelvic floor, the head rotates about 45 degrees anteriorly to the midline under the symphysis.
- Extension: fetal head extend so that it can pass under the pubic arch. The anterior fontanel, brow, nose, mouth, and chin are born successively.
- External Rotation (Restitution): occiput to move about 45 degrees back to its original left or right position (restitution).
- Expulsion of the rest of the body occurs more smoothly after the birth of the head and the anterior and posterior shoulders

Cardinal movements of labor



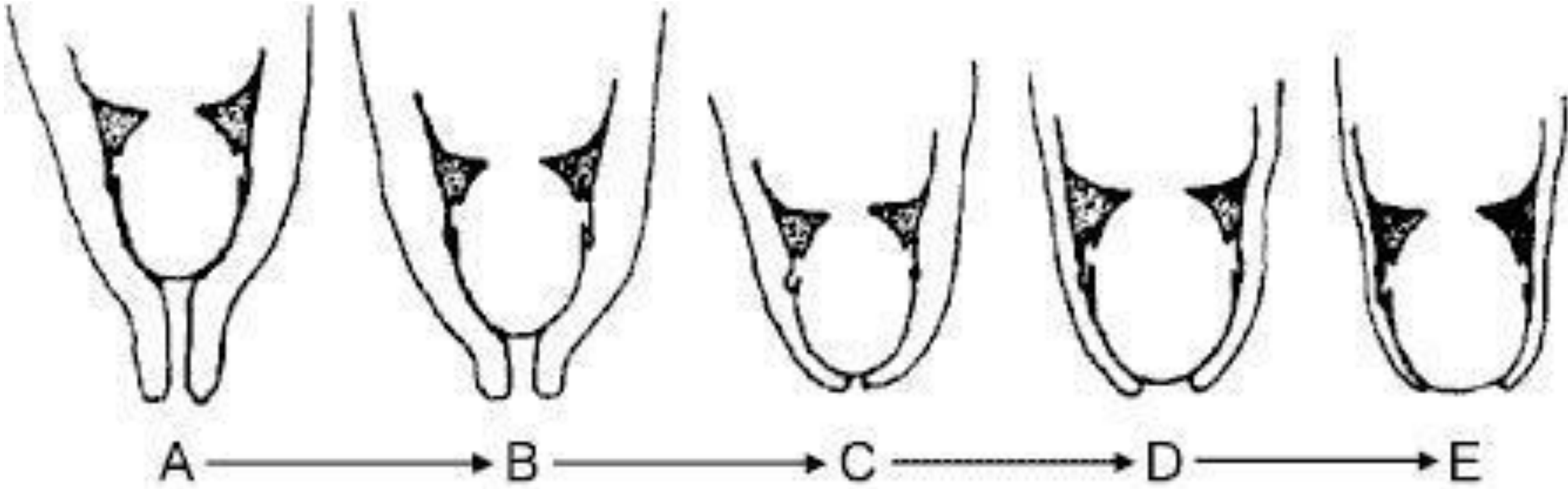


Powers

- The primary stimulus powering labor is uterine contractions.
- Contractions cause complete dilation and effacement of the cervix during the first stage of labor.
- The secondary powers in labor involve the use of intra-abdominal pressure (voluntary muscle contractions) exerted by the woman as she pushes and bears down during the second stage of labor.

Uterine Contractions

- Uterine contractions are involuntary and therefore cannot be controlled by the woman experiencing them, regardless of whether they are spontaneous or induced.
- Uterine contractions are rhythmic and intermittent, with a period of relaxation between contractions.
- Uterine contractions are responsible for thinning and dilating the cervix and the cervix become longer, thinner, and more distended.
- The cervical canal reduces in length from 2 cm to a paper-thin entity and is described in terms of percentages from 0 to 100%. On clinical examination the following may be assessed: • Cervical canal 2 cm in length (0% effaced)• 1 cm in length (50% effaced), Cervical canal 0 cm in length (100% effaced).



A	B	C	D	E
Cervix not effaced. Length of cervical canal = 4 cm	Cervix partly effaced. Length of cervical canal = 2 cm	Cervix fully effaced	Cervix dilated 3 cm	Cervix dilated 8 cm

Uterine Contractions

- Dilation is dependent on the pressure of the presenting part and the contraction and retraction of the uterus.
- The diameter of the cervical os increases from less than 1 cm to approximately 10 cm
- Descriptions may include the following: • External cervical os closed: 0 cm dilated • External cervical os half open: 5 cm dilated • External cervical os fully open: 10 cm dilated
- During early labor, uterine contractions are described as mild, they last about 30 seconds, and they occur about every 5 to 7 minutes. As labor progresses, contractions last longer (60 seconds), occur more frequently (2 to 3 minutes apart), and are described as being moderate to high in intensity.

- Each contraction has three phases:
  - increment (buildup of the contraction),
  - acme (peak or highest intensity), and
  - decrement (descent or relaxation of the uterine muscle fibers).
- Uterine contractions are monitored and assessed according to three parameters: frequency, duration, and intensity.

1. Frequency refers to how often the contractions occur and is measured from the increment of one contraction to the increment of the next contraction.
2. Duration refers to how long a contraction lasts
3. Intensity refers to the strength of the contraction determined by manual palpation or measured by an internal intrauterine catheter (IUPC).

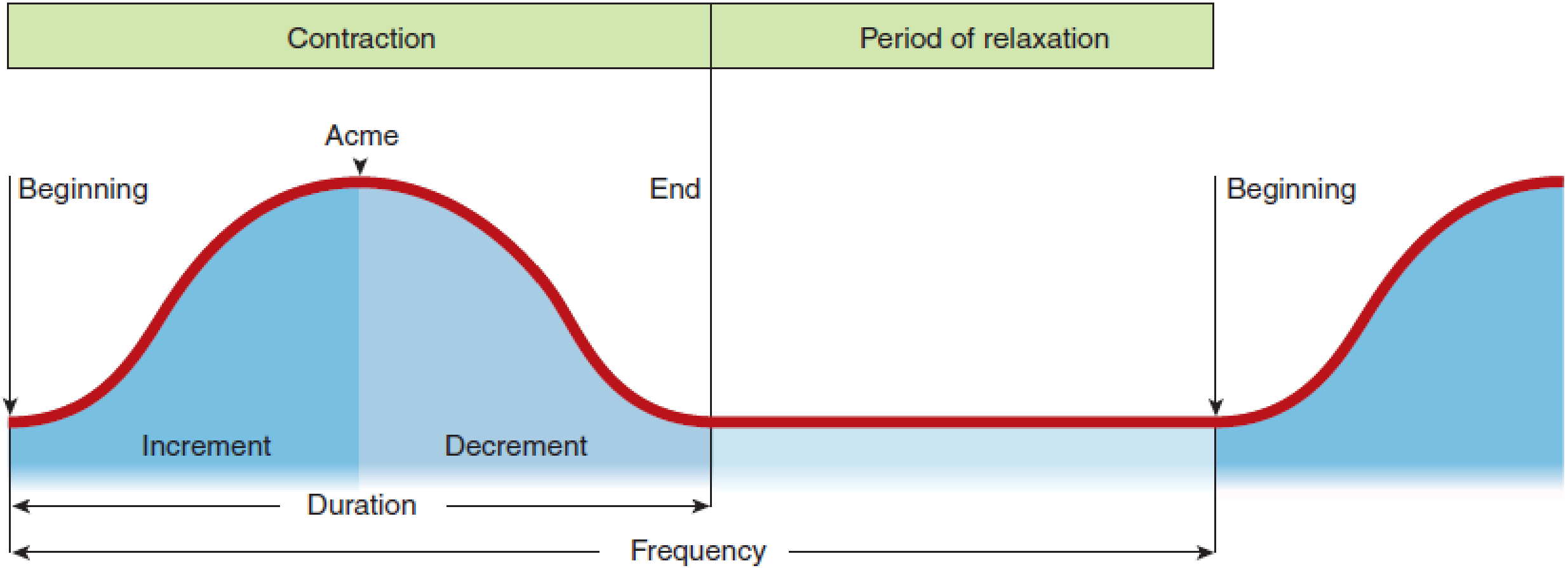


FIGURE 13.12 The three phases of a uterine contraction.

Intra-Abdominal Pressure

- Increased intra-abdominal pressure (voluntary muscle contractions) compresses the uterus and adds to the power of the expulsion forces of the uterine contractions
- Coordination of these forces in unison promotes birth of the fetus and expulsion of the fetal membranes and placenta from the uterus.
- Factors promoting a positive birth experience include:
  - Clear information about procedures
  - Support; not being alone
  - Trust in staff caring for her
  - Personal control over breathing

Position (Maternal)

- Scientific evidence has shown that nonmoving, back-lying positions during labor are not healthy (Hanson 2006). However, despite this evidence to the contrary, most women continue to lie flat on their backs during labor.
- Some of the reasons why this practice continues include the following:
  - Belief that laboring women need to conserve their energy and not tire themselves
  - Belief that nurses cannot keep track of the whereabouts of ambulating women
  - Belief that the supine position facilitates vaginal examinations and external belt adjustment
  - Belief that a bed is “where one is supposed to be” in a hospital setting
  - Belief that the position is more convenient for the delivering health professional

- The use of any upright or lateral position, compared with supine or lithotomy positions, may:
  - Reduce the length of the first stage of labor
  - Reduce the duration of the second stage of labor
  - Reduce the number of assisted deliveries (vacuum and forceps)
  - Reduce episiotomies and perineal tears
  - Contribute to fewer abnormal fetal heart rate patterns
  - Increase comfort/reduce requests for pain medication
  - Enhance a sense of control by the mother
  - Alter the shape and size of the pelvis, which assists in descent
  - Assist gravity to move the fetus downward

Position (Maternal)

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  - Belief that a bed is “where one is supposed to be” in a hospital setting
  - Belief that the position is more convenient for the delivering health professional

Physiologic Responses to Labor

- Labor is the physiologic process by which the uterus expels the fetus and placenta from the body.
- During pregnancy, progesterone secreted from the placenta suppresses the spontaneous contractions of a typical uterus, keeping the fetus within the uterus.
- During labor, the mother and fetus make several physiologic adaptations.
- The labor process stresses several of the woman’s body systems, which react through numerous compensatory mechanisms.

Maternal physiologic responses include:

- Heart rate increases by 10 to 20
- Cardiac output increases 10% to 15% during 1st stage and by 30% to 50% during the 2nd stage of labor.
- Blood pressure increases by 10 to 30
- The white blood cell count increases to 25,000 to 30,000
- Respiratory rate increases and more oxygen is consumed related to the increase in metabolism.
- Gastric motility and food absorption decrease, which may increase the risk of nausea and vomiting during the transition stage of labor.
- Gastric emptying and gastric pH decrease, increasing the risk of vomiting with aspiration.
- Temperature rises slightly, possibly due to an increase in muscle activity.
- Muscular aches/cramps occur as a result of the stressed musculoskeletal system.
- Basal metabolic rate increases and blood glucose levels decrease because of the stress of labor

factors that affect her coping ability are

- Previous birth experiences and their outcomes
- Current pregnancy experience (planned versus unplanned, discomforts experienced, age, risk status of pregnancy, chronic illness, weight gain)
- Support system (presence and support of a valued partner during labor)
- Childbirth preparation (attended childbirth classes and has practiced paced breathing techniques)
- Exercise during pregnancy
- Expectations of the birthing experience and Anxiety level
- Fear of labor and loss of control
- Fatigue and weariness

Fetal Responses

Fetal responses to labor include:

- Periodic fetal heart rate accelerations and slight decelerations related to fetal movement, fundal pressure, and uterine contractions
- Decrease in circulation and perfusion to the fetus secondary to uterine contractions (a healthy fetus is able to compensate for this drop)
- Increase in arterial carbon dioxide pressure (PCO2)
- Decrease in fetal breathing movements throughout labor
- Decrease in fetal oxygen pressure with a decrease in the partial pressure of oxygen (PO2)

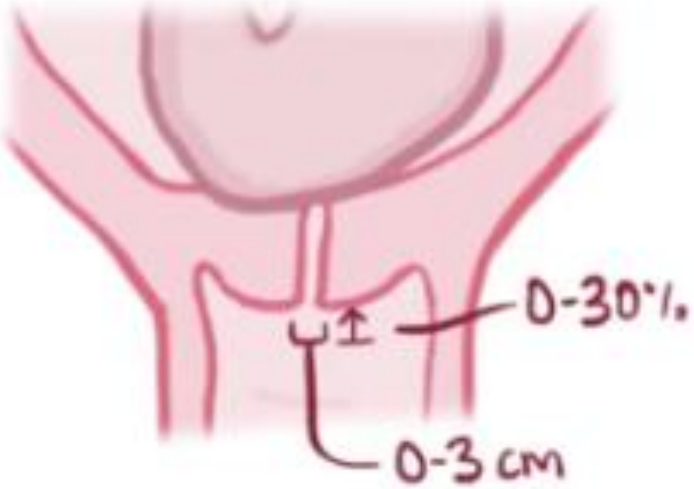
STAGES AND PHASES OF LABOR

Labor is typically divided into four stages: dilation, expulsive, placental, and restorative.

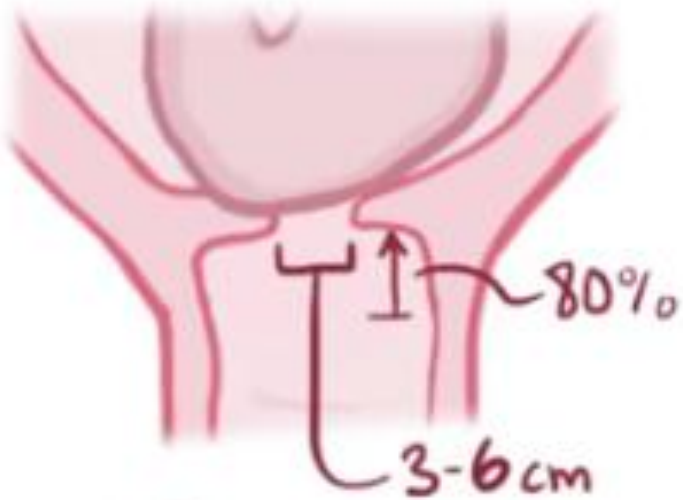
- First stage: From 0–10 cm dilation; consists of three phases(early and active)
- Second stage: From complete dilation (10 cm) to birth of the newborn; lasts up to 1 hour
- Third stage: Separation and delivery of the placenta
- Fourth stage: 1–4 hours after the birth of the newborn; time of maternal Physiologic adjustment

FIRST STAGE

(LATENT) EARLY



* 20 HOURS
* IRREGULAR CONTRACTIONS
  - Every 5-30 min
  - Lasts ~30 sec



* REGULAR CONTRACTIONS
  - Every 3-5 min
  - Lasts 1+ min

ACTIVE



* INTENSE CONTRACTIONS
  - 60-90 sec
  - Every 0.5-2 min

Performing Leopold's Maneuvers



- Leopold's maneuversare a method for determining the presentation, position, and lie of the fetus.
- This method involves inspection and palpation of the maternal screening assessment for malpresentation.
- A longitudinal lie is expected, and the presentation can be cephalic, breech, or shoulder. Each maneuver answers a question:
  - What fetal part (head or buttocks) is located in the fundus (top of the uterus)?
  - On which maternal side is the fetal back located? (Fetal heart tones are best auscultated through the back of the fetus.)
  - What is the presenting part?
  - Is the fetal head flexed and engaged in the pelvis? see Page 365-366

Analysis of the FHR

Analysis of the FHR is one of the primary evaluation tools used to determine fetal oxygen status indirectly.

FHR assessment can be done intermittently using a fetoscope (a modified stethoscope attached to a headpiece) or a Doppler (ultrasound) device, or continuously with an electronic fetal monitor applied externally or internally.

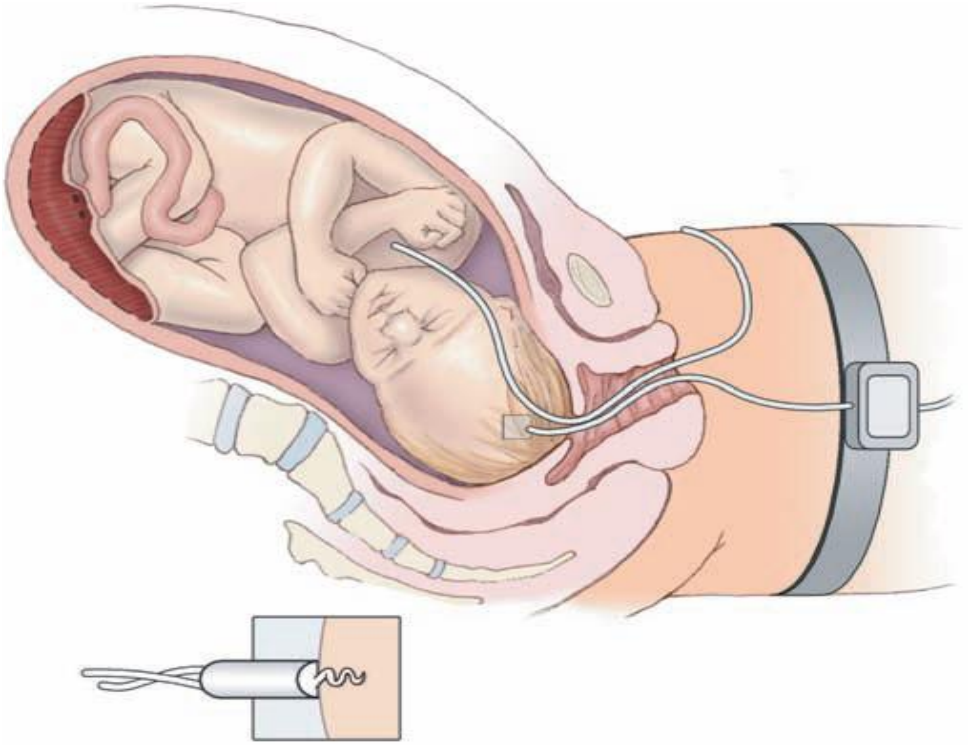


Continuous Electronic Fetal Monitoring Electronic fetal monitoring uses a machine to produce a continuous tracing of

Continuous external electronic fetal monitoring device applied to the woman in labor.



Continuous internal electronic fetal monitoring.



The FHR can demonstrate patterns of acceleration or deceleration in response to most stimuli.
Accelerations are transitory increases in the FHR above the baseline associated with sympathetic nervous stimuli.
A deceleration is a transient fall in FHR caused by stimulation of the parasympathetic nervous system.

Promoting Comfort and Providing Pain Management During Labor

Non-pharmacologic: The techniques used to manage the pain of labor vary according to geography and culture. It include continuous labor support, hydrotherapy, ambulation and position changes, acupuncture and acupressure, attention focusing and imagery, therapeutic touch and massage, breathing techniques, and effleurage (massage)

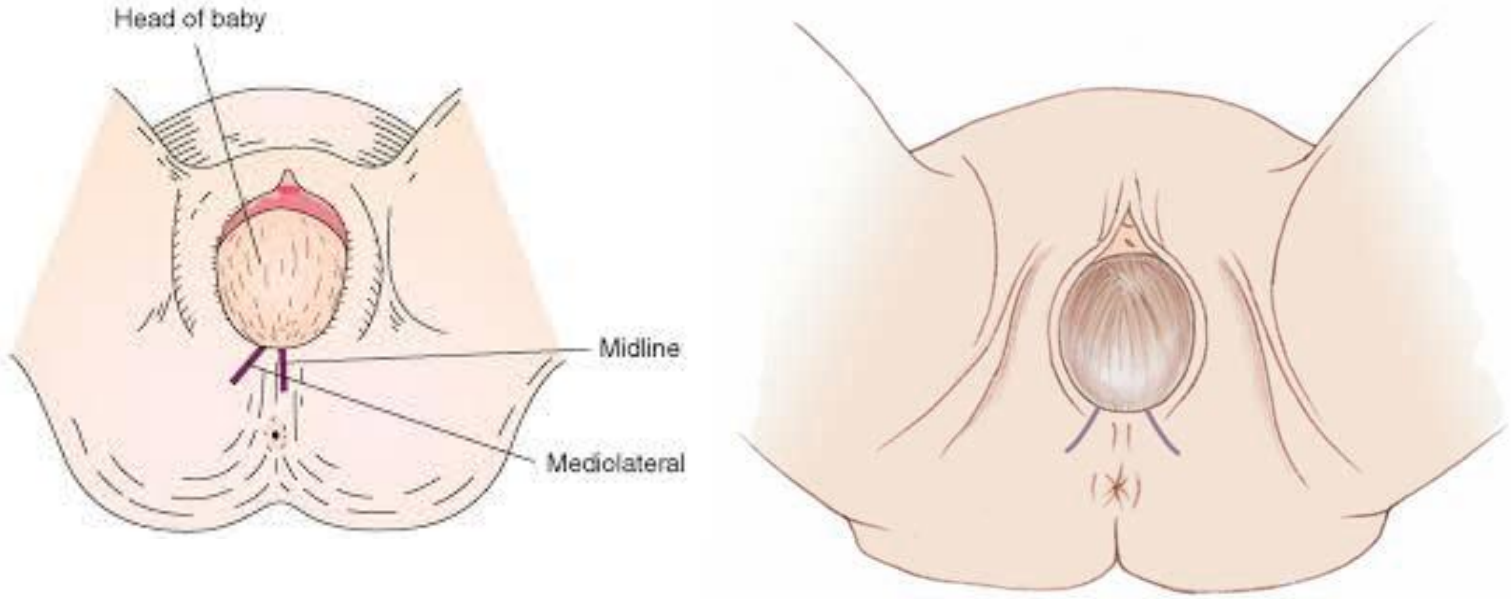
Nonpharmacologic measures are usually simple, safe, and inexpensive to use.

Pharmacologic : includes systemic analgesia and regional or local anesthesia.

Incising the perineum area to provide more space for the presenting part is called an **episiotomy**.

Location of an episiotomy

(**A**) Midline episiotomy. (**B**) Right and left mediolateral episiotomies.



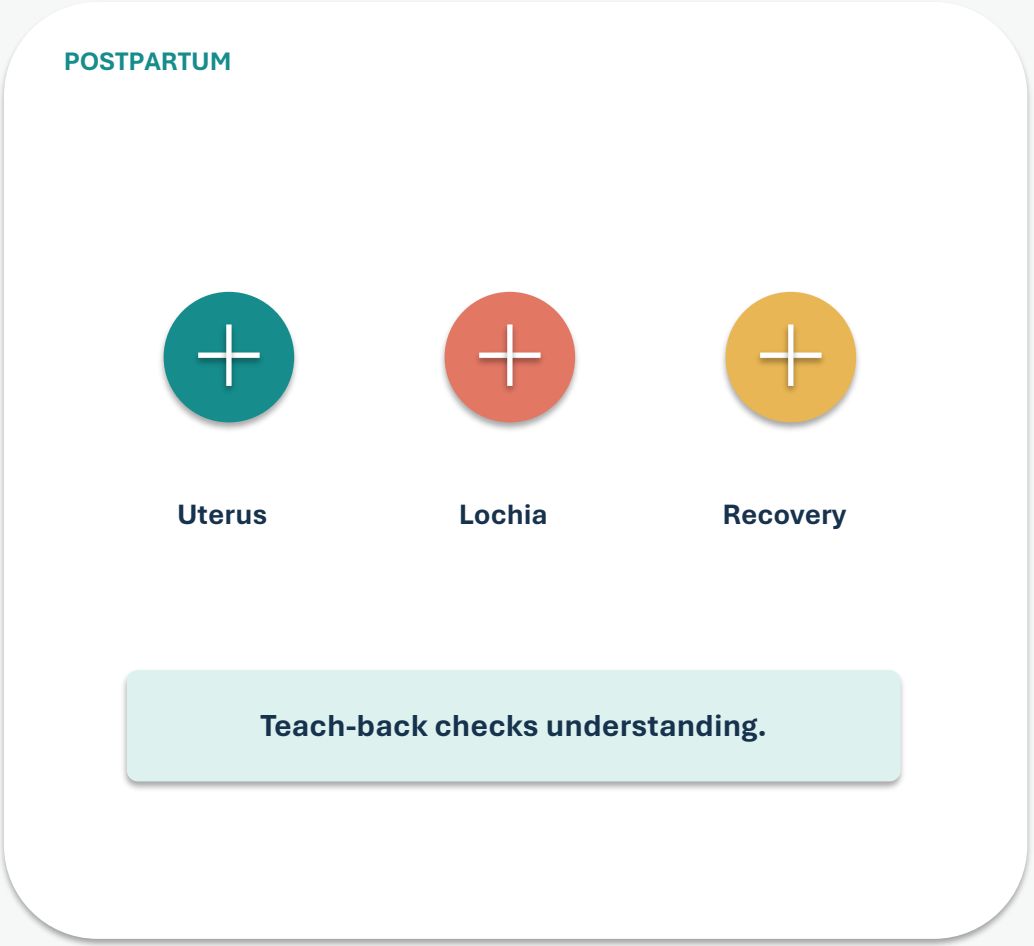
Postpartum recovery



10 focused lessons • assessment, teaching, and clinical reasoning

Normal postpartum physiology

- The uterus contracts and involutes over weeks after birth.
- Lochia changes over time; amount and odor should be assessed.
- Cardiovascular, urinary, breast, and emotional recovery vary.



Sources: WHO 2022

Immediate recovery: first hours

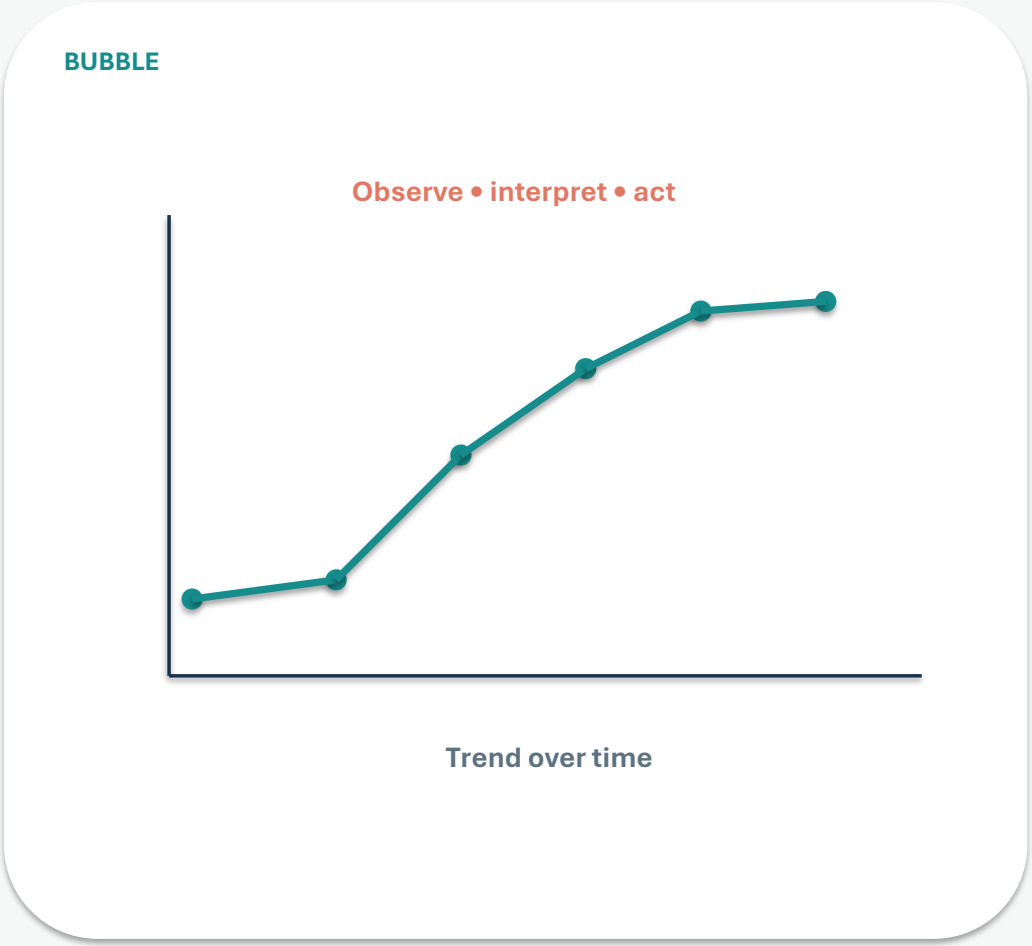
- Assess uterine tone, bleeding, vital signs, pain, bladder, and perineum.
- Promote skin-to-skin and feeding when parent and newborn are stable.
- Use frequent checks based on risk and local protocol.

FOURTH

Sources: ACOG

Postpartum assessment: BUBBLE-HE

- Breasts, uterus, bladder, bowel, lochia, episiotomy/incision, and emotional status.
- Add vital signs, mobility, pain, feeding, and support needs.
- Use the framework flexibly; it does not replace clinical judgment.



Sources: Nursing education

Fundus and lochia assessment

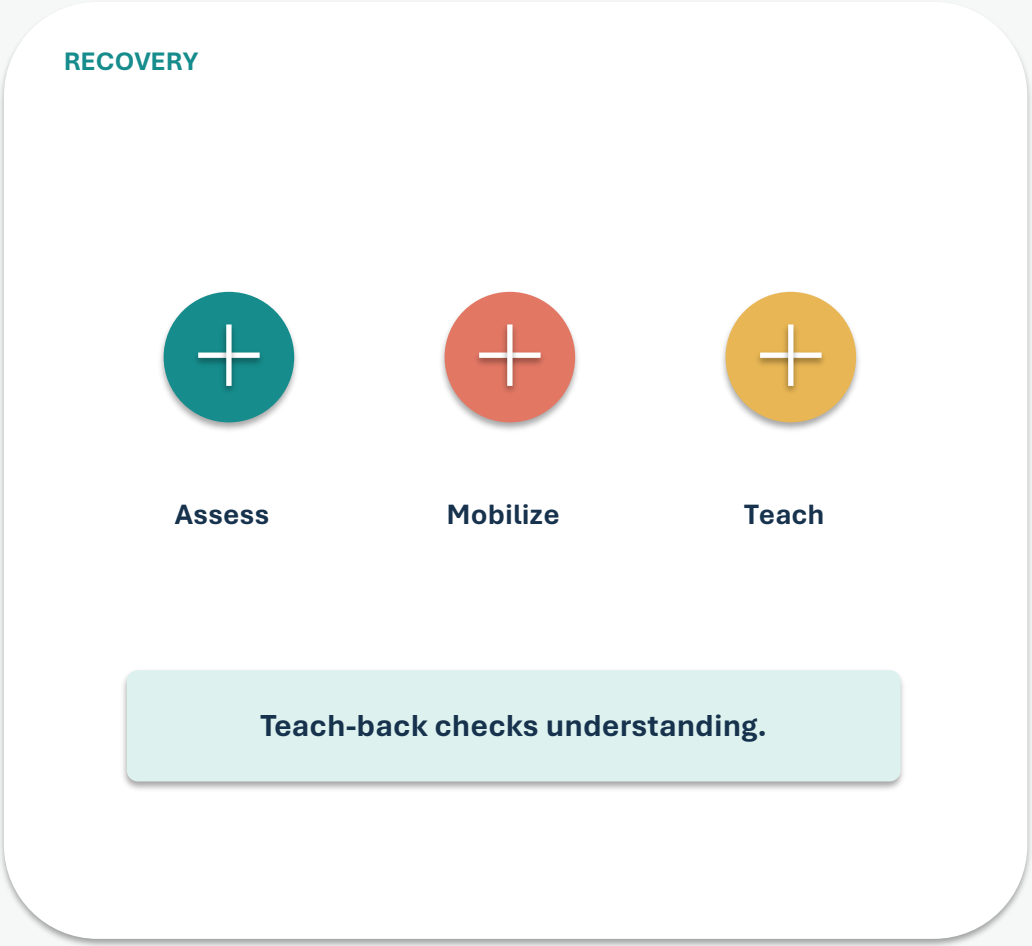
- A firm, midline uterus supports normal hemostasis.
- A boggy or displaced fundus prompts assessment and action under protocol.
- Quantify bleeding when possible and communicate changes promptly.



Sources: ACOG

Bladder, bowel, and mobility

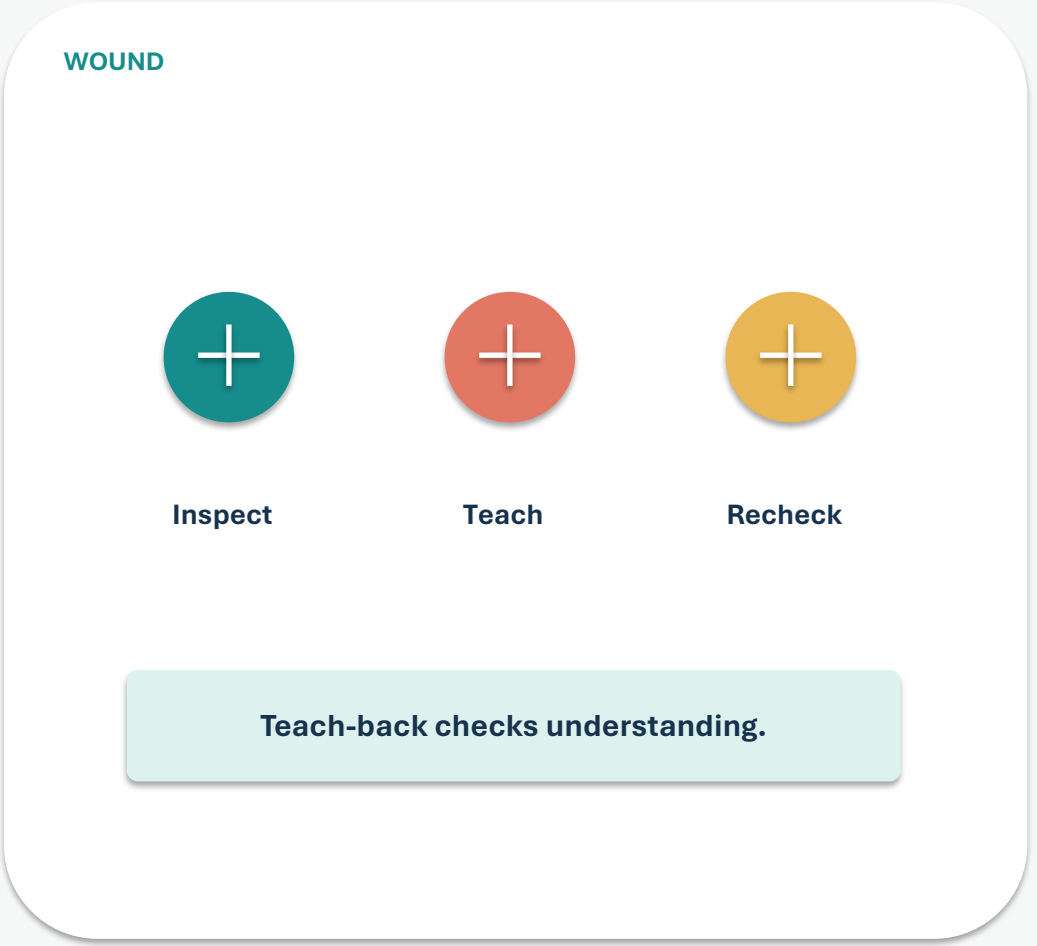
- Assess voiding, retention, constipation, and ability to ambulate safely.
- Regional anesthesia, edema, pain, and fatigue can delay recovery.
- Encourage early safe mobility and use fall precautions.



Sources: ACOG

Perineal and cesarean wound care

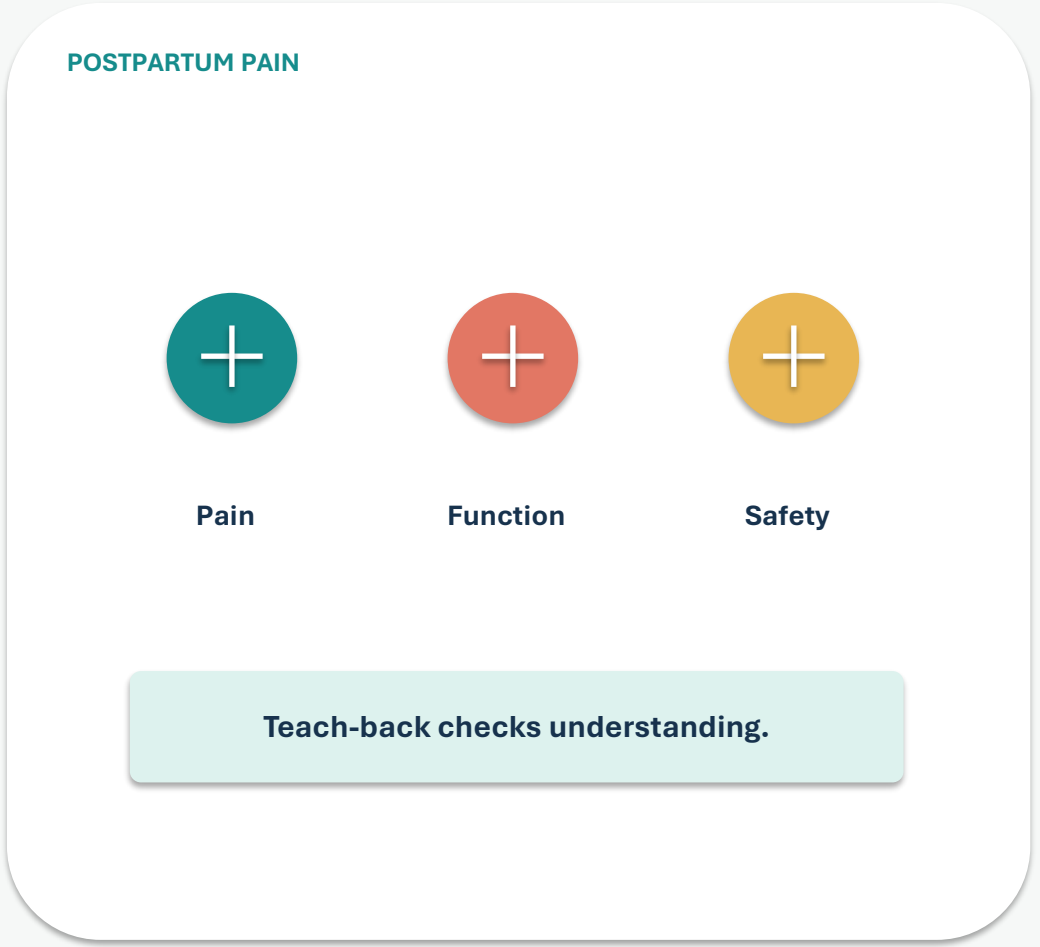
- Assess pain, edema, bruising, drainage, and wound approximation.
- Teach hygiene, wound care, and infection warning signs.
- Offer pain management that supports mobility and infant care.



Sources: ACOG

Postpartum pain and recovery

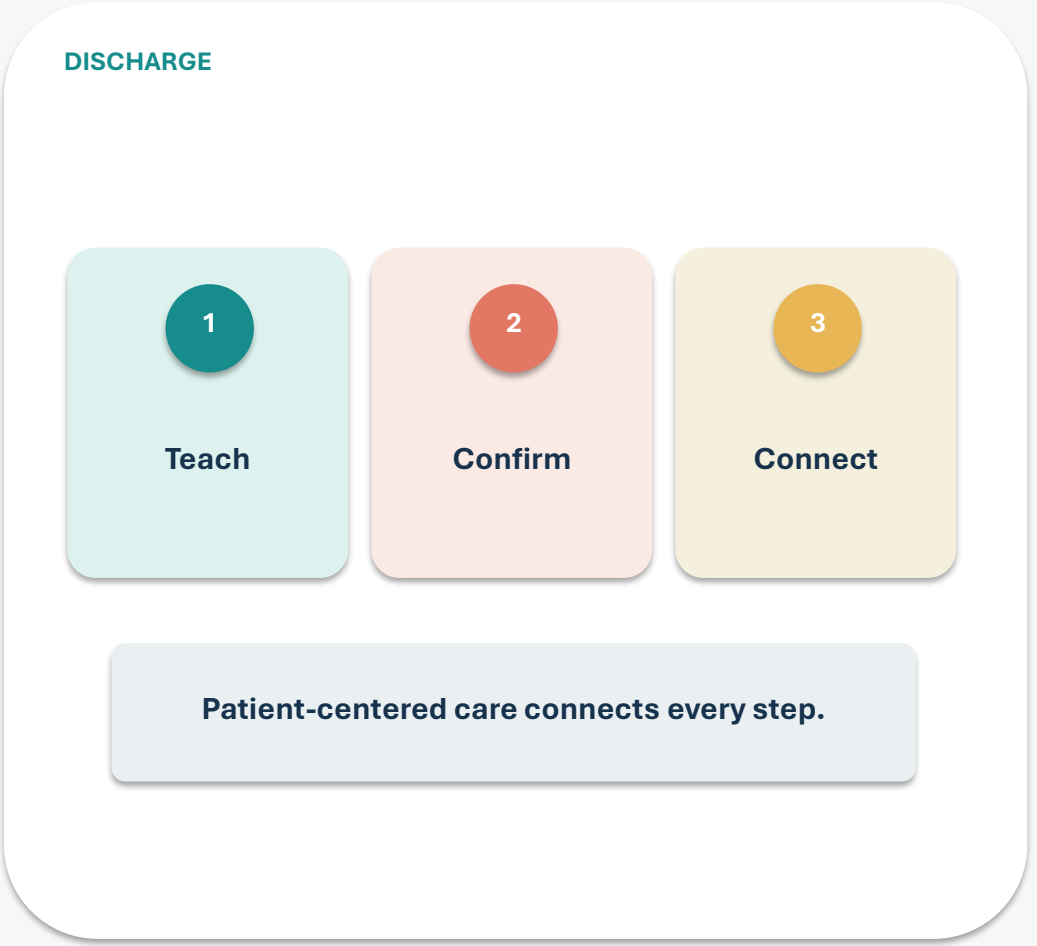
- Use multimodal pain relief based on clinical history and feeding goals.
- Reassess pain, function, sedation, and adverse effects.
- Severe or worsening pain warrants evaluation for a complication.



Sources: ACOG

Postpartum discharge readiness

- Review bleeding, BP concerns, pain, voiding, mobility, and feeding.
- Give written warning signs and a direct contact pathway.
- Confirm follow-up for both parent and newborn.



Sources: WHO 2022

Postpartum follow-up is ongoing

- Early contact should occur within the first weeks, with comprehensive care by 12 weeks.
- Timing may be sooner for hypertension, mood concerns, wound issues, or chronic disease.
- A single visit does not meet every patient’s needs.

FOLLOWUP



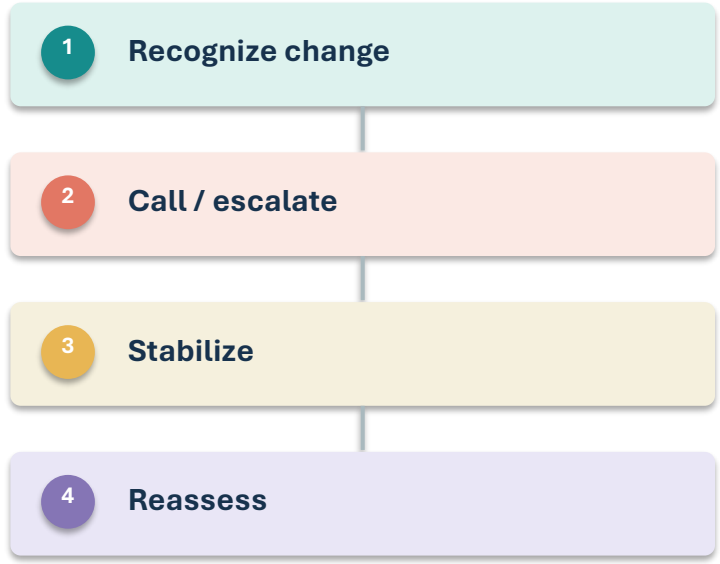
Timing and response depend on clinical context.

Sources: ACOG 2018

Case: postpartum dizziness

- A patient feels faint when standing and reports heavier bleeding.
- Assess vital signs, fundus, lochia, pain, and perfusion; call for help.
- Do not attribute symptoms to fatigue without assessment.

CASE



Use the local emergency pathway.

Sources: ACOG

Postpartum complications



10 focused lessons • assessment, teaching, and clinical reasoning

Postpartum hemorrhage: recognize early

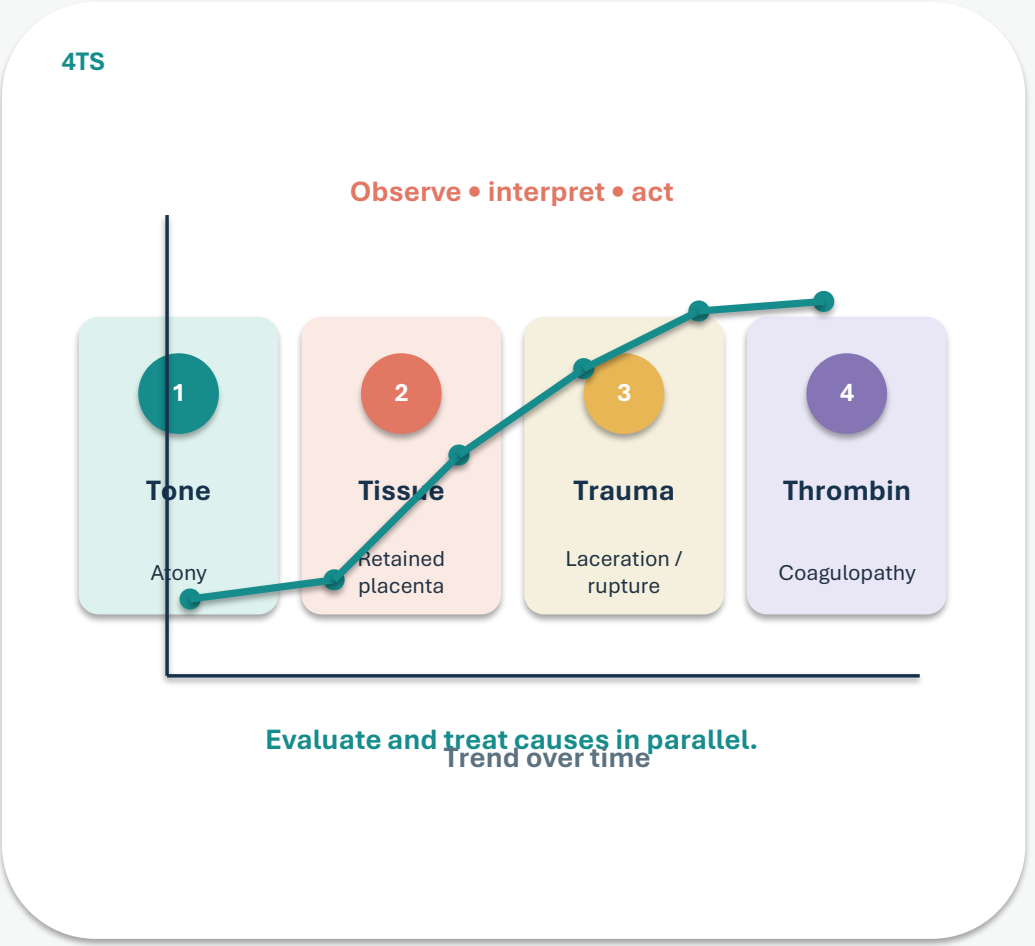
- Concern is based on cumulative blood loss and signs of hypovolemia.
- Tachycardia, pallor, dizziness, and falling BP can indicate deterioration.
- Quantify blood loss and activate the hemorrhage pathway early.



Sources: ACOG 2025

The 4 Ts of postpartum hemorrhage

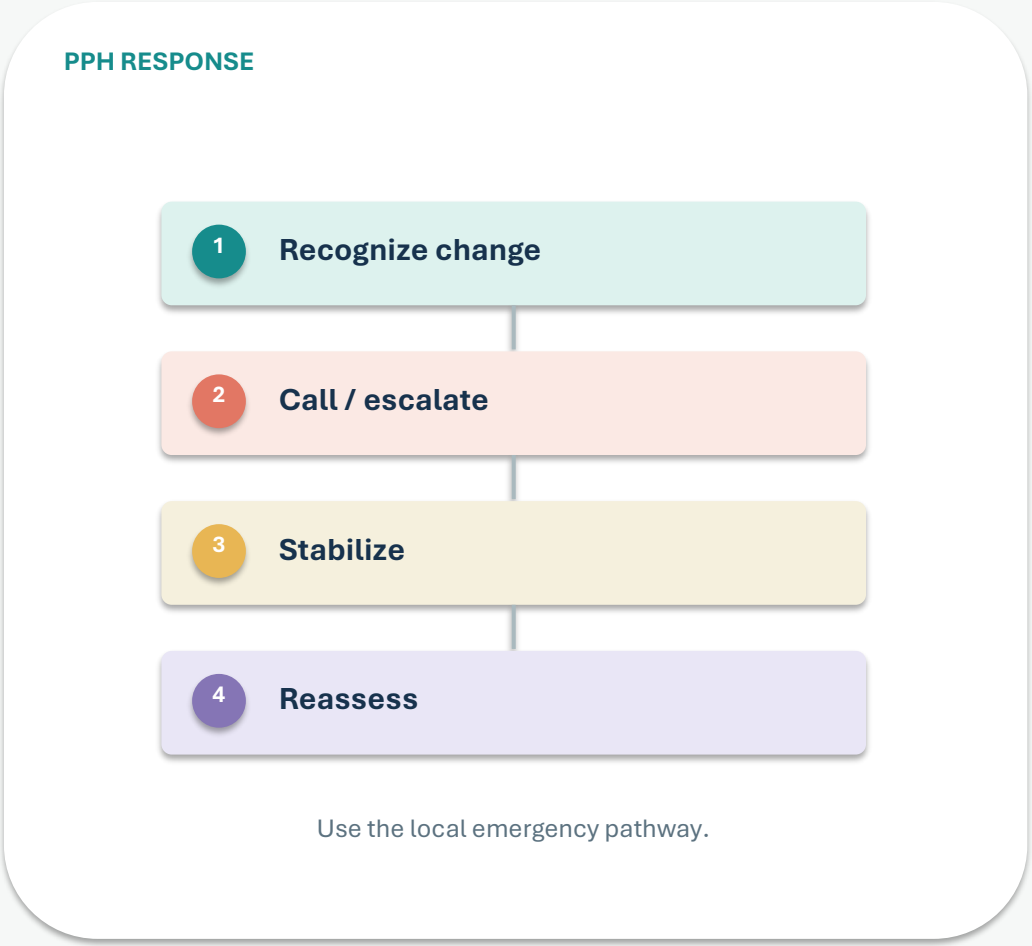
- Tone: uterine atony.
- Tissue: retained placenta or products; Trauma: laceration or rupture.
- Thrombin: coagulopathy; investigate and treat causes in parallel.



Sources: ACOG 2025

Postpartum hemorrhage response

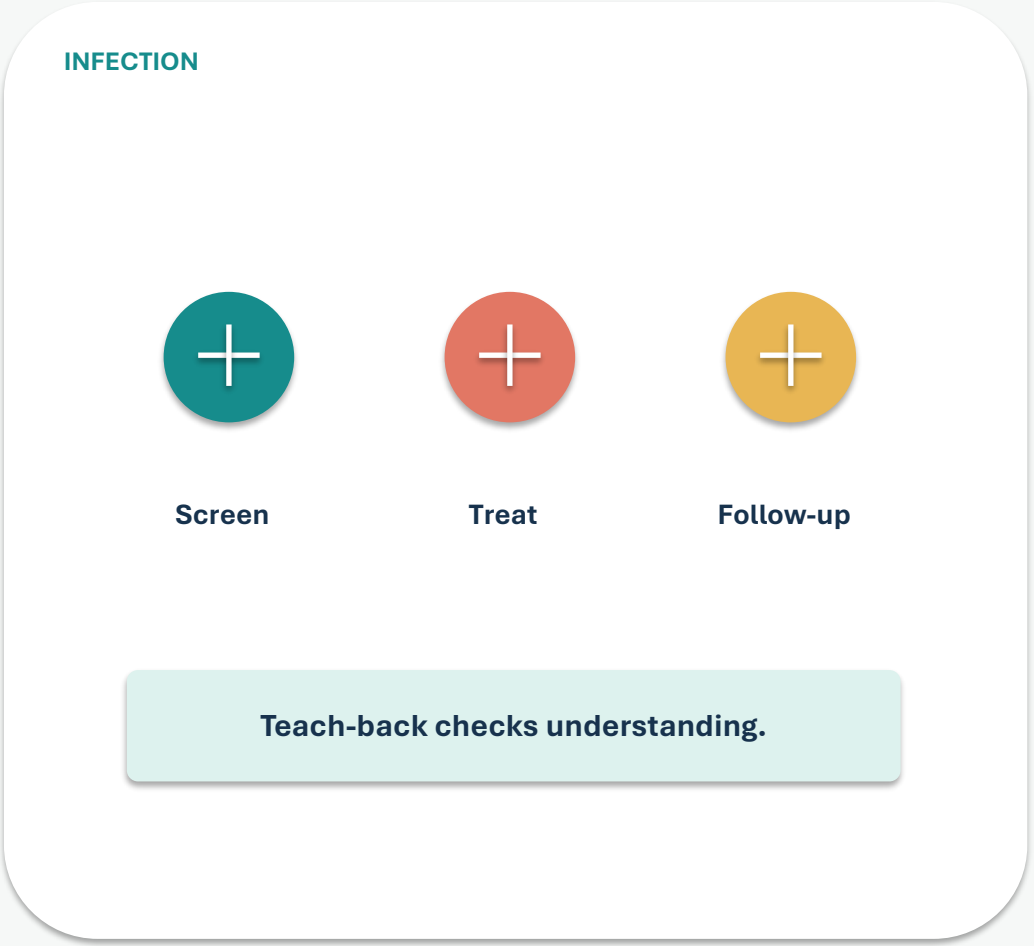
- Call the team, assess airway and circulation, and quantify bleeding.
- Massage a boggy uterus and give uterotonics under protocol.
- Prepare IV access, labs, blood products, and escalation as indicated.



Sources: ACOG 2025

Postpartum infection and sepsis

- Watch for fever, uterine tenderness, foul lochia, wound changes, or tachycardia.
- Sepsis can progress rapidly; activate local screening and treatment pathways.
- Obtain cultures and start indicated treatment without avoidable delay.



Sources: WHO/ACOG

Puerperium: A Recovery Timeline

- Puerperium commonly describes approximately the first 6 weeks after delivery.
- Immediate period: first 24 hours—close observation for bleeding and deterioration.
- Recovery continues over weeks; complications can occur after discharge.
- Postnatal care is individualized; NICE routine guidance extends to 8 weeks.

0–24 HOURS

DAYS 2–7

WEEKS 2–6+

Sources: WHO, NICE, MSD | Details and links in notes

Normal Changes: An Overview

- Uterus: contraction and progressive involution.
- Lochia: red → pink/brown → yellow-white, usually with decreasing flow.
- Fluid balance: increased urine output and sweating may occur.
- Breasts: milk production and feeding-related changes.
- Recovery varies; combine findings rather than using one observation alone.

UTERUS

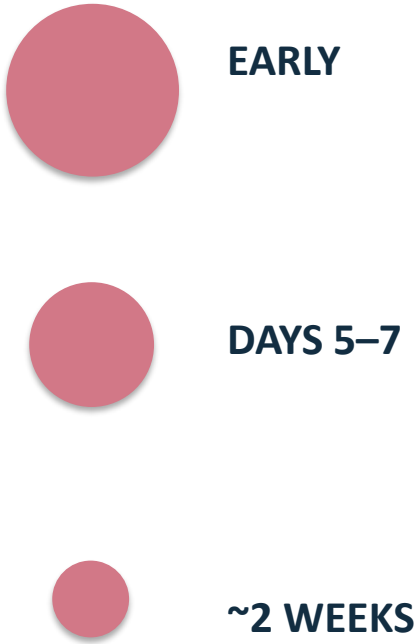
LOCHIA

WHOLE PERSON

Sources: MSD | Details and links in notes

Uterine Involution: Size & Position

- Assess the fundus gently with consent and appropriate support.
- Fundal height falls progressively; document its relation to the umbilicus and the trend.
- Around days 5–7: fundus is midway between umbilicus and symphysis.
- By about 2 weeks: uterus is no longer palpable abdominally.
- By approximately 4–6 weeks: approaches pre-pregnancy size.



Schematic—not anatomical scale

Sources: MSD | Details and links in notes

Fundal Assessment: Tone & Midline

- Assess firmness, position, tenderness and lochia together.

- A firm, central uterus is generally reassuring in the appropriate context.

- A boggy uterus with heavy bleeding suggests poor uterine contraction: urgent action.

- A full bladder can interfere with contraction; assess voiding and bladder fullness.

- Do not allow fundal assessment to delay emergency response.

FIRM?

MIDLINE?

BLEEDING?

Sources: WHO, MSD, ACOG | Details and links in notes

Lochia: What Is It?

- Postpartum vaginal discharge containing blood, mucus and uterine tissue.
- Occurs after vaginal birth and caesarean birth.
- Record type/colour, amount over time, clots and odour.
- Also assess uterine tone, pain, temperature and overall condition.
- Use pads to observe discharge; explain hygiene and warning signs.

COLOUR

AMOUNT

ODOUR / CLOTS

Sources: LOCHIA, MSD | Details and links in notes

Lochia Stages: Colour & Timing

- Rubra: dark or bright red; approximately days 1–3/4.

- Serosa: pinkish-brown, thinner and less bloody; approximately days 4–10/12.

- Alba: yellowish-white, light discharge; from around days 10–14, may continue to 6 weeks.

- Timing overlaps and differs between individuals: assess the trend and associated symptoms.



Sources: LOCHIA, MSD | Details and links in notes

Lochia Rubra: Early Days

- Predominantly bloody discharge with mucus and shed uterine tissue.
- Small clots may occur, but assess their size, frequency and associated bleeding.
- Amount should decrease rather than become progressively heavier.
- Persistent red lochia beyond the expected period or a return to heavy red bleeding needs review.

RED

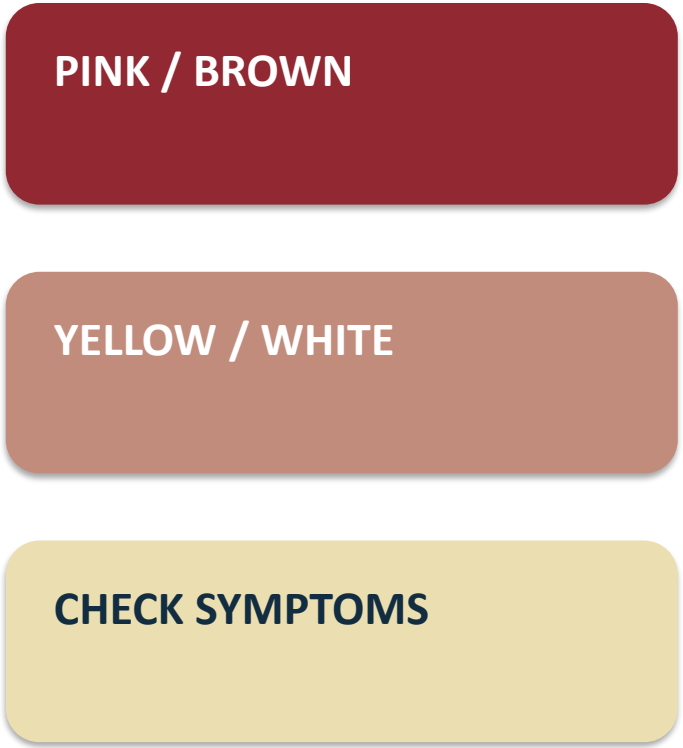
DAYS 1–3/4

DECREASING FLOW

Sources: LOCHIA, MSD | Details and links in notes

Lochia Serosa & Alba

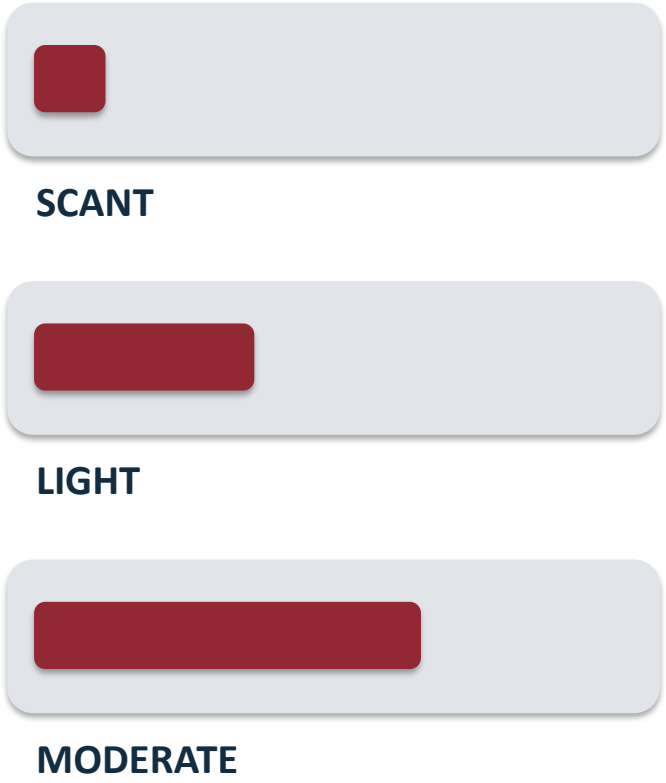
- Serosa: pink/brown discharge, less blood, usually fewer or no clots.
- Alba: yellow-white discharge with little/no blood; light flow or spotting.
- Do not confuse yellow-white alba with foul-smelling or green discharge.
- Offensive odour, fever, worsening pelvic pain or increased bleeding is abnormal.



Sources: LOCHIA | Details and links in notes

Lochia Amount: Pad-Stain Teaching Guide

- Scant: stain less than approximately 2.5 cm.
- Light: approximately 2.5 to less than 10 cm.
- Moderate: approximately 10 to less than 15 cm.
- These are approximate teaching categories, not accurate blood volumes.
- Record pad type and time interval; definitions vary by policy and pad absorbency.



Sources: PAD | Details and links in notes

Heavy Bleeding & Clot Size

- Soaking through one or more pads in 1 hour is an urgent maternal warning sign.
- Egg-sized or larger clots, passing tissue, dizziness or fainting require urgent assessment.
- Rapid bleeding, collapse or shock symptoms: activate the emergency response immediately.
- Do not wait for a pad threshold if the woman is deteriorating.
- Do not reassure solely because the fundus feels firm.

≥1 PAD / HOUR

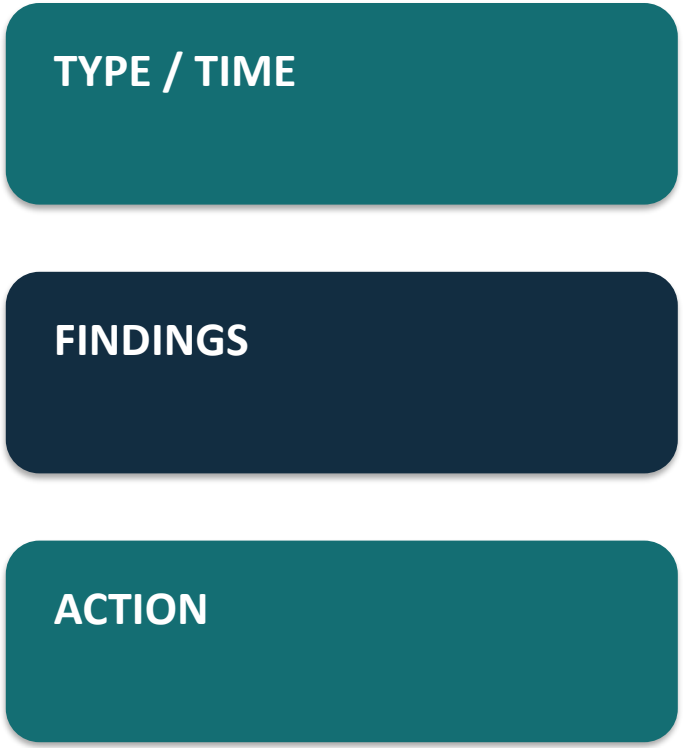
LARGE CLOTS

URGENT ACTION

Sources: CDC, ACOG | Details and links in notes

Documenting Lochia: Worked Example

- Example: “Day 2 postpartum; rubra; light stain over 1 hour on maternity pad.”
- Add odour, presence/absence and size of clots, fundal tone and symptoms.
- Record observations and escalation when findings are abnormal.
- Example abnormal entry: “Pad saturated in 40 minutes; dizzy; urgent obstetric review requested.”
- Documentation must describe the actual findings, not copy a template.



Sources: NICE, CDC, PAD | Details and links in notes

Temperature After Labour: Expected vs Fever

- Temperature can be slightly elevated during the first 24 hours.
- A temperature of 38°C (100.4°F) or higher is an urgent maternal warning sign.
- Do not dismiss fever as dehydration, milk production or normal recovery without assessment.
- Assess chills, uterine pain, discharge, wounds, urinary/breast symptoms and vital signs.
- Sepsis can occur without fever: worsening illness still needs urgent review.



Sources: MSD, CDC | Details and links in notes

Postpartum Fever: Look for the Source

- Uterine infection: fever, uterine/pelvic tenderness, offensive discharge.
- Urinary infection: dysuria, frequency, flank pain or systemic illness.
- Wound infection: increasing redness, pain, swelling, discharge or separation.
- Breast infection: localized breast pain/redness with systemic symptoms.
- Consider other causes with the clinical team; do not diagnose by timing alone.

UTERUS / URINE

WOUND / BREAST

SYSTEMIC SIGNS

Sources: MSD, NICE, ACOG | Details and links in notes

Other Vital Signs: Interpret in Context

- Pulse tends to decrease after birth; persistent tachycardia warrants assessment.
- Measure blood pressure and consider symptoms, history and trend.
- Assess respiratory rate, oxygenation when indicated, consciousness and perfusion.
- Pain, bleeding and infection can alter observations.
- Use the local maternal early-warning system and escalation thresholds.

PULSE / BP

BREATHING

TREND

Sources: WHO, MSD, NICE | Details and links in notes

Afterpains, Perineal Pain & Wounds

- Uterine cramps can accompany involution and breastfeeding.
- Assess pain location, severity, progression and response to analgesia.
- Inspect perineum or caesarean wound with consent.
- Increasing pain, offensive discharge, wound opening or fever needs review.
- Severe localized pain/pressure with little external bleeding may indicate hematoma.

LOCATION

SEVERITY

PROGRESSION

Sources: MSD, NICE | Details and links in notes

Bladder & Bowel Assessment

- Document passing urine within 6 hours after birth, as recommended by WHO.
- Assess inability to void, bladder distension, discomfort or incomplete emptying.
- Ask about dysuria, urinary leakage, constipation and bowel control.
- Support fluids, appropriate mobility, toileting and prescribed treatment.
- Persistent retention, incontinence or bowel-control problems require follow-up.

VOIDING

BOWEL

FOLLOW-UP

Sources: WHO, NICE | Details and links in notes

Breast Changes, Engorgement & Mastitis

- Ask about feeding preferences and provide individualized support.
- Assess fullness, pain, nipple trauma, attachment and effective feeding.
- Engorgement and mastitis are not the same: systemic illness requires assessment.
- Increasing localized redness/pain, fever or flu-like illness needs clinical review.
- Arrange skilled feeding support and prescribed treatment; reassess progress.

FEEDING

LOCAL SIGNS

SYSTEMIC SIGNS

Sources: NICE | Details and links in notes

Postpartum Hemorrhage: The 4 Ts

- Tone: inadequate uterine contraction.
- Trauma: tears, operative injury or hematoma.
- Tissue: retained placental tissue.
- Thrombin: coagulation disorder.
- Use this framework to support urgent team assessment; more than one cause may coexist.



Sources: ACOG, MSD | Details and links in notes

Hemorrhage: Immediate Nursing Actions

- Call the obstetric emergency team; stay with the woman.
- Assess airway, breathing, circulation and consciousness; monitor frequently.
- Quantify loss; prepare IV access, investigations and prescribed fluids/blood/medicines.
- Assess uterine tone; trained staff perform massage when indicated by protocol.
- Use closed-loop communication, document events and reassess continuously.



Sources: ACOG, MSD | Details and links in notes

Endometritis & Sepsis

- Suspect endometritis with fever, uterine tenderness and abnormal discharge.
- Deterioration, confusion, breathing difficulty, poor perfusion or low urine output is an emergency.
- Activate the local sepsis pathway and urgent medical review.
- Prepare cultures/investigations and prescribed treatment without avoidable delay.
- Monitor response; absence of a high temperature does not exclude serious infection.



Sources: ACOG, MSD | Details and links in notes

Postpartum Preeclampsia

- Can first develop after delivery, even without known antenatal hypertension.
- Warning symptoms: persistent severe headache, visual changes, upper abdominal pain or breathlessness.
- Blood pressure ≥160 systolic or ≥110 diastolic is severe: urgent assessment and escalation.
- Repeat/confirm observations promptly under protocol, but do not delay help when symptomatic.
- Prepare prescribed treatment; monitor neurological status, urine output and vital signs.

HEADACHE / VISION

SEVERE BP

URGENT REVIEW

Sources: ACOG, CDC | Details and links in notes

DVT & Pulmonary Embolism

- DVT: unilateral leg swelling, tenderness, warmth or pain.
- Possible PE: sudden breathlessness, chest pain, rapid heartbeat, collapse.
- Activate emergency assessment for chest pain, respiratory symptoms or collapse.
- Do not massage a suspected DVT or use Homan’s sign to exclude it.
- Support risk assessment, safe mobility and prescribed thromboprophylaxis.

LEG SIGNS

CHEST / BREATHING

ESCALATE

Sources: NICE, CDC | Details and links in notes

Mood, Depression & Postpartum Psychosis

- Ask about mood, anxiety, coping, sleep, support and safety.
- Persistent/worsening symptoms or impaired functioning require assessment and referral.
- Use validated screening within a clear follow-up pathway; screening is not diagnosis.
- Hallucinations, delusions, severe confusion or thoughts of harm need immediate emergency assessment.
- Stay with an unsafe patient and protect mother and infant while arranging urgent help.

LISTEN

ASSESS SAFETY

REFER

Sources: NICE, CDC | Details and links in notes

What Is Family Planning?

- Supports informed decisions about whether and when to have children.
- Includes contraception, pregnancy planning and referral for fertility concerns.
- Method choice must be voluntary and free of coercion.
- The most suitable option depends on health, preferences and access.

ASSESS

EXPLAIN

SUPPORT

Sources: CDC | Full links and teaching notes in slide notes

Pregnancy Prevention: Where Methods Act

- Some methods primarily prevent ovulation.

- Others interfere with sperm movement or prevent sperm reaching the egg.

- Barrier methods block sperm; fertility awareness avoids fertile-day exposure.

- Different mechanisms do not mean equal effectiveness.

OVULATION

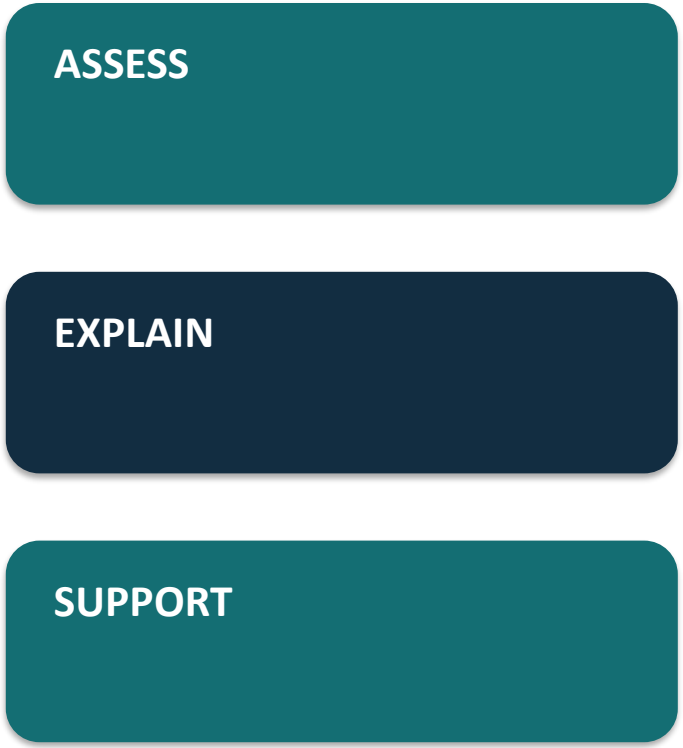
SPERM / MUCUS

BARRIER

Sources: CDC | Full links and teaching notes in slide notes

Method Groups

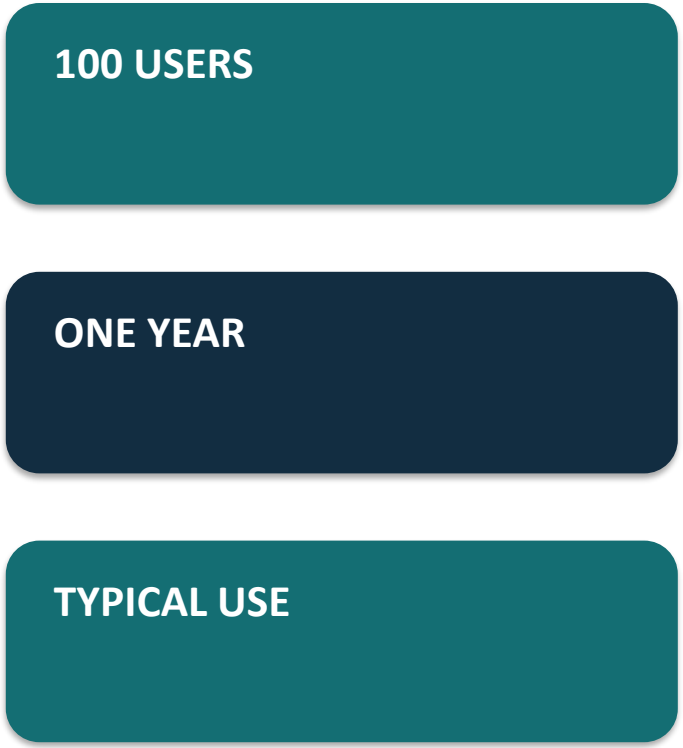
- Long-acting reversible: implant, hormonal IUD and copper IUD.
- Short-acting hormonal: pills, injection, patch and ring.
- Barrier: external/internal condoms and other barriers.
- Behaviour-based: fertility awareness and withdrawal; temporary LAM.
- Permanent contraception: tubal surgery and vasectomy.



Sources: CDC | Full links and teaching notes in slide notes

Effectiveness: Typical vs Perfect Use

- Typical use includes missed pills, delayed injections and inconsistent use.
- Perfect use means correct use every time.
- Compare pregnancies per 100 users in the first year—not per sexual act.
- A 7% typical-use failure rate means about 7 pregnancies per 100 users in one year.



Sources: CDC | Full links and teaching notes in slide notes

Dual Protection & STI Prevention

- Most contraceptive methods do not protect against sexually transmitted infections.
- Condoms reduce STI risk when used correctly and consistently.
- Consider condoms plus another method for pregnancy and STI prevention.
- Offer STI testing/referral based on history, risk and local guidance.

CONDOM

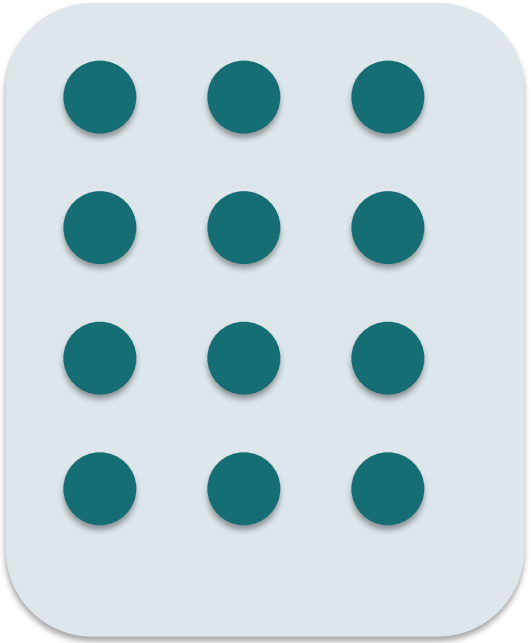
ANOTHER METHOD

DUAL PROTECTION

Sources: CDC | Full links and teaching notes in slide notes

Combined Pill: How It Works

- Contains estrogen and progestin; mainly prevents ovulation.
- Can improve cycle predictability and reduce menstrual symptoms for some users.
- Take according to the prescribed formulation and schedule.
- Check blood pressure and relevant medical history before starting.
- Typical-use failure: about 7 pregnancies per 100 users/year.



Illustrative blister pack
Follow exact product schedule

Sources: CHC, CDC | Full links and teaching notes in slide notes

Combined Methods: Safety Screening

- Estrogen-containing methods are unsuitable in some medical conditions.
- Key concerns include migraine with aura, severe hypertension and certain VTE histories.
- Smoking at age 35 or older requires specific eligibility assessment.
- Early postpartum use is restricted because of increased clot risk.
- Use a current eligibility table; a brief checklist is not sufficient for prescribing.

ASSESS

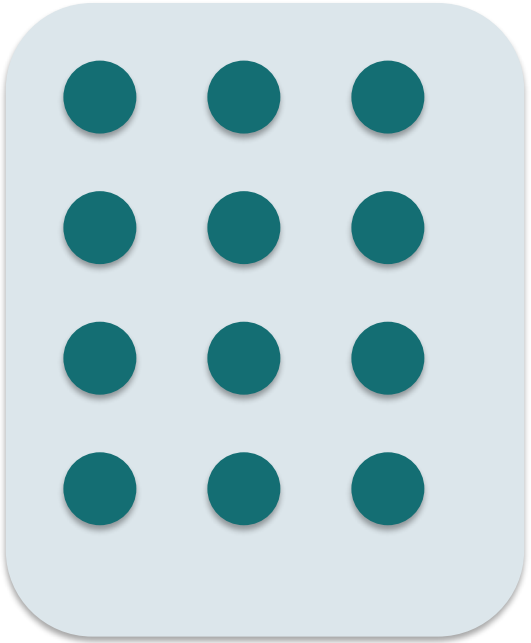
EXPLAIN

SUPPORT

Sources: CHC, WHO MEC, ACOG | Full links and teaching notes in slide notes

Progestin-Only Pills

- Contain no estrogen; formulations differ in how they work and are taken.
- Often useful when estrogen is unsuitable, but eligibility still needs assessment.
- Irregular bleeding or amenorrhea may occur.
- Timing windows and missed-pill instructions differ by product.
- Identify the exact pill before advising on missed doses or backup contraception.



Illustrative blister pack
Follow exact product schedule

Sources: CDC, WHO MEC | Full links and teaching notes in slide notes

Patch & Vaginal Ring

- Combined hormonal methods containing estrogen and progestin.
- Share important medical-eligibility concerns with the combined pill.
- Schedules vary by formulation; demonstrate the prescribed product instructions.
- Discuss delayed changes, detachment/expulsion and backup requirements.
- They do not protect against STIs.

ASSESS

EXPLAIN

SUPPORT

Sources: CHC, CDC | Full links and teaching notes in slide notes

Injectable Contraception

- Progestin injection; interval depends on product—DMPA is generally every 3 months.
- Discuss bleeding changes, amenorrhea and possible weight changes.
- Return to fertility can be delayed after the last DMPA injection.
- Explain repeat-visit timing and what to do if late.
- Assess eligibility, including bone-health considerations where relevant.

ASSESS

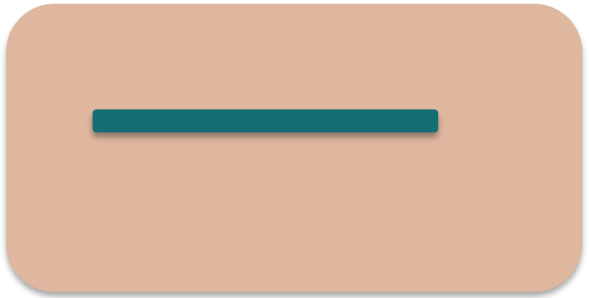
EXPLAIN

SUPPORT

Sources: CDC, WHO MEC | Full links and teaching notes in slide notes

Contraceptive Implant

- Small progestin-releasing rod placed under the upper-arm skin by a trained provider.
- Highly effective; duration depends on the product and approved local guidance.
- Irregular bleeding is common and should be discussed before insertion.
- Check insertion-site problems and arrange removal whenever requested.
- No STI protection; fertility generally returns rapidly after removal.

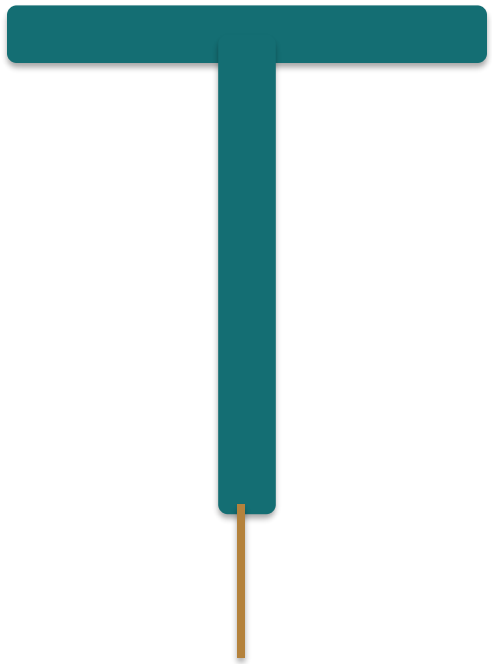


Upper-arm implant
Conceptual illustration

Sources: CDC | Full links and teaching notes in slide notes

Hormonal IUD

- Releases levonorgestrel in the uterus; duration depends on device.
- Very effective without daily action.
- Early irregular bleeding may be followed by lighter bleeding or amenorrhea.
- Insertion/removal requires a trained provider and informed consent.
- Assess new pelvic pain, fever or pregnancy symptoms promptly.

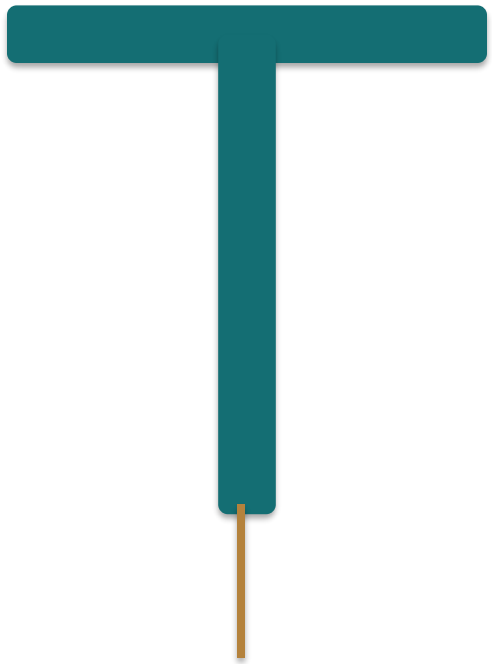


T-shaped device
Schematic—not to scale

Sources: CDC | Full links and teaching notes in slide notes

Copper IUD

- Hormone-free, highly effective and long acting; duration varies by device.
- Copper impairs sperm function and prevents fertilization.
- May increase menstrual bleeding/cramps, particularly initially.
- Can also provide emergency contraception when medically eligible.
- It does not protect against STIs.

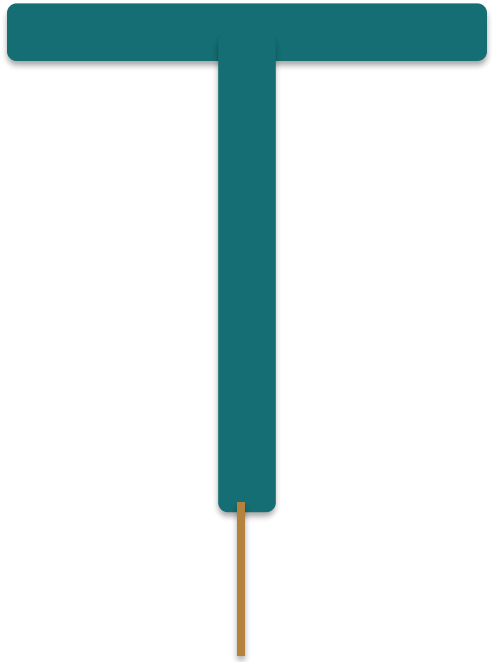


T-shaped device
Schematic—not to scale

Sources: CDC, WHO EC | Full links and teaching notes in slide notes

IUD Safety & Follow-Up

- Assess pregnancy possibility, infection symptoms and medical eligibility.
- Current pelvic infection or postpartum sepsis can preclude insertion.
- Discuss insertion discomfort, expulsion and rare perforation.
- Severe pelvic pain, fever, heavy bleeding or pregnancy symptoms need review.
- Postpartum insertion timing affects expulsion risk; use a trained service and local protocol.



T-shaped device
Schematic—not to scale

Sources: WHO MEC, WHO EC, ACOG | Full links and teaching notes in slide notes

Fertility Awareness & Withdrawal

- Fertility awareness uses cycle-related signs to identify fertile days.
- Requires training, consistent observations and abstinence/barrier use during fertile days.
- Effectiveness varies widely by method and correct use.
- Withdrawal is user-dependent and does not protect against STIs.
- Irregular cycles and postpartum changes can complicate fertility awareness.

ASSESS

EXPLAIN

SUPPORT

Sources: CDC | Full links and teaching notes in slide notes

LAM: All Three Criteria Are Required

- Lactational amenorrhea method is a temporary postpartum method.
- 1. Baby is younger than 6 months.
- 2. Menstruation has not returned.
- 3. Fully or nearly fully breastfeeding.
- If any criterion is no longer met, start another method promptly.

BABY <6 MONTHS

AMENORRHEA

FULL / NEAR-FULL FEEDING

Sources: CDC, ACOG | Full links and teaching notes in slide notes

Permanent Contraception

- Tubal surgery and vasectomy are intended to be permanent.
- Require voluntary informed consent and discussion of reversible alternatives.
- Vasectomy is not immediately effective: use backup until clearance under the semen-testing protocol.
- Neither method protects against STIs.
- Do not promise successful reversal.

ASSESS

EXPLAIN

SUPPORT

Sources: CDC | Full links and teaching notes in slide notes

Emergency Contraception: When & Options

- Offer after unprotected intercourse, contraceptive failure or sexual assault.
- Copper IUD: within 5 days (120 hours), when medically eligible; most effective option.
- Ulipristal acetate: up to 120 hours.
- Levonorgestrel: use as early as possible; local labels often specify 72 hours, while WHO supports use within 120 hours with reduced later effectiveness.
- Do not delay access; refer promptly and follow local protocols.

ACT EARLY

PILLS / COPPER IUD

FOLLOW-UP

Sources: WHO EC, NHS EC | Full links and teaching notes in slide notes

After Emergency Contraception

- Emergency pills delay/prevent ovulation; they do not end an established pregnancy.
- After levonorgestrel, regular contraception may be started immediately.
- After ulipristal, start progestin-containing contraception on day 6.
- Use condoms/abstinence as instructed until the ongoing method is effective.
- Explain pregnancy testing and follow-up; severe pain or pregnancy concerns need review.

ASSESS

EXPLAIN

SUPPORT

Postpartum & Breastfeeding Choices

- Ovulation may precede the first menstrual period.

- Discuss a contraceptive plan before discharge and revisit preferences later.

- IUDs, implants and progestin-only methods may be options; eligibility and timing differ.

- Combined hormonal contraception must not be used in the first 21 days under CDC guidance.

- Later estrogen eligibility depends on breastfeeding and VTE risk; check the specific guideline.

ASSESS

EXPLAIN

SUPPORT

Sources: ACOG, CHC, WHO MEC | Full links and teaching notes in slide notes

Medical Eligibility Categories

- MEC 1: no restriction for use.
- MEC 2: advantages generally outweigh theoretical/proven risks.
- MEC 3: risks generally outweigh advantages—usually choose an alternative.
- MEC 4: unacceptable health risk; do not use the method.
- Categories are condition- and method-specific; check current guidance.

1 / 2: GENERALLY USE

3: USUALLY AVOID

4: DO NOT USE

Sources: WHO MEC | Full links and teaching notes in slide notes

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